

Kube-OVN Document

v1.16.0

Kube-OVN Team

2025 Kube-OVN Team

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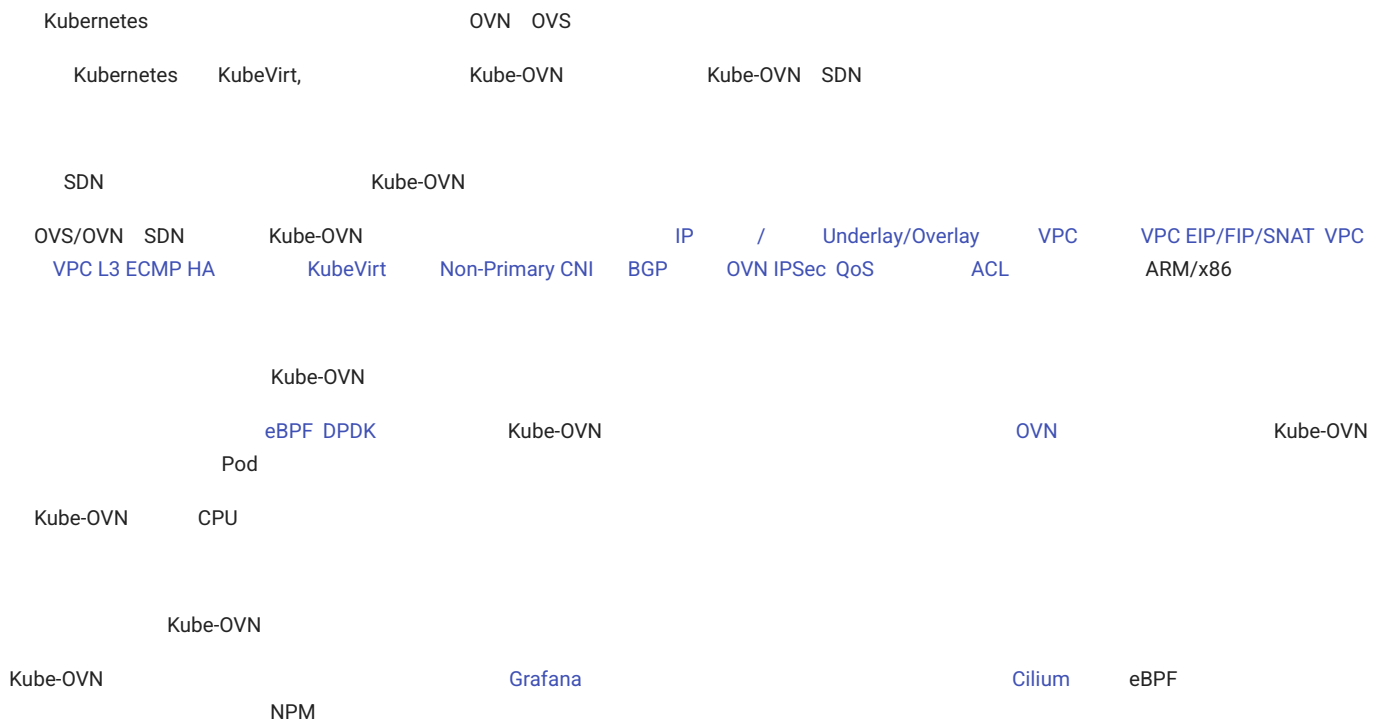
1. Kube-OVN



1.1 What is Kube-OVN?



1.2 Why Kube-OVN?



1.3 CNI

Kubernetes	CNI		
1.			
2.			
Kube-OVN	CNI	Kube-OVN	Kube-OVN

1.3.1 eBPF

[Cilium](#) [Calico eBPF](#)
[Kube-OVN](#) [Open vSwitch](#)

1.3.2 CNI, Ingress, Service Mesh Observability All in One

Cilium
Kube-OVN CNI

1.3.3 OpenShift

OpenShift CNI [Cluster Network Operator](#) Kube-OVN RedHat Kubernetes

1.3.4 Kubernetes EKS/AKS/GKE

Kubernetes	CNI
------------	-----

1.3.5 AI

Hostnetwork [host-device](#)

AI

1.4 OVN/ovn-kubernetes/Kube-OVN

1.4.1 OVN

OVN	Open vSwitch	OpenStack, Kubernetes	CMS	ovn-kubernetes	Kube-OVN	OVN
Kubernetes						

1.4.2 ovn-kubernetes

ovn-kubernetes Acceleration	OVN	OVN	Kubernetes	CNI	OpenShift	OpenShift	UDN, Multihoming, Hardware
--------------------------------	-----	-----	------------	-----	-----------	-----------	----------------------------

1.4.3 Kube-OVN

Kube-OVN	IP	Namespace	OVN	ovn-kubernetes	annotation	Pod	join
		Underlay VPC KubeVirt					

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1.5

2.

2.1

Kube-OVN CNI Kubernetes

2.1.1

- Kubernetes >= 1.29
- Docker >= 1.12.6, Containerd >= 1.3.4
- : CentOS 7/8, Ubuntu 16.04/18.04/20.04
- Linux `geneve`, `openvswitch`, `ip_tables` `iptables_nat` Kube-OVN

1. 3.10.0-862 `netfilter` bug Kube-OVN CentOS bug [Floating IPs broken after kernel upgrade to Centos/RHEL 7.5 - DNAT not working](#)
2. Rocky Linux 8.6 4.18.0-372.9.1.el8.x86_64 TCP [TCP connection failed in Rocky Linux 8.6](#) 4.18.0-372.13.1.el8_6.x86_64
3. 4.4 `openvswitch` `openvswitch`
4. Geneve IPv6 `cat /proc/cmdline` bug [Geneve tunnels don't work when ipv6 is disabled](#)

2.1.2

- IPv6 `ipv6.disable=1` 0
- `kube-proxy` Kube-OVN Service ClusterIP `kube-apiserver`
- `kubelet` CNI , `kubelet` `--network-plugin=cni --cni-bin-dir=/opt/cni/bin --cni-conf-dir=/etc/cni/net.d`
- `/etc/cni/net.d/`

2.1.3

ovn-central	6641/tcp	ovn nb db server
ovn-central	6642/tcp	ovn sb db server
ovn-central	6643/tcp	ovn northd server
ovn-central	6644/tcp	ovn raft server
ovn-ic	6645/tcp	ovn ic nb db server
ovn-ic	6646/tcp	ovn ic sb db server
ovs-ovn	Geneve 6081/udp, STT 7471/tcp, Vxlan 4789/udp	
kube-ovn-controller	10660/tcp	
kube-ovn-daemon	10665/tcp	
kube-ovn-monitor	10661/tcp	

firewalld Packet Forwarding Masquerade

```
# Packet Forwarding
firewall-cmd --add-forward --permanent
```

```
# IPv4 Masquerade
firewall-cmd --add-masquerade --permanent
# Kube-OVN IPv6/      Masquerade
firewall-cmd --permanent --add-rich-rule 'rule family="ipv6" source address="fd00:10:16::/112" masquerade'

firewall-cmd --reload
```

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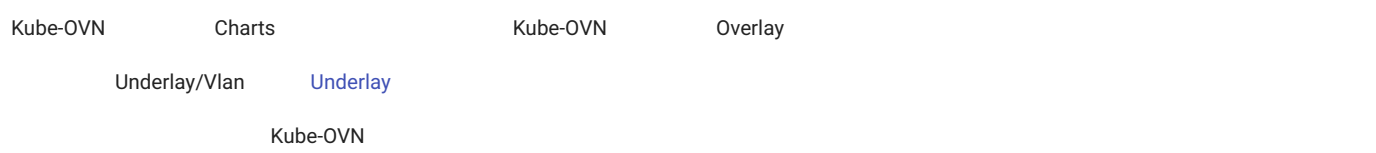


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2.1.4

2.2



2.2.1

release

wget https://raw.githubusercontent.com/kubeovn/kube-ovn/refs/tags/v1.16.5/dist/images/install.sh

master

wget https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/images/install.sh

```
REGISTRY="docker.io/kubeovn"      #
VERSION="v1.16.5"                # /Tag
POD_CIDR="10.16.0.0/16"           # CIDR      SVC/NODE/JOIN CIDR
SVC_CIDR="10.96.0.0/12"           # apiserver service-cluster-ip-range
JOIN_CIDR="100.64.0.0/16"         # Pod      CIDR      SVC/NODE/POD CIDR
LABEL="node-role.kubernetes.io/control-plane" # OVN DB
IFACE=""                          #          Kubernetes  Node IP
TUNNEL_TYPE="geneve"             #          geneve, vxlan stt stt      ovs
```

IFACE=enp6s0f0,eth.*

root

bash install.sh

- Kube-OVN
1.

[Step 4/6]

Pod
2.

Kube-OVN

2.2.2 Helm Chart

Kube-OVN

Helm

Kube-OVN

IP

```
# kubectl get node -o wide
NAME                STATUS    ROLES    AGE   VERSION   INTERNAL-IP   EXTERNAL-IP   OS-IMAGE             KERNEL-VERSION   CONTAINER-RUNTIME
kube-ovn-control-plane NotReady control-plane 20h   v1.26.0   172.18.0.3    <none>        Ubuntu 22.04.1 LTS   5.10.104-linuxkit containerd://1.6.9
kube-ovn-worker      NotReady <none>      20h   v1.26.0   172.18.0.2    <none>        Ubuntu 22.04.1 LTS   5.10.104-linuxkit containerd://1.6.9
```

label

```
# kubectl label node -lbeta.kubernetes.io/os=linux kubernetes.io/os=linux --overwrite
node/kube-ovn-control-plane not labeled
node/kube-ovn-worker not labeled

# kubectl label node -lnode-role.kubernetes.io/control-plane kube-ovn/role=master --overwrite
node/kube-ovn-control-plane labeled

# label dpdk dpdk
# kubectl label node -lovn.kubernetes.io/ovs_dp_type!=userspace ovn.kubernetes.io/ovs_dp_type=kernel --overwrite
node/kube-ovn-control-plane labeled
node/kube-ovn-worker labeled
```

Helm Repo

```
# helm repo add kubeovn https://kubeovn.github.io/kube-ovn/
"kubeovn" has been added to your repositories

# helm repo list
NAME      URL
kubeovn   https://kubeovn.github.io/kube-ovn/

# helm repo update kubeovn
Hang tight while we grab the latest from your chart repositories...
...Successfully got an update from the "kubeovn" chart repository
Update Complete. *Happy Helming!*
```

NAME	CHART VERSION	APP VERSION	DESCRIPTION
kubeovn/kube-ovn	v1.16.5	v1.16.5	Helm chart for Kube-OVN

helm install Kube-OVN

Chart values.yaml

```
# helm install kube-ovn kubeovn/kube-ovn --wait -n kube-system --version v1.16.5
NAME: kube-ovn
LAST DEPLOYED: Thu Apr 24 08:30:13 2025
NAMESPACE: kube-system
STATUS: deployed
REVISION: 1
TEST SUITE: None
```

Helm values.yaml

```
helm upgrade -f values.yaml kube-ovn kubeovn/kube-ovn --wait -n kube-system --version v1.16.5
```

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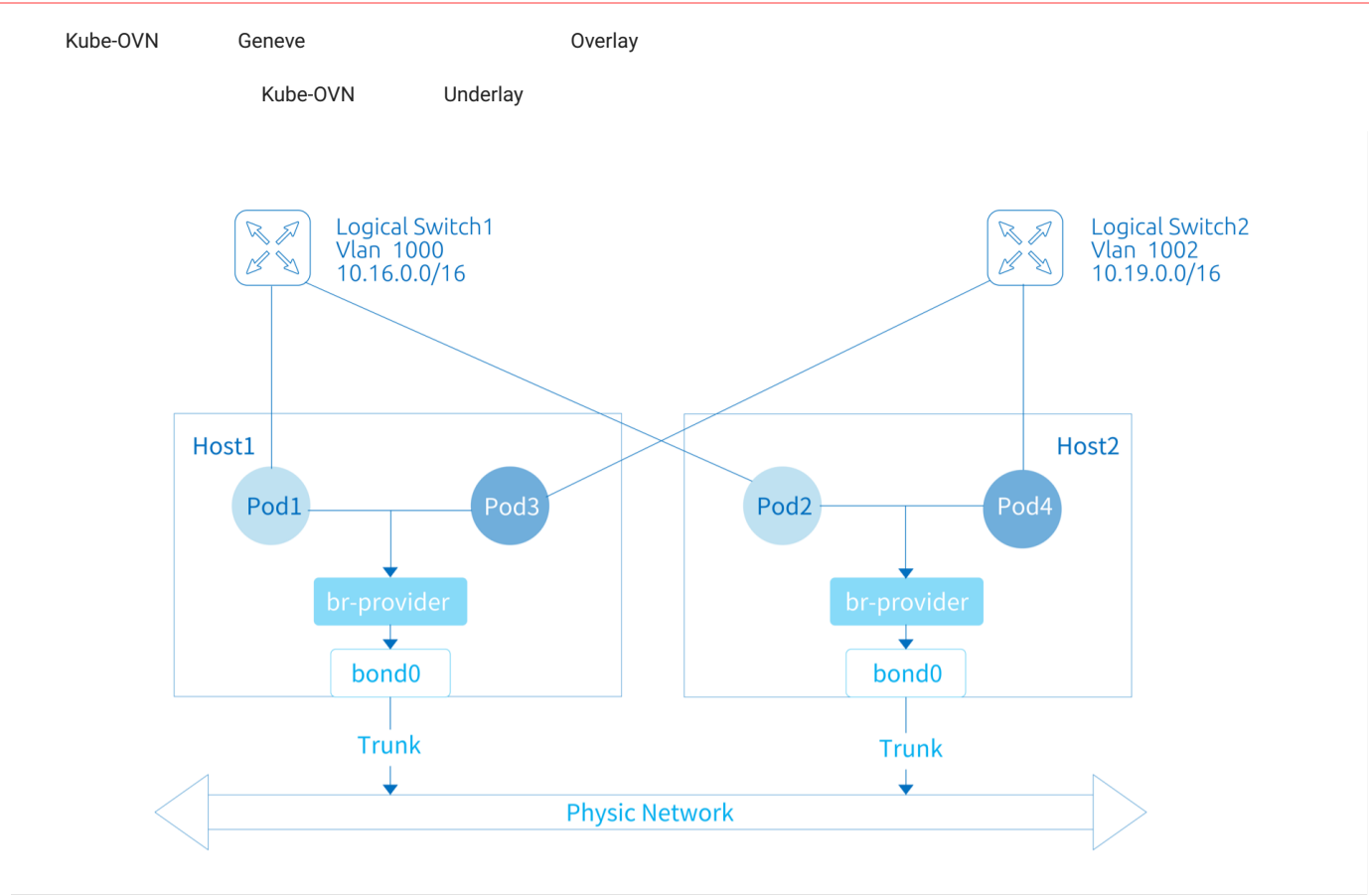






GitHub

2.3 Underlay



2.3.1

Overlay	SNAT/EIP	/	L3	VPC	Underlay
---------	----------	---	----	-----	----------

2.3.2 Macvlan

Kube-OVN	Underlay	Macvlan							
1.	Macvlan	OVS	Macvlan						
2.	Kube-OVN	arp-proxy	arp						
3.	Macvlan	netfilter	Service	NetworkPolicy	Kube-OVN	OVS	Service	NetworkPolicy	
4.	Kube-OVN	Underlay	Macvlan	IP	QoS				

2.3.3

Underlay	OVS	OVS	L2/L3		Vlan			
1. OpenStack VM		PortSecurity						
2. VMware vSwitch		MAC Address Changes, Forged Transmits	Promiscuous Mode Operation		allow			
3. VMware NSX-T		Underlay						
4. Hyper-V		MAC Address Spoofing						
5. AWS GCE		Mac	Underlay	Underlay	VPC-CNI			
6. Linux Bridge								
	Kube-OVN Underlay			Underlay				
	Underlay							
	Kube-OVN	Mac	IP	MTU	OVS Bridge	Underlay	OVS Bridge	br-PROVIDER_NAME
PROVIDER_NAME	ProviderNetwork		provider					
1. OVS Bridge		NetworkManager			DHCP			
2. OVS Bridge		OVS	Kube-OVN		OVS			
	Underlay							

2.3.4

Underlay

Pod

Underlay

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/dist/images/install.sh
```

```
ENABLE_ARP_DETECT_IP_CONFLICT #          vlan  arp
NETWORK_TYPE                  #          vlan
VLAN_INTERFACE_NAME           #          eth1
VLAN_ID                        #          VLAN Tag  0      VLAN
POD_CIDR                      #          CIDR    192.168.1.0/24
POD_GATEWAY                    #          192.168.1.1
EXCLUDE_IPS                    #          IP      192.168.1.1..192.168.1.100
ENABLE_LB                      #  Underlay  Service true
EXCHANGE_LINK_NAME             #  provider-network OVS      false
LS_DNAT_MOD_DL_DST             # DNAT      MAC      Service true
```

```
bash install.sh
```

2.3.5 CRD Underlay

Underlay

Pod

ProviderNetwork

Vlan

Subnet

ProviderNetwork

ProviderNetwork

ProviderNetwork :

```
apiVersion: kubeovn.io/v1
kind: ProviderNetwork
metadata:
  name: net1
```

```
spec:
  defaultInterface: eth1
  customInterfaces:
    - interface: eth2
      nodes:
        - node1
  nodeSelector:
    matchLabels:
      kubernetes.io/arch: amd64
      network-type: underlay
  matchExpressions:
    - key: kubernetes.io/hostname
      operator: In
      values:
        - node1
        - node2
```

ProviderNetwork 12

- defaultInterface: Bond Vlan ProviderNetwork excludeNodes br-net1 br-NAME OVS
- customInterfaces:
- nodeSelector: OVS matchLabels matchExpressions
- excludeNodes: net1.provider-network.ovn.kubernetes.io/exclude=true nodeSelector
- autoCreateVlanSubinterfaces: false true VLAN <interface>.<vlanid>

Key	Value
net1.provider-network.ovn.kubernetes.io/ready	true ProviderNetwork
net1.provider-network.ovn.kubernetes.io/interface	eth1
net1.provider-network.ovn.kubernetes.io/mtu	1500 MTU

IP IP OVS

VLAN

ProviderNetwork autoCreateVlanSubinterfaces: true VLAN VLAN

```
apiVersion: kubeovn.io/v1
kind: ProviderNetwork
metadata:
  name: vlan-provider
spec:
  defaultInterface: eth0.341
  autoCreateVlanSubinterfaces: true
  customInterfaces:
    - interface: eth2.341
      nodes:
        - worker-node-1
        - worker-node-2
```

<interface>.<vlanid> VLAN VLAN ProviderNetwork

VLAN

Vlan Vlan Tag ProviderNetwork

VLAN

```
apiVersion: kubeovn.io/v1
kind: Vlan
metadata:
```

```
name: vlan1
spec:
  id: 0
  provider: net1
```

- id: VLAN ID/Tag Kube-OVN Vlan Vlan 0 vlan localnet
- provider: ProviderNetwork VLAN ProviderNetwork

Subnet

Vlan

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: subnet1
spec:
  protocol: IPv4
  cidrBlock: 172.17.0.0/16
  gateway: 172.17.0.1
  vlan: vlan1
  disableGatewayCheck: false
```

- vlan VLAN Subnet VLAN
- disableGatewayCheck Underlay true

2.3.6

IP

IP Pod IP Mac

2.3.7

Kube-OVN Underlay spec.logicalGateway true

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: subnet1
spec:
  protocol: IPv4
  cidrBlock: 172.17.0.0/16
  gateway: 172.17.0.1
  vlan: vlan1
  logicalGateway: true
```

Pod Kube-OVN Logical Router

2.3.8 Underlay Overlay

Underlay Overlay Overlay Pod NAT Underlay Pod IP Underlay Pod Overlay
 Overlay Underlay Pod Pod IP Overlay Pod
Underlay Overlay u2oInterconnection true Kube-OVN Underlay IP Underlay ovn-cluster
 Kube-OVN Underlay Overlay

Warning

Vlan Underlay Underlay u2o

IP

subnet IP Underlay Subnet u2oInterconnectionIP

Underlay Subnet	VPC						
Underlay Subnet	VPC	Overlay Subnet	VPC	u2oInterconnection	true	subnet.spec.vpc	VPC

2.3.9

IP	Netplan	Ubuntu	Netplan	renderer	NetworkManager	IP	DHCP
<pre>network: renderer: NetworkManager ethernet: eth0: dhcp4: no addresses: - 172.16.143.129/24 version: 2</pre>							
IP	netplan						
<pre>netplan generate nmcli connection reload netplan-eth0 nmcli device set eth0 managed yes</pre>							
Kube-OVN	IP	OVS					
NetworkManager	CentOS						
<pre>nmcli connection reload eth0 nmcli device set eth0 managed yes nmcli -t -f GENERAL.STATE device show eth0 grep -qw unmanaged nmcli device reapply eth0</pre>							
IP	MAC						

2.3.10

hairpin Pod				
hairpin	Pod	Pod	OVS	MAC
hairpin	Kube-OVN			
Pod	Pod			
Pod	300	ARP	OVS	resubmit
<pre>2022-11-13T08:43:46.782Z 00222 ofproto_dpif_upcall(handler5) WARN Flow: arp,in_port=331,vlan_tci=0x0000,d1_src=00:00:00:25:eb:39,d1_dst=ff:ff:ff:ff:ff:ff,arp_spa=10.213.131.240,arp_tpa=10.213.159.254,arp_op=1,arp_sha=00:00:00:25:eb:39,arp_tha=ff:ff:ff:ff:ff:ff bridge("br-int") ----- 0. No match. >>>> received packet on unknown port 331 <<<< drop Final flow: unchanged MegafLOW: recirc_id=0,eth,arp,in_port=331,d1_src=00:00:00:25:eb:39 Datapath actions: drop 2022-11-13T08:44:34.077Z 00224 ofproto_dpif_xlate(handler5) WARN over 4096 resubmit actions on bridge br-int while processing arp,in_port=13483,vlan_tci=0x0000,d1_src=00:00:00:59:ef:13,d1_dst=ff:ff:ff:ff:ff:ff,arp_spa=10.213.152.3,arp_tpa=10.213.159.254,arp_op=1,arp_sha=00:00:00:59:ef:13,arp_tha=ff:ff:ff:ff:ff:ff</pre>				
OVN NB	bcast_arp_req_flood	false		
<pre>kubectl ko nbctl set NB_Global . options:bcast_arp_req_flood=false</pre>				
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2.3.11

2.4 CNI

Kube-OVN CNI CNI Cilium Calico Flannel

2.4.1

- CNI Kube-OVN CNI Pod CNI eth0
- CNI Kube-OVN VPC
 -
 - Pod
 - Kube-OVN VPC NAT

2.4.2

1. CNI Kubernetes CNI Cilium Calico Flannel
2. Multus CNI
3. Kube-OVN

1. CNI
2. Multus CNI
3. Kube-OVN

2.4.3

Helm Chart v2

```
# values.yaml
cni:
  nonPrimaryCNI: true
```

Helm

```
helm install kube-ovn ./charts/kube-ovn-v2 \
--namespace kube-system \
--set cni.nonPrimaryCNI=true
```

Helm Chart v1

```
# values.yaml
cni_conf:
  NON_PRIMARY_CNI: true
```

Helm

```
helm install kube-ovn ./charts/kube-ovn \
--namespace kube-system \
--set cni_conf.NON_PRIMARY_CNI=true
```

kube-ovn-controller Deployment kube-ovn-cni DaemonSet Pod

```
# kube-ovn-controller Deployment
containers:
- name: kube-ovn-controller
  args:
  - --non-primary-cni-mode=true
```

```
# kube-ovn-cni DaemonSet
containers:
- name: kube-ovn-cni
  args:
  - --non-primary-cni-mode=true
```

2.4.4

NADs

NAD

```
apiVersion: k8s.cni.cncf.io/v1
kind: NetworkAttachmentDefinition
metadata:
  name: kube-ovn-vpc-network
  namespace: default
spec:
  config: |
    {
      "cniVersion": "0.3.1",
      "type": "kube-ovn",
      "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
      "provider": "kube-ovn-vpc-network.default.ovn"
    }
```

VPC

VPC

```
apiVersion: kubeovn.io/v1
kind: Vpc
metadata:
  name: vpc-secondary
spec:
  namespaces:
  - default
---
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: subnet-secondary
spec:
  vpc: vpc-secondary
  cidr: "10.100.0.0/16"
  gateway: "10.100.0.1"
  provider: kube-ovn-vpc-network.default.ovn
```

Pod

Pod

```
apiVersion: v1
kind: Pod
metadata:
  name: multi-network-pod
  annotations:
    k8s.v1.cni.cncf.io/networks: default/kube-ovn-vpc-network
spec:
  containers:
  - name: app
    image: nginx
```

2.4.5

eth0

- CNI Cilium Calico
-

- Kubernetes

net1 net2...

- Kube-OVN
-
- VPC
- Kube-OVN QoS NAT

2.4.6

1.

```
# Pod
apiVersion: v1
kind: Pod
metadata:
  name: frontend
  annotations:
    k8s.v1.cni.cncf.io/networks: |
    [
      {"name": "default/public-network"},
      {"name": "default/internal-network"}
    ]
```

2.

```
# A Pod
apiVersion: v1
kind: Pod
metadata:
  name: tenant-a-app
  annotations:
    k8s.v1.cni.cncf.io/networks: default/tenant-a-network

---
# B Pod
apiVersion: v1
kind: Pod
metadata:
  name: tenant-b-app
  annotations:
    k8s.v1.cni.cncf.io/networks: default/tenant-b-network
```

3. VPC NAT

```
apiVersion: kubeovn.io/v1
kind: VpcNatGateway
metadata:
  name: vpc-nat-gw
spec:
  vpc: vpc-secondary
  subnet: subnet-secondary
  lanIp: "10.100.0.254"
```

2.4.7

Pod

```
metadata:
  annotations:
    k8s.v1.cni.cncf.io/networks: |
    [
```



```
{ "name": "default/kube-ovn-network-1", "interface": "net1"},  
  { "name": "default/kube-ovn-network-2", "interface": "net2"}  
]
```

IP

IP

```
# IP  
metadata:  
  annotations:  
    k8s.v1.cni.cncf.io/networks: |  
    [{  
      "name": "default/kube-ovn-network",  
      "ips": ["10.100.0.100/16"]  
    }]
```

QoS

QoS

```
metadata:  
  annotations:  
    ovn.kubernetes.io/ingress_rate: "1000"  
    ovn.kubernetes.io/egress_rate: "1000"
```

2.4.8

- 1. **Pod** - NAD - kube-ovn-cni - Multus
- 2. - VPC - Pod -
- 3. **IP** - CIDR - IP - IPAM

```
# Pod  
kubectl get pod <pod-name> -o yaml | grep -A 10 "networks-status"  
  
# Pod  
kubectl exec <pod-name> -- ip addr show  
  
#  
kubectl exec <pod-name> -- ip route show  
  
# kube-ovn-cni  
kubectl logs -n kube-system daemonset/kube-ovn-cni
```

2.4.9

- 1. Kubernetes
- 2.
- 3. **DNS** Pod DNS

2.4.10

- 1. IP
- 2.
- 3.
- 4.
- 5.

2.4.11

- Multus CNI
-
- Kube-OVN

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2.4.12

2.5 Talos

Talos Linux Kubernetes Linux

2.5.1 Helm Chart Kube-OVN

Talos Linux Kube-OVN

```
helm install kube-ovn kubeovn/kube-ovn --wait \
-n kube-system \
--version v1.16.5 \
--set OVN_DIR=/var/lib/ovn \
--set OPENVSWITCH_DIR=/var/lib/openvswitch \
--set DISABLE_MODULES_MANAGEMENT=true \
--set cni_conf.MOUNT_LOCAL_BIN_DIR=false
```

Underlay Helm Chart

```
helm install kubeovn kubeovn/kube-ovn --wait \
-n kube-system \
--version v1.16.5 \
--set OVN_DIR=/var/lib/ovn \
--set OPENVSWITCH_DIR=/var/lib/openvswitch \
--set DISABLE_MODULES_MANAGEMENT=true \
--set cni_conf.MOUNT_LOCAL_BIN_DIR=false \
--set networking.NETWORK_TYPE=vlan \
--set networking.vlan.VLAN_INTERFACE_NAME=enp0s5f1 \
--set networking.vlan.VLAN_ID=0 \
--set networking.NET_STACK=ipv4 \
--set-json networking.EXCLUDE_IPS="'172.99.99.11..172.99.99.99'" \
--set-json ipv4.POD_CIDR="'172.99.99.0/24'" \
--set-json ipv4.POD_GATEWAY="'172.99.99.1'"
```

 Note

VLAN Bond Bridge Underlay Underlay Talos ignore=true

```
machine:
  network:
    interfaces:
      - interface: enp0s5f1
        ignore: true
```

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2.5.2

2.6

Kube-OVN	pre-v0.0.1	Kube-OVN
----------	------------	----------

2.6.1

- [Linkerd](#)
- [Elasticsearch](#)
- [EMQX](#)
- [KubeSphere](#)

2.6.2 OpenVswitch/OVN

Kube-OVN	OpenVswitch	OVN
• OVN • OpenVswitch		
ovn-architecture		

2.6.3 Kube-OVN

Kube-OVN		
Kube-OVN	Kube-OVN	kubectl bash

2.6.4

Kube-OVN	Kube-OVN	Kube-OVN E2E
----------	----------	--------------

2.6.5

Kube-OVN	OpenTelemetry	DeepFlow
----------	-------------------------------	--------------------------

2.6.6

Kube-OVN	7*24	Github Issue
Github Issue	AI	AI
 PDF	 Slack	 Support

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 GitHub 

2.6.7

2.7

Kube-OVN

Kube-OVN

OVS

issue

Kube-OVN

2.7.1 Kubernetes

Script Uninstall

Helm Uninstall

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/dist/images/cleanup.sh
bash cleanup.sh

helm uninstall kube-ovn -n kube-system
```

2.7.2

ovsdb

openvswitch

```
rm -rf /var/run/openvswitch
rm -rf /var/run/ovn
rm -rf /etc/origin/openvswitch/
rm -rf /etc/origin/ovn/
rm -rf /etc/cni/net.d/01-kube-ovn.conflist
#      v1.10      00-kube-ovn.conflist
rm -rf /etc/cni/net.d/00-kube-ovn.conflist
rm -rf /var/log/openvswitch
rm -rf /var/log/ovn
rm -fr /var/log/kube-ovn
```

2.7.3

iptables/ipset

reboot

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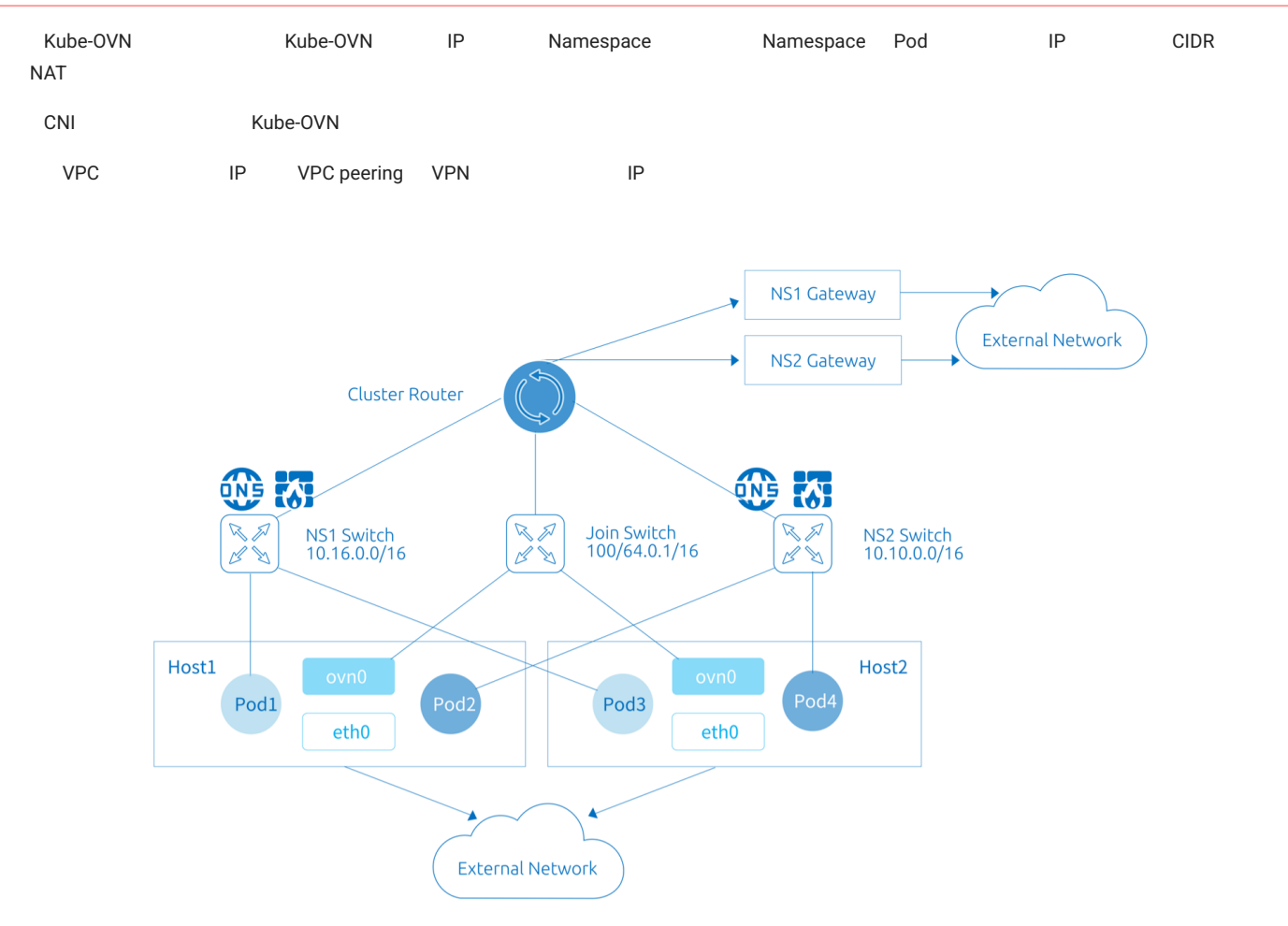
 GitHub



2.7.4

3.

3.1



Overlay Underlay

3.1.1

Kube-OVN	Namespace	IP	CIDR
Overlay	NAT	Flannel	
Underlay	arping		

spec	default	true	ovn-default
------	---------	------	-------------

```
# kubectl get subnet ovn-default -o yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  creationTimestamp: "2019-08-06T09:33:43Z"
  generation: 1
  name: ovn-default
  resourceVersion: "1571334"
```

```

selfLink: /apis/kubeovn.io/v1/subnets/ovn-default
uid: 7e2451f8-fb44-4f7f-b3e0-cfd27f6fd5d6
spec:
  cidrBlock: 10.16.0.0/16
  default: true
  excludeIps:
    - 10.16.0.1
  gateway: 10.16.0.1
  gatewayType: distributed
  natOutgoing: true
  private: false
  protocol: IPv4

```

3.1.2 Join

Kubernetes	Node	Pod	Overlay	Kube-OVN	join	Node	ovn0	join
Pods	Nodes	ovn0	Node	Pod	ovn0	ovn0		
		join	CIDR	Join				

Join

Pod	hostport	externalTrafficPolicy: Local	NodePort	Service
join	CIDR			

```

# kubectl get subnet join -o yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  creationTimestamp: "2019-08-06T09:33:43Z"
  generation: 1
  name: join
  resourceVersion: "1571333"
  selfLink: /apis/kubeovn.io/v1/subnets/join
  uid: 9c744810-c678-4d50-8a7d-b8ec12ef91b8
spec:
  cidrBlock: 100.64.0.0/16
  default: false
  excludeIps:
    - 100.64.0.1
  gateway: 100.64.0.1
  gatewayNode: ""
  gatewayType: ""
  natOutgoing: false
  private: false
  protocol: IPv4

```

Node ovn0

```

# ifconfig ovn0
ovn0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1420
    inet 100.64.0.4 netmask 255.255.0.0 broadcast 100.64.255.255
    inet6 fe80::800:ff:fe40:5 prefixlen 64 scopeid 0x20<link>
    ether 0a:00:00:40:00:05 txqueuelen 1000 (Ethernet)
    RX packets 18 bytes 1428 (1.3 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 19 bytes 1810 (1.7 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

```

3.1.3

Namespace

```

cat <<EOF | kubectl create -f -
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: subnet1
spec:
  protocol: IPv4
  cidrBlock: 10.66.0.0/16
  excludeIps:

```



```
- 10.66.0.1..10.66.0.10
- 10.66.0.101..10.66.0.151
gateway: 10.66.0.1
gatewayType: distributed
natOutgoing: true
routeTable: ""
namespaces:
- ns1
- ns2
```

- | | | | | |
|---------------|-----------|-----------|-------------|----------|
| • cidrBlock: | CIDR | VPC | Subnet CIDR | |
| • excludeIps: | | | IP | Underlay |
| • gateway | Overlay | Kube-OVN | | Underlay |
| • namespaces: | Namespace | Namespace | | Pod |
| • routeTable: | | | | |

```
# kubectl create ns ns1
namespace/ns1 created

# kubectl run nginx --image=docker.io/library/nginx:alpine -n ns1
deployment.apps/nginx created

# kubectl get pod -n ns1 -o wide
```

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED	NODE	READINESS	GATES
nginx-74d5899f46-n8wta	1/1	Running	0	10s	10.66.0.11	node1	<none>		<none>	

Workload

Pod	Namespace	IP	Namespace	Workload	Pod	Annotation
						ovn.kubernetes.io/logical_switch

```
apiVersion: v1
kind: Pod
metadata:
  name: another-subnet
  annotations:
    ovn.kubernetes.io/logical_switch: subnet1
spec:
  containers:
  - name: another-subnet
    image: docker.io/library/nginx:alpine
```

```
Workload      Deployment StatefulSet      ovn.kubernetes.io/logical_switch Annotation
spec.template.metadata.annotations
```

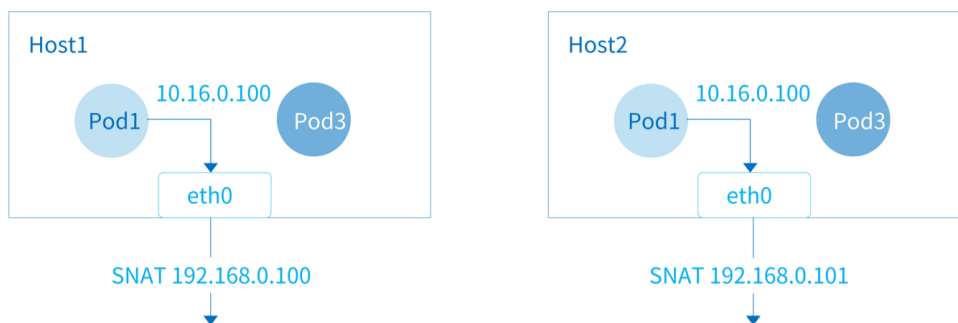
3.1.4 Overlay

The diagram illustrates a network architecture with the following components and connections:

- Top Layer:** VPC, Overlay, Underlay, VPC.
- Second Layer:** Overlay, Pod, Kube-OVN.
- Third Layer:** natOutgoing, Pod, snat.
- Bottom Layer:** Node, Node, Pod, ovn0, natOutgoing, true, Pod, IP.

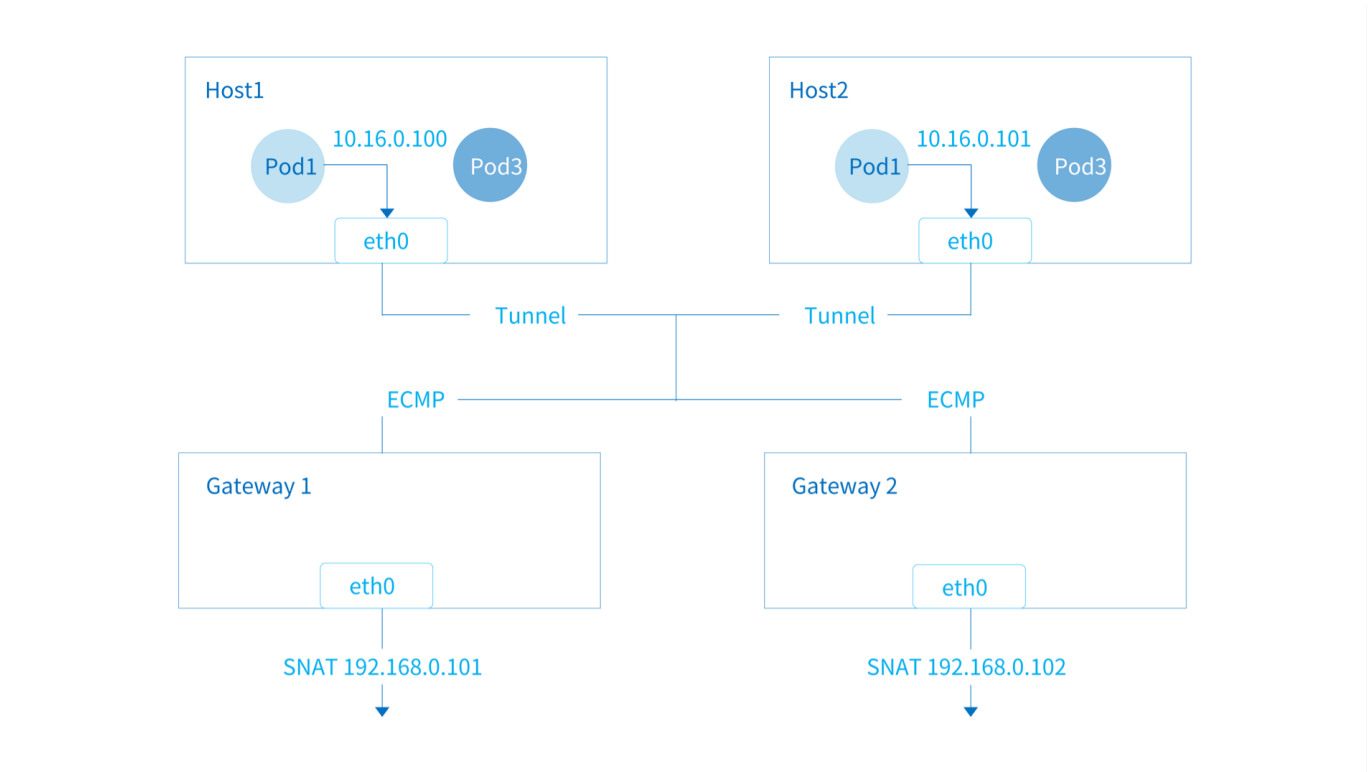
Connections are shown as follows:

- A vertical line on the far left connects the top VPC to the bottom Node.
- A horizontal line connects the top VPC to the top Overlay.
- A horizontal line connects the top Overlay to the top Underlay.
- A horizontal line connects the top Underlay to the top VPC.
- A horizontal line connects the top Overlay to the second-layer Overlay.
- A horizontal line connects the top Underlay to the second-layer Pod.
- A horizontal line connects the second-layer Pod to the second-layer Kube-OVN.
- A horizontal line connects the second-layer Overlay to the third-layer natOutgoing.
- A horizontal line connects the second-layer Pod to the third-layer Pod.
- A horizontal line connects the second-layer Kube-OVN to the third-layer snat.
- A horizontal line connects the third-layer natOutgoing to the bottom Node.
- A horizontal line connects the third-layer Pod to the bottom Pod.
- A horizontal line connects the third-layer snat to the bottom ovn0.
- A horizontal line connects the bottom Node to the bottom Node.
- A horizontal line connects the bottom Pod to the bottom ovn0.
- A horizontal line connects the bottom ovn0 to the bottom natOutgoing.
- A horizontal line connects the bottom natOutgoing to the bottom true.
- A horizontal line connects the bottom true to the bottom Pod.
- A horizontal line connects the bottom Pod to the bottom IP.



gatewayType distributed

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: distributed
spec:
  protocol: IPv4
  cidrBlock: 10.166.0.0/16
  default: false
  excludeIps:
    - 10.166.0.1
  gateway: 10.166.0.1
  gatewayType: distributed
  natOutgoing: true
```



Pod	IP	Pod	ovn0	natOutgoing	true
	IP				
gatewayType	centralized	gatewayNode	Kubernetes	nodeName	gatewayNode

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: centralized
spec:
  protocol: IPv4
  cidrBlock: 10.166.0.0/16
  default: false
  excludeIps:
    - 10.166.0.1
  gateway: 10.166.0.1
  gatewayType: centralized
  gatewayNode: "node1,node2"
  natOutgoing: true
```

- gatewayNode kube-ovn-worker:172.18.0.2, kube-ovn-control-plane:172.18.0.3
- ECMP ECMP
- Kube-OVN v1.12.0 subnet crd spec enableEcmp ECMP ECMP kube-ovn-
controller Deployment enable-ecmp v1.12.0

ECMP	kube-ovn-controller	ping	5s	5s-10s
Node Ready				

gatewayNodeSelectors

 gatewayNodeSelectors

- gatewayNode gatewayNode gatewayNodeSelectors
- OR
-

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: centralized-selector
spec:
  protocol: IPv4
  cidrBlock: 10.166.0.0/16
  default: false
  excludeIps:
    - 10.166.0.1
  gateway: 10.166.0.1
  gatewayType: centralized
  gatewayNodeSelectors:
    - matchLabels:
        role: gateway
    - matchExpressions:
        - key: node-type
          operator: In
          values: ["gateway", "egress"]
  natOutgoing: true
```

3.1.5 ACL

 Warning

Kube-OVN NetworkPolicy Network Policy API Subnet ACL Security Group OVN ACL NetworkPolicy NetworkPolicy API

ACL Kube-OVN Subnet ACL

Subnet ACL OVN ACL ovn-nb ACL Table match ovn-sb Logical Flow Table

IP 10.10.0.2 Pod ACL

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: acl
spec:
  allowEWTraffic: false
  acls:
    - action: drop
      direction: to-lport
      match: ip4.dst == 10.10.0.2 && ip
      priority: 1002
    - action: allow-related
      direction: from-lport
      match: ip4.src == 10.10.0.2 && ip
      priority: 1002
  cidrBlock: 10.10.0.0/24
```

ACL allowEWTraffic: true

3.1.6

ACL ACL

Kube-OVN Pod

CRD private true allowSubnets allowSubnets

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: private
spec:
  protocol: IPv4
  default: false
  namespaces:
    - ns1
    - ns2
  cidrBlock: 10.69.0.0/16
  private: true
  allowSubnets:
    - 10.16.0.0/16
    - 10.18.0.0/16
```

3.1.7 Underlay

```

      Underlay
    • vlan:      Underlay      Subnet      Vlan CR      Underlay
    • logicalGateway: Underlay      OVN      Underlay  Overlay      false
  
```

3.1.8

kube-ovn-cni	Pod	ICMP	ARP	Underlay	ICMP
--------------	-----	------	-----	----------	------

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: disable-gw-check
spec:
  disableGatewayCheck: true
```

3.1.9 Multicast-Snoop

subnet	Pod	OVN	Pod	subnet	multicast snoop	OVN	South Database	Multicast_Group
--------	-----	-----	-----	--------	-----------------	-----	----------------	-----------------

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: sample1
spec:
  enableMulticastSnoop: true
```

3.1.10 Subnet MTU

Subnet	Pod	MTU	Subnet	Pod
--------	-----	-----	--------	-----

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: sample1
spec:
  mtu: 1300
```

3.1.11

- IP
- VPC NAT
- QoS
-
- DHCP
-
-

• IP

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3.1.12

3.2

Kube-OVN	Pod	Namespace	IP	Mac	Kube-OVN
	• Pod	IP/Mac			
	•	IP/MAC			
	• Workload	IP Pool			
	• StatefulSet				
	• KubeVirt VM				
	• Multus				

3.2.1 Pod IP Mac

Pod	annotation	Pod	IP/Mac, kube-ovn-controller
<pre>apiVersion: v1 kind: Pod metadata: name: static-ip annotations: ovn.kubernetes.io/ip_address: 10.16.0.15 # 10.16.0.15, fd00:10:16::15 ovn.kubernetes.io/mac_address: 00:00:00:53:6B:B6 spec: containers: - name: static-ip image: docker.io/library/nginx:alpine</pre>			
	annotation	Pod IP/Mac	
1.	IP/Mac	IP/Mac	
2.	IP	CIDR	
3.	IP	Mac	

3.2.2 IP/MAC

Pod	VM	Multus	IP/MAC	Multus	k8s.v1.cni.cncf.io/networks	interface
annotation						
• <nadName>.<nadNamespace>.kubernetes.io/ip_address.<interfaceName>			IP			
• <nadName>.<nadNamespace>.kubernetes.io/mac_address.<interfaceName>			MAC			
<nadName>	<nadNamespace>	NetworkAttachmentDefinition	<interfaceName>	Multus	interface	annotation
annotation ovn.kubernetes.io/ip_address / ovn.kubernetes.io/mac_address						
NAD		IP/MAC				

```
apiVersion: v1
kind: Pod
metadata:
  name: multi-if-static
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/networks: '[{"name": "attachnet", "namespace": "default", "interface": "net1"}, {"name": "attachnet", "namespace": "default", "interface": "net2"}]'
    attachnet.default.kubernetes.io/ip_address.net1: 172.17.0.100
    attachnet.default.kubernetes.io/mac_address.net1: 00:00:00:53:6B:B1
    attachnet.default.kubernetes.io/ip_address.net2: 172.17.0.101
    attachnet.default.kubernetes.io/mac_address.net2: 00:00:00:53:6B:B2
spec:
  containers:
    - name: multi-if-static
      image: docker.io/library/nginx:alpine
```

3.2.3 Workload IP Pool

Kube-OVN	annotation	ovn.kubernetes.io/ip_pool	Workload	Deployment/StatefulSet/DaemonSet/Job/CronJob	IP	kube-ovn-
controller		ovn.kubernetes.io/ip_pool	IP			
IP Pool	Annotation	template	annotation	Kubernetes	Workload	Workload

Deployment IP

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: ippool
  labels:
    app: ippool
spec:
  replicas: 2
  selector:
    matchLabels:
      app: ippool
  template:
    metadata:
      labels:
        app: ippool
      annotations:
        ovn.kubernetes.io/ip_pool: 10.16.0.15,10.16.0.16,10.16.0.17 # 10.16.0.15, fd00:10:16::000E;10.16.0.16, fd00:10:16::000F;
        10.16.0.17, fd00:10:16::0010
    spec:
      containers:
        - name: ippool
          image: docker.io/library/nginx:alpine
```

Workload	IP								
1.	ovn.kubernetes.io/ip_pool	IP	CIDR						
2.	ovn.kubernetes.io/ip_pool	IP	IP						
3.	ovn.kubernetes.io/ip_pool	IP	replicas	Pod	Workload	ovn.kubernetes.io/ip_pool	IP		

3.2.4 StatefulSet

StatefulSet	IP	Workload	ovn.kubernetes.io/ip_pool	Pod	IP				
StatefulSet		Kube-OVN							
1.	Pod	ovn.kubernetes.io/ip_pool	IP	StatefulSet	web	web-0	ovn.kubernetes.io/ip_pool	IP	web-1
2.	StatefulSet Pod	OVN	logical_switch_port	Pod	interface	Pod	IP/Mac	StatefulSet	Volume
3.	2	ovn.kubernetes.io/ip_pool	StatefulSet Pod		IP/Mac	StatefulSet			

StatefulSet

```
apiVersion: apps/v1
kind: StatefulSet
metadata:
  name: web
spec:
  serviceName: "nginx"
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: docker.io/library/nginx:alpine
          ports:
            - containerPort: 80
              name: web
```


StatefulSet Pod Pod IP

StatefulSet Pod IP

StatefulSet	IP	Pod Name	Statefulset	ovn.kubernetes.io/ip_pool	Annotation	Pod	IP
StatefulSet	Pod IP	StatefulSet	scale	0	Annotation	StatefulSet	

3.2.5 KubeVirt VM

KubeVirt	VM	kube-ovn-controller	StatefulSet Pod	IP	VM
VM	IP				

3.2.6 Multus

Multus	Pod	Kube-OVN	annotation	Kube-OVN	CNI	Kube-OVN	IPAM	CNI
--------	-----	----------	------------	----------	-----	----------	------	-----

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3.2.7

3.3 IP

IP	IPPool	Subnet	IPAM	IP	Namespace	Workload
----	--------	--------	------	----	-----------	----------

3.3.1

```
apiVersion: kubeovn.io/v1
kind: IPPool
metadata:
  name: pool-1
spec:
  subnet: ovn-default
  ips:
    - "10.16.0.201"
    - "10.16.0.210/30"
    - "10.16.0.220..10.16.0.230"
  namespaces:
    - ns-1
  enableAddressSet: true
```

Workload

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: ippool
  labels:
    app: ippool
spec:
  replicas: 2
  selector:
    matchLabels:
      app: ippool
  template:
    metadata:
      labels:
        app: ippool
      annotations:
        ovn.kubernetes.io/ip_pool: pool-1
    spec:
      containers:
        - name: ippool
          image: docker.io/library/nginx:alpine
```

subnet						
ips	IP	..	IPv6			
namespaces				Pod	IP	IP
enableAddressSet	AddressSet	false	true	ACL	AddressSet	

3.3.2

1.	Workload	IP Pool	IP	IP		
2.	IP	.spec.ips	IP	IP	.spec.ips	CIDR
3.	IP	IP				
4.	IP	.spec.ips				
5.	IP	IP	IP	IP	IP	
6.	IP	IP				
7.	IP	Namespace				

8. IP `.spec.enableAddressSet` `false` `true` IP OVN NB AddressSet IP IP AddressSet
NetworkPolicy VPC AddressSet

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3.3.3

3.4

Pod Annotations

```
apiVersion: v1
kind: Pod
metadata:
  name: custom-routes
  annotations:
    ovn.kubernetes.io/routes: |
      [{
        "dst": "192.168.0.101/24",
        "gw": "10.16.0.254"
      }, {
        "gw": "10.16.0.254"
      }]
spec:
  containers:
    - name: nginx
      image: docker.io/library/nginx:alpine
```

dst

Deployment DaemonSet StatefulSet Annotation .spec.template.metadata.annotations

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: custom-routes
  labels:
    app: nginx
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
      annotations:
        ovn.kubernetes.io/routes: |
          [{
            "dst": "192.168.0.101/24",
            "gw": "10.16.0.254"
          }, {
            "gw": "10.16.0.254"
          }]
    spec:
      containers:
        - name: nginx
          image: docker.io/library/nginx:alpine
```

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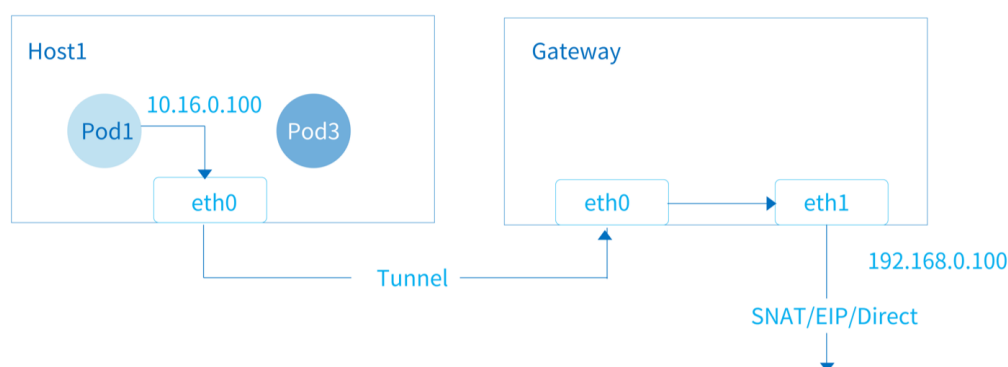
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3.4.1

3.5 EIP SNAT

[illegible]

3.5.1

- OVN L3 Gateway OVS Overlay Underlay
- NAT Underlay
- EIP SNAT

3.5.2

kube-system ConfigMap ovn-external-gw-config

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: ovn-external-gw-config
  namespace: kube-system
data:
  type: "centralized"
  enable-external-gw: "true"
  external-gw-nodes: "kube-ovn-worker"
  external-gw-nic: "eth1"
  external-gw-addr: "172.56.0.1/16"
```

```
nic-ip: "172.56.0.254/16"
nic-mac: "16:52:f3:13:6a:25"
```

- enable-external-gw: SNAT EIP
- type: centralized distributed centralized distributed
- external-gw-nodes: centralized
- external-gw-nic:
- external-gw-addr: IP
- nic-ip, nic-mac: IP Mac IP Mac

3.5.3 OVN OVS

OVN-NB ovn-external ovn-cluster-ovn-external chassis

```
# kubectl ko nbctl show
switch 3de4cea7-1a71-43f3-8b62-435a57ef16a6 (external)
  port localnet.external
    type: localnet
    addresses: ["unknown"]
  port external-ovn-cluster
    type: router
    router-port: ovn-cluster-external
router e1eb83ad-34be-4ed5-9a02-fcc8b1d357c4 (ovn-cluster)
  port ovn-cluster-external
    mac: "ac:1f:6b:2d:33:f1"
    networks: ["172.56.0.100/16"]
    gateway chassis: [a5682814-2e2c-46dd-9c1c-6803ef0dab66]
```

OVS br-external

```
# kubectl ko vsctl ${gateway node name} show
e7d81150-7743-4d6e-9e6f-5c688232e130
Bridge br-external
  Port br-external
    Interface br-external
      type: internal
  Port eth1
    Interface eth1
  Port patch-localnet.external-to-br-int
    Interface patch-localnet.external-to-br-int
      type: patch
      options: {peer=patch-br-int-to-localnet.external}
```

3.5.4 Pod EIP SNAT

Pod ovn.kubernetes.io/snat ovn.kubernetes.io/eip annotation SNAT EIP

```
apiVersion: v1
kind: Pod
metadata:
  name: pod-snat
  annotations:
    ovn.kubernetes.io/snat: 172.56.0.200
spec:
  containers:
  - name: pod-snat
    image: docker.io/library/nginx:alpine
---
apiVersion: v1
kind: Pod
metadata:
  name: pod-eip
  annotations:
    ovn.kubernetes.io/eip: 172.56.0.233
spec:
  containers:
  - name: pod-eip
    image: docker.io/library/nginx:alpine
```

kubectl Pod EIP SNAT ovn.kubernetes.io/routed annotation

```
kubectl annotate pod pod-eip ovn.kubernetes.io/eip=172.56.0.221 --overwrite
kubectl annotate pod pod-eip ovn.kubernetes.io/routed-
```

EIP SNAT ovn.kubernetes.io/routed annotation

3.5.5

```
kube-ovn-controller          SNAT  EIP

• --external-gateway-config-ns: Configmap  ovn-external-gw-config  Namespace  kube-system
• --external-gateway-net:          external
• --external-gateway-vlanid:      Vlan Tag      0      Vlan
```

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3.5.6

3.6 QoS

Kube-OVN	Pod	QoS
• QoS		
• linux-netem QoS		
	Pod	QoS
	Namespace	Subnet
	QoS	

3.6.1 QoS

QoS	Pod annotation	Pod	Mbit/s
-----	----------------	-----	--------

```
apiVersion: v1
kind: Pod
metadata:
  name: qos
  namespace: ls1
  annotations:
    ovn.kubernetes.io/ingress_rate: "3"
    ovn.kubernetes.io/egress_rate: "1"
spec:
  containers:
  - name: qos
    image: docker.io/library/nginx:alpine
```

annotation	QoS
------------	-----

```
kubectl annotate --overwrite pod nginx-74d5899f46-d7qkn ovn.kubernetes.io/ingress_rate=3
```

QoS

```
kind: DaemonSet
apiVersion: apps/v1
metadata:
  name: perf
  namespace: ls1
  labels:
    app: perf
spec:
  selector:
    matchLabels:
      app: perf
  template:
    metadata:
      labels:
        app: perf
    spec:
      containers:
      - name: nginx
        image: docker.io/kubeovn/perf
```

Pod	iperf3 server
-----	---------------

```
# kubectl exec -it perf-4n4gt -n ls1 sh
# iperf3 -s
-----
Server listening on 5201
-----
```

Pod	Pod
-----	-----

```
# kubectl exec -it perf-d4mqc -n ls1 sh
# iperf3 -c 10.66.0.12
Connecting to host 10.66.0.12, port 5201
[ 4] local 10.66.0.14 port 51544 connected to 10.66.0.12 port 5201
[ ID] Interval      Transfer    Bandwidth  Retr  Cwnd
[ 4] 0.00-1.00 sec   86.4 MBytes 725 Mbits/sec  3    350 KBytes
[ 4] 1.00-2.00 sec   89.9 MBytes 754 Mbits/sec 118   473 KBytes
[ 4] 2.00-3.00 sec   101 MBytes 848 Mbits/sec 184   586 KBytes
[ 4] 3.00-4.00 sec   104 MBytes 875 Mbits/sec 217   671 KBytes
[ 4] 4.00-5.00 sec   111 MBytes 935 Mbits/sec 175   772 KBytes
```



```
[ 4] 5.00-6.00 sec 100 MBytes 840 Mbits/sec 658 598 KBytes
[ 4] 6.00-7.00 sec 106 MBytes 890 Mbits/sec 742 668 KBytes
[ 4] 7.00-8.00 sec 102 MBytes 857 Mbits/sec 764 724 KBytes
[ 4] 8.00-9.00 sec 97.4 MBytes 817 Mbits/sec 1175 764 KBytes
[ 4] 9.00-10.00 sec 111 MBytes 934 Mbits/sec 1083 838 KBytes
-----
[ ID] Interval      Transfer    Bandwidth    Retr
[ 4] 0.00-10.00 sec 1010 MBytes 848 Mbits/sec 5119          sender
[ 4] 0.00-10.00 sec 1008 MBytes 846 Mbits/sec          receiver

iperf Done.
```

Pod QoS

```
kubectl annotate --overwrite pod perf-4n4gt -n ls1 ovn.kubernetes.io/ingress_rate=30
```

Pod Pod

```
# iperf3 -c 10.66.0.12
Connecting to host 10.66.0.12, port 5201
[ 4] local 10.66.0.14 port 52372 connected to 10.66.0.12 port 5201
[ ID] Interval      Transfer    Bandwidth    Retr  Cwnd
[ 4] 0.00-1.00 sec 3.66 MBytes 30.7 Mbits/sec 2 76.1 KBytes
[ 4] 1.00-2.00 sec 3.43 MBytes 28.8 Mbits/sec 0 104 KBytes
[ 4] 2.00-3.00 sec 3.50 MBytes 29.4 Mbits/sec 0 126 KBytes
[ 4] 3.00-4.00 sec 3.50 MBytes 29.3 Mbits/sec 0 144 KBytes
[ 4] 4.00-5.00 sec 3.43 MBytes 28.8 Mbits/sec 0 160 KBytes
[ 4] 5.00-6.00 sec 3.43 MBytes 28.8 Mbits/sec 0 175 KBytes
[ 4] 6.00-7.00 sec 3.50 MBytes 29.3 Mbits/sec 0 212 KBytes
[ 4] 7.00-8.00 sec 3.68 MBytes 30.9 Mbits/sec 0 294 KBytes
[ 4] 8.00-9.00 sec 3.74 MBytes 31.4 Mbits/sec 0 398 KBytes
[ 4] 9.00-10.00 sec 3.80 MBytes 31.9 Mbits/sec 0 526 KBytes
-----
[ ID] Interval      Transfer    Bandwidth    Retr
[ 4] 0.00-10.00 sec 35.7 MBytes 29.9 Mbits/sec 2          sender
[ 4] 0.00-10.00 sec 34.5 MBytes 29.0 Mbits/sec          receiver

iperf Done.
```

3.6.2 linux-netem QoS

RHEL netem yum install -y kernel-modules-extra && modprobe sch_netem

Pod annotation linux-netem QoS netem QoS Pod Ingress

- ovn.kubernetes.io/latency Pod ms
- ovn.kubernetes.io/jitter Pod ms
- ovn.kubernetes.io/limit qdisc 1000
- ovn.kubernetes.io/loss float 20 20%

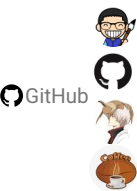
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3.6.3

3.7

Kube-OVN	IPv4 IPv6
	Kubernetes Kubernetes

3.7.1

CIDR

cidr=<IPv4 CIDR>,<IPv6 CIDR>

CIDR

IPv4

IPv6

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: ovn-test
spec:
  cidrBlock: 10.16.0.0/16,fd00:10:16::/64
  excludeIps:
    - 10.16.0.1
    - fd00:10:16::1
  gateway: 10.16.0.1,fd00:10:16::1
```

install.sh

/112

ULA

/64

/Service CIDR

```
POD_CIDR="10.16.0.0/16,fd00:10:16::/112"
JOIN_CIDR="100.64.0.0/16,fd00:100:64::/112"
```

3.7.2 Pod

Pod

IPv4 IPv6

Pod annotation :

```
apiVersion: v1
kind: Pod
metadata:
  annotations:
    ovn.kubernetes.io/allocated: "true"
    ovn.kubernetes.io/cidr: 10.16.0.0/16,fd00:10:16::/64
    ovn.kubernetes.io/gateway: 10.16.0.1,fd00:10:16::1
    ovn.kubernetes.io/ip_address: 10.16.0.9,fd00:10:16::9
    ovn.kubernetes.io/logical_switch: ovn-default
    ovn.kubernetes.io/mac_address: 00:00:00:14:88:09
    ovn.kubernetes.io/routed: "true"
...
podIP: 10.16.0.9
podIPs:
  - ip: 10.16.0.9
  - ip: fd00:10:16::9
```

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3.7.3

3.8 Webhook

Webhook	Kube-OVN	CRD	Webhook	IP	Subnet CIDR
Webhook	Subnet	Pod	Kube-OVN	Webhook	Pod

3.8.1 Cert-Manager

Webhookcert-managerWebhookcert-manager

cert-manager:

kubectl apply -f https://github.com/cert-manager/cert-manager/releases/download/v1.8.0/cert-manager.yaml

cert-managercert-manager

3.8.2 Webhook

Webhookyaml:

kubectl apply -f https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/yamls/webhook.yaml
deployment.apps/kube-ovn-webhook created
service/kube-ovn-webhook created
validatingwebhookconfiguration.admissionregistration.k8s.io/kube-ovn-webhook created
certificate.cert-manager.io/kube-ovn-webhook-serving-cert created
issuer.cert-manager.io/kube-ovn-webhook-selfsigned-issuer created

3.8.3 Webhook

PodPod IP 10.16.0.15

kubectl get pod -o wide
NAME READY STATUS RESTARTS AGE IP NODE NOMINATED NODE READINESS GATES
static-7584848b74-fw9dm 1/1 Running 0 2d13h 10.16.0.15 kube-ovn-worker <none>

yamlIPPod

apiVersion: v1
kind: Pod
metadata:
 annotations:
 ovn.kubernetes.io/ip_address: 10.16.0.15
 ovn.kubernetes.io/mac_address: 00:00:00:53:6B:B6
 labels:
 app: static
 managedFields:
 name: staticip-pod
 namespace: default
spec:
 containers:
 - image: docker.io/library/nginx:alpine
 imagePullPolicy: IfNotPresent
 name: qatest

yamlPodIP

kubectl apply -f pod-static.yaml
Error from server (annotation static-ip 10.16.0.15 is conflict with ip crd static-7584848b74-fw9dm.default, ip 10.16.0.15): error when creating "pod-static.yaml": admission webhook "pod-ip-validating.kube-ovn.io" denied the request: annotation static-ip 10.16.0.15 is conflict with ip crd static-7584848b74-fw9dm.default, ip 10.16.0.15

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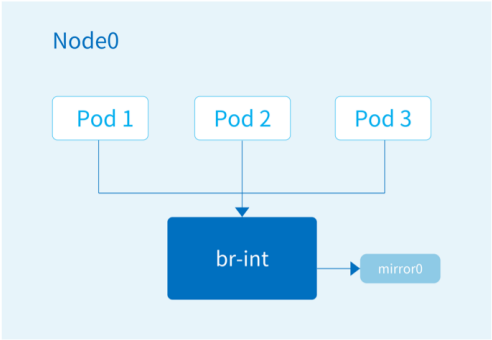
 GitHub 

3.8.4

3.9

NPM

CPU 5%~10% CPU



3.9.1

kube-ovn-cni DaemonSet

- --enable-mirror=true
- --mirror-iface=mirror0:

br-int

Kube-OVN

tcpdump mirror0

```
tcpdump -ni mirror0
```

3.9.2 Pod

Pod

Pod

ovn.kubernetes.io/mirror annotation

Pod

```
apiVersion: v1
kind: Pod
metadata:
  name: mirror-pod
  namespace: ls1
  annotations:
    ovn.kubernetes.io/mirror: "true"
spec:
  containers:
    - name: mirror-pod
      image: docker.io/library/nginx:alpine
```

3.9.3

1. Pod to Pod in the same Nodes

Size	TCP Latency	TCP Bandwidth	UDP Latency	UDP Lost Rate	UDP Bandwidth
64	12.7 us	289 Mbits/sec	12.6 us	(1.8%)	77.9 Mbits/sec
128	15.5 us	517 Mbits/sec	12.7 us	(0%)	155 Mbits/sec
512	12.2 us	1.64 Gbits/sec	12.4 us	(0%)	624 Mbits/sec
1k	13 us	2.96 Gbits/sec	11.4 us	(0.53%)	1.22 Gbits/sec
4k	18 us	7.67 Gbits/sec	25.7 us	(0.41%)	1.50 Gbits/sec

Size	TCP Latency	TCP Bandwidth	UDP Latency	UDP Lost Rate	UDP Bandwidth
64	11.9 us	324 Mbits/sec	12.2 us	(0.22%)	102 Mbits/sec
128	10.5 us	582 Mbits/sec	9.5 us	(0.21%)	198 Mbits/sec
512	11.6 us	1.84 Gbits/sec	9.32 us	(0.091%)	827 Mbits/sec
1k	10.5 us	3.44 Gbits/sec	10 us	(1.2%)	1.52 Gbits/sec
4k	16.7 us	8.52 Gbits/sec	18.2 us	(1.3%)	2.42 Gbits/sec

2. Pod to Pod in the different Nodes

Size	TCP Latency	TCP Bandwidth	UDP Latency	UDP Lost Rate	UDP Bandwidth
64	258 us	143 Mbits/sec	237 us	(61%)	28.5 Mbits/sec
128	240 us	252 Mbits/sec	231 us	(64%)	54.9 Mbits/sec
512	236 us	763 Mbits/sec	256 us	(68%)	194 Mbits/sec
1k	242 us	969 Mbits/sec	225 us	(62%)	449 Mbits/sec
4k	352 us	1.12 Gbits/sec	382 us	(0.71%)	21.4 Mbits/sec

Size	TCP Latency	TCP Bandwidth	UDP Latency	UDP Lost Rate	UDP Bandwidth
64	278 us	140 Mbits/sec	227 us	(24%)	59.6 Mbits/sec
128	249 us	265 Mbits/sec	265 us	(23%)	114 Mbits/sec
512	233 us	914 Mbits/sec	235 us	(21%)	468 Mbits/sec
1k	238 us	1.14 Gbits/sec	240 us	(15%)	891 Mbits/sec
4k	370 us	1.25 Gbits/sec	361 us	(0.43%)	7.54 Mbits/sec

3. Node to Node

Size	TCP Latency	TCP Bandwidth	UDP Latency	UDP Lost Rate	UDP Bandwidth
64	205 us	162 Mbits/sec	183 us	(11%)	74.2 Mbits/sec
128	222 us	280 Mbits/sec	206 us	(6.3%)	155 Mbits/sec
512	220 us	1.04 Gbits/sec	177 us	(20%)	503 Mbits/sec
1k	213 us	2.06 Gbits/sec	201 us	(8.6%)	1.14 Gbits/sec
4k	280 us	5.01 Gbits/sec	315 us	(37%)	1.20 Gbits/sec

Size	TCP Latency	TCP Bandwidth	UDP Latency	UDP Lost Rate	UDP Bandwidth
64	204 us	157 Mbits/sec	204 us	(8.8%)	81.9 Mbits/sec
128	213 us	262 Mbits/sec	225 us	(19%)	136 Mbits/sec
512	220 us	1.02 Gbits/sec	227 us	(21%)	486 Mbits/sec
1k	217 us	1.79 Gbits/sec	218 us	(29%)	845 Mbits/sec
4k	275 us	5.27 Gbits/sec	336 us	(34%)	1.21 Gbits/sec

4. Pod to the Node where the Pod is located

Size	TCP Latency	TCP Bandwidth	UDP Latency	UDP Lost Rate	UDP Bandwidth
64	12.2 us	295 Mbits/sec	12.7 us	(0.27%)	74.1 Mbits/sec
128	14.1 us	549 Mbits/sec	10.6 us	(0.41%)	153 Mbits/sec
512	13.5 us	1.83 Gbits/sec	12.7 us	(0.23%)	586 Mbits/sec
1k	12 us	2.69 Gbits/sec	13 us	(1%)	1.16 Gbits/sec
4k	18.9 us	4.51 Gbits/sec	21.8 us	(0.42%)	1.81 Gbits/sec

Size	TCP Latency	TCP Bandwidth	UDP Latency	UDP Lost Rate	UDP Bandwidth
64	10.4 us	335 Mbits/sec	12.2 us	(0.75%)	95.4 Mbits/sec
128	12.1 us	561 Mbits/sec	11.3 us	(0.25%)	194 Mbits/sec
512	11.6 us	1.87 Gbits/sec	10.7 us	(0.66%)	745 Mbits/sec
1k	12.7 us	3.12 Gbits/sec	10.9 us	(1.2%)	1.46 Gbits/sec
4k	16.5 us	8.23 Gbits/sec	17.9 us	(1.5%)	2.51 Gbits/sec

5. Pod to the Node where the Pod is not located

Size	TCP Latency	TCP Bandwidth	UDP Latency	UDP Lost Rate	UDP Bandwidth
64	234 us	153 Mbits/sec	232 us	(63%)	29.4 Mbits/sec
128	237 us	261 Mbits/sec	238 us	(49%)	76.1 Mbits/sec
512	231 us	701 Mbits/sec	238 us	(57%)	279 Mbits/sec
1k	256 us	1.05 Gbits/sec	228 us	(56%)	524 Mbits/sec
4k	330 us	1.08 Gbits/sec	359 us	(1.5%)	35.7 Mbits/sec

Size	TCP Latency	TCP Bandwidth	UDP Latency	UDP Lost Rate	UDP Bandwidth
64	283 us	141 Mbits/sec	230 us	(26%)	55.8 Mbits/sec
128	234 us	255 Mbits/sec	234 us	(25%)	113 Mbits/sec
512	246 us	760 Mbits/sec	234 us	(22%)	458 Mbits/sec
1k	268 us	1.23 Gbits/sec	242 us	(20%)	879 Mbits/sec
4k	326 us	1.20 Gbits/sec	369 us	(0.5%)	7.87 Mbits/sec

6. Pod to the cluster ip service

Size	TCP Latency	TCP Bandwidth	UDP Latency	UDP Lost Rate	UDP Bandwidth
64	237 us	133 Mbits/sec	213 us	(65%)	25.5 Mbits/sec
128	232 us	271 Mbits/sec	222 us	(62%)	54.8 Mbits/sec
512	266 us	800 Mbits/sec	234 us	(60%)	232 Mbits/sec
1k	248 us	986 Mbits/sec	239 us	(50%)	511 Mbits/sec
4k	314 us	1.03 Gbits/sec	367 us	(0.6%)	13.2 Mbits/sec

TCP-Conn-Number	QPS	Avg-Resp-Time	Stdev-Resp-Time	Max-Resp-Time
10	14305.17	0.87ms	1.48ms	24.46ms
100	29082.07	3.87ms	4.35ms	102.85ms

Size	TCP Latency	TCP Bandwidth	UDP Latency	UDP Lost Rate	UDP Bandwidth
64	241 us	145 Mbits/sec	225 us	(19%)	60.2 Mbits/sec
128	245 us	261 Mbits/sec	212 us	(15%)	123 Mbits/sec
512	252 us	821 Mbits/sec	219 us	(14%)	499 Mbits/sec
1k	253 us	1.08 Gbits/sec	242 us	(16%)	852 Mbits/sec
4k	320 us	1.32 Gbits/sec	360 us	(0.47%)	6.70 Mbits/sec

TCP-Conn-Number	QPS	Avg-Resp-Time	Stdev-Resp-Time	Max-Resp-Time
10	13634.07	0.96ms	1.72ms	30.07ms
100	30215.23	3.59ms	3.20ms	77.56ms

7. Host to the Node port service where the Pod is not located on the target Node

TCP-Conn-Number	QPS	Avg-Resp-Time	Stdev-Resp-Time	Max-Resp-Time
10	14802.73	0.88ms	1.66ms	31.49ms
100	29809.58	3.78ms	4.12ms	105.34ms

TCP-Conn-Number	QPS	Avg-Resp-Time	Stdev-Resp-Time	Max-Resp-Time
10	14273.33	0.90ms	1.60ms	37.16ms
100	30757.81	3.62ms	3.41ms	59.78ms

8. Host to the Node port service where the Pod is located on the target Node

TCP-Conn-Number	QPS	Avg-Resp-Time	Stdev-Resp-Time	Max-Resp-Time
10	15402.39	802.50us	1.42ms	30.91ms
100	29424.66	4.05ms	4.31ms	90.60ms

TCP-Conn-Number	QPS	Avg-Resp-Time	Stdev-Resp-Time	Max-Resp-Time
10	14649.21	0.91ms	1.72ms	43.92ms
100	32143.61	3.66ms	3.76ms	67.02ms

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3.9.4

3.10 LoadBalancer Service

Kube-OVN VPC VPC VPC
 VPC VPC VPC LoadBalancer Service LoadBalancerIP VPC Service

1. multus-cni macvlan cni
2. LoadBalancer Service VPC vpc-nat-gw macvlan
3. VPC VPC LoadBalancer VPC VPC

3.10.1 VPC LoadBalancer Service

kube-system namespace deployment kube-ovn-controller args --enable-lb-svc=true false

```
containers:
- args:
  - /kube-ovn/start-controller.sh
  - --default-cidr=10.16.0.0/16
  - --default-gateway=10.16.0.1
  - --default-gateway-check=true
  - --enable-lb-svc=true # true
```

NetworkAttachmentDefinition CRD

yaml net-attach-def

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: lb-svc-attachment
  namespace: kube-system
spec:
  # master eth0 / ens192 / bond0
  config: '{
    "cniVersion": "0.3.0",
    "type": "macvlan",
    "master": "eth0",
    "mode": "bridge"
  }'
```

eth0 master

Subnet

Subnet LoadBalancer Service LoadBalancerIP Underlay Subnet

yaml

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: attach-subnet
spec:
  protocol: IPv4
  provider: lb-svc-attachment.kube-system # provider net-attach-def Name.Namespace
  cidrBlock: 172.18.0.0/16
  gateway: 172.18.0.1
  excludeIps:
  - 172.18.0.0..172.18.0.10
```

Subnet provider ovn .ovn Kube-OVN logical switch

provider ovn .ovn Kube-OVN IPAM IP

LoadBalancer Service

yaml LoadBalancer Service

```
apiVersion: v1
kind: Service
metadata:
  annotations:
    lb-svc-attachment.kube-system.kubernetes.io/logical_switch: attach-subnet #
    ovn.kubernetes.io/attachmentprovider: lb-svc-attachment.kube-system #
  labels:
    app: dynamic
    name: test-service
    namespace: default
spec:
  loadBalancerIP: 172.18.0.18 #
  ports:
    - name: test
      protocol: TCP
      port: 80
      targetPort: 80
  selector:
    app: dynamic
  sessionAffinity: None
  type: LoadBalancer
```

yaml annotation ovn.kubernetes.io/attachmentprovider net-attach-def Name.Namespace net-attach-def Pod

annotation annotation key net-attach-def Name.Namespace.kubernetes.io/logical_switch LoadBalancerIP LoadBalancerIP LoadBalancerIP spec.loadBalancerIP

yaml Service Service Namespace Pod

```
# kubectl get pod
NAME lb-svc-test-service-6869d98dd8-cjv1l READY STATUS RESTARTS AGE
lb-svc-test-service-6869d98dd8-cjv1l 1/1 Running 0 107m

# kubectl get svc
NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE
test-service LoadBalancer 10.109.201.193 172.18.0.18 80:30056/TCP 107m
```

service.spec.loadBalancerIP service external-ip

Pod yaml

```
# kubectl get pod -o yaml lb-svc-test-service-6869d98dd8-cjv1l
apiVersion: v1
kind: Pod
metadata:
  annotations:
    k8s.v1.cni.cncf.io/network-status: |-
      [{"name": "kube-ovn", "ips": ["10.16.0.2"], "default": true, "dns": {}}, {"name": "default/test-service", "interface": "net1", "mac": "ba:85:f7:02:9f:42", "dns": {}}]
    k8s.v1.cni.cncf.io/networks: default/test-service
    k8s.v1.cni.cncf.io/networks-status: |-
      [{"name": "kube-ovn", "ips": ["10.16.0.2"], "default": true, "dns": {}}, {"name": "default/test-service", "interface": "net1", "mac": "ba:85:f7:02:9f:42", "dns": {}}]
  ovn.kubernetes.io/allocated: "true"
  ovn.kubernetes.io/cidr: 10.16.0.0/16
  ovn.kubernetes.io/gateway: 10.16.0.1
```

```

ovn.kubernetes.io/ip_address: 10.16.0.2
ovn.kubernetes.io/logical_router: ovn-cluster
ovn.kubernetes.io/logical_switch: ovn-default
ovn.kubernetes.io/mac_address: 00:00:00:45:F4:29
ovn.kubernetes.io/pod_nic_type: veth-pair
ovn.kubernetes.io/routed: "true"
test-service.default.kubernetes.io/allocated: "true"
test-service.default.kubernetes.io/cidr: 172.18.0.0/16
test-service.default.kubernetes.io/gateway: 172.18.0.1
test-service.default.kubernetes.io/ip_address: 172.18.0.18
test-service.default.kubernetes.io/logical_switch: attach-subnet
test-service.default.kubernetes.io/mac_address: 00:00:00:AF:AA:BF
test-service.default.kubernetes.io/pod_nic_type: veth-pair

```

Service

```

# kubectl get svc -o yaml test-service
apiVersion: v1
kind: Service
metadata:
  annotations:
    kubectl.kubernetes.io/last-applied-configuration: |
      {"apiVersion":"v1","kind":"Service","metadata":{"annotations":{"test-service.default.kubernetes.io/logical_switch":"attach-subnet"},"labels":{"app":"dynamic"},"name":"test-service","namespace":"default"},"spec":{"ports":[{"name":"test","port":80,"protocol":"TCP","targetPort":80}],"selector":{"app":"dynamic"},"sessionAffinity":"None","type":"LoadBalancer"}}
    ovn.kubernetes.io/vpc: ovn-cluster
    test-service.default.kubernetes.io/logical_switch: attach-subnet
  creationTimestamp: "2022-06-15T09:01:58Z"
  labels:
    app: dynamic
  name: test-service
  namespace: default
  resourceVersion: "38485"
  uid: 161ede1-7f6e-40f5-9e09-5a52c44267d0
spec:
  allocateLoadBalancerNodePorts: true
  clusterIP: 10.109.201.193
  clusterIPs:
    - 10.109.201.193
  externalTrafficPolicy: Cluster
  internalTrafficPolicy: Cluster
  ipFamilies:
    - IPv4
  ipFamilyPolicy: SingleStack
  ports:
    - name: test
      nodePort: 30056
      port: 80
      protocol: TCP
      targetPort: 80
  selector:
    app: dynamic
  sessionAffinity: None
  type: LoadBalancer
status:
  loadBalancer:
    ingress:
      - ip: 172.18.0.18

```

3.10.2 LoadBalancerIP

yaml, Pod Service Endpoints

```

apiVersion: apps/v1
kind: Deployment
metadata:
  labels:
    app: dynamic
  name: dynamic
  namespace: default
spec:
  replicas: 2
  selector:
    matchLabels:
      app: dynamic
  strategy:
    rollingUpdate:
      maxSurge: 25%
      maxUnavailable: 25%
    type: RollingUpdate
  template:
    metadata:
      creationTimestamp: null
      labels:
        app: dynamic
    spec:
      containers:
        - image: docker.io/library/nginx:alpine

```

```

imagePullPolicy: IfNotPresent
name: nginx
dnsPolicy: ClusterFirst
restartPolicy: Always

```

Service LoadBalancerIP:Port

```

# curl 172.18.0.18:80
<html>
<head>
  <title>Hello World!</title>
  <link href="//fonts.googleapis.com/css?family=Open+Sans:400,700" rel="stylesheet" type="text/css">
  <style>
    body {
      background-color: white;
      text-align: center;
      padding: 50px;
      font-family: "Open Sans", "Helvetica Neue", Helvetica, Arial, sans-serif;
    }
    #logo {
      margin-bottom: 40px;
    }
  </style>
</head>
<body>
  <h1>Hello World!</h1>
  <h3>Links found</h3>
  <h3>I am on dynamic-7d8d7874f5-hsgc4</h3>
  <h3>Cookie =</h3>
  <b>KUBERNETES</b> listening in 443 available at tcp://10.96.0.1:443<br />
  <h3>my name is hanhouchao!</h3>
  <h3> RequestURI='/'</h3>
</body>
</html>

```

Service Pod

```

# ip a
4: net1@if62: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether ba:85:f7:02:9f:42 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 172.18.0.18/16 scope global net1
        valid_lft forever preferred_lft forever
    inet6 fe80::b885:f7ff:fe02:9f42/64 scope link
        valid_lft forever preferred_lft forever
36: eth0@if37: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc noqueue state UP group default
    link/ether 00:00:00:45:f4:29 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 10.16.0.2/16 brd 10.16.255.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::200:ff:fe45:f429/64 scope link
        valid_lft forever preferred_lft forever

# ip rule
0: from all lookup local
32764: from all iif eth0 lookup 100
32765: from all iif net1 lookup 100
32766: from all lookup main
32767: from all lookup default

# ip route show table 100
default via 172.18.0.1 dev net1
10.109.201.193 via 10.16.0.1 dev eth0
172.18.0.0/16 dev net1 scope link

# iptables -t nat -L -n -v
Chain PREROUTING (policy ACCEPT 0 packets, 0 bytes)
  pkts bytes target    prot opt in     out     source    destination
    0     0 DNAT      tcp -- *      *        0.0.0.0/0 172.18.0.18      tcp dpt:80 to:10.109.201.193:80

Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
  pkts bytes target    prot opt in     out     source    destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
  pkts bytes target    prot opt in     out     source    destination

Chain POSTROUTING (policy ACCEPT 0 packets, 0 bytes)
  pkts bytes target    prot opt in     out     source    destination
    0     0 MASQUERADE all -- *      *        0.0.0.0/0 10.109.201.193

```

lb service Pod nodeSelector

```

ovn-vpc-nat-config ConfigMap nodeSelector LoadBalancer service Pod

```

```

apiVersion: v1
data:
  image: docker.io/kubeovn/vpc-nat-gateway:v1.16.5
  nodeSelector: |
    kubernetes.io/hostname: kube-ovn-control-plane

```

```
kubernetes.io/os: linux
kind: ConfigMap
metadata:
  name: ovn-vpc-nat-config
  namespace: kube-system
```

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3.10.3

3.11

Kube-OVN	Prometheus					
kube-prometheus	CRD	Prometheus	kube-prometheus	CRD	Kube-OVN	Kube-OVN

3.11.1 Prometheus Monitor

Kube-OVN Prometheus Monitor CRD

```
#
kubectl apply -f https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/pinger-monitor.yaml
# kube-ovn-controller
kubectl apply -f https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/controller-monitor.yaml
# kube-ovn-cni
kubectl apply -f https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/cni-monitor.yaml
# ovn
kubectl apply -f https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/ovn-monitor.yaml
```

Prometheus 15s yml interval

3.11.2 Grafana

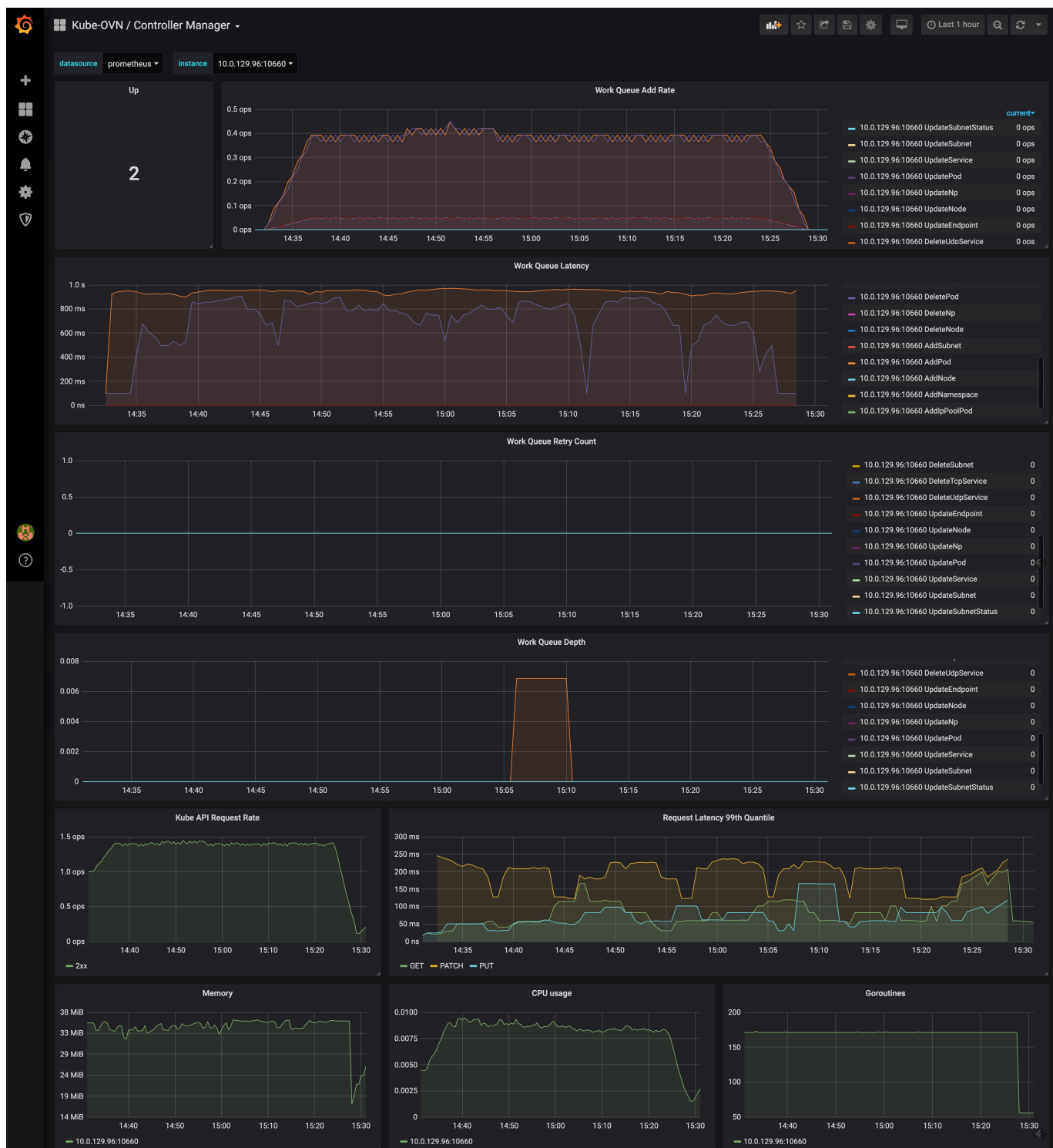
Kube-OVN Grafana Dashboard

Dashboard

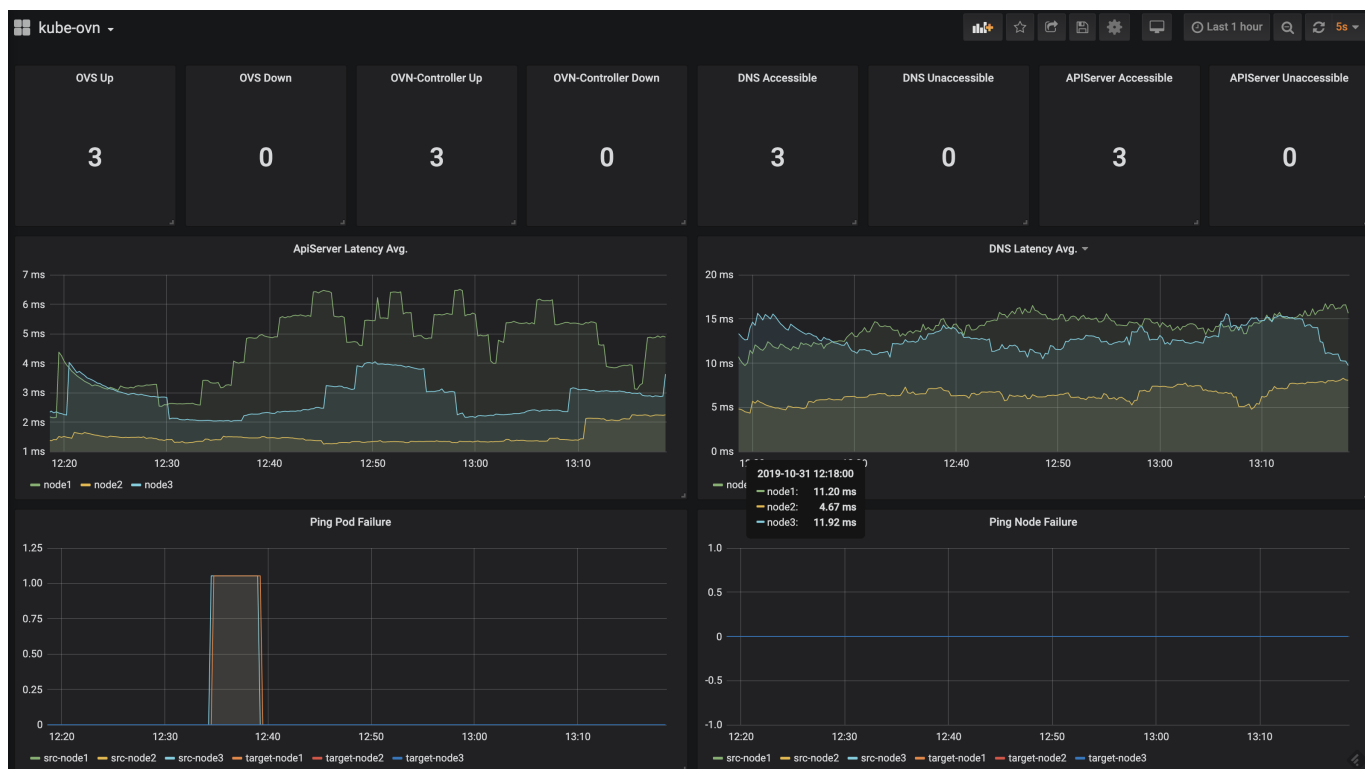
```
#
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/pinger-grafana.json
# kube-ovn-controller
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/controller-grafana.json
# kube-ovn-cni
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/cni-grafana.json
# ovn
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/ovn-grafana.json
# ovs
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/ovs-grafana.json
```

Grafana Prometheus Dashboard

kube-ovn-controller



kube-ovn-pinger



kube-ovn-cni



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3.11.3

3.12

3.12.1 NetworkPolicy

NetworkPolicy	Kubernetes	Pod	Pod	Kube-OVN	OVN	ACL	Kubernetes NetworkPolicy
	NetworkPolicy						

Kube-OVN

Kube-OVN	OVN	Open Virtual Network	NetworkPolicy	OVN
----------	-----	----------------------	---------------	-----

PORT GROUP

NetworkPolicy	Kube-OVN	Port Group	podSelector	Pod	Port Group	<	>	<	>	-	.
---------------	----------	------------	-------------	-----	------------	---	---	---	---	---	---

ADDRESS SET

Address Set	NetworkPolicy	IP	NetworkPolicy	Kube-OVN	podSelector	namespaceSelector	ipBlock
IP	Address Set						

NetworkPolicy	Ingress	Egress	Allow	Except	Address Set
• < > . < > . ingress . allow			IP		
• < > . < > . ingress . except			IP		
• < > . < > . egress . allow			IP		
• < > . < > . egress . except			IP		

ACL

ACL	OVN	Kube-OVN	NetworkPolicy	OVN ACL	Port Group	ACL
-----	-----	----------	---------------	---------	------------	-----

Kube-OVN	NetworkPolicy	ACL
----------	---------------	-----

• Ingress Allow	2001		
• Egress Allow	2001		
• Default Deny	2000		
Kube-OVN	Kubernetes NetworkPolicy	OVN	ACL

Kube-OVN

• NetworkPolicy	Kubernetes	
• Network Policy API	AdminNetworkPolicy	BaselineAdminNetworkPolicy
• Subnet ACL		
• Security Group		
	OVN ACL	NetworkPolicy Network Policy API

NAMED PORT

NetworkPolicy	Named Port
---------------	------------

```
ports:
- protocol: TCP
  port: http
```

Kube-OVN	Named Port		Named Port							
	Pod	Named Port	NetworkPolicy		Pod A	http	8080	Pod B	http	8081
	port: http	NetworkPolicy								
	Named Port	Pod								

IPBLOCK EXCEPT

NetworkPolicy ipBlock except IP

```
egress:
- to:
  - ipBlock:
      cidr: 10.0.0.0/8
      except:
        - 10.0.1.0/24
        - 10.0.2.0/24
```

OVN except except ACL OVN

except IP

- CIDR
- podSelector namespaceSelector IP

NetworkPolicy

Kube-OVN NetworkPolicy

Warning

NetworkPolicy CPU

ovn.kubernetes.io/acl_log_meter_rate annotation ACL

NetworkPolicy annotation ovn.kubernetes.io/enable_log

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: default-deny-ingress
  namespace: default
  annotations:
    ovn.kubernetes.io/enable_log: "true"
spec:
  podSelector: {}
  policyTypes:
  - Ingress
```

Drop

Pod /var/log/ovn/ovn-controller.log

```
# tail -f /var/log/ovn/ovn-controller.log
2022-07-20T05:55:03.229Z|00394|acl_log(ovn_pinctrl0)|INFO|name="np/default-deny-ingress.default/IPv4/0", verdict=drop, severity=warning, direction=to-lport:
udp,vlan_tci=0x0000,dl_src=00:00:00:21:b7:d1,dl_dst=00:00:00:8d:0b:86,nw_src=10.16.0.10,nw_dst=10.16.0.7,nw_tos=0,nw_ecn=0,nw_ttl=63,tp_src=54343,tp_dst=53
```

IP IP

Kube-OVN v1.13.0 ovn.kubernetes.io/log_acl_actions annotation Allow

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: allow-from-client
  namespace: default
  annotations:
    ovn.kubernetes.io/enable_log: "true"
    ovn.kubernetes.io/log_acl_actions: "allow"
spec:
  podSelector:
    matchLabels:
      app: web
  policyTypes:
    - Ingress
  ingress:
    - from:
      - podSelector:
          matchLabels:
            app: client
```

ovn.kubernetes.io/log_acl_actions

- drop
- allow
- allow, drop

```
# tail -f /var/log/ovn/ovn-controller.log
2024-08-14T09:27:49.590Z|00004|acl_log(ovn_pinctrl0)|INFO|name="np/allow-from-client.default/ingress/IPv4/0", verdict=allow, severity=info, direction=to-
lport: icmp, vlan_tci=0x0000, dl_src=96:7b:b0:2f:a0:1a, dl_dst=a6:e5:1b:c2:1b:f8, nw_src=10.16.0.7, nw_dst=10.
16.0.12, nw_tos=0, nw_ecn=0, nw_ttl=64, nw_frag=no, icmp_type=8, icmp_code=0
```

ovn.kubernetes.io/acl_log_meter_rate annotation

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: allow-from-client
  namespace: default
  annotations:
    ovn.kubernetes.io/enable_log: "true"
    ovn.kubernetes.io/log_acl_actions: "allow"
    ovn.kubernetes.io/acl_log_meter_rate: "100"
spec:
  podSelector:
    matchLabels:
      app: web
  policyTypes:
    - Ingress
  ingress:
    - from:
      - podSelector:
          matchLabels:
            app: client
```

ovn.kubernetes.io/acl_log_meter_rate / 100 100

annotation ovn.kubernetes.io/enable_log false

```
kubectctl annotate networkpolicy -n default allow-from-client ovn.kubernetes.io/enable_log=false --overwrite
```

Kube-OVN

• standard	NetworkPolicy	IP						
• lax		TCP/UDP/SCTP	ICMP	L4	IP	DHCP	UDP	
	NetworkPolicy	annotation						

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: example-policy
  namespace: default
  annotations:
    ovn.kubernetes.io/network_policy_enforcement: "lax"
spec:
  podSelector: {}
  policyTypes:
  - Ingress
```

Kube-OVN

`--np-enforcement`

- [NetworkPolicy](#)

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3.12.2 NetworkPolicy



```
metadata:
  annotations:
    ovn.kubernetes.io/network_policy_for: "ovn,default/net-a,default/net-b"
```

- ovn OVN Provider
- <namespace>/<net-attach-def>
- ovn
- default/net-a
- ovn,default/net-a

Provider

- OVN Provider
- Provider
- <namespace>/<net-attach-def> Kube-OVN Provider
- <nad-name>.<nad-namespace>.ovn

- Pod
- OVN Provider ovn
 - default/net-a
 - default/net-b

- `network_policy_for`
- `ovn net-a net-b`
- `network_policy_for: default/net-a`
- `net-a`
- `network_policy_for: ovn,default/net-b`
- `ovn net-b`

- Provider/ Kubernetes NetworkPolicy
-

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3.12.3

Kubernetes	NetworkPolicy	L3	L4	AdminNetworkPolicy (ANP)					Pod
DNSNameResolver		CoreDNS							
NetworkPolicy		OVN	AddressSet	IP	IP	OVN	AddressSet	DNS	
1. kube-ovn-controller		AdminNetworkPolicy		DNSNameResolver CR					
2. CoreDNS		DNSNameResolver CR		IP	DNSNameResolver status				
3. kube-ovn-controller		DNSNameResolver CR		status	AddressSet				
IP		Deny		Allow		Allow		Deny	

ANP BANP CRD

AdminNetworkPolicyCRD

```
kubectl apply -f https://raw.githubusercontent.com/kubernetes-sigs/network-policy-api/refs/heads/main/config/crd/experimental/policy.networking.k8s.io_adminnetworkpolicies.yaml
kubectl apply -f https://raw.githubusercontent.com/kubernetes-sigs/network-policy-api/refs/heads/main/config/crd/experimental/policy.networking.k8s.io_baselineadminnetworkpolicies.yaml
```

DNSNAMERESOLVER

DNSNameResolver

```
kubectl apply -f https://raw.githubusercontent.com/kubeovn/dnsresolver/refs/heads/main/manifest/crd.yaml
kubectl apply -f https://raw.githubusercontent.com/kubeovn/dnsresolver/refs/heads/main/manifest/rbac.yaml
kubectl apply -f https://raw.githubusercontent.com/kubeovn/dnsresolver/refs/heads/main/manifest/cm.yaml
```

COREDNS

DNSNameResolverCoreDNStagdevkubeovn/dnsresolver releases

```
kubectl set image deployment/coredns coredns=kubeovn/dnsresolver:<release-tag> -n kube-system
```

CoreDNS

```
kubectl get pod -n kube-system -l k8s-app=kube-dns
```

ANP

kube-ovn-controller

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: kube-ovn-controller
spec:
  template:
    spec:
      containers:
        - name: kube-ovn-controller
          args:
            - --enable-anp=true
            - --enable-dns-name-resolver=true
            # ...
```

```
apiVersion: policy.networking.k8s.io/v1alpha1
kind: AdminNetworkPolicy
metadata:
  name: deny-external-domains
spec:
  priority: 55
  subject:
    namespaces:
      matchLabels:
        kubernetes.io/metadata.name: kube-system
  egress:
  - action: Deny
    name: deny-baidu-google
    to:
      - domainNames:
        - '*.baidu.com.'
        - '*.google.com.'
```

priority		
subject	Pod	
egress		
action	Allow Deny Pass	
domainNames	.	

kube-ovn-pinger

```
#
kubectl exec -it -n kube-system kube-ovn-pinger-xxxxx -- ping baidu.com
```

DNS ACL

DNSNameResolver

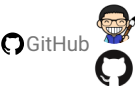
kubectl get dnsnameresolver

NAME	DNS NAME	RESOLVED IPS
anp-deny-external-domains-88dc32ab	*.google.com.	
anp-deny-external-domains-fb3029ce	*.baidu.com.	220.181.7.203

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3.12.4

	Pod	Pod	Kube-OVN IPAM	MAC	IP
•	Pod	IP			
•	Pod	MAC	ARP		
•					

	OVN	Port Security	Pod	Kube-OVN	OVN
•		MAC	IP		
•	OVN	MAC	IP		
•	IPAM				
•	OVN				
	OVN				

Pod `ovn.kubernetes.io/port_security` annotation

```
apiVersion: v1
kind: Pod
metadata:
  name: secure-pod
  annotations:
    ovn.kubernetes.io/port_security: "true"
spec:
  containers:
    - name: nginx
      image: docker.io/library/nginx:alpine
```

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4. KubeVirt

4.1 VM IP

IP	VM	VM	IP
CNI			
• IP VM VM IP			
• IP Node VM IP			
• IP VM IP			
KubeVirt masquerade iptables VM VM IP masquerade bridge			
• Pod IP VM IP Pod IP			
• masquerade iptables bridge			
• masquerade			
• masquerade conntrack			
Kube-OVN KubeVirt bridge managedTap IP VM IP VM annotation VM IP			
Kube-OVN Multus			

4.1.1 IP VM

VM	IP	IP	VM	Kube-OVN	IPAM	VM	VM	IP
bridge		VM	IP					

1. VM

```
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: testvm
spec:
  runStrategy: Always
  template:
    metadata:
      labels:
        kubevirt.io/size: small
        kubevirt.io/domain: testvm
      annotations:
        kubevirt.io/allow-pod-bridge-network-live-migration: "true"
    spec:
      domain:
        devices:
          disks:
            - name: containerdisk
              disk:
                bus: virtio
            - name: cloudinitdisk
              disk:
                bus: virtio
          interfaces:
            - name: default
              bridge: {}
        resources:
          requests:
            memory: 64M
      networks:
        - name: default
          pod: {}
      volumes:
        - name: containerdisk
          containerDisk:
            image: quay.io/kubevirt/cirros-container-disk-demo
        - name: cloudinitdisk
```

```
cloudInitNoCloud:
  userDataBase64: SGkuXG4=
```

1. VM

```
kubecttl get vmi testvm
```

1. VM

```
virtctl restart testvm
```

1. VM

```
virtctl migrate testvm
```

bridge VM IP

4.1.2 IP/Mac

VM	IP/Mac	VM	annotation	VM	IP	KubeVirt	VM	IP/Mac
<nadName>.<nadNamespace>.kubernetes.io/ip_address.<interfaceName>		<nadName>.<nadNamespace>.kubernetes.io/mac_address.<interfaceName>		-	IP/MAC			

```
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: testvm
spec:
  runStrategy: Always
  template:
    metadata:
      labels:
        kubevirt.io/size: small
        kubevirt.io/domain: testvm
    annotations:
      ovn.kubernetes.io/ip_address: 10.16.0.15 #(1)
      ovn.kubernetes.io/mac_address: 00:00:00:53:68:B6 #(2)
      kubevirt.io/allow-pod-bridge-network-live-migration: "true"
  spec:
    domain:
      devices:
        disks:
          - name: containerdisk
            disk:
              bus: virtio
          - name: cloudinitdisk
            disk:
              bus: virtio
        interfaces:
          - name: default
            bridge: {}
        resources:
          requests:
            memory: 64M
    networks:
      - name: default
        pod: {}
    volumes:
      - name: containerdisk
        containerDisk:
          image: quay.io/kubevirt/cirros-container-disk-demo
      - name: cloudinitdisk
        cloudInitNoCloud:
          userDataBase64: SGkuXG4=
```

- 1. IP
- 2. Mac

 **Warning**

KubeVirt VM API	spec.template.spec.domain.devices.interfaces.macAddress	Mac	Mac	Kube-OVN
	annotation	Mac	Mac	

4.1.3 VM IP

- Kube-OVNVM IPIPVM
- VM IP
1. VM AnnotationIP

2. virtctl restart <vm name>VMIP

4.1.4

VMIPVMovn.kubernetes.io/logical_switch

```
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: testvm
spec:
  runStrategy: Always
  template:
    metadata:
      labels:
        kubevirt.io/size: small
        kubevirt.io/domain: testvm
    annotations:
      ovn.kubernetes.io/logical_switch: subnet1 #(1)
      kubevirt.io/allow-pod-bridge-network-live-migration: "true"
  spec:
    domain:
      devices:
        disks:
          - name: containerdisk
            disk:
              bus: virtio
          - name: cloudinitdisk
            disk:
              bus: virtio
        interfaces:
          - name: default
            bridge: {}
        resources:
          requests:
            memory: 64M
    networks:
      - name: default
        pod: {}
    volumes:
      - name: containerdisk
        containerDisk:
          image: quay.io/kubevirt/cirros-container-disk-demo
      - name: cloudinitdisk
        cloudInitNoCloud:
          userDataBase64: SGkuXG4=
```

1. 🧑 VM

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4.1.5

4.2

KubeVirt Clone

Label Annotation

Kube-OVN Annotation

IP MAC

4.2.1 Annotation

VirtualMachineClone

Kube-OVN

Annotation

```
kind: VirtualMachineClone
apiVersion: "clone.kubevirt.io/v1beta1"
metadata:
  name: testclone
spec:
  source:
    apiGroup: kubevirt.io
    kind: VirtualMachine
    name: vm-source
  target:
    apiGroup: kubevirt.io
    kind: VirtualMachine
    name: vm-target
  template:
    annotationFilters:
      - "ovn.kubernetes.io/*"
```

4.2.2

Note

patches KubeVirt 1.6

IP

```
kind: VirtualMachineClone
apiVersion: "clone.kubevirt.io/v1beta1"
metadata:
  name: testclone
spec:
  source:
    apiGroup: kubevirt.io
    kind: VirtualMachine
    name: vm-source
  target:
    apiGroup: kubevirt.io
    kind: VirtualMachine
    name: vm-target
  patches:
    - '{"op": "replace", "path": "/spec/template/metadata/annotations/ovn.kubernetes.io~1ip_address", "value": "10.16.0.15"}'
```

```
kind: VirtualMachineClone
apiVersion: "clone.kubevirt.io/v1beta1"
metadata:
  name: testclone
spec:
  source:
    apiGroup: kubevirt.io
    kind: VirtualMachine
    name: vm-source
  target:
    apiGroup: kubevirt.io
    kind: VirtualMachine
    name: vm-target
  patches:
    - '{"op": "remove", "path": "/spec/template/metadata/annotations/ovn.kubernetes.io~1ip_address"}'
```

Annotation

KubeVirt Clone API

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4.2.3

4.3

KubeVirt	Bridge	DHCP	virt-launcher	KubeVirt	IPv4	DHCP	Bridge	KubeVirt VM	RA	IPv6	Kube-
OVN	DHCP RA	KubeVirt	DHCP/RA								
KubeVirt 1.4.0		Network Binding Plugin	Bridge	managedTap		KubeVirt	DHCP	managedTap		Kube-OVN	
DHCP/RA	VM										

4.3.1 DHCP

Kube-OVN Subnet DHCP IPv6 RA YAML

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: dual-stack-subnet
spec:
  cidrBlock: "10.244.0.0/16,fd00:10:244::/64"
  enableDHCP: true
  enableIPv6RA: true
```

4.3.2 managedTap

KubeVirt managedTap Network Binding Plugin:

```
# kubectl patch kubevirt -n kubevirt kubevirt --type=json -p=\
'[{ "op": "add", "path": "/spec/configuration/network", "value": {
  "binding": {
    "managedtap": {
      "domainAttachmentType": "managedTap"
    }
  }
}]'
```

4.3.3 managedTap

```
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: dual-stack-vm
  namespace: default
spec:
  running: false
  template:
    spec:
      domain:
        devices:
          interfaces:
            - name: default
              binding:
                name: managedtap
      networks:
        - name: default
      pod: {}
```

VM DHCP IPv6 RA IPv4/IPv6

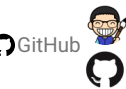
PDF

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Support

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4.3.4

4.4

KubeVirt

- KubeVirt Bridge
- KubeVirt
- IP
-

Kube-OVN

0.5

TCP

4.4.1

VM Spec `kubevirt.io/allow-pod-bridge-network-live-migration: "true"` annotation Kube-OVN

1. VM

```
kubectl apply -f - <<EOF
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: testvm
spec:
  runStrategy: Always
  template:
    metadata:
      labels:
        kubevirt.io/size: small
        kubevirt.io/domain: testvm
      annotations:
        kubevirt.io/allow-pod-bridge-network-live-migration: "true"
    spec:
      domain:
        devices:
          disks:
            - name: containerdisk
              disk:
                bus: virtio
            - name: cloudinitdisk
              disk:
                bus: virtio
          interfaces:
            - name: default
              bridge: {}
          resources:
            requests:
              memory: 64M
        networks:
          - name: default
            pod: {}
        volumes:
          - name: containerdisk
            containerDisk:
              image: quay.io/kubevirt/cirros-container-disk-demo
          - name: cloudinitdisk
            cloudInitNoCloud:
              userDataBase64: SGkuXG4=
EOF
```

1. SSH

```
# password: gocubsgo
virtctl ssh cirros@testvm
ping 8.8.8.8
```

1.

```
virtctl migrate testvm
```

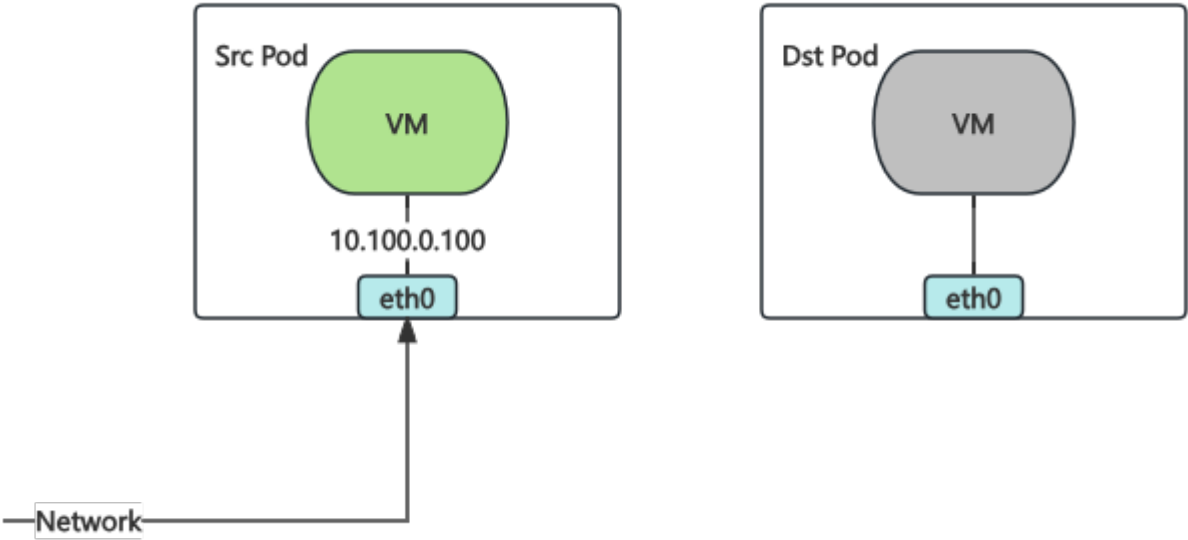
VM SSH ping

4.4.2

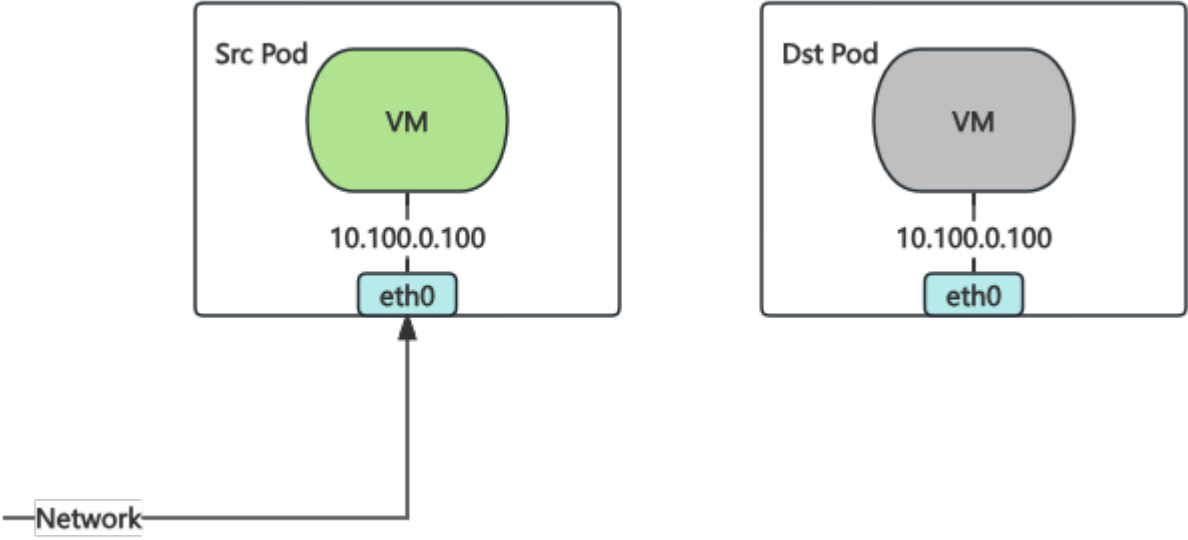
Kube-OVN [Live migration - Reducing downtime with multichassis port bindings](#)

IP

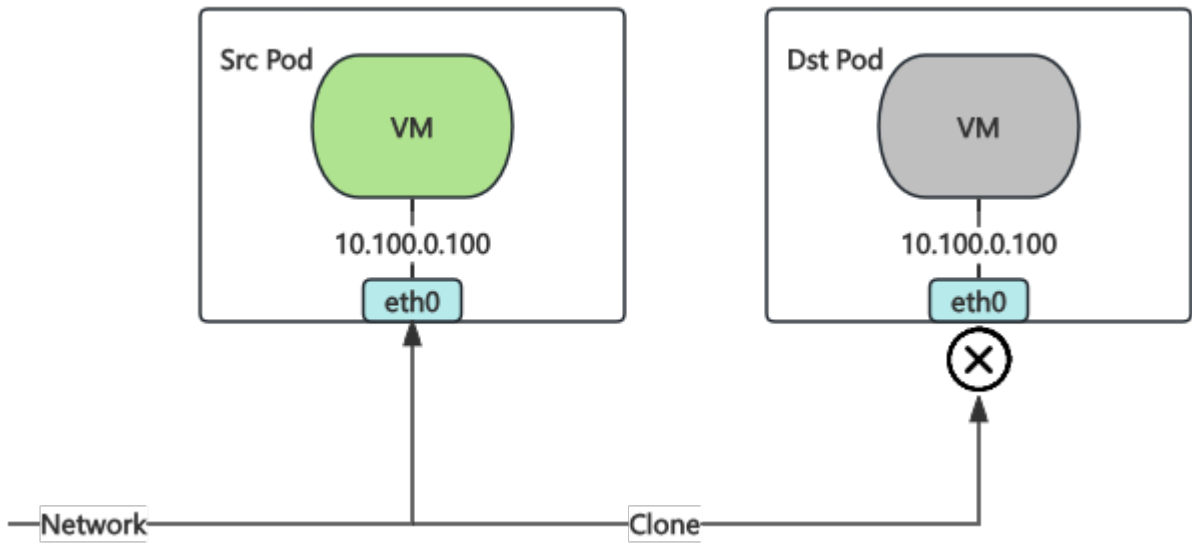
1. KubeVirt Pod



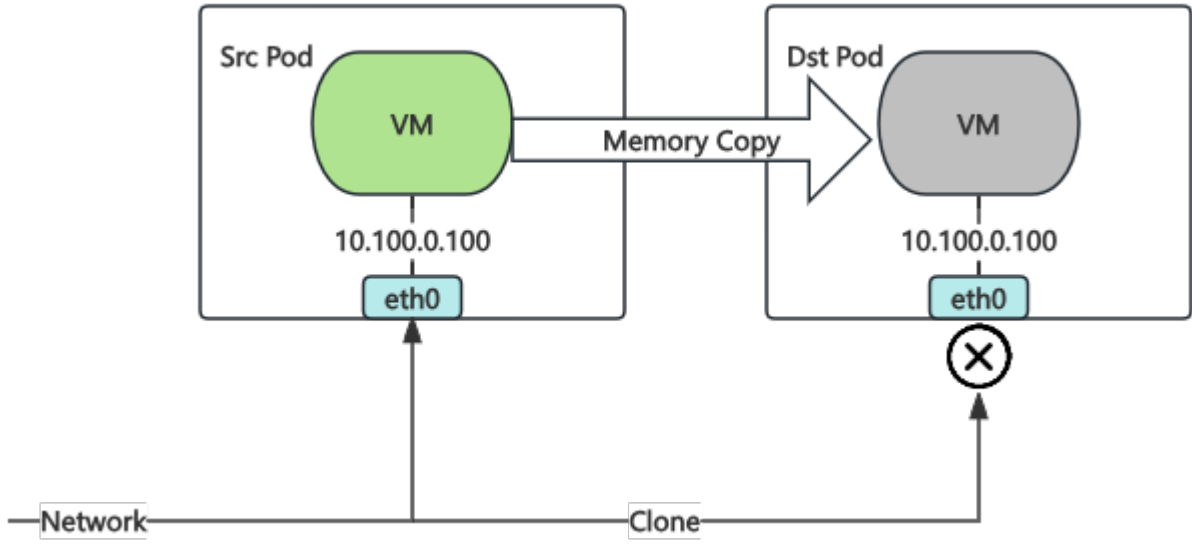
1. Kube-OVN Pod Pod Pod



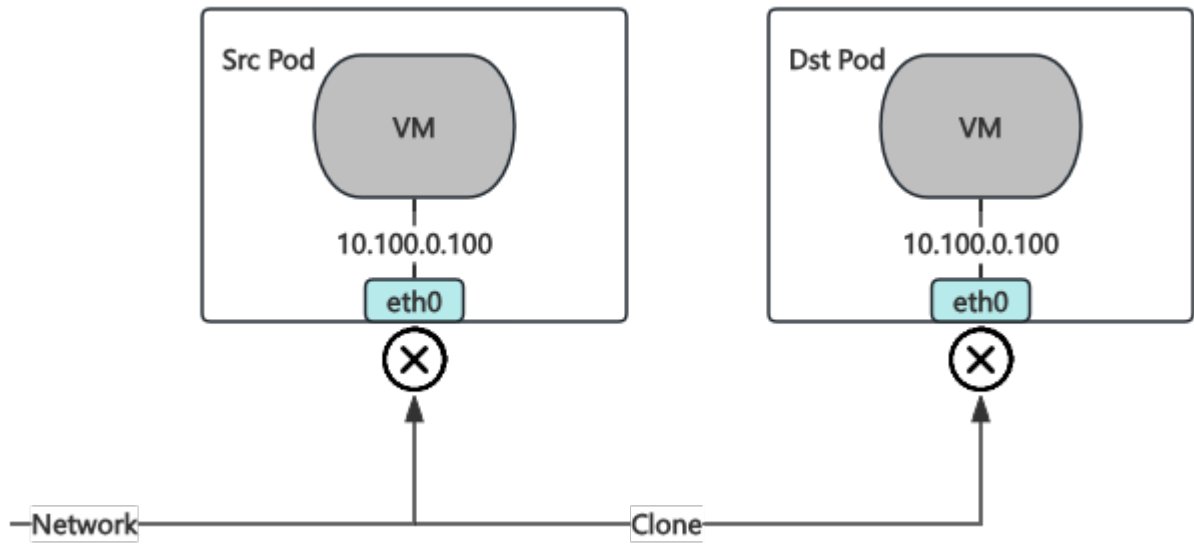
1. Kube-OVN Pod Pod Pod Pod Pod



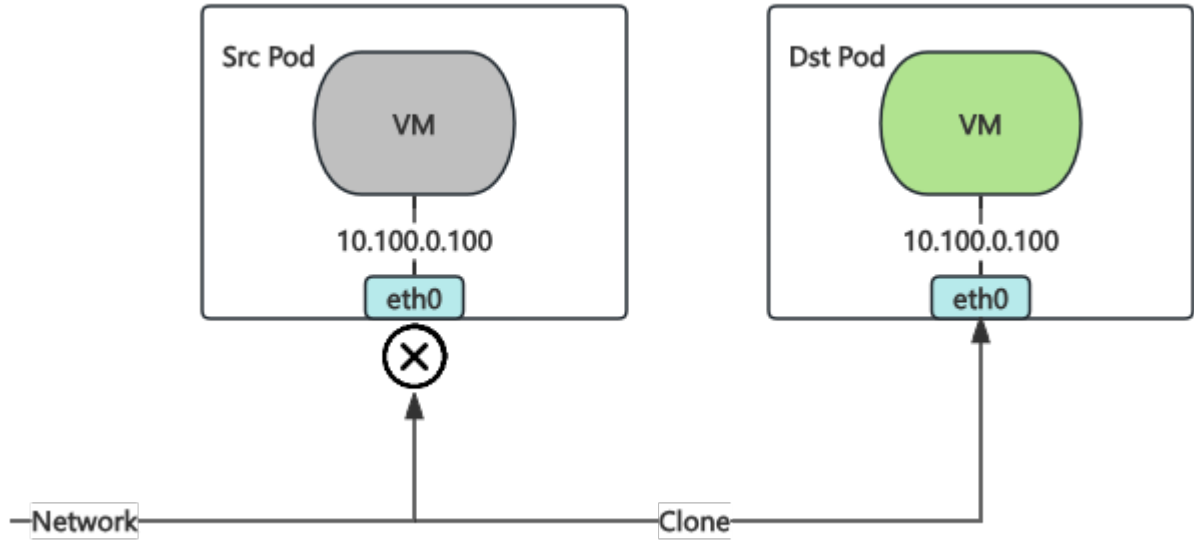
1. KubeVirt VM



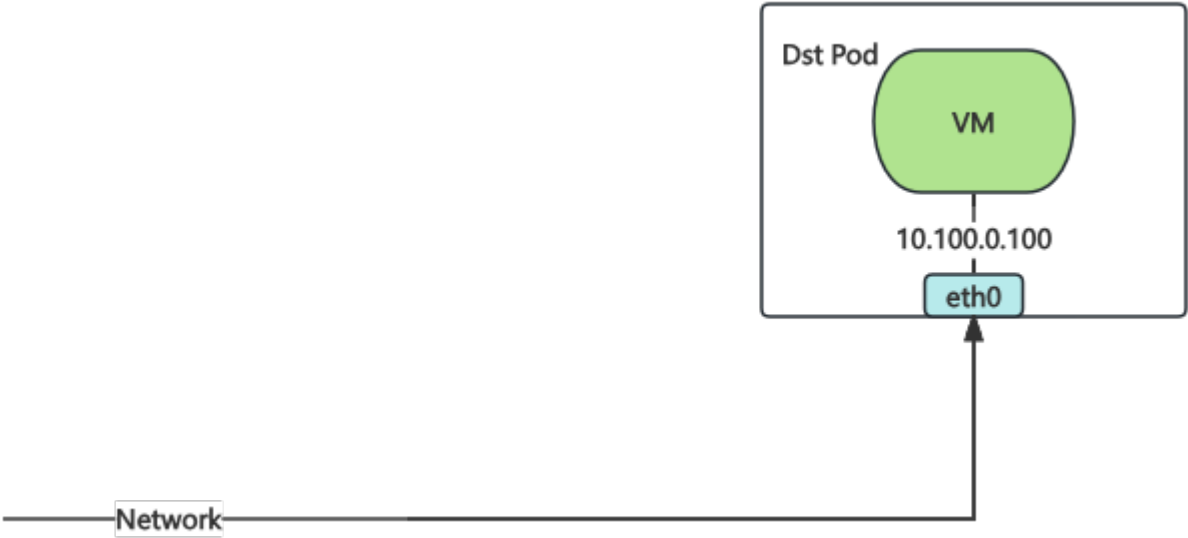
1. KubeVirt Pod Pod



1. KubeVirt Pod libvirt RARP Pod Pod



1. KubeVirt Pod Kube-OVN Watch Migration CR



5 6 libvirt RARP 0.5 TCP

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4.4.3

4.5

Kube-OVN

Multus Dynamic Networks Controller

KubeVirt

v1.4.0

VM

4.5.1

Thick

Multus

```
kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/multus-cni/refs/heads/master/deployments/multus-daemonset-thick.yaml
```

Multus Dynamic Networks Controller

```
kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/multus-dynamic-networks-controller/refs/heads/main/manifests/dynamic-networks-controller.yaml
```

4.5.2

NetworkAttachmentDefinition

provider

ovn

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: attachnet
  namespace: default
spec:
  config: '{
    "cniVersion": "0.3.0",
    "type": "kube-ovn",
    "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
    "provider": "attachnet.default.ovn"
  }'
```

- spec.config.type: kube-ovn CNI Kube-OVN
- server_socket: Kube-OVN socket /run/openvswitch/kube-ovn-daemon.sock
- provider: NetworkAttachmentDefinition <name>.<namespace>.ovn , Kube-OVN Subnet ovn

Kube-OVN Subnet

Kube-OVN

provider

NetworkAttachmentDefinition

<name>.<namespace>.ovn

ovn

Kube-OVN

Subnet

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: attachnet
spec:
  protocol: IPv4
  provider: attachnet.default.ovn
  cidrBlock: 172.17.0.0/16
  gateway: 172.17.0.1
  excludeIps:
    - 172.17.0.0..172.17.0.10
```

4.5.3 VM

yaml

VM

```
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: vm-fedora
spec:
  runStrategy: Always
  template:
    spec:
      domain:
```

```

    devices:
      disks:
        - disk:
            bus: virtio
            name: containerdisk
      interfaces:
        - masquerade: {}
          name: defaultnetwork
      rng: {}
    resources:
      requests:
        memory: 1024M
    networks:
      - name: defaultnetwork
        pod: {}
    terminationGracePeriodSeconds: 0
    volumes:
      - containerDisk:
          image: quay.io/kubevirt/fedora-with-test-tooling-container-disk:devel
          name: containerdisk

```

VM Spec

```

apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: vm-fedora
template:
  spec:
    domain:
      devices:
        interfaces:
          - name: defaultnetwork
            masquerade: {}
          # new interface
          - name: dyniface1
            bridge: {}
        networks:
          - name: defaultnetwork
            pod: {}
          # new network
          - name: dyniface1
        multus:
          networkName: attachnet

```

interface state absent

```

apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: vm-fedora
template:
  spec:
    domain:
      devices:
        interfaces:
          - name: defaultnetwork
            masquerade: {}
          # set the interface state to absent
          - name: dyniface1
            state: absent
            bridge: {}
        networks:
          - name: defaultnetwork
            pod: {}
          - name: dyniface1
        multus:
          networkName: attachnet

```


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4.5.4

4.6 DHCP

managedTapDHCP

SR-IOVIP

DPDKKube-OVN

KubeVirtDHCP

DHCPDHCPv6, IPv6RA, DNS

Kube-OVNTFTP

OVNDHCP

PodDHCP

DHCP

KubeVirt

Warning

bridgeKubeVirtDHCPKube-OVNDHCPKube-OVNDHCPKube-OVNbridgemanagedTapmanagedTap

4.6.1 DHCP

DHCP

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: sn-dual
spec:
  cidrBlock: "10.0.0.0/24,240e::a00/120"
  default: false
  disableGatewayCheck: true
  disableInterConnection: false
  excludeIps:
    - 10.0.0.1
    - 240e::a01
  gateway: 10.0.0.1,240e::a01
  gatewayNode: ''
  gatewayType: distributed
  natOutgoing: false
  private: false
  protocol: Dual
  provider: ovn
  vpc: vpc-test
  enableDHCP: true
  dhcpV4Options: "lease_time=3600,router=10.0.0.1,server_id=169.254.0.254,server_mac=00:00:00:2E:2F:B8"
  dhcpV6Options: "server_id=00:00:00:2E:2F:C5"
  enableIPv6RA: true
  ipv6RAConfigs: "address_mode=dhcpv6_stateful,max_interval=30,min_interval=5,send_periodic=true"
```

enableDHCP:

DHCP

dhcpV4Options, dhcpV6Options:

ovn-nb

DHCP

DHCP Options

"lease_time=3600, router=\$ipv4_gateway, server_id=169.254.0.254, server_mac=\$random_mac"

server_id=\$random_mac

enableIPv6RA:

DHCPv6

ipv6RAConfigs

ovn-nb

Logical_Router_Port

Logical Router Port

address_mode=dhcpv6_stateful, max_interval=30, min_interval=5, send_periodic=true

4.6.2 Pod DHCP

DHCP

Kube-OVN

Pod

Pod

DHCP

Pod

DHCP

DHCP

enableDHCP

```
# provider "ovn"
ovn.kubernetes.io/dhcp-v4-options: "lease_time=3600,router=10.0.0.1,dns_server=8.8.8.8"
ovn.kubernetes.io/dhcp-v6-options: "server_id=00:00:00:00:00:01"

# provider "net1.ns1.ovn"
net1.ns1.ovn.kubernetes.io/dhcp-v4-options: "lease_time=7200"
```

key

<provider>.kubernetes.io/dhcp-v4-options

<provider>.kubernetes.io/dhcp-v6-options

<provider>

provider

ovn

Multus

provider

<net-attach-def-name>.<namespace>.ovn

Pod DHCPv4

```
apiVersion: v1
kind: Pod
metadata:
  name: dhcp-pod
  annotations:
    ovn.kubernetes.io/dhcp-v4-options: "lease_time=3600,router=10.0.0.1,dns_server=8.8.8.8"
    ovn.kubernetes.io/dhcp-v6-options: "server_id=00:00:00:00:00:01"
spec:
  containers:
    - name: test
      image: docker.io/library/nginx:alpine
```

Multus provider DHCP

```
apiVersion: v1
kind: Pod
metadata:
  name: multi-nic-dhcp-pod
  annotations:
    k8s.v1.cni.cncf.io/networks: '[{"name": "net1", "namespace": "ns1"}]'
    ovn.kubernetes.io/dhcp-v4-options: "lease_time=3600,router=10.0.0.1"
    net1.ns1.ovn.kubernetes.io/dhcp-v4-options: "lease_time=7200,router=10.0.1.1"
spec:
  containers:
    - name: test
      image: docker.io/library/nginx:alpine
```

DHCP

1. Pod <provider>.kubernetes.io/dhcp-v4-options <provider>.kubernetes.io/dhcp-v6-options Pod DHCP
2. Subnet CRD dhcpV4Options dhcpV6Options DHCP
- Pod DHCP Pod DHCP

Note

Pod DHCP DHCP Pod

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Support

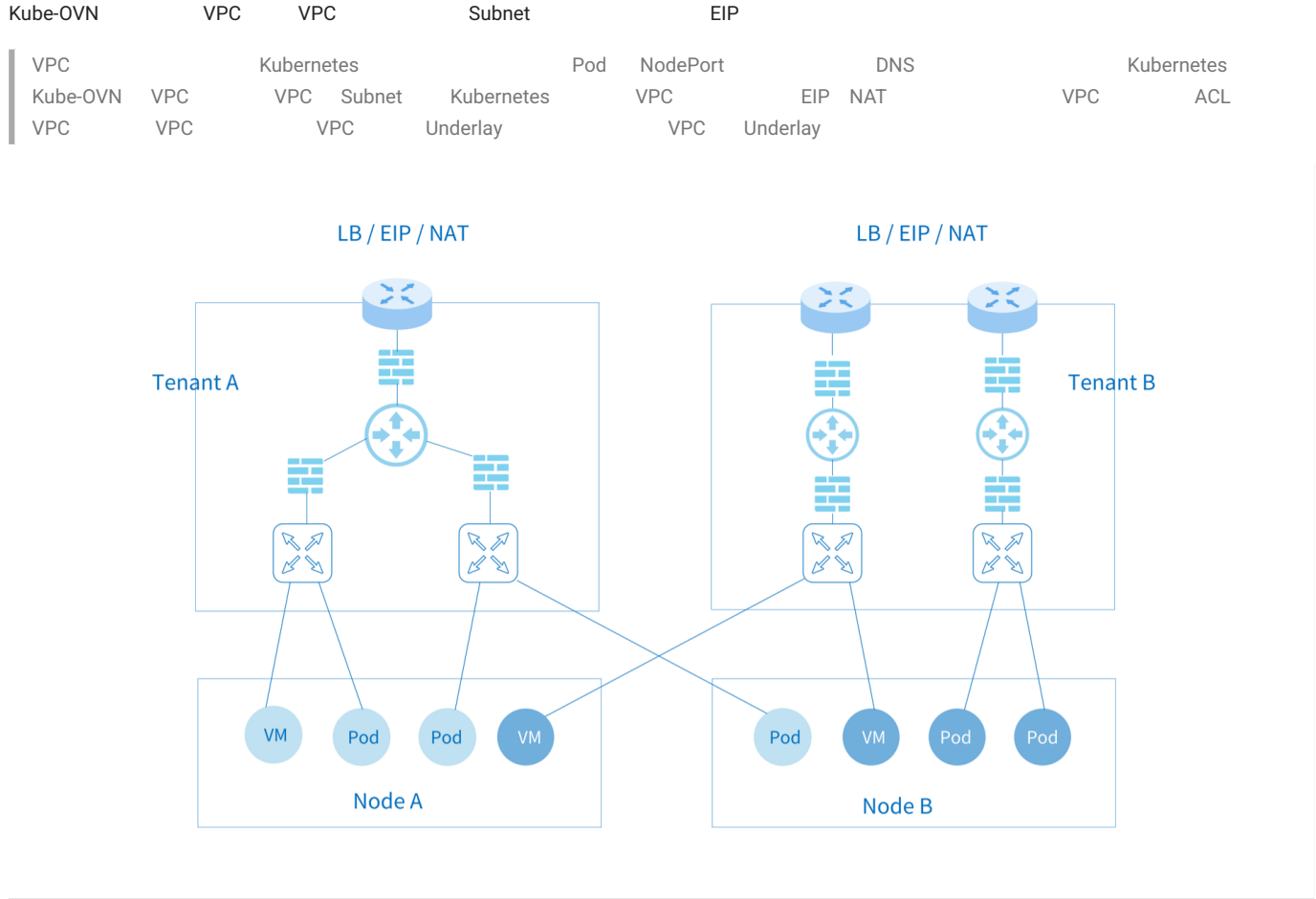
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5. VPC

5.1 VPC



5.1.1



5.1.2 VPC

VPC

```
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: test-vpc-1
spec:
  namespaces:
    - ns1
---
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: test-vpc-2
spec:
  namespaces:
    - ns2
```

namespaces

Namespace

VPC

VPC CIDR:

```
kind: Subnet
apiVersion: kubeovn.io/v1
metadata:
  name: net1
spec:
  vpc: test-vpc-1
  cidrBlock: 10.0.1.0/24
  protocol: IPv4
  namespaces:
    - ns1
---
kind: Subnet
apiVersion: kubeovn.io/v1
metadata:
  name: net2
spec:
  vpc: test-vpc-2
  cidrBlock: 10.0.1.0/24
  protocol: IPv4
  namespaces:
    - ns2
```

Namespace Pod:

```
apiVersion: v1
kind: Pod
metadata:
  namespace: ns1
  name: vpc1-pod
spec:
  containers:
    - name: vpc1-pod
      image: docker.io/library/nginx:alpine
---
apiVersion: v1
kind: Pod
metadata:
  namespace: ns2
  name: vpc2-pod
spec:
  containers:
    - name: vpc2-pod
      image: docker.io/library/nginx:alpine
```

Pod CIDR VPC Pod

5.1.3 VPC

VPC	VPC	VPC	IP SNAT	DNAT
VPC	Multus-CNI	multus-cni		

 **Note**

VPC	VPC								
VPC	VPC NAT	OVN	Egress Gateway	VPC NAT	Kube-OVN	VPC NAT	Pod	VPC	
Macvlan	Pod	iptables							
OVN	OVN	NAT	OVN	BFD	OVN	OVN			
Egress Gateway	VPC NAT								

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: ovn-vpc-external-network
spec:
  protocol: IPv4
  provider: ovn-vpc-external-network.kube-system
  cidrBlock: 192.168.0.0/24
  gateway: 192.168.0.1 # IP address of the physical gateway
  excludeIps:
```

```
- 192.168.0.1..192.168.0.10
---
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: ovn-vpc-external-network
  namespace: kube-system
spec:
  config: '{
    "cniVersion": "0.3.0",
    "type": "macvlan",
    "master": "eth1",
    "mode": "bridge",
    "ipam": {
      "type": "kube-ovn",
      "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
      "provider": "ovn-vpc-external-network.kube-system"
    }
  }'

      'ipam'      VPC      net1      Kube-OVN      IP

---
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: ovn-vpc-external-network
  namespace: kube-system
spec:
  config: '{
    "cniVersion": "0.3.0",
    "type": "macvlan",
    "master": "eth1",
    "mode": "bridge"
  }'
```

- Subnet Macvlan VPC IP
 - VPC Macvlan NetworkAttachmentDefinition master
 - name
- | Macvlan | L2/L3 | Vlan |
|-------------------------|---------------------------------------|-------------------------------------|
| 1. OpenStack VM | PortSecurity | |
| 2. VMware vSwitch | MAC Address Changes, Forged Transmits | Promiscuous Mode Operation allow |
| 3. Hyper-V | MAC Address Spoofing | |
| 4. AWS GCE | Mac Macvlan | |
| 5. Macvlan Macvlan | | VpcNATGateway Pod Pod |
| 6. Trunk | Macvlan | |

VPC

VPC kube-system ovn-vpc-nat-gw-config ovn-vpc-nat-config ConfigMap image Pod

nodeSelector nodeSelector LoadBalancer Service lb-svc VpcNATGateway Pod VpcNatGateway.spec.selector

```
---
kind: ConfigMap
apiVersion: v1
metadata:
  name: ovn-vpc-nat-config
  namespace: kube-system
data:
  image: 'docker.io/kubeovn/vpc-nat-gateway:v1.16.5'
  nodeSelector: |
    kubernetes.io/hostname: kube-ovn-control-plane
---
kind: ConfigMap
apiVersion: v1
metadata:
  name: ovn-vpc-nat-gw-config
  namespace: kube-system
data:
  enable-vpc-nat-gw: 'true'
```

- image: Pod
- enable-vpc-nat-gw VPC

VPC

```
kind: VpcNatGateway
apiVersion: kubeovn.io/v1
metadata:
  name: gw1
spec:
  vpc: test-vpc-1
  subnet: net1
  lanIp: 10.0.1.254
  selector:
    - "kubernetes.io/hostname: kube-ovn-worker"
    - "kubernetes.io/os: linux"
  externalSubnets:
    - ovn-vpc-external-network
  noDefaultEIP: false
```

- vpc VpcNatGateway VPC
- subnet VPC Subnet VPC Pod lanIp
- lanIp subnet IP VPC Pod IP VPC VpcNatGateway nextHopIP lanIp
- selector VpcNatGateway Pod Kubernetes NodeSelector
- externalSubnets VPC ovn-vpc-external-network
- noDefaultEIP VPC EIP false v1.15 BGP true Underlay
- tolerations VPC
- affinity VPC Pod

VPC-NAT-GW

1. nat gw pod net1 arp ping eip arp ping

EIP

EIP IP VPC DNAT SNAT IP

EIP

```
kind: IptablesEIP
apiVersion: kubeovn.io/v1
metadata:
  name: eip-random
spec:
  natGwDp: gw1
```

EIP

```
kind: IptablesEIP
apiVersion: kubeovn.io/v1
metadata:
  name: eip-static
spec:
  natGwDp: gw1
  v4ip: 192.168.0.100
```

EIP

```
kind: IptablesEIP
apiVersion: kubeovn.io/v1
metadata:
  name: eip-random
spec:
  natGwDp: gw1
  externalSubnet: ovn-vpc-external-network
```

- externalSubnet EIP ovn-vpc-external-network IptablesEIP VpcNatGateway externalSubnets EIP

DNAT

DNATEIPVPCIP

```
kind: IptablesEIP
apiVersion: kubeovn.io/v1
metadata:
  name: eipd01
spec:
  natGwDp: gw1

---
kind: IptablesDnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: dnat01
spec:
  eip: eipd01
  externalPort: '8888'
  internalIp: 10.0.1.10
  internalPort: '80'
  protocol: tcp
```

SNAT

SNATVPCPodEIPSNAT

```
---
kind: IptablesEIP
apiVersion: kubeovn.io/v1
metadata:
  name: eips01
spec:
  natGwDp: gw1
---
kind: IptablesSnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: snat01
spec:
  eip: eips01
  internalCIDR: 10.0.1.0/24
```

IP

IPVPCIP EIP EIPVPCIP VPCIP SNATEIP

```
---
kind: IptablesEIP
apiVersion: kubeovn.io/v1
metadata:
  name: eipf01
spec:
  natGwDp: gw1

---
kind: IptablesFIPRule
apiVersion: kubeovn.io/v1
metadata:
  name: fip01
spec:
  eip: eipf01
  internalIp: 10.0.1.5
```

5.1.4

VCKube-OVN

```
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: test-vpc-1
spec:
  staticRoutes:
    - cidr: 0.0.0.0/0
      nextHopIP: 10.0.1.254
      policy: policyDst
```

```
- cidr: 172.31.0.0/24
  nextHopIP: 10.0.1.253
  policy: policySrc
  routeTable: "rtb1"
```

- policy: policyDst policySrc
- CIDR
- routeTable:

OVN Logical Router Policy

```
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: test-vpc-1
spec:
  policyRoutes:
    - action: drop
      match: ip4.src==10.0.1.0/24 && ip4.dst==10.0.1.250
      priority: 11
    - action: reroute
      match: ip4.src==10.0.1.0/24
      nextHopIP: 10.0.1.252
      priority: 10
```

5.1.5

Kubernetes	Service	Kubernetes	Service IP	VPC	VPC
Kubernetes	Service				

Kube-OVN SwitchLBRule

SwitchLBRule

```
apiVersion: kubeovn.io/v1
kind: SwitchLBRule
metadata:
  name: cjh-slr-nginx
spec:
  vip: 1.1.1.1
  sessionAffinity: ClientIP
  namespace: default
  selector:
    - app: nginx
  ports:
    - name: dns
      port: 8888
      targetPort: 80
      protocol: TCP
```

- vip
- namespace Pod Namespace
- sessionAffinity Service sessionAffinity
- selector Service selector
- ports Service port

```
# kubectl get slr
NAME          VIP      PORT(S)          SERVICE          AGE
vpc-dns-test-cjh2 10.96.0.3 53/UDP, 53/TCP, 9153/TCP kube-system/slr-vpc-dns-test-cjh2 88m
```

5.1.6 vpc-dns

VPC	VPC	VPC	Pod	coredns	VPC	CoreDNS	Service	Kube-OVN	vpc-dns
-----	-----	-----	-----	---------	-----	---------	---------	----------	---------

```

apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: ovn-nad
  namespace: default
spec:
  config: '{
    "cniVersion": "0.3.0",
    "type": "kube-ovn",
    "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
    "provider": "ovn-nad.default.ovn"
  }'

```

vpc-dns ConfigMap

kube-system configmap vpc-dns vpc-dns

```

apiVersion: v1
kind: ConfigMap
metadata:
  name: vpc-dns-config
  namespace: kube-system
data:
  coredns-vip: 10.96.0.3
  enable-vpc-dns: "true"
  nad-name: ovn-nad
  nad-provider: ovn-nad.default.ovn

```

- enable-vpc-dns true false true
- coredns-image dns coredns
- coredns-vip coredns lb vip
- nad-name network-attachment-definitions
- nad-provider provider
- k8s-service-host coredns k8s apiserver ip
- k8s-service-port coredns k8s apiserver port

vpc-dns

```

apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
  labels:
    kubernetes.io/bootstrapping: rbac-defaults
  name: system:vpc-dns
rules:
- apiGroups:
  - ""
  resources:
  - endpoints
  - services
  - pods
  - namespaces
  verbs:
  - list
  - watch
- apiGroups:
  - discovery.k8s.io
  resources:
  - endpointslices
  verbs:
  - list
  - watch
---
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRoleBinding
metadata:
  annotations:
    rbac.authorization.kubernetes.io/autoupdate: "true"
  labels:
    kubernetes.io/bootstrapping: rbac-defaults
  name: vpc-dns
roleRef:
  apiGroup: rbac.authorization.k8s.io
  kind: ClusterRole
  name: system:vpc-dns
subjects:

```

```
- kind: ServiceAccount
  name: vpc-dns
  namespace: kube-system
---
apiVersion: v1
kind: ServiceAccount
metadata:
  name: vpc-dns
  namespace: kube-system
---
apiVersion: v1
kind: ConfigMap
metadata:
  name: vpc-dns-corefile
  namespace: kube-system
data:
  Corefile: |
    .:53 {
      errors
      health {
        lameduck 5s
      }
      ready
      kubernetes cluster.local in-addr.arpa ip6.arpa {
        pods insecure
        fallthrough in-addr.arpa ip6.arpa
      }
      prometheus :9153
      forward . /etc/resolv.conf {
        prefer_udp
      }
      cache 30
      loop
      reload
      loadbalance
    }
  }
```

vpc-dns

```
kind: VpcDns
apiVersion: kubeovn.io/v1
metadata:
  name: test-cjh1
spec:
  vpc: cjh-vpc-1
  subnet: cjh-subnet-1
```

- vpc dns vpc
- subnet dns

```
[root@hci-dev-mst-1 kubeovn]# kubectl get vpc-dns
NAME      ACTIVE  VPC      SUBNET
test-cjh1 false   cjh-vpc-1 cjh-subnet-1
test-cjh2 true    cjh-vpc-1 cjh-subnet-2
```

- ACTIVE: true dns false
- VPC DNS
- VPC vpc-dns VPC subnet vpc-dns true false
- true vpc-dns false vpc-dns

5.1.7

VPC VPC

```
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: test-vpc-1
spec:
  namespaces:
```

```
- ns1
defaultSubnet: test
```

• defaultSubnet VPC

Namespace ovn.kubernetes.io/logical_switch ovn.kubernetes.io/logical_switch Pod

VPC Pod livenessProbe readinessProbe

VPC Pod kubelet VPC Pod Kube-OVN TProxy kubelet VPC Pod

DaemonSet kube-ovn-cni --enable-tproxy=true

```
spec:
  template:
    spec:
      containers:
      - args:
        - --enable-tproxy=true
```

- 1. VPC Pod IP
- 2. tcpSocket httpGet

 PDF

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5.1.8

5.2 VPC Egress Gateway

 **Note**

VPC	VPC								
VPC		VPC NAT	OVN	Egress Gateway	VPC NAT	Kube-OVN	VPC NAT	Pod	VPC
Macvlan		Pod	iptables						
OVN	OVN	NAT		OVN	BFD	OVN	OVN		
Egress Gateway	VPC NAT								

VPC Egress Gateway VPC VPC Pod

- ECMP Active-Active
- BFD <1s
- IPv6
- Namespace Pod
- Node Egress Gateway

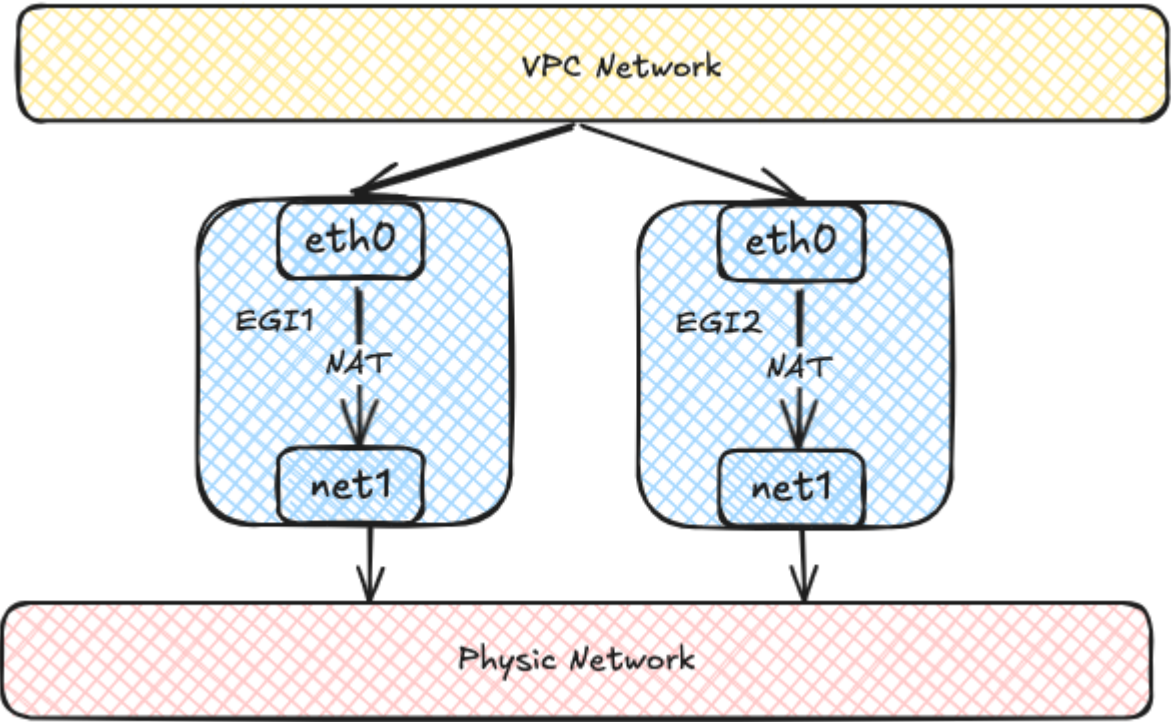
VPC Egress Gateway BGP EVPN/VXLAN [Egress Gateway BGP/EVPN](#)

VPC Egress Gateway

- Macvlan [Underlay](#)
- Gateway Egress IP
- SNAT EIP DNAT
-

5.2.1

Egress Gateway Pod Pod VPC Macvlan Egress Gateway NAT



Egress Gateway

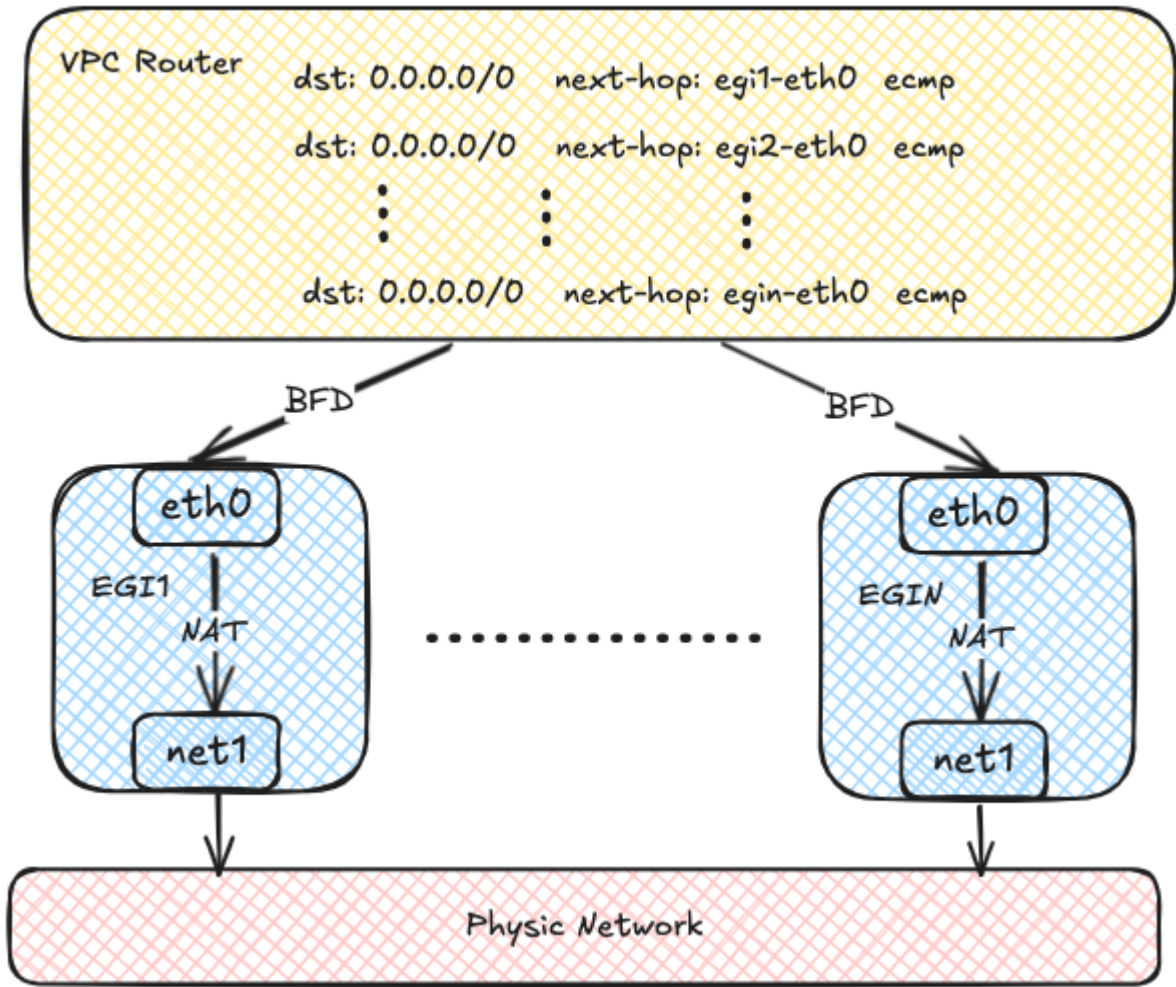
OVN

VPC Pod

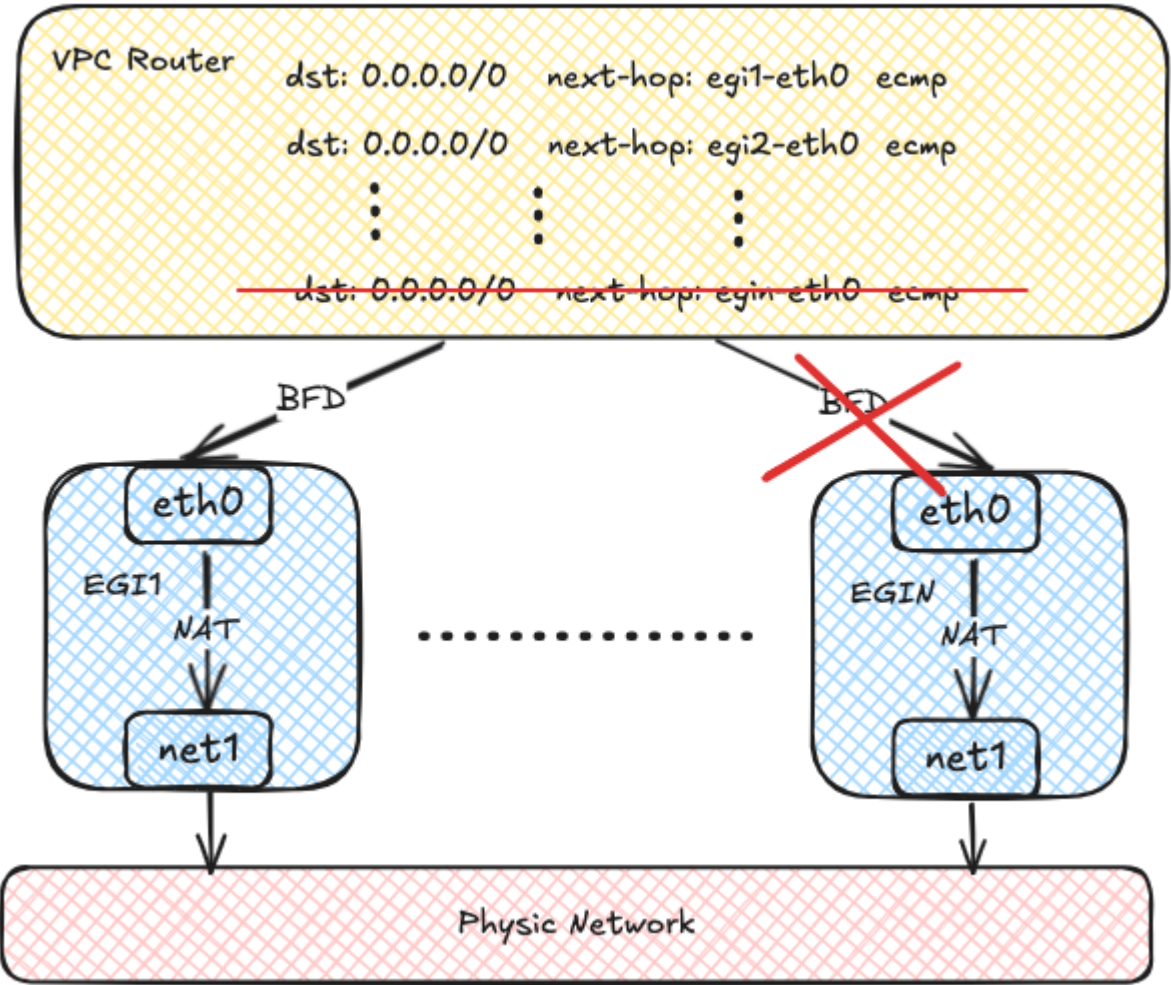
OVN

Egress Gateway

Egress Gateway



OVN BFD Egress Gateway Egress Gateway OVN



5.2.2

VPC Egress Gateway VPC NAT Gateway [Multus-CNI](#)

VPC Egress Gateway ConfigMap

5.2.3

NetworkAttachmentDefinition

VPC Egress Gateway VPC NetworkAttachmentDefinition [macvlan](#) Kube-OVN IPAM

```
apiVersion: k8s.cni.cncf.io/v1
kind: NetworkAttachmentDefinition
metadata:
  name: eth1
  namespace: default
spec:
  config: '{
    "cniVersion": "0.3.0",
    "type": "macvlan",
    "master": "eth1",
    "mode": "bridge",
    "ipam": {
      "type": "kube-ovn",
      "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
      "provider": "eth1.default"
    }
  }'
```

```
---
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: macvlan1
```

```
spec:
  protocol: IPv4
  provider: eth1.default
  cidrBlock: 172.17.0.0/16
  gateway: 172.17.0.1
  excludeIps:
    - 172.17.0.0..172.17.0.10
```

CNI NetworkAttachmentDefinition VPC Egress Gateway

VPC Egress Gateway

VPC Egress Gateway

```
apiVersion: kubeovn.io/v1
kind: VpcEgressGateway
metadata:
  name: gateway1
  namespace: default
spec:
  vpc: ovn-cluster
  replicas: 1
  externalSubnet: macvlan1
  policies:
    - snat: true
      subnets:
        - ovn-default
```

default VPC ovn-cluster gateway1 VPC Egress Gateway ovn-cluster ovn-default 10.16.0.0/16 Pod
macvlan1 SNAT

VPC Egress Gateway

```
$ kubectl get veg gateway1
NAME      VPC      REPLICAS  BFD ENABLED  EXTERNAL SUBNET  PHASE      READY  AGE
gateway1  ovn-cluster  1          false        macvlan1         Completed  true   13s
```

```
kubectl get veg gateway1 -o wide
NAME      VPC      REPLICAS  BFD ENABLED  EXTERNAL SUBNET  PHASE      READY  INTERNAL IPS  EXTERNAL IPS  WORKING NODES  AGE
gateway1  ovn-cluster  1          false        macvlan1         Completed  true   ["10.16.0.12"]  ["172.17.0.11"]  ["kube-ovn-worker"]  82s
```

```
$ kubectl get deployment -l ovn.kubernetes.io/vpc-egress-gateway=gateway1
NAME      READY  UP-TO-DATE  AVAILABLE  AGE
gateway1  1/1    1            1           4m40s

$ kubectl get pod -l ovn.kubernetes.io/vpc-egress-gateway=gateway1 -o wide
NAME                        READY  STATUS    RESTARTS  AGE  IP          NODE           NOMINATED NODE  READINESS GATES
gateway1-b9f8b4448-76lhm    1/1    Running   0          4m48s  10.16.0.12  kube-ovn-worker  <none>          <none>
```

Pod IP iptables

```
$ kubectl exec gateway1-b9f8b4448-76lhm -c gateway -- ip address show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: net1@if13: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 62:d8:71:90:7b:86 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 172.17.0.11/16 brd 172.17.255.255 scope global net1
        valid_lft forever preferred_lft forever
    inet6 fe80::60d8:71ff:fe90:7b86/64 scope link
        valid_lft forever preferred_lft forever
17: eth0@if18: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc noqueue state UP group default
    link/ether 36:7c:6b:c7:82:6b brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 10.16.0.12/16 brd 10.16.255.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::347c:6bff:fec7:826b/64 scope link
        valid_lft forever preferred_lft forever

$ kubectl exec gateway1-b9f8b4448-76lhm -c gateway -- ip route show
default via 172.17.0.1 dev net1
10.16.0.0/16 dev eth0 proto kernel scope link src 10.16.0.12
```

```
172.17.0.0/16 dev net1 proto kernel scope link src 172.17.0.11

$ kubectl exec gateway1-b9f8b4448-76lhm -c gateway -- iptables -t nat -S
-P PREROUTING ACCEPT
-P INPUT ACCEPT
-P OUTPUT ACCEPT
-P POSTROUTING ACCEPT
-A POSTROUTING -s 10.16.0.0/16 -j MASQUERADE --random-fully
```

Gateway Pod

```
$ kubectl exec -ti gateway1-b9f8b4448-76lhm -c gateway -- bash
nobody@gateway1-b9f8b4448-76lhm:/kube-ovn$ tcpdump -i any -nnve icmp and host 172.17.0.1
tcpdump: data link type LINUX_SLL2
tcpdump: listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144 bytes
06:50:58.936528 eth0 In ifindex 17 92:26:b8:9e:f2:1c ethertype IPv4 (0x0800), length 104: (tos 0x0, ttl 63, id 30481, offset 0, flags [DF], proto ICMP (1), length 84)
    10.16.0.9 > 172.17.0.1: ICMP echo request, id 37989, seq 0, length 64
06:50:58.936574 net1 Out ifindex 2 62:d8:71:90:7b:86 ethertype IPv4 (0x0800), length 104: (tos 0x0, ttl 62, id 30481, offset 0, flags [DF], proto ICMP (1), length 84)
    172.17.0.11 > 172.17.0.1: ICMP echo request, id 39449, seq 0, length 64
06:50:58.936613 net1 In ifindex 2 02:42:39:79:7f:08 ethertype IPv4 (0x0800), length 104: (tos 0x0, ttl 64, id 26701, offset 0, flags [none], proto ICMP (1), length 84)
    172.17.0.1 > 172.17.0.11: ICMP echo reply, id 39449, seq 0, length 64
06:50:58.936621 eth0 Out ifindex 17 36:7c:6b:c7:82:6b ethertype IPv4 (0x0800), length 104: (tos 0x0, ttl 63, id 26701, offset 0, flags [none], proto ICMP (1), length 84)
    172.17.0.1 > 10.16.0.9: ICMP echo reply, id 37989, seq 0, length 64
```

OVN Logical Router

VPC

```
$ kubectl ko nbctl lr-policy-list ovn-cluster
Routing Policies
31000 ip4.dst == 10.16.0.0/16 allow
31000 ip4.dst == 100.64.0.0/16 allow
30000 ip4.dst == 172.18.0.2 reroute 100.64.0.3
30000 ip4.dst == 172.18.0.3 reroute 100.64.0.2
30000 ip4.dst == 172.18.0.4 reroute 100.64.0.4
29100 ip4.src == 10.16.0.0/16 reroute 10.16.0.12
29000 ip4.src == $ovn.default.kube.ovn.control.plane_ip4 reroute 100.64.0.2
29000 ip4.src == $ovn.default.kube.ovn.worker2_ip4 reroute 100.64.0.4
29000 ip4.src == $ovn.default.kube.ovn.worker_ip4 reroute 100.64.0.3
```

.spec.replicas

```
$ kubectl scale veg gateway1 --replicas=2
vpcgressgateway.kubeovn.io/gateway1 scaled

$ kubectl get veg gateway1
NAME VPC REPLICAS BFD ENABLED EXTERNAL SUBNET PHASE READY AGE
gateway1 ovn-cluster 2 false macvlan Completed true 39m

$ kubectl get pod -l ovn.kubernetes.io/vpc-egress-gateway=gateway1 -o wide
NAME READY STATUS RESTARTS AGE IP NODE NOMINATED NODE READINESS GATES
gateway1-b9f8b4448-76lhm 1/1 Running 0 40m 10.16.0.12 kube-ovn-worker <none> <none>
gateway1-b9f8b4448-zd4dl 1/1 Running 0 64s 10.16.0.13 kube-ovn-worker2 <none> <none>

$ kubectl ko nbctl lr-policy-list ovn-cluster
Routing Policies
31000 ip4.dst == 10.16.0.0/16 allow
31000 ip4.dst == 100.64.0.0/16 allow
30000 ip4.dst == 172.18.0.2 reroute 100.64.0.3
30000 ip4.dst == 172.18.0.3 reroute 100.64.0.2
30000 ip4.dst == 172.18.0.4 reroute 100.64.0.4
29100 ip4.src == 10.16.0.0/16 reroute 10.16.0.12, 10.16.0.13
29000 ip4.src == $ovn.default.kube.ovn.control.plane_ip4 reroute 100.64.0.2
29000 ip4.src == $ovn.default.kube.ovn.worker2_ip4 reroute 100.64.0.4
29000 ip4.src == $ovn.default.kube.ovn.worker_ip4 reroute 100.64.0.3
```

Egress Gateway IP

externalIPs nodeSelector Egress Gateway Pods Egress IP

```
apiVersion: kubeovn.io/v1
kind: VpcEgressGateway
metadata:
  name: gateway1
  namespace: default
spec:
  vpc: ovn-cluster
  replicas: 2
  externalSubnet: macvlan1
  policies:
    - snat: true
      subnets:
        - ovn-default
```

```
externalIPs:
  - 172.17.0.10
  - 172.17.0.11
nodeSelector:
  - matchLabels:
      kubernetes.io/hostname: kube-ovn-worker
```

IP internalIPPool externalIPPool IPPool IPPool internalSubnet externalSubnet

internalIPs externalIPs

```
spec:
  internalIPPool: veg-internal
  externalIPPool: veg-external
```

BFD

BFD VPC BFD LRP VPC BFD Port

```
apiVersion: kubeovn.io/v1
kind: Vpc
metadata:
  name: vpc1
spec:
  bfdPort:
    enabled: true
    ip: 10.255.255.255
---
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: subnet1
spec:
  vpc: vpc1
  protocol: IPv4
  cidrBlock: 192.168.0.0/24
```

BFD Port OVN LR BFD LRP

```
$ kubectl ko nbctl show vpc1
router 0c1d1e8f-4c86-4d96-88b2-c4171c7ff824 (vpc1)
  port bfd@vpc1
    mac: "8e:51:4b:16:3c:90"
    networks: ["10.255.255.255"]
  port vpc1-subnet1
    mac: "de:c9:5c:38:7a:61"
    networks: ["192.168.0.1/24"]
```

VPC Egress Gateway .spec.bfd.enabled true

```
apiVersion: kubeovn.io/v1
kind: VpcEgressGateway
metadata:
  name: gateway2
  namespace: default
spec:
  vpc: vpc1
  replicas: 2
  internalSubnet: subnet1
  externalSubnet: macvlan
  bfd:
    enabled: true
  policies:
    - snat: true
      ipBlocks:
        - 192.168.0.0/24
```

VPC Egress Gateway

```
$ kubectl get veg gateway2 -o wide
NAME      VPC  REPLICAS  BFD  ENABLED  EXTERNAL SUBNET  PHASE  READY  INTERNAL IPS  EXTERNAL IPS  WORKING
gateway2  vpc1  2         true  macvlan  Completed  true   ["192.168.0.2","192.168.0.3"]  ["172.17.0.13","172.17.0.14"]  ["kube-ovn-worker","kube-ovn-worker2"]  58s

$ kubectl get pod -l ovn.kubernetes.io/vpc-egress-gateway=gateway2 -o wide
NAME      READY  STATUS  RESTARTS  AGE  IP  NODE  NOMINATED NODE  READINESS GATES
gateway2-fcc6b8b87-8lgvx  1/1    Running  0         2m18s  192.168.0.3  kube-ovn-worker2  <none>  <none>
gateway2-fcc6b8b87-wmmw6  1/1    Running  0         2m18s  192.168.0.2  kube-ovn-worker  <none>  <none>

$ kubectl ko nbctl lr-route-list vpc1
IPv4 Routes
Route Table <main>:
```

```
192.168.0.0/24      192.168.0.2 src-ip ecmp ecmp-symmetric-reply bfd
192.168.0.0/24      192.168.0.3 src-ip ecmp ecmp-symmetric-reply bfd

$ kubectl ko nbctl list bfd
 _uuid      : 223ede10-9169-4c7d-9524-a546e24bfab5
detect_mult : 3
dst_ip      : "192.168.0.2"
external_ids : {af="4", vendor=kube-ovn, vpc-egress-gateway="default/gateway2"}
logical_port : "bfd@vpc1"
min_rx      : 1000
min_tx      : 1000
options     : {}
status      : up

 _uuid      : b050c75e-2462-470b-b89c-7bd38889b758
detect_mult : 3
dst_ip      : "192.168.0.3"
external_ids : {af="4", vendor=kube-ovn, vpc-egress-gateway="default/gateway2"}
logical_port : "bfd@vpc1"
min_rx      : 1000
min_tx      : 1000
options     : {}
status      : up
```

Pod BFD

```
$ kubectl exec gateway2-fcc6b8b87-8lgvx -c bfdd -- bfdd-control status
There are 1 sessions:
Session 1
id=1 local=192.168.0.3 (p) remote=10.255.255.255 state=Up

$ kubectl exec gateway2-fcc6b8b87-wmww6 -c bfdd -- bfdd-control status
There are 1 sessions:
Session 1
id=1 local=192.168.0.2 (p) remote=10.255.255.255 state=Up
```

VPC BFD PORT

enabled	boolean	false	BFD Port		true
ip	string	-	BFD Port	IP IPv6	169.255.255.2 169.255.255.2
nodeSelector	object	-	BFD Port	BFD Port	-
				OVN HA Chassis Group Active/ Backup Active nodeSelector Kube-OVN	
				kubectl ko nbctl list ha_chassis_group OVN HA Chassis Group	
nodeSelector.matchLabels	dict/map	-			-
nodeSelector.matchExpressions	object array	-			-

VPC EGRESS GATEWAY

Spec

vpc	string	VPC cluster	ovn-	VPC	vpc1
replicas	integer/int32	1			2
prefix	string	-		Deployment	veg-
image	string	-		Deployment	docker.io/kubeovn/kube-
internalSubnet	string	VPC		VPC	subnet1
externalSubnet	string	-			ext1
internalIPs	string array	-		VPC IPv6 IP IP <replicas> + 1 Pod	10.16.0.101 / fd00::11
externalIPs	string array	-		IP IPv6 IP <replicas> + 1 Pod	10.16.0.101 / fd00::11
internalIPPool	string	-		IPPool IPPool internalSubnet internalIPs	veg-internal
externalIPPool	string	-		IPPool IPPool externalSubnet externalIPs	veg-external
bfd	object	-		BFD	-
policies	object array	-		Egress selectors	-
selectors	object array	-		Namespace Selector Pod Selector Egress Pod SNAT/ MASQUERADE policies	-
nodeSelector	object array	-		Deployment/Pod	-
trafficPolicy	string	Cluster		Cluster / Local BFD Local Egress VPC	Local

		Egress Gateway VPC Egress Gateway Egress	
bandwidth	object	-	-
bandwidth.ingress	integer string	-	VPC 1024 / 1Gi
bandwidth.egress	integer string	-	VPC 1024 / 1Gi

MbpsM Mi G Giquantitybit/sMbps100M100Mi1G1Gi10010510001074 Mbps

9223372036854 Mbps

ingressegress

```
spec:
  bandwidth:
    ingress: 1024
    egress: 1Gi
```



0 9223372036854MbpsKube-OVNvpc-egress-gateways.kubeovn.io CRDx-kubernetes-int-or-

string: true

HelmChartcrds/CRDkube-ovn-v2 ChartKUBE_OVN_VERSIONKube-OVNCRD bundle

kubectl apply -f "https://raw.githubusercontent.com/kubeovn/kube-ovn/\${KUBE_OVN_VERSION}/charts/kube-ovn-v2/crds/kube-ovn-crd.yaml"

quantityMbps

BFD

enabled	boolean	false	BFD	true
minRX	integer/int32	1000	BFD minRX ms	500
minTX	integer/int32	1000	BFD minTX ms	500
multiplier	integer/int32	3	BFD multiplier	1

Egress

snat	boolean	false	SNAT/ MASQUERADE	true
ipBlocks	string array	-	Gateway IP IPv6	192.168.0.1 / 192.168.0.0/24
subnets	string array	-	Gateway VPC IPv6	subnet1

Selectors

namespaceSelector	object	-	Namespace	-
			Namespace	
namespaceSelector.matchLabels	dict/map	-		-
namespaceSelector.matchExpressions	object array	-		-
podSelector	object	-	Pod Pod	-
podSelector.matchLabels	dict/map	-		-
podSelector.matchExpressions	object array	-		-

matchLabels	dict/map	-	-
matchExpressions	object array	-	-
matchFields	object array	-	-

Status

ready	boolean	Gateway	true
phase	string	Gateway	Pending / Processing / Completed
internalIPs	string array	VPC IP	-
externalIPs	string array	IP	-
workload	object		-
workload.apiVersion	string	API	apps/v1
workload.kind	string		Deployment
workload.name	string		gateway1
workload.nodes	string array		-
conditions	object array	-	-

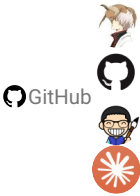
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5.2.4

5.3 Egress Gateway BGP EVPN



BGP EVPN

L3VPN

L2VPN

FRR

Pod

BGP BFD

5.3.1

VPC Egress Gateway

BGP EVPN/VXLAN

Egress Gateway Pod

FRR Free Range Routing

BGP EVPN

Gateway init

BgpConf EvpnConf

FRR

BGP Egress Gateway

BGP

EVPN init

Pod

Linux VRF bridge VXLAN

L2VPN EVPN

VXLAN

Egress Gateway Pod

Kubernetes

Geneve

BGP

kube-ovn-speaker

Pod/ /Service

5.3.2

BGP/EVPN

VPC Egress Gateway

Multus-CNI

BGP

EVPN

BGP

5.3.3

BgpConf

BgpConf

BGP

```
apiVersion: kubeovn.io/v1
kind: BgpConf
metadata:
  name: bgp-conf-6502
spec:
  localASN: 65002
  peerASN: 65001
  neighbours:
    - 10.0.1.1
  holdTime: 90s
  keepaliveTime: 30s
  connectTime: 10s
  ebgpMultiHop: true
```

AS

65002

AS

65001

BGP

10.0.1.1

EBGP Multi-Hop

EvpnConf

EvpnConf

EVPN

EvpnConf

BgpConf

```
apiVersion: kubeovn.io/v1
kind: EvpnConf
metadata:
  name: evpn-conf-1016
spec:
  vni: 1016
  routeTargets:
    - "65000:1016"
```

VNI 1016 EVPN Route Target 65000:1016

BGP/EVPN VPC Egress Gateway

VPC Egress Gateway NetworkAttachmentDefinition

VpcEgressGateway .spec bgpConf evpnConf

```
apiVersion: kubeovn.io/v1
kind: VpcEgressGateway
metadata:
  name: gateway1
  namespace: default
spec:
  vpc: ovn-cluster
  replicas: 1
  externalIPs:
    - 10.0.1.13
  internalIPs:
    - 10.16.0.13
  externalSubnet: macvlan1
  bgpConf: bgp-conf-6502
  evpnConf: evpn-conf-1016
  policies:
    - snat: false
      subnets:
        - ovn-default
    - snat: false
      ipBlocks:
        - 10.17.0.0/16
```

	BGP	EVPN	VPC Egress Gateway	Gateway Pod	FRR	BGP EVPN	VXLAN	BGP/EVPN
snat	false							

BGP EVPN

BGP EVPN/VXLAN bgpConf evpnConf FRR BGP IPv4 Unicast

```
apiVersion: kubeovn.io/v1
kind: VpcEgressGateway
metadata:
  name: gateway-bgp-only
  namespace: default
spec:
  vpc: ovn-cluster
  replicas: 1
  externalSubnet: macvlan1
  bgpConf: bgp-conf-6502
  policies:
    - snat: true
      subnets:
        - ovn-default
```

5.3.4

BgpConf

localASN	uint32	-	AS	65002
peerASN	uint32	-	AS	65001
routerId	string	Pod IP	BGP Router ID	10.0.1.13
neighbours	string array	-	BGP IP	10.0.1.1
password	string	-	BGP	secret
holdTime	string (duration)	-	BGP Hold	90s
keepaliveTime	string (duration)	-	BGP Keepalive	30s
connectTime	string (duration)	-	BGP	10s
ebgpMultiHop	boolean	false	EBGP Multi-Hop	true

EvpnConf

vni	uint32	-	VXLAN Network Identifier	1016
routeTargets	string array	-	Route Target	65000:1016

VpcEgressGateway

VpcEgressGateway .spec BGP/EVPN VpcEgressGateway Egress Gateway

bgpConf	string	-	BgpConf Egress Gateway BGP	bgp-conf-6502
evpnConf	string	-	EvpnConf	evpn-conf-1016
			bgpConf Egress Gateway EVPN/VXLAN	

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5.3.5

5.4 VPC QoS

Kube-OVN QoSPolicy CRD VPC

5.4.1

QoSPolicy priority QoS

EIP QoS		1		EIP
NATGW	QoS	2	NATGW	IP
NATGW net1	QoS	3		

- shared=false QoS shared=true
- EIP QoS NATGW QoS

5.4.2 EIP QoS

EIP 1Mbps 1 shared=false QoSPolicy EIP QoSPolicy QoS

QoSPolicy

```
apiVersion: kubeovn.io/v1
kind: QoSPolicy
metadata:
  name: qos-eip-example
spec:
  shared: false
  bindingType: EIP
  bandwidthLimitRules:
    - name: eip-ingress
      rateMax: "1" # Mbps
      burstMax: "1" # Mbps
      priority: 1
      direction: ingress
    - name: eip-egress
      rateMax: "1" # Mbps
      burstMax: "1" # Mbps
      priority: 1
      direction: egress
```

IptablesEIP

```
kind: IptablesEIP
apiVersion: kubeovn.io/v1
metadata:
  name: eip-1
spec:
  natGwDp: gw1
  qosPolicy: qos-eip-example
```

.spec.qosPolicy

5.4.3 QoS EIP

label qos eip

```
# kubectl get eip -l ovn.kubernetes.io/qos=qos-eip-example
NAME IP MAC NAT NATGWDP READY
eip-1 172.18.11.24 00:00:00:34:41:0B fip gw1 true
```


5.4.4 VPC NATGW net1 QoS

VPC NATGW	net1	10Mbps	3	shared=true	QoSPolicy	QoSPolicy
-----------	------	--------	---	-------------	-----------	-----------

QoSPolicy

```
apiVersion: kubeovn.io/v1
kind: QoSPolicy
metadata:
  name: qos-natgw-example
spec:
  shared: true
  bindingType: NATGW
  bandwidthLimitRules:
  - name: net1-ingress
    interface: net1
    rateMax: "10" # Mbps
    burstMax: "10" # Mbps
    priority: 3
    direction: ingress
  - name: net1-egress
    interface: net1
    rateMax: "10" # Mbps
    burstMax: "10" # Mbps
    priority: 3
    direction: egress
```

VpcNatGateway

```
kind: VpcNatGateway
apiVersion: kubeovn.io/v1
metadata:
  name: gw1
spec:
  vpc: test-vpc-1
  subnet: net1
  lanIp: 10.0.1.254
  qosPolicy: qos-natgw-example
  selector:
  - "kubernetes.io/hostname: kube-ovn-worker"
  - "kubernetes.io/os: linux"
```

```
.spec.qosPolicy
```

5.4.5 net1 QoS

net1	5Mbps	2	shared=true	QoSPolicy	QoSPolicy
------	-------	---	-------------	-----------	-----------

QoSPolicy

```
apiVersion: kubeovn.io/v1
kind: QoSPolicy
metadata:
  name: qos-natgw-example
spec:
  shared: true
  bindingType: NATGW
  bandwidthLimitRules:
  - name: net1-extip-ingress
    interface: net1
    rateMax: "5" # Mbps
    burstMax: "5" # Mbps
    priority: 2
    direction: ingress
    matchType: ip
    matchValue: src 172.18.11.22/32
  - name: net1-extip-egress
    interface: net1
    rateMax: "5" # Mbps
    burstMax: "5" # Mbps
    priority: 2
    direction: egress
    matchType: ip
    matchValue: dst 172.18.11.23/32
```

VpcNatGateway

```
kind: VpcNatGateway
apiVersion: kubeovn.io/v1
metadata:
  name: gw1
spec:
```

```
vpc: test-vpc-1
subnet: net1
lanIp: 10.0.1.254
qosPolicy: qos-natgw-example
selector:
  - "kubernetes.io/hostname: kube-ovn-worker"
  - "kubernetes.io/os: linux"
```

5.4.6 QoS NATGW

label qos eip

kubectl get vpc-nat-gw -l ovn.kubernetes.io/qos=qos-natgw-example

NAME	VPC	SUBNET	LANIP
gw1	test-vpc-1	net1	10.0.1.254

5.4.7 qos

```
# kubectl get qos -A
NAME          SHARED  BINDINGTYPE
qos-eip-example  false   EIP
qos-natgw-example true     NATGW
```

5.4.8

• QoS QoS QoS EIP NATGW spec.qosPolicy

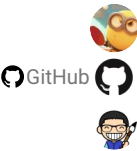
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5.4.9

5.5 VPC

Kubernetes	Service	VPC	Service
1. Service IP	VPC		
2.	IP		
Kube-OVN	1.11	SwitchLBRule CRD	VPC
SwitchLBRule	VPC		

5.5.1 Selector

selector label pod

SwitchLBRule

```
apiVersion: kubeovn.io/v1
kind: SwitchLBRule
metadata:
  name: cjh-slr-nginx
spec:
  vip: 1.1.1.1
  sessionAffinity: ClientIP
  namespace: default
  selector:
    - app: nginx
  ports:
    - name: dns
      port: 8888
      targetPort: 80
      protocol: TCP
```

- selector, sessionAffinity port Kubernetes Service
- vip IP
- namespace selector Pod

Kube-OVN SwitchLBRule Pod Pod VPC L2 LB

5.5.2 Endpoints

endpoints selector kubevirt vm

SwitchLBRule

```
apiVersion: kubeovn.io/v1
kind: SwitchLBRule
metadata:
  name: cjh-slr-nginx
spec:
  vip: 1.1.1.1
  sessionAffinity: ClientIP
  namespace: default
  endpoints:
    - 192.168.0.101
    - 192.168.0.102
    - 192.168.0.103
  ports:
    - name: dns
      port: 8888
      targetPort: 80
      protocol: TCP
```

- sessionAffinity port Kubernetes Service
- vip IP
- namespace
- endpoints IP

selector endpoints, selector

5.5.3

OVN IPv4

[Health Checks](https://www.ovn.org/support/dist-docs/ovn-nb.5.html)

ovn SwitchLBRule SwitchLBRule VPC subnet vip ip_port_mappings

load_balancer_health_check

- vip subnet subnet SwitchLBRule
- Selector

```
root@server:~# kubectl get po -o wide -n vulpecula
NAME          READY   STATUS    RESTARTS   AGE   IP           NODE   NOMINATED NODE   READINESS GATES
nginx-78d9578975-f4qn4  1/1     Running   3           4d16h  10.16.0.4    worker  <none>           <none>
nginx-78d9578975-t8tm5  1/1     Running   3           4d16h  10.16.0.6    worker  <none>           <none>
```

```
# slr
root@server:~# cat << END > slr.yaml
apiVersion: kubeovn.io/v1
kind: SwitchLBRule
metadata:
  name: nginx
  namespace: vulpecula
spec:
  vip: 1.1.1.1
  sessionAffinity: ClientIP
  namespace: default
  selector:
    - app: nginx
  ports:
    - name: dns
      port: 8888
      targetPort: 80
      protocol: TCP
END
root@server:~# kubectl apply -f slr.yaml
root@server:~# kubectl get slr
NAME          VIP      PORT(S)   SERVICE                     AGE
vulpecula-nginx  1.1.1.1  8888/TCP  default/slr-vulpecula-nginx  3d21h
```

subnet vip

```
# vip
root@server:~# kubectl get vip
NAME          NS      V4IP      MAC              V6IP      PMAC      SUBNET      READY   TYPE
vulpecula-subnet  10.16.0.2  00:00:00:39:95:C1 <nil>        vulpecula-subnet  true
```

Load_Balancer_Health_Check Service_Monitor

```
root@server:~# kubectl ko nbctl list Load_Balancer
 _uuid      : 3cbb6d43-44aa-4028-962f-30d2dba9f0b8
external_ids : {}
health_check : [5bee3f12-6b54-411c-9cc8-c9def8f67356]
ip_port_mappings : {"10.16.0.4"="nginx-78d9578975-f4qn4.default:10.16.0.2", "10.16.0.6"="nginx-78d9578975-t8tm5.default:10.16.0.2"}
name        : cluster-tcp-session-loadbalancer
options     : {affinity_timeout="10800"}
protocol    : tcp
selection_fields : [ip_src]
vips       : {"1.1.1.1:8888"="10.16.0.4:80,10.16.0.6:80"}
```

```
root@server:~# kubectl ko nbctl list Load_Balancer_Health_Check
 _uuid      : 5bee3f12-6b54-411c-9cc8-c9def8f67356
external_ids : {switch_lb_subnet=vulpecula-subnet}
options     : {failure_count="3", interval="5", success_count="3", timeout="20"}
vip        : "1.1.1.1:8888"
```

```
root@server:~# kubectl ko sbctl list Service_Monitor
 _uuid      : 1bddc541-cc49-44ea-9935-a4208f627a91
external_ids : {}
ip          : "10.16.0.4"
logical_port : nginx-78d9578975-f4qn4.default
options     : {failure_count="3", interval="5", success_count="3", timeout="20"}
port       : 80
protocol    : tcp
```

```

src_ip      : "10.16.0.2"
src_mac     : "c6:d4:b8:08:54:e7"
status      : online

_uuid       : 84dd24c5-e1b4-4e97-9daa-13687ed59785
external_ids : {}
ip          : "10.16.0.6"
logical_port : nginx-78d9578975-t8tm5.default
options     : {failure_count="3", interval="5", success_count="3", timeout="20"}
port        : 80
protocol    : tcp
src_ip      : "10.16.0.2"
src_mac     : "c6:d4:b8:08:54:e7"
status      : online

```

vip

```

root@server:~# kubectl exec -it -n vulpecula nginx-78d9578975-t8tm5 -- curl 1.1.1.1:8888
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<p><em>Thank you for using nginx.</em></p>
</body>
</html>

```

pod

```

kubectl delete po nginx-78d9578975-f4qn4
kubectl get po -o wide -n vulpecula

```

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED NODE	READINESS GATES
nginx-78d9578975-lxmvh	1/1	Running	0	31s	10.16.0.8	worker	<none>	<none>
nginx-78d9578975-t8tm5	1/1	Running	3	4d16h	10.16.0.6	worker	<none>	<none>

Load_Balancer_Health_Check Service_Monitor

```

root@server:~# kubectl ko nbctl list Load_Balancer
_uuid       : 3cbb6d43-44aa-4028-962f-30d2dba9f0b8
external_ids : {}
health_check : [5bee3f12-6b54-411c-9cc8-c9def8f67356]
ip_port_mappings : {"10.16.0.4"="nginx-78d9578975-f4qn4.default:10.16.0.2", "10.16.0.6"="nginx-78d9578975-t8tm5.default:10.16.0.2", "10.16.0.8"="nginx-78d9578975-lxmvh.default:10.16.0.2"}
name        : cluster-tcp-session-loadbalancer
options     : {affinity_timeout="10800"}
protocol    : tcp
selection_fields : [ip_src]
vips        : {"1.1.1.1:8888"="10.16.0.6:80,10.16.0.8:80"}

root@server:~# kubectl ko nbctl list Load_Balancer_Health_Check
_uuid       : 5bee3f12-6b54-411c-9cc8-c9def8f67356
external_ids : {switch_lb_subnet=vulpecula-subnet}
options     : {failure_count="3", interval="5", success_count="3", timeout="20"}
vip         : "1.1.1.1:8888"

root@server:~# kubectl ko sbctl list Service_Monitor
_uuid       : 84dd24c5-e1b4-4e97-9daa-13687ed59785
external_ids : {}
ip          : "10.16.0.6"
logical_port : nginx-78d9578975-t8tm5.default
options     : {failure_count="3", interval="5", success_count="3", timeout="20"}
port        : 80
protocol    : tcp
src_ip      : "10.16.0.2"
src_mac     : "c6:d4:b8:08:54:e7"
status      : online

_uuid       : 5917b7b7-a999-49f2-a42d-da81f1eeb28f
external_ids : {}
ip          : "10.16.0.8"
logical_port : nginx-78d9578975-lxmvh.default
options     : {failure_count="3", interval="5", success_count="3", timeout="20"}
port        : 80
protocol    : tcp
src_ip      : "10.16.0.2"
src_mac     : "c6:d4:b8:08:54:e7"
status      : online

```

SwitchLBRule

Load_Balancer_Health_Check

Service_Monitor

vip

```

root@server:~# kubectl delete -f slr.yaml
switchlbrule.kubeovn.io "vulpecula-nginx" deleted
root@server:~# kubectl get vip
No resources found

```

```
root@server:~# kubectl ko sbctl list Service_Monitor
root@server:~#
root@server:~# kubectl ko nbctl list Load_Balancer_Health_Check
root@server:~#
```



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5.5.4

5.6 VPC DNS

VPC	VPC	VPC	VPC	coredns	VPC	Kubernetes	vpc-dns	CRD
CRD	coredns	Pod	VPC	VPC	VPC	VPC		
 Note DNSPod VMWebhook/etc/resolv.conf								

5.6.1 vpc-dns

```
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
  labels:
    kubernetes.io/bootstrapping: rbac-defaults
  name: system:vpc-dns
rules:
- apiGroups:
  - ""
  resources:
  - endpoints
  - services
  - pods
  - namespaces
  verbs:
  - list
  - watch
- apiGroups:
  - discovery.k8s.io
  resources:
  - endpointslices
  verbs:
  - list
  - watch
---
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRoleBinding
metadata:
  annotations:
    rbac.authorization.kubernetes.io/autoupdate: "true"
  labels:
    kubernetes.io/bootstrapping: rbac-defaults
  name: vpc-dns
roleRef:
  apiGroup: rbac.authorization.k8s.io
  kind: ClusterRole
  name: system:vpc-dns
subjects:
- kind: ServiceAccount
  name: vpc-dns
  namespace: kube-system
---
apiVersion: v1
kind: ServiceAccount
metadata:
  name: vpc-dns
  namespace: kube-system
---
apiVersion: v1
kind: ConfigMap
metadata:
  name: vpc-dns-corefile
  namespace: kube-system
data:
  Corefile: |
    .:53 {
      errors
      health {
        lameduck 5s
      }
      ready
      kubernetes cluster.local in-addr.arpa ip6.arpa {
        pods insecure
        fallthrough in-addr.arpa ip6.arpa
      }
      prometheus :9153
      forward . /etc/resolv.conf {
        prefer_udp
      }
      cache 30
```

```

    loop
    reload
    loadbalance
  }

```

nat-gw-pod

5.6.2

```

apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: ovn-nad
  namespace: default
spec:
  config: '{
    "cniVersion": "0.3.0",
    "type": "kube-ovn",
    "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
    "provider": "ovn-nad.default.ovn"
  }'

```

5.6.3 vpc-dns Configmap

kube-system configmap vpc-dns vpc-dns

```

apiVersion: v1
kind: ConfigMap
metadata:
  name: vpc-dns-config
  namespace: kube-system
data:
  coredns-vip: 10.96.0.3
  enable-vpc-dns: "true"
  nad-name: ovn-nad
  nad-provider: ovn-nad.default.ovn

```

- enable-vpc-dns true
- coredns-image dns coredns
- coredns-vip coredns lb vip
- nad-name network-attachment-definitions
- nad-provider provider
- k8s-service-host coredns k8s apiserver ip apiserver
- k8s-service-port coredns k8s apiserver port apiserver

5.6.4 vpc-dns

vpc-dns yaml

```

kind: VpcDns
apiVersion: kubeovn.io/v1
metadata:
  name: test-cjh1
spec:
  vpc: cjh-vpc-1
  subnet: cjh-subnet-1
  replicas: 2

```

- vpc dns vpc
- subnet dns
- replicas : vpc dns deployment replicas

```

# kubectl get vpc-dns
NAME      ACTIVE  VPC      SUBNET
test-cjh1 false   cjh-vpc-1  cjh-subnet-1
test-cjh2 true    cjh-vpc-1  cjh-subnet-2

```



```
ACTIVE : true      dns      false

      VPC      DNS

•   VPC      vpc-dns      VPC      subnet      vpc-dns      true      false

•   true      vpc-dns      false      vpc-dns
```

5.6.5

vpc-dns Pod label app=vpc-dns vpc-dns Pod

```
# kubectl -n kube-system get pods -l app=vpc-dns
NAME                                READY   STATUS    RESTARTS   AGE
vpc-dns-test-cjh1-7b878d96b4-g5979 1/1     Running   0           28s
vpc-dns-test-cjh1-7b878d96b4-ltmf9 1/1     Running   0           28s
```

slr

```
# kubectl -n kube-system get slr
NAME      VIP      PORT(S)      SERVICE      AGE
vpc-dns-test-cjh1 10.96.0.3 53/UDP, 53/TCP, 9153/TCP  kube-system/slr-vpc-dns-test-cjh1 113s
```

VPC Pod dns :

```
nslookup kubernetes.default.svc.cluster.local 10.96.0.3
```

VPC switch lb rule VPC Pod

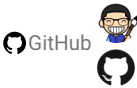
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5.6.6

5.7 SecurityGroup

Kube-OVN Pod

 **Warning**

Kube-OVN

[NetworkPolicy](#) [Network Policy API](#) [Subnet ACL](#) [Security Group](#) [OVN ACL](#) [NetworkPolicy](#) [NetworkPolicy API](#)

5.7.1

```
apiVersion: kubeovn.io/v1
kind: SecurityGroup
metadata:
  name: sg-example
spec:
  allowSameGroupTraffic: true
  egressRules:
    - ipVersion: ipv4
      policy: allow
      priority: 1
      protocol: all
      remoteAddress: 10.16.0.13 # 10.16.0.0/16
      remoteType: address
  ingressRules:
    - ipVersion: ipv4
      policy: deny
      priority: 1
      protocol: icmp
      remoteAddress: 10.16.0.14
      remoteType: address
```

[Kube-OVN](#)

Pod `ovn.kubernetes.io/security_groups` annotation

```
ovn.kubernetes.io/security_groups: sg-example
```

Port Security

5.7.2



```
apiVersion: kubeovn.io/v1
kind: SecurityGroup
metadata:
  name: sg-tier0
spec:
  tier: 0
  allowSameGroupTraffic: true
  ingressRules:
    - ipVersion: ipv4
      policy: pass
      priority: 1
      protocol: tcp
      remoteAddress: 10.16.0.0/16
```

```
remoteType: address
- ipVersion: ipv4
  policy: deny
  priority: 2
  protocol: all
  remoteAddress: 0.0.0.0/0
  remoteType: address
---
apiVersion: kubeovn.io/v1
kind: SecurityGroup
metadata:
  name: sg-tier1
spec:
  tier: 1
  allowSameGroupTraffic: true
  ingressRules:
  - ipVersion: ipv4
    policy: allow
    priority: 1
    protocol: tcp
    remoteAddress: 10.16.0.0/16
    remoteType: address
    portRangeMin: 80
    portRangeMax: 443
```

sg-tier0	10.16.0.0/16	TCP	tier 1	sg-tier1	80-443	TCP
----------	--------------	-----	--------	----------	--------	-----

Pod annotation

```
apiVersion: v1
kind: Pod
metadata:
  labels:
    app: web
  annotations:
    ovn.kubernetes.io/security_groups: 'sg-tier0,sg-tier1'
  name: multi-tier-pod
  namespace: default
spec:
  nodeName: kube-ovn-worker
  containers:
  - image: docker.io/library/nginx:alpine
    imagePullPolicy: IfNotPresent
    name: nginx
```

5.7.3

localAddress						
--------------	--	--	--	--	--	--

- localAddress IP CIDR
- sourcePortRangeMin / sourcePortRangeMax 1-65535 TCP UDP

```
apiVersion: kubeovn.io/v1
kind: SecurityGroup
metadata:
  name: sg-local-filter
spec:
  allowSameGroupTraffic: true
  ingressRules:
  - ipVersion: ipv4
    policy: allow
    priority: 1
    protocol: tcp
    remoteAddress: 10.16.0.0/16
    remoteType: address
    portRangeMin: 8080
    portRangeMax: 8080
    localAddress: 10.16.0.100
    sourcePortRangeMin: 1024
    sourcePortRangeMax: 65535
```

10.16.0.0/16	10.16.0.100	8080	TCP	1024-65535
--------------	-------------	------	-----	------------

5.7.4

- ACL OVN ACL ACL

•	priority	1-16384	ACL	ACL	ACL	= 18484	-	ACL
•	tier	0	1	policy: pass	tier 0	tier 1		
•		Kube-OVN	CNI	Pod	Pod	Pod	ContainerCreating	Running

5.7.5

YAML Pod annotation

```
apiVersion: v1
kind: Pod
metadata:
  labels:
    app: static
  annotations:
    ovn.kubernetes.io/security_groups: 'sg-example'
  name: sg-test-pod
  namespace: default
spec:
  nodeName: kube-ovn-worker
  containers:
  - image: docker.io/library/nginx:alpine
    imagePullPolicy: IfNotPresent
    name: qatest
```

```
# kubectl get pod -o wide
```

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED NODE	READINESS GATES
sg-test-pod	0/1	ContainerCreating	0	5h32m	<none>	kube-ovn-worker	<none>	<none>
test-99ffff7f86-52h9r	1/1	Running	0	5h41m	10.16.0.14	kube-ovn-control-plane	<none>	<none>
test-99ffff7f86-qcgjw	1/1	Running	0	5h43m	10.16.0.13	kube-ovn-worker	<none>	<none>

kubectl describe pod Pod

```
# kubectl describe pod sg-test-pod
Name: sg-test-pod
Namespace: default
Priority: 0
Node: kube-ovn-worker/172.18.0.2
Start Time: Tue, 28 Feb 2023 10:29:36 +0800
Labels: app=static
Annotations: ovn.kubernetes.io/allocated: true
              ovn.kubernetes.io/cidr: 10.16.0.0/16
              ovn.kubernetes.io/gateway: 10.16.0.1
              ovn.kubernetes.io/ip_address: 10.16.0.15
              ovn.kubernetes.io/logical_router: ovn-cluster
              ovn.kubernetes.io/logical_switch: ovn-default
              ovn.kubernetes.io/mac_address: 00:00:00:FA:17:97
              ovn.kubernetes.io/pod_nic_type: veth-pair
              ovn.kubernetes.io/port_security: true
              ovn.kubernetes.io/routed: true
              ovn.kubernetes.io/security_groups: sg-allow-reject
Status: Pending
IP:
IPs: <none>
.
.
.
Events:
  Type      Reason      Age      From      Message
  ----      -
Warning    FailedCreatePodSandBox    5m3s (x70 over 4h59m) kubelet (combined from similar events): Failed to create pod sandbox: rpc error: code = Unknown desc = failed to setup network for sandbox "40636e0c7f1ade5500fa958486163d74f2e2300051a71522a9afd7ba0538afb6": plugin type="kube-ovn" failed (add): RPC failed; request ip return 500 configure nic failed 10.16.0.15 network not ready after 200 ping 10.16.0.1
```

```
apiVersion: kubeovn.io/v1
kind: SecurityGroup
metadata:
  name: sg-gw-both
spec:
  allowSameGroupTraffic: true
  egressRules:
  - ipVersion: ipv4
    policy: allow
    priority: 2
    protocol: all
    remoteAddress: 10.16.0.13
    remoteType: address
```

```
- ipVersion: ipv4
  policy: allow
  priority: 1
  protocol: all
  remoteAddress: 10.16.0.1
  remoteType: address
ingressRules:
- ipVersion: ipv4
  policy: deny
  priority: 2
  protocol: icmp
  remoteAddress: 10.16.0.14
  remoteType: address
- ipVersion: ipv4
  policy: allow
  priority: 1
  protocol: icmp
  remoteAddress: 10.16.0.1
  remoteType: address
```

yamlPodPod

```
apiVersion: v1
kind: Pod
metadata:
  labels:
    app: static
  annotations:
    ovn.kubernetes.io/security_groups: 'sg-gw-both'
  name: sg-gw-both
  namespace: default
spec:
  nodeName: kube-ovn-worker
  containers:
  - image: docker.io/library/nginx:alpine
    imagePullPolicy: IfNotPresent
    name: qatest
```

Pod

# kubectl get pod -o wide									
NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED	NODE	READINESS GATES
sg-test-pod	0/1	ContainerCreating	0	5h41m	<none>	kube-ovn-worker	<none>		<none>
sg-gw-both	1/1	Running	0	5h37m	10.16.0.19	kube-ovn-worker	<none>		<none>

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5.7.6

5.8 OVN EIP FIP SNAT DNAT

 **Note**

VPC	VPC								
VPC		VPC NAT	OVN	Egress Gateway	VPC NAT	Kube-OVN	VPC NAT	Pod	VPC
Macvlan		Pod	iptables						
OVN	OVN	NAT		OVN	BFD	OVN	OVN		
Egress Gateway	VPC NAT								

VPC OVN NAT provider-network vlan (external) subnet VPC EIP/SNAT

5.8.1

- kube-ovn-controller kube-ovn-cni ovn-external-gw-config VPC spec enableExternal
- CRD provider-network vlan subnet VPC spec extraExternalSubnets ovn-eip ovn-dnat ovn-fip ovn-snat CRD

```
graph LR
pod-->subnet-->vpc-->lrp--bind-->gw-chassis-->snat-->lsp-->external-subnet
lrp-.peer-.lsp
```

Pod SNAT Pod Fip

```
graph LR
pod-->subnet-->vpc-->lrp--bind-->local-chassis-->snat-->lsp-->external-subnet
lrp-.peer-.lsp
```

Pod FIP (dnat_and_snat)

- CRD iptables nat gw
- ovn eip: ip underlay provider network vlan subnet
 - ovn fip dnat snat VPC ip vip
 - ovn snat VPC ip snat
 - ovn dnat router lb , ip + VPC endpoints

5.8.2 1.

OpenStack Neutron ovn provider network VPC EIP/SNAT

vlan vlan 0 vlan id

```
#
# 1. kube-ovn-controller
- --external-gateway-vlanid=204
- --external-gateway-switch=external204
# 2. kube-ovn-cni :
- --external-gateway-switch=external204
```

```
###          vlan id          underlay
```

- provider network vlan subnet
- VPC enable_eip_snat vlan subnet ip ipam
- VPC enable_eip_snat , pod annotation fip snat
- VPC enable_eip_snat vlan subnet VPC eip snat

1.1 underlay

```
# provider-network vlan subnet
# cat 01-provider-network.yaml

apiVersion: kubeovn.io/v1
kind: ProviderNetwork
metadata:
  name: external204
spec:
  defaultInterface: vlan

# cat 02-vlan.yaml

apiVersion: kubeovn.io/v1
kind: Vlan
metadata:
  name: vlan204
spec:
  id: 204
  provider: external204

# cat 03-vlan-subnet.yaml

apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: external204
spec:
  protocol: IPv4
  cidrBlock: 10.5.204.0/24
  gateway: 10.5.204.254
  vlan: vlan204
  excludeIps:
    - 10.5.204.1..10.5.204.100
```

1.2 VPC eip_snat

```
# VPC underlay provider subnet
# cat 00-centralized-external-gw-no-ip.yaml

apiVersion: v1
kind: ConfigMap
metadata:
  name: ovn-external-gw-config
  namespace: kube-system
data:
  enable-external-gw: "true"
  external-gw-nodes: "pc-node-1,pc-node-2,pc-node-3"
  type: "centralized"
  external-gw-nic: "vlan" # ovs
  external-gw-addr: "10.5.204.254/24" # underlay ip
```

```
logical router port (lrp) ip mac          underlay          lrp          ovn eip
ip          lrp          ovn-eip          lrp          ovn eip
```

1.3 VPC eip snat fip

node

```
# external-gw-nodes
kubectl label nodes pc-node-1 pc-node-2 pc-node-3 ovn.kubernetes.io/external-gw=true

# cat 00-ns.yaml

apiVersion: v1
```

```
kind: Namespace
metadata:
  name: vpc1

# cat 01-vpc-ecmp-enable-external-bfd.yml
```

```
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: vpc1
spec:
  namespaces:
    - vpc1
  enableExternal: true
  staticRoutes:
    - cidr: 0.0.0.0/0
      nextHopIP: 10.5.204.254
      policy: policyDst
```

```
# VPC    enableExternal    lrp
```

```
# cat 02-subnet.yml
```

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: vpc1-subnet1
spec:
  cidrBlock: 192.168.0.0/24
  default: false
  disableGatewayCheck: false
  disableInterConnection: true
  enableEcmp: true
  gatewayNode: ""
  gatewayType: distributed
  #gatewayType: centralized
  natOutgoing: false
  private: false
  protocol: IPv4
  provider: ovn
  vpc: vpc1
  namespaces:
    - vpc1
```

```
# subnet
```

```
# kubectl ko nbctl show vpc1
```

```
router 87ad06fd-71d5-4ff8-a1f0-54fa3bba1a7f (vpc1)
  port vpc1-vpc1-subnet1
    mac: "00:00:00:ED:8E:C7"
    networks: ["192.168.0.1/24"]
  port vpc1-external204
    mac: "00:00:00:EF:05:C7"
    networks: ["10.5.204.105/24"]
    gateway chassis: [7cedd14f-265b-42e5-ac17-e03e7a1f2342 276baccb-fe9c-4476-b41d-05872a94976d fd9f140c-c45d-43db-a6c0-0d4f8ea298dd]
  nat 21d853b0-f7b4-40bd-9a53-31d2e2745739
    external ip: "10.5.204.115"
    logical ip: "192.168.0.0/24"
    type: "snat"
```

```
# kubectl ko nbctl lr-route-list vpc1
```

```
IPv4 Routes
Route Table <main>:
      0.0.0.0/0          10.5.204.254 dst-ip
# VPC CRD
```

enableExternal

VPC CRD

1.4

1.4.1 UNDERLAY

eip snat fip

1

eip snat fip

```
# provider-network vlan subnet
# cat 01-extra-provider-network.yaml
apiVersion: kubeovn.io/v1
kind: ProviderNetwork
metadata:
  name: extra
spec:
  defaultInterface: vlan
# cat 02-extra-vlan.yaml
```



```
apiVersion: kubeovn.io/v1
kind: Vlan
metadata:
  name: vlan0
spec:
  id: 0
  provider: extra
# cat 03-extra-vlan-subnet.yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: extra
spec:
  protocol: IPv4
  cidrBlock: 10.10.204.0/24
  gateway: 10.10.204.254
  vlan: vlan0
  excludeIps:
    - 10.10.204.1..10.10.204.100
```

1.4.2 VPC

```
apiVersion: kubeovn.io/v1
kind: Vpc
metadata:
  name: vpc1
spec:
  namespaces:
    - vpc1
  enableExternal: true # enableExternal VPC external ls
  extraExternalSubnets: # extraExternalSubnets
    - extra

# kubectl ko nbctl show vpc1
router 87ad06fd-71d5-4ff8-a1f0-54fa3bba1a7f (vpc1)
  port vpc1-vpc1-subnet1
    mac: "00:00:00:ED:8E:C7"
    networks: ["192.168.0.1/24"]
  port vpc1-external204
    mac: "00:00:00:EF:05:C7"
    networks: ["10.5.204.105/24"]
  gateway chassis: [7cedd14f-265b-42e5-ac17-e03e7a1f2342 276baccb-fe9c-4476-b41d-05872a94976d fd9f140c-c45d-43db-a6c0-0d4f8ea298dd]
  port vpc1-extra
    mac: "00:00:00:EF:6A:C7"
    networks: ["10.10.204.105/24"]
  gateway chassis: [7cedd14f-265b-42e5-ac17-e03e7a1f2342 276baccb-fe9c-4476-b41d-05872a94976d fd9f140c-c45d-43db-a6c0-0d4f8ea298dd]
```

5.8.3 2. ovn-eip

iptables-eip		ovn-eip		type	
• nat:	ovn dn timer, snat	nat			
• lrp:	underlay	lrp	ip	dnat snat	
• lsp:	ovn bfd ecmp			ovs internal port	ecmp

```
---
kind: OvnEip
apiVersion: kubeovn.io/v1
metadata:
  name: eip-static
spec:
  externalSubnet: external204
  type: nat

# eip fip
```

externalSubnet	external204
externalSubnet	extra

2.1 ovn-fip pod fip

```
# kubectl get po -o wide -n vpc1 vpc-1-busybox01
NAME READY STATUS RESTARTS AGE IP NODE
vpc-1-busybox01 1/1 Running 0 3d15h 192.168.0.2 pc-node-2

# kubectl get ip vpc-1-busybox01.vpc1
```

```

NAME          V4IP          V6IP          MAC          NODE          SUBNET
vpc-1-busybox01.vpc1  192.168.0.2          00:00:00:0A:DD:27  pc-node-2    vpc1-subnet1

```

```
---
```

```

kind: OvnEip
apiVersion: kubeovn.io/v1
metadata:
  name: eip-static
spec:
  externalSubnet: external204
  type: nat

```

```
---
```

```

kind: OvnFip
apiVersion: kubeovn.io/v1
metadata:
  name: eip-static
spec:
  ovnEip: eip-static
  ipName: vpc-1-busybox01.vpc1 # ip crd
  type: "centralized" # centralized distributed

```

```
--
```

```

#          VPC          ip

kind: OvnFip
apiVersion: kubeovn.io/v1
metadata:
  name: eip-static
spec:
  ovnEip: eip-static
  vpc: vpc1
  v4Ip: 192.168.0.2
  type: "centralized" # centralized distributed

```

```

# kubectl get ofip
NAME      VPC      V4EIP          V4IP          READY  IPTYPE  IPNAME
eip-for-vip vpc1    10.5.204.106  192.168.0.3  true   vip     test-fip-vip
eip-static vpc1    10.5.204.101  192.168.0.2  true   vip     vpc-1-busybox01.vpc1
# kubectl get ofip eip-static
NAME      VPC      V4EIP          V4IP          READY  IPTYPE  IPNAME
eip-static vpc1    10.5.204.101  192.168.0.2  true   vip     vpc-1-busybox01.vpc1

```

```

[root@pc-node-1 03-cust-vpc]# ping 10.5.204.101
PING 10.5.204.101 (10.5.204.101) 56(84) bytes of data.
64 bytes from 10.5.204.101: icmp_seq=2 ttl=62 time=1.21 ms
64 bytes from 10.5.204.101: icmp_seq=3 ttl=62 time=0.624 ms
64 bytes from 10.5.204.101: icmp_seq=4 ttl=62 time=0.368 ms
^C
--- 10.5.204.101 ping statistics ---
4 packets transmitted, 3 received, 25% packet loss, time 3049ms
rtt min/avg/max/mdev = 0.368/0.734/1.210/0.352 ms
[root@pc-node-1 03-cust-vpc]#

```

```
#          node ping      VPC      pod      ip
```

```

#          ip
# kubectl ko nbctl show vpc1
router 87ad06fd-71d5-4ff8-a1f0-54fa3bba1a7f (vpc1)
  port vpc1-vpc1-subnet1
    mac: "00:00:00:ED:8E:C7"
    networks: ["192.168.0.1/24"]
  port vpc1-external204
    mac: "00:00:00:EF:05:C7"
    networks: ["10.5.204.105/24"]
  gateway chassis: [7cedd14f-265b-42e5-ac17-e03e7a1f2342 276baccb-fe9c-4476-b41d-05872a94976d fd9f140c-c45d-43db-a6c0-0d4f8ea298dd]
  nat 813523e7-c68c-408f-bd8c-cba30cb2e4f4
    external ip: "10.5.204.101"
    logical ip: "192.168.0.2"
    type: "dnat_and_snat"

```

2.2 ovn-fip vip fip

```
vip          kubevirt          vip          keepalived kube-vip
```

```
fip VPC      vip          vip
```

```

#          vip eip      eip      vip
# cat vip.yaml

apiVersion: kubeovn.io/v1
kind: Vip
metadata:
  name: test-fip-vip
spec:
  subnet: vpc1-subnet1

```

```
# cat 04-fip.yaml

---
kind: OvnEip
apiVersion: kubeovn.io/v1
metadata:
  name: eip-for-vip
spec:
  externalSubnet: external204
  type: nat

---
kind: OvnFip
apiVersion: kubeovn.io/v1
metadata:
  name: eip-for-vip
spec:
  ovnEip: eip-for-vip
  ipType: vip # fip pod ip vip
  ipName: test-fip-vip

---
# VPC ip

kind: OvnFip
apiVersion: kubeovn.io/v1
metadata:
  name: eip-for-vip
spec:
  ovnEip: eip-for-vip
  ipType: vip # fip pod ip vip
  vpc: vpc1
  v4Ip: 192.168.0.3
```

```
# kubectl get ofip
NAME      VPC    V4EIP      V4IP      READY  IPTYPE  IPNAME
eip-for-vip vpc1    10.5.204.106 192.168.0.3 true    vip     test-fip-vip

[root@pc-node-1 fip-vip]# ping 10.5.204.106
PING 10.5.204.106 (10.5.204.106) 56(84) bytes of data.
64 bytes from 10.5.204.106: icmp_seq=1 ttl=62 time=0.694 ms
64 bytes from 10.5.204.106: icmp_seq=2 ttl=62 time=0.436 ms

# node ping

# pod ip

[root@pc-node-1 fip-vip]# kubectl -n vpc1 exec -it vpc1-busybox03 -- bash
[root@vpc1-busybox03 /]#
[root@vpc1-busybox03 /]#
[root@vpc1-busybox03 /]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
1568: eth0@if1569: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 00:00:00:56:40:e5 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 192.168.0.5/24 brd 192.168.0.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet 192.168.0.3/24 scope global secondary eth0 # vip
        valid_lft forever preferred_lft forever
    inet6 fe80::200:ff:fe56:40e5/64 scope link
        valid_lft forever preferred_lft forever

[root@vpc1-busybox03 /]# tcpdump -i eth0 host 192.168.0.3 -netvv
tcpdump: listening on eth0, link-type EN10MB (Ethernet), capture size 262144 bytes
00:00:00:ed:8e:c7 > 00:00:00:56:40:e5, ethertype IPv4 (0x0800), length 98: (tos 0x0, ttl 62, id 44830, offset 0, flags [DF], proto ICMP (1), length 84)
    10.5.32.51 > 192.168.0.3: ICMP echo request, id 177, seq 1, length 64
00:00:00:56:40:e5 > 00:00:00:ed:8e:c7, ethertype IPv4 (0x0800), length 98: (tos 0x0, ttl 64, id 43962, offset 0, flags [none], proto ICMP (1), length 84)
    192.168.0.3 > 10.5.32.51: ICMP echo reply, id 177, seq 1, length 64

# pod fip icmp
```

5.8.4.3. ovn-snat

3.1 ovn-snat subnet cidr

iptables-snat

```
# cat 03-subnet-snat.yaml

---
kind: OvnEip
apiVersion: kubeovn.io/v1
metadata:
```

```

name: snat-for-subnet-in-vpc
spec:
  externalSubnet: external204
  type: nat
---
kind: OvnSnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: snat-for-subnet-in-vpc
spec:
  ovnEip: snat-for-subnet-in-vpc
  vpcSubnet: vpc1-subnet1 # eip
---
#           VPC      subnet cidr

kind: OvnSnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: snat-for-subnet-in-vpc
spec:
  ovnEip: snat-for-subnet-in-vpc
  vpc: vpc1
  v4IpCidr: 192.168.0.0/24 #      cidr      ip

```

externalSubnet

extra

3.2 ovn-snat pod ip

iptables-snat

```

# cat 03-pod-snat.yaml
---
kind: OvnEip
apiVersion: kubeovn.io/v1
metadata:
  name: snat-for-pod-vpc-ip
spec:
  externalSubnet: external204
  type: nat
---
kind: OvnSnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: snat01
spec:
  ovnEip: snat-for-pod-vpc-ip
  ipName: vpc-1-busybox02.vpc1 # eip      pod ip
---
#           VPC      ip

kind: OvnSnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: snat-for-subnet-in-vpc
spec:
  ovnEip: snat-for-subnet-in-vpc
  vpc: vpc1
  v4IpCidr: 192.168.0.4

```

externalSubnet

extra

snat

```

# kubectl ko nbctl show vpc1
router 87ad06fd-71d5-4ff8-a1f0-54fa3bba1a7f (vpc1)
  port vpc1-vpc1-subnet1
    mac: "00:00:00:ED:8E:C7"
    networks: ["192.168.0.1/24"]
  port vpc1-external204
    mac: "00:00:00:EF:05:C7"
    networks: ["10.5.204.105/24"]
  gateway chassis: [7cedd14f-265b-42e5-ac17-e03e7a1f2342 276baccb-fe9c-4476-b41d-05872a94976d fd9f140c-c45d-43db-a6c0-0d4f8ea298dd]
  nat 21d853b0-f7b4-40bd-9a53-31d2e2745739
    external ip: "10.5.204.115"
    logical ip: "192.168.0.0/24"
    type: "snat"
  nat da77a11f-c523-439c-b1d1-72c664196a0f
    external ip: "10.5.204.116"
    logical ip: "192.168.0.4"
    type: "snat"

```

```
[root@pc-node-1 03-cust-vpc]# kubectl get po -A -o wide | grep busy
vpc1          vpc-1-busybox01          1/1    Running    0          3d15h    192.168.0.2    pc-node-2    <none>      <none>
vpc1          vpc-1-busybox02          1/1    Running    0          17h      192.168.0.4    pc-node-1    <none>      <none>
vpc1          vpc-1-busybox03          1/1    Running    0          17h      192.168.0.5    pc-node-1    <none>      <none>
vpc1          vpc-1-busybox04          1/1    Running    0          17h      192.168.0.6    pc-node-3    <none>      <none>
vpc1          vpc-1-busybox05          1/1    Running    0          17h      192.168.0.7    pc-node-1    <none>      <none>

# kubectl exec -it -n vpc1          vpc-1-busybox04    bash
kubectl exec [POD] [COMMAND] is DEPRECATED and will be removed in a future version. Use kubectl exec [POD] -- [COMMAND] instead.
[root@vpc-1-busybox04 /]#
[root@vpc-1-busybox04 /]#
[root@vpc-1-busybox04 /]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
17895: eth0@if17895: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 00:00:00:76:94:55 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 192.168.0.6/24 brd 192.168.0.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::200:ff:fe76:9455/64 scope link
        valid_lft forever preferred_lft forever
[root@vpc-1-busybox04 /]# ping 223.5.5.5
PING 223.5.5.5 (223.5.5.5) 56(84) bytes of data.
64 bytes from 223.5.5.5: icmp_seq=1 ttl=114 time=22.2 ms
64 bytes from 223.5.5.5: icmp_seq=2 ttl=114 time=21.8 ms

[root@pc-node-1 03-cust-vpc]# kubectl exec -it -n vpc1          vpc-1-busybox02    bash
kubectl exec [POD] [COMMAND] is DEPRECATED and will be removed in a future version. Use kubectl exec [POD] -- [COMMAND] instead.
[root@vpc-1-busybox02 /]#
[root@vpc-1-busybox02 /]#
[root@vpc-1-busybox02 /]#
[root@vpc-1-busybox02 /]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
1566: eth0@if1567: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 00:00:00:b:e9:d0 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 192.168.0.4/24 brd 192.168.0.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::200:ff:fe0b:e9d0/64 scope link
        valid_lft forever preferred_lft forever
[root@vpc-1-busybox02 /]# ping 223.5.5.5
PING 223.5.5.5 (223.5.5.5) 56(84) bytes of data.
64 bytes from 223.5.5.5: icmp_seq=2 ttl=114 time=22.7 ms
64 bytes from 223.5.5.5: icmp_seq=3 ttl=114 time=22.6 ms
64 bytes from 223.5.5.5: icmp_seq=4 ttl=114 time=22.1 ms
^C
--- 223.5.5.5 ping statistics ---
4 packets transmitted, 3 received, 25% packet loss, time 3064ms
rtt min/avg/max/mdev = 22.126/22.518/22.741/0.278 ms

#          pod          snat
```

5.8.5 4. ovn-dnat

4.1 ovn-dnat pod dnat

```
kind: OvnEip
apiVersion: kubeovn.io/v1
metadata:
  name: eip-dnat
spec:
  externalSubnet: underlay

---
kind: OvnDnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: eip-dnat
spec:
  ovnEip: eip-dnat
  ipName: vpc-1-busybox01.vpc1 #          pod ip crd
  protocol: tcp
  internalPort: "22"
  externalPort: "22"

---
#          VPC          ip

kind: OvnDnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: eip-dnat
```

```
spec:
  ovnEip: eip-dnat
  protocol: tcp
  internalPort: "22"
  externalPort: "22"
  vpc: vpc1
  v4Ip: 192.168.0.3
```

externalSubnet

extra

OvnDnatRule

IptablesDnatRule

```
# kubectl get eip eip-dnat
NAME      V4IP      V6IP      MAC          TYPE      READY
eip-dnat  10.5.49.4  10.5.49.4  00:00:00:4D:CE:49  dnat      true

# kubectl get odnat
NAME      EIP      EIP-dnat      PROTOCOL      V4EIP      V4IP      INTERNALPORT      EXTERNALPORT      IPNAME      READY
eip-dnat  10.5.49.4  10.5.49.4      tcp           10.5.49.4  192.168.0.3  22              22              vpc-1-busybox01.vpc1      true
```

4.2 ovn-dnat vip dn

```
kind: OvnDnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: eip-dnat
spec:
  ipType: vip #          dnat          pod ip          vip
  ovnEip: eip-dnat
  ipName: test-dnat-vip
  protocol: tcp
  internalPort: "22"
  externalPort: "22"

---
#          VPC          ip

kind: OvnDnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: eip-dnat
spec:
  ipType: vip #          dnat          pod ip          vip
  ovnEip: eip-dnat
  ipName: test-dnat-vip
  protocol: tcp
  internalPort: "22"
  externalPort: "22"
  vpc: vpc1
  v4Ip: 192.168.0.4
```

OvnDnatRule

IptablesDnatRule

```
# kubectl get vip test-dnat-vip
NAME          V4IP          PV4IP          MAC          PMAC          V6IP          PV6IP          SUBNET          READY
test-dnat-vip 192.168.0.4    192.168.0.4    00:00:00:D0:C0:B5 00:00:00:D0:C0:B5 00:00:00:D0:C0:B5 00:00:00:D0:C0:B5 vpc1-subnet1    true

# kubectl get oeip eip-dnat
NAME          V4IP          V6IP          MAC          TYPE          READY
eip-dnat      10.5.49.4     10.5.49.4     00:00:00:4D:CE:49 00:00:00:4D:CE:49 dnat          true

# kubectl get odnat eip-dnat
NAME          EIP          PROTOCOL          V4EIP          V4IP          INTERNALPORT          EXTERNALPORT          IPNAME          READY
eip-dnat      eip-dnat     tcp              10.5.49.4     192.168.0.4    22                  22                  test-dnat-vip    true
```



PDF



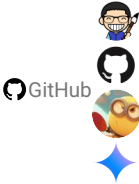
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5.8.6

5.9 OVN SNAT ECMP BFD L3 HA

VPC OVN SNAT ECMP

Gateway Node ovnext0

- bfd
- hash

```
graph LR
pod-->vpc-subnet-->vpc-->snat-->ecmp-->external-subnet-->gw-node1-ovnext0--> node1-external-switch
external-subnet-->gw-node2-ovnext0--> node2-external-switch
external-subnet-->gw-node3-ovnext0--> node3-external-switch
```

ovn-eip-fip-snat.md

install.sh

provider-network

vlan

subnet

lsp

ovn-eip

vpc enable_bfd

bfd

ecmp

5.9.1 1.

1.1 underlay

1.2 vpc eip_snat

1.3 VPC eip snat fip

ovn-eip-fip-snat.md

VPC ecmp bfd

VPC

2

ovn-eip

```
# cat gw-node-eip.yaml
---
kind: OvnEip
apiVersion: kubeovn.io/v1
metadata:
  name: pc-node-1
spec:
  externalSubnet: external204
  type: lsp

---
kind: OvnEip
apiVersion: kubeovn.io/v1
metadata:
  name: pc-node-2
spec:
  externalSubnet: external204
  type: lsp

---
kind: OvnEip
apiVersion: kubeovn.io/v1
metadata:
  name: pc-node-3
spec:
  externalSubnet: external204
  type: lsp
```

vpc ecmp

vpc bfd

enable bfd

lsp ovn eip

bfd

5.9.2 2. vpc ecmp bfd L3 HA

```
# cat 01-vpc-ecmp-enable-external-bfd.yml
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: vpc1
spec:
  namespaces:
    - vpc1
    enableExternal: true
    enableBfd: true # bfd
    #enableBfd: false
```

bfd ecmp


```
# cat 02-subnet.yml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: vpc1-subnet1
spec:
  cidrBlock: 192.168.0.0/24
  default: false
  disableGatewayCheck: false
  disableInterConnection: true
  enableEcmp: true # ecmp
  gatewayNode: ""
  gatewayType: distributed
  #gatewayType: centralized
  natOutgoing: false
  private: false
  protocol: IPv4
  provider: ovn
  vpc: vpc1
  namespaces:
  - vpc1
```

:

1. vpc ecmp ecmp bfd vpc enableBfd subnet enableEcmp ecmp bfd
- 2.
3. VPC VPC VPC snat
4. vpc subnet enableEcmp gatewayType
5. EnableExternal vpc
6. EnableExternal EnableBfd

```
# ovn
# vpc
# k get vpc
NAME      ENABLEEXTERNAL  ENABLEBFD  STANDBY  SUBNETS                                NAMESPACES
ovn-cluster true            true       true     ["external1204", "join", "ovn-default"] ["vpc1"]
vpc1      true            true       true     ["vpc1-subnet1"]

# vpc ENABLEBFD
# vpc

# 1. bfd
# k ko nbctl list bfd
_uuid      : be7df545-2c4c-4751-878f-b3507987f050
detect_mult : 3
dst_ip      : "10.5.204.121"
external_ids : {}
logical_port : vpc1-external1204
min_rx      : 100
min_tx      : 100
options     : {}
status      : up

_uuid      : 684c4489-5b59-4693-8d8c-3beab93f8093
detect_mult : 3
dst_ip      : "10.5.204.109"
external_ids : {}
logical_port : vpc1-external1204
min_rx      : 100
min_tx      : 100
options     : {}
status      : up

_uuid      : f0f62077-2ae9-4e79-b4f8-a446ec6e784c
detect_mult : 3
dst_ip      : "10.5.204.108"
external_ids : {}
logical_port : vpc1-external1204
min_rx      : 100
min_tx      : 100
options     : {}
status      : up

###      status      up

# 2. bfd
# k ko nbctl lr-route-list vpc1
IPv4 Routes
Route Table <main>:
      192.168.0.0/24      10.5.204.108 src-ip ecmp ecmp-symmetric-reply bfd
      192.168.0.0/24      10.5.204.109 src-ip ecmp ecmp-symmetric-reply bfd
      192.168.0.0/24      10.5.204.121 src-ip ecmp ecmp-symmetric-reply bfd

# 3.
```

```
# k ko nbctl find Logical_Router_Static_Route policy=src-ip options=ecmp_symmetric_reply="true"
_uuid      : 3aacb384-d5ee-4b14-aebf-59e8c11717ba
bfd        : 684c4489-5b59-4693-8d8c-3beab93f8093
external_ids : {}
ip_prefix   : "192.168.0.0/24"
nexthop     : "10.5.204.109"
options     : {ecmp_symmetric_reply="true"}
output_port : []
policy      : src-ip
route_table : ""

_uuid      : 18bcc585-bc05-430b-925b-ef673c8e1aef
bfd        : f0f62077-2ae9-4e79-b4f8-a446ec6e784c
external_ids : {}
ip_prefix   : "192.168.0.0/24"
nexthop     : "10.5.204.108"
options     : {ecmp_symmetric_reply="true"}
output_port : []
policy      : src-ip
route_table : ""

_uuid      : 7d0a4e6b-cde0-4110-8176-fbaf19738498
bfd        : be7df545-2c4c-4751-878f-b3507987f050
external_ids : {}
ip_prefix   : "192.168.0.0/24"
nexthop     : "10.5.204.121"
options     : {ecmp_symmetric_reply="true"}
output_port : []
policy      : src-ip
route_table : ""
```

```
#
[root@pc-node-1 ~]# ip netns exec ovnext ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
1541: ovnext0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc noqueue state UNKNOWN group default qlen 1000
    link/ether 00:00:00:ab:bd:87 brd ff:ff:ff:ff:ff:ff
    inet 10.5.204.108/24 brd 10.5.204.255 scope global ovnext0
        valid_lft forever preferred_lft forever
    inet6 fe80::200:ff:feab:bd87/64 scope link
        valid_lft forever preferred_lft forever
[root@pc-node-1 ~]#
[root@pc-node-1 ~]# ip netns exec ovnext route -n
Kernel IP routing table
Destination Gateway Genmask Flags Metric Ref Use Iface
0.0.0.0 10.5.204.254 0.0.0.0 UG 0 0 0 ovnext0
10.5.204.0 0.0.0.0 255.255.255.0 U 0 0 0 ovnext0

## internal port unerlay pod ns ns
[root@pc-node-1 ~]# ip netns exec ovnext bfd-control status
There are 1 sessions:
Session 1
id=1 local=10.5.204.108 (p) remote=10.5.204.122 state=Up

## lrp bfd lrp ecmp
[root@pc-node-1 ~]# ip netns exec ovnext ping -c1 223.5.5.5
PING 223.5.5.5 (223.5.5.5) 56(84) bytes of data.
64 bytes from 223.5.5.5: icmp_seq=1 ttl=115 time=21.6 ms

#
```

ovnext ns

```
[root@pc-node-3 ~]# ip netns exec ovnext bash
[root@pc-node-3 ~]# tcpdump -i ovnext0 host 223.5.5.5 -netvv
dropped privs to tcpdump
tcpdump: listening on ovnext0, link-type EN10MB (Ethernet), capture size 262144 bytes
00:00:00:2d:f8:ce > 00:00:00:fd:b2:a4, ethertype IPv4 (0x0800), length 98: (tos 0x0, ttl 63, id 57978, offset 0, flags [DF], proto ICMP (1), length 84)
    10.5.204.102 > 223.5.5.5: ICMP echo request, id 22, seq 71, length 64
00:00:00:fd:b2:a4 > dc:ef:80:5a:44:1a, ethertype IPv4 (0x0800), length 98: (tos 0x0, ttl 62, id 57978, offset 0, flags [DF], proto ICMP (1), length 84)
    10.5.204.102 > 223.5.5.5: ICMP echo request, id 22, seq 71, length 64
^C
2 packets captured
2 packets received by filter
0 packets dropped by kernel
[root@pc-node-3 ~]#

# down pod
# 3
```

5.9.3.3. bfd

vpc enable_eip_snat

```
# cat 01-vpc-ecmp-enable-external-bfd.yml
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: vpc2
spec:
  namespaces:
  - vpc2
  enableExternal: true
  #enableBfd: true
  enableBfd: false

##  bfd

# k ko nbctl lr-route-list vpc2
IPv4 Routes
Route Table <main>:
      0.0.0.0/0                10.5.204.254 dst-ip

#
# nbctl list bfd      lrp      bfd
# ovnext ns      bfd
#      vpc subnet      ping      ( )
#      ( )
```

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5.10 VPC

VPC	VPC	VPC	NAT
-----	-----	-----	-----

5.10.1

1.	VPC	
2.	VPC	CIDR
3.	VPC	VPC

5.10.2

VPC	VPC	Subnet	Subnet	CIDR
-----	-----	--------	--------	------

```
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: vpc-1
spec: {}
---
kind: Subnet
apiVersion: kubeovn.io/v1
metadata:
  name: net1
spec:
  vpc: vpc-1
  cidrBlock: 10.0.0.0/16
---
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: vpc-2
spec: {}
---
kind: Subnet
apiVersion: kubeovn.io/v1
metadata:
  name: net2
spec:
  vpc: vpc-2
  cidrBlock: 172.31.0.0/16
```

VPC	vpcPeerings
-----	-------------

```
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: vpc-1
spec:
  vpcPeerings:
    - remoteVpc: vpc-2
      localConnectIP: 169.254.0.1/30
  staticRoutes:
    - cidr: 172.31.0.0/16
      nextHopIP: 169.254.0.2
      policy: policyDst
---
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: vpc-2
spec:
  vpcPeerings:
    - remoteVpc: vpc-1
      localConnectIP: 169.254.0.2/30
  staticRoutes:
    - cidr: 10.0.0.0/16
      nextHopIP: 169.254.0.1
      policy: policyDst
```

• remoteVpc:	VPC			
• localConnectIP:	IP	CIDR	IP	CIDR
• cidr	Subnet	CIDR		
• nextHopIP	VPC	localConnectIP		

Subnet Pod

```

apiVersion: v1
kind: Pod
metadata:
  annotations:
    ovn.kubernetes.io/logical_switch: net1
  name: vpc-1-pod
spec:
  containers:
    - name: vpc-1-pod
      image: docker.io/library/nginx:alpine
---
apiVersion: v1
kind: Pod
metadata:
  annotations:
    ovn.kubernetes.io/logical_switch: net2
  name: vpc-2-pod
spec:
  containers:
    - name: vpc-2-pod
      image: docker.io/library/nginx:alpine

```

```

# kubectl exec -it vpc-1-pod -- ping $(kubectl get pod vpc-2-pod -o jsonpath='{.status.podIP}')
PING 172.31.0.2 (172.31.0.2): 56 data bytes
64 bytes from 172.31.0.2: seq=0 ttl=62 time=0.655 ms
64 bytes from 172.31.0.2: seq=1 ttl=62 time=0.086 ms
64 bytes from 172.31.0.2: seq=2 ttl=62 time=0.098 ms
^C
--- 172.31.0.2 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.086/0.279/0.655 ms
# kubectl exec -it vpc-2-pod -- ping $(kubectl get pod vpc-1-pod -o jsonpath='{.status.podIP}')
PING 10.0.0.2 (10.0.0.2): 56 data bytes
64 bytes from 10.0.0.2: seq=0 ttl=62 time=0.594 ms
64 bytes from 10.0.0.2: seq=1 ttl=62 time=0.093 ms
64 bytes from 10.0.0.2: seq=2 ttl=62 time=0.088 ms
^C
--- 10.0.0.2 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.088/0.258/0.594 ms

```



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5.10.3

6.

6.1 kubectl

Kube-OVN	kubectl	OVN	OVN	OVS	tcpdump
----------	---------	-----	-----	-----	---------

6.1.1

Kube-OVN

kubectl

`kubectl-ko`

`wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/dist/images/kubectl-ko`

`$PATH`

`mv kubectl-ko /usr/local/bin/kubectl-ko`

`chmod +x /usr/local/bin/kubectl-ko`

`# kubectl plugin list`
`The following compatible plugins are available:`
`/usr/local/bin/kubectl-ko`

6.1.2

`kubectl ko`

`# kubectl ko`
`kubectl ko {subcommand} [option...]`
`Available Subcommands:`
`[nb|sb] [status|kick|backup|dbstatus|restore] ovn-db operations show cluster status, kick stale server, backup database, get db consistency status or`
`restore ovn nb db when met 'inconsistent data' error`
`nbctl [ovn-nbctl options ...] invoke ovn-nbctl`
`sbctl [ovn-sbctl options ...] invoke ovn-sbctl`
`vsctl {nodeName} [ovs-vsctl options ...] invoke ovs-vsctl on the specified node`
`ofctl {nodeName} [ovs-ofctl options ...] invoke ovs-ofctl on the specified node`
`dpctl {nodeName} [ovs-dpctl options ...] invoke ovs-dpctl on the specified node`
`appctl {nodeName} [ovs-appctl options ...] invoke ovs-appctl on the specified node`
`tcpdump {namespace/podname} [tcpdump options ...] capture pod traffic`
`{trace|ovn-trace} ... trace ovn microflow of specific packet`
`{trace|ovn-trace} {namespace/podname} {target ip address} [target mac address] {icmp|tcp|udp} [target tcp/udp port] trace ICMP/TCP/UDP`
`{trace|ovn-trace} {namespace/podname} {target ip address} [target mac address] arp {request|reply} trace ARP request/reply`
`{trace|ovn-trace} {node//nodename} {target ip address} [target mac address] {icmp|tcp|udp} [target tcp/udp port] trace ICMP/TCP/UDP`
`{trace|ovn-trace} {node//nodename} {target ip address} [target mac address] arp {request|reply} trace ARP request/reply`
`diagnose {all|node|subnet|IPPorts} [nodename|subnetName|{proto1}-{IP1}-{Port1},{proto2}-{IP2}-{Port2}] diagnose connectivity of all nodes or a specific`
`node or specify subnet's ds pod or IPPorts like 'tcp-172.18.0.2-53,udp-172.18.0.3-53'`
`env-check check the environment configuration`
`reload restart all kube-ovn components`
`log {kube-ovn|ovn|ovs|linux|all} save log to ./kubectl-ko-log/`
`perf [image] performance test default image is docker.io/kubeovn/test:v1.13.0`
`icnbctl [ovn-nbctl options ...] invoke ovn-ic-nbctl`
`icsbctl [ovn-sbctl options ...] invoke ovn-ic-sbctl`

[nb | sb] [status | kick | backup | dbstatus | restore]

OVN

OVN leader ovs-appctl cluster/status :

```
# kubectl ko nb status
306b
Name: OVN_Northbound
Cluster ID: 9a87 (9a872522-3e7d-47ca-83a3-d74333e1a7ca)
Server ID: 306b (306b256b-b5e1-4eb0-be91-4ca96adf6bad)
Address: tcp:[172.18.0.2]:6643
Status: cluster member
Role: leader
Term: 1
Leader: self
Vote: self

Last Election started 280309 ms ago, reason: timeout
Last Election won: 280309 ms ago
Election timer: 5000
Log: [139, 139]
Entries not yet committed: 0
Entries not yet applied: 0
Connections: <-8723 ->8723 <-85d6 ->85d6
Disconnections: 0
Servers:
  85d6 (85d6 at tcp:[172.18.0.4]:6643) next_index=139 match_index=138 last msg 763 ms ago
  8723 (8723 at tcp:[172.18.0.3]:6643) next_index=139 match_index=138 last msg 763 ms ago
  306b (306b at tcp:[172.18.0.2]:6643) (self) next_index=2 match_index=138
status: ok
```

Server	match_index	last msg	Server
--------	-------------	----------	--------

OVN 172.18.0.3 :

```
# kubectl ko nb kick 8723
started removal
```

```
# kubectl ko nb status
306b
Name: OVN_Northbound
Cluster ID: 9a87 (9a872522-3e7d-47ca-83a3-d74333e1a7ca)
Server ID: 306b (306b256b-b5e1-4eb0-be91-4ca96adf6bad)
Address: tcp:[172.18.0.2]:6643
Status: cluster member
Role: leader
Term: 1
Leader: self
Vote: self

Last Election started 324356 ms ago, reason: timeout
Last Election won: 324356 ms ago
Election timer: 5000
Log: [140, 140]
Entries not yet committed: 0
Entries not yet applied: 0
Connections: <-85d6 ->85d6
Disconnections: 2
Servers:
  85d6 (85d6 at tcp:[172.18.0.4]:6643) next_index=140 match_index=139 last msg 848 ms ago
  306b (306b at tcp:[172.18.0.2]:6643) (self) next_index=2 match_index=139
status: ok
```

OVN

```
# kubectl ko nb backup
tar: Removing leading '/' from member names
backup ovn-nb db to /root/ovnnb_db.060223191654183154.backup
```

```
# kubectl ko nb dbstatus
status: ok
```

inconsistent data

inconsistent data

```
# kubectl ko nb restore
deployment.apps/ovn-central scaled
ovn-central original replicas is 3
first nodeIP is 172.18.0.5
ovs-ovn pod on node 172.18.0.5 is ovs-ovn-8jxv9
ovs-ovn pod on node 172.18.0.3 is ovs-ovn-sjzb6
ovs-ovn pod on node 172.18.0.4 is ovs-ovn-t87zk
backup nb db file
restore nb db file, operate in pod ovs-ovn-8jxv9
deployment.apps/ovn-central scaled
finish restore nb db file and ovn-central replicas
recreate ovs-ovn pods
pod "ovs-ovn-8jxv9" deleted
pod "ovs-ovn-sjzb6" deleted
pod "ovs-ovn-t87zk" deleted
```

[nbctl | sbctl] [options ...]

OVN

leader

ovn-nbctl

ovn-sbctl

OVN

ovn-nbctl(8)

ovn-sbctl(8)

```
# kubectl ko nbctl show
switch c7cd17e8-ceee-4a91-9bb3-e5a313fe1ece (snat)
  port snat-ovn-cluster
    type: router
    router-port: ovn-cluster-snat
switch 20e0c6d0-023a-4756-aec5-200e0c60f95d (join)
  port node-liumengxin-ovn3-192.168.137.178
    addresses: ["00:00:00:64:FF:A8 100.64.0.4"]
  port node-liumengxin-ovn1-192.168.137.176
    addresses: ["00:00:00:AF:98:62 100.64.0.2"]
  port node-liumengxin-ovn2-192.168.137.177
    addresses: ["00:00:00:D9:58:B8 100.64.0.3"]
  port join-ovn-cluster
    type: router
    router-port: ovn-cluster-join
switch 0191705c-f827-427b-9de3-3c3b7d971ba5 (central)
  port central-ovn-cluster
    type: router
    router-port: ovn-cluster-central
switch 2a45ff05-388d-4f85-9daf-e6fccd5833dc (ovn-default)
  port alertmanager-main-0.monitoring
    addresses: ["00:00:00:6C:DF:A3 10.16.0.19"]
  port kube-state-metrics-5d6885d89-4nf8h.monitoring
    addresses: ["00:00:00:6F:02:1C 10.16.0.15"]
  port fake-kubelet-67c55dfd89-pv86k.kube-system
    addresses: ["00:00:00:5C:12:E8 10.16.19.177"]
  port ovn-default-ovn-cluster
    type: router
    router-port: ovn-cluster-ovn-default
router 212f73dd-d63d-4d72-864b-a537e9afbee1 (ovn-cluster)
  port ovn-cluster-snat
    mac: "00:00:00:7A:82:8F"
    networks: ["172.22.0.1/16"]
  port ovn-cluster-join
    mac: "00:00:00:F8:18:5A"
    networks: ["100.64.0.1/16"]
  port ovn-cluster-central
    mac: "00:00:00:4D:8C:F5"
    networks: ["192.101.0.1/16"]
  port ovn-cluster-ovn-default
    mac: "00:00:00:A3:F8:18"
    networks: ["10.16.0.1/16"]
```

vsctl {nodeName} [options ...]

nodeName

ovs-ovn

ovs-vsctl

vswitchd

OVS

ovs-vsctl(8)

```
# kubectl ko vsctl kube-ovn-01 show
0d4c4675-c9cc-440a-8c1a-878e17f81b88
  Bridge br-int
    fail_mode: secure
    datapath_type: system
    Port a2c1a8a8b83a_h
      Interface a2c1a8a8b83a_h
      Port "4fa5c4cbb1a5_h"
      Interface "4fa5c4cbb1a5_h"
    Port ovn-eef07d-0
      Interface ovn-eef07d-0
      type: stt
      options: {csum="true", key=flow, remote_ip="192.168.137.178"}
    Port ovn0
```



```

    Interface ovn0
      type: internal
    Port mirror0
      Interface mirror0
        type: internal
    Port ovn-efa253-0
      Interface ovn-efa253-0
        type: stt
        options: {csum="true", key=flow, remote_ip="192.168.137.177"}
    Port br-int
      Interface br-int
        type: internal
    ovs_version: "2.17.2"

```

ofctl {nodeName} [options ...]

nodeName ovs-ovn ovs-ofctl OpenFlow OVS ovs-ofctl(8)

```

# kubectl ko ofctl kube-ovn-01 dump-flows br-int
NXST_FLOW reply (xid=0x4): flags=[more]
cookie=0xcfc3429e6, duration=671791.432s, table=0, n_packets=0, n_bytes=0, idle_age=65534, hard_age=65534, priority=100,in_port=2 actions=load:0x4->NXM_NX_REG13[],load:0x9->NXM_NX_REG11[],load:0xb->NXM_NX_REG12[],load:0x4->OXM_OF_METADATA[],load:0x1->NXM_NX_REG14[],resubmit(,8)
cookie=0xc91413c6, duration=671791.431s, table=0, n_packets=907489, n_bytes=99978275, idle_age=0, hard_age=65534, priority=100,in_port=7 actions=load:0x1->NXM_NX_REG13[],load:0x9->NXM_NX_REG11[],load:0xb->NXM_NX_REG12[],load:0x4->OXM_OF_METADATA[],load:0x4->NXM_NX_REG14[],resubmit(,8)
cookie=0xf180459, duration=671791.431s, table=0, n_packets=17348582, n_bytes=2667811214, idle_age=0, hard_age=65534, priority=100,in_port=6317 actions=load:0xa->NXM_NX_REG13[],load:0x9->NXM_NX_REG11[],load:0xb->NXM_NX_REG12[],load:0x4->OXM_OF_METADATA[],load:0x9->NXM_NX_REG14[],resubmit(,8)
cookie=0x7806dd90, duration=671791.431s, table=0, n_packets=3235428, n_bytes=833821312, idle_age=0, hard_age=65534, priority=100,in_port=1 actions=load:0xd->NXM_NX_REG13[],load:0x9->NXM_NX_REG11[],load:0xb->NXM_NX_REG12[],load:0x4->OXM_OF_METADATA[],load:0x3->NXM_NX_REG14[],resubmit(,8)
...

```

dpctl {nodeName} [options ...]

nodeName ovs-ovn ovs-dpctl OVS datapath OVS ovs-dpctl(8)

```

# kubectl ko dpctl kube-ovn-01 show
system@ovs-system:
  lookups: hit:350805055 missed:21983648 lost:73
  flows: 105
  masks: hit:1970748791 total:22 hit/pkt:5.29
  port 0: ovs-system (internal)
  port 1: ovn0 (internal)
  port 2: mirror0 (internal)
  port 3: br-int (internal)
  port 4: stt_sys_7471 (stt: packet_type=ptap)
  port 5: eeb4d9e51b5d_h
  port 6: a2c1a8a8b83a_h
  port 7: 4fa5c4cbb1a5_h

```

apctl {nodeName} [options ...]

nodeName ovs-ovn ovs-apctl daemon OVS ovs-apctl(8)

```

# kubectl ko apctl kube-ovn-01 vlog/list
      console      syslog      file
      -----
backtrace      OFF      ERR      INFO
bfd            OFF      ERR      INFO
bond           OFF      ERR      INFO
bridge         OFF      ERR      INFO
bundle         OFF      ERR      INFO
bundles        OFF      ERR      INFO
...

```

tcpdump {namespace/podname} [tcpdump options ...]

namespace/podname kube-ovn-cni tcpdump veth

```

# kubectl ko tcpdump default/ds1-l6n7p icmp
+ kubectl exec -it kube-ovn-cni-wlg4s -n kube-ovn -- tcpdump -nn -i d7176fe7b4e0_h icmp
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on d7176fe7b4e0_h, link-type EN10MB (Ethernet), capture size 262144 bytes
06:52:36.619688 IP 100.64.0.3 > 10.16.0.4: ICMP echo request, id 2, seq 1, length 64
06:52:36.619746 IP 10.16.0.4 > 100.64.0.3: ICMP echo reply, id 2, seq 1, length 64
06:52:37.619588 IP 100.64.0.3 > 10.16.0.4: ICMP echo request, id 2, seq 2, length 64
06:52:37.619630 IP 10.16.0.4 > 100.64.0.3: ICMP echo reply, id 2, seq 2, length 64
06:52:38.619933 IP 100.64.0.3 > 10.16.0.4: ICMP echo request, id 2, seq 3, length 64
06:52:38.619973 IP 10.16.0.4 > 100.64.0.3: ICMP echo reply, id 2, seq 3, length 64

```

trace [arguments ...]

Pod	OVN	Openflow
-----	-----	----------

```
kubectl ko trace {namespace/podname} {target ip address} [target mac address] {icmp|tcp|udp} [target tcp/udp port]
kubectl ko trace {namespace/podname} {target ip address} [target mac address] arp {request|reply}
kubectl ko trace {node//nodename} {target ip address} [target mac address] {icmp|tcp|udp} [target tcp/udp port]
kubectl ko trace {node//nodename} {target ip address} [target mac address] arp {request|reply}
```

```
# kubectl ko trace default/ds1-l6n7p 8.8.8.8 icmp
+ kubectl exec ovn-central-5bc494cb5-np9hm -n kube-ovn -- ovn-trace --ct=new ovn-default 'inport == "ds1-l6n7p.default" && ip.ttl == 64 && icmp && eth.src ==
0a:00:00:10:00:05 && ip4.src == 10.16.0.4 && eth.dst == 00:00:00:b8:ca:43 && ip4.dst == 8.8.8.8'
# icmp, reg14=0xf, vlan_tci=0x0000, dl_src=0a:00:00:10:00:05, dl_dst=00:00:00:b8:ca:
43, nw_src=10.16.0.4, nw_dst=8.8.8.8, nw_tos=0, nw_ecn=0, nw_ttl=64, icmp_type=0, icmp_code=0

ingress(dp="ovn-default", inport="ds1-l6n7p.default")
-----
0. ls_in_port_sec_l2 (ovn-northd.c:4143): inport == "ds1-l6n7p.default" && eth.src == {0a:00:00:10:00:05}, priority 50, uuid 39453393
  next;
1. ls_in_port_sec_ip (ovn-northd.c:2898): inport == "ds1-l6n7p.default" && eth.src == 0a:00:00:10:00:05 && ip4.src == {10.16.0.4}, priority 90, uuid 81bcd485
  next;
3. ls_in_pre_acl (ovn-northd.c:3269): ip, priority 100, uuid 7b4f4971
  reg0[0] = 1;
  next;
5. ls_in_pre_stateful (ovn-northd.c:3396): reg0[0] == 1, priority 100, uuid 36cdd577
  ct_next;

ct_next(ct_state=new|trk)
-----
6. ls_in_acl (ovn-northd.c:3759): ip && (!ct.est || (ct.est && ct_label.blocked == 1)), priority 1, uuid 7608af5b
  reg0[1] = 1;
  next;
10. ls_in_stateful (ovn-northd.c:3995): reg0[1] == 1, priority 100, uuid 2aba1b90
  ct_commit(ct_label=0/0x1);
  next;
16. ls_in_l2_lkup (ovn-northd.c:4470): eth.dst == 00:00:00:b8:ca:43, priority 50, uuid 5c9c3c9f
  output = "ovn-default-ovn-cluster";
  output;

...
```

trace	Underlay	Mac
-------	----------	-----

```
kubectl ko trace default/virt-handler-7lvm1 8.8.8.8 82:7c:9f:83:8c:01 icmp
```

diagnose {all|node|subnet|IPPorts} [nodename|subnetName|{proto1}-{IP1}-{Port1},{proto2}-{IP2}-{Port2}]

```
kube-ovn-pinger
```

```
# kubectl ko diagnose all
switch c7cd17e8-ceee-4a91-9bb3-e5a313fe1ece (snat)
  port snat-ovn-cluster
    type: router
    router-port: ovn-cluster-snat
switch 20e0c6d0-023a-4756-aec5-200e0c60f95d (join)
  port node-liumengxin-ovn3-192.168.137.178
    addresses: ["00:00:00:64:FF:A8 100.64.0.4"]
  port node-liumengxin-ovn1-192.168.137.176
    addresses: ["00:00:00:AF:98:62 100.64.0.2"]
  port join-ovn-cluster
    type: router
    router-port: ovn-cluster-join
switch 0191705c-f827-427b-9de3-3c3b7d971ba5 (central)
  port central-ovn-cluster
    type: router
    router-port: ovn-cluster-central
switch 2a45ff05-388d-4f85-9daf-e6fccd5833dc (ovn-default)
  port ovn-default-ovn-cluster
    type: router
    router-port: ovn-cluster-ovn-default
  port prometheus-k8s-1.monitoring
    addresses: ["00:00:00:AA:37:DF 10.16.0.23"]
router 212f73dd-d63d-4d72-864b-a537e9afbee1 (ovn-cluster)
  port ovn-cluster-snat
    mac: "00:00:00:7A:82:8F"
    networks: ["172.22.0.1/16"]
  port ovn-cluster-join
    mac: "00:00:00:F8:18:5A"
    networks: ["100.64.0.1/16"]
  port ovn-cluster-central
```

```

    mac: "00:00:00:4D:8C:F5"
    networks: ["192.101.0.1/16"]
  port ovn-cluster-ovn-default
    mac: "00:00:00:A3:F8:18"
    networks: ["10.16.0.1/16"]
Routing Policies
31000 ip4.dst == 10.16.0.0/16 allow
31000 ip4.dst == 100.64.0.0/16 allow
30000 ip4.dst == 192.168.137.177 reroute 100.64.0.3
30000 ip4.dst == 192.168.137.178 reroute 100.64.0.4
29000 ip4.src == $ovn.default.fake.6_ip4 reroute 100.64.0.22
29000 ip4.src == $ovn.default.fake.7_ip4 reroute 100.64.0.21
29000 ip4.src == $ovn.default.fake.8_ip4 reroute 100.64.0.23
20000 ip4.src == $ovn.default.liumengxin.ovn3.192.168.137.178_ip4 reroute 100.64.0.4
20000 ip4.src == $ovn.default.liumengxin.ovn1.192.168.137.176_ip4 && ip4.dst != $ovn.cluster.overlay.subnets.IPV4 reroute 100.
64.0.2
20000 ip4.src == $ovn.default.liumengxin.ovn2.192.168.137.177_ip4 && ip4.dst != $ovn.cluster.overlay.subnets.IPV4 reroute 100.
64.0.3
20000 ip4.src == $ovn.default.liumengxin.ovn3.192.168.137.178_ip4 && ip4.dst != $ovn.cluster.overlay.subnets.IPV4 reroute 100.
64.0.4
IPv4 Routes
Route Table <main>:
0.0.0.0/0 100.64.0.1 dst-ip
UUID LB PROTO VIP IPs
e9bcbfd9d-793e-4431-9073-6dec96b75d71 cluster-tcp-load tcp 10.100.209.132:10660 192.168.137.176:10660
tcp 10.101.239.192:6641 192.168.137.177:6641
tcp 10.101.240.101:3000 10.16.0.7:3000
tcp 10.103.184.186:6642 192.168.137.177:6642
35d2b7a5-e3a7-485a-a4b7-b4970eb0e63b cluster-tcp-sess tcp 10.100.158.128:8080 10.16.0.10:8080,10.16.0.5:8080,10.16.63.30:8080
tcp 10.107.26.215:8080 10.16.0.19:8080,10.16.0.20:8080,10.16.0.21:8080
tcp 10.107.26.215:9093 10.16.0.19:9093,10.16.0.20:9093,10.16.0.21:9093
tcp 10.98.187.99:8080 10.16.0.22:8080,10.16.0.23:8080
tcp 10.98.187.99:9090 10.16.0.22:9090,10.16.0.23:9090
f43303e4-89aa-4d3e-a3dc-278a552fe27b cluster-udp-load udp 10.96.0.10:53 10.16.0.4:53,10.16.0.9:53
_uuid : 06776304-5a96-43ed-90c4-c4854c251699
addresses : []
external_ids : {vendor=kube-ovn}
name : node_liumengxin_ovn2_192.168.137.177_underlay_v6
_uuid : 62690625-87d5-491c-8675-9fd83b1f433c
addresses : []
external_ids : {vendor=kube-ovn}
name : node_liumengxin_ovn1_192.168.137.176_underlay_v6
_uuid : b03a9bae-94d5-4562-b34c-b5f6198e180b
addresses : ["10.16.0.0/16", "100.64.0.0/16", "172.22.0.0/16", "192.101.0.0/16"]
external_ids : {vendor=kube-ovn}
name : ovn.cluster.overlay.subnets.IPV4
_uuid : e1056f3a-24cc-4666-8a91-75ee6c3c2426
addresses : []
external_ids : {vendor=kube-ovn}
name : ovn.cluster.overlay.subnets.IPV6
_uuid : 3e5d5fff-e670-47b2-a2f5-a39f4698a8c5
addresses : []
external_ids : {vendor=kube-ovn}
name : node_liumengxin_ovn3_192.168.137.178_underlay_v6
_uuid : 2d85dbdc-d0db-4abe-b19e-cc806d32b492
action : drop
direction : from-lport
external_ids : {}
label : 0
log : false
match : "inport==@ovn.sg.kubeovn_deny_all && ip"
meter : []
name : []
options : {}
priority : 2003
severity : []
_uuid : de790cc8-f155-405f-bb32-5a51f30c545f
action : drop
direction : to-lport
external_ids : {}
label : 0
log : false
match : "outport==@ovn.sg.kubeovn_deny_all && ip"
meter : []
name : []
options : {}
priority : 2003
severity : []
Chassis "e15ed4d4-1780-4d50-b09e-ea8372ed48b8"
  hostname: liumengxin-ovn1-192.168.137.176
  Encap stt
    ip: "192.168.137.176"
    options: {csum="true"}
  Port_Binding node-liumengxin-ovn1-192.168.137.176
  Port_Binding perf-6vxkn.default
  Port_Binding kube-state-metrics-5d6885d89-4nf8h.monitoring
  Port_Binding alertmanager-main-0.monitoring
  Port_Binding kube-ovn-pinger-6ftdf.kube-system
  Port_Binding fake-kubelet-67c55dfd89-pv86k.kube-system

```

```

    Port_Binding prometheus-k8s-0.monitoring
Chassis "eef07da1-f8ad-4775-b14d-bd6a3b4eb0d5"
  hostname: liumengxin-ovn3-192.168.137.178
  Encap stt
    ip: "192.168.137.178"
    options: {csum="true"}
  Port_Binding kube-ovn-pinger-7twb4.kube-system
  Port_Binding prometheus-adapter-86df476d87-r188g.monitoring
  Port_Binding prometheus-k8s-1.monitoring
  Port_Binding node-liumengxin-ovn3-192.168.137.178
  Port_Binding perf-ff475.default
  Port_Binding alertmanager-main-1.monitoring
  Port_Binding blackbox-exporter-676d976865-tvsjd.monitoring
Chassis "efa253c9-494d-4719-83ae-b48ab0f11c03"
  hostname: liumengxin-ovn2-192.168.137.177
  Encap stt
    ip: "192.168.137.177"
    options: {csum="true"}
  Port_Binding grafana-6c4c6b8fb7-pzd2c.monitoring
  Port_Binding node-liumengxin-ovn2-192.168.137.177
  Port_Binding alertmanager-main-2.monitoring
  Port_Binding coredns-6789c94dd8-9jqs.kube-system
  Port_Binding coredns-6789c94dd8-25d4r.kube-system
  Port_Binding prometheus-operator-7bbc99fc8b-wgjm4.monitoring
  Port_Binding prometheus-adapter-86df476d87-gdxmc.monitoring
  Port_Binding perf-fjnws.default
  Port_Binding kube-ovn-pinger-vh2xg.kube-system
ds kube-proxy ready
kube-proxy ready
deployment ovn-central ready
deployment kube-ovn-controller ready
ds kube-ovn-cni ready
ds ovs-ovn ready
deployment coredns ready
ovn-nb leader check ok
ovn-sb leader check ok
ovn-northd leader check ok
### kube-ovn-controller recent log

### start to diagnose node liumengxin-ovn1-192.168.137.176
#### ovn-controller log:
2022-06-03T00:56:44.897Z|16722|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T01:06:44.912Z|16723|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T01:16:44.925Z|16724|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T01:26:44.936Z|16725|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T01:36:44.959Z|16726|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T01:46:44.974Z|16727|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T01:56:44.988Z|16728|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T02:06:45.001Z|16729|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T02:16:45.025Z|16730|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T02:26:45.040Z|16731|inc_proc_eng|INFO|User triggered force recompute.

#### ovs-vswnitchd log:
2022-06-02T23:03:00.137Z|00079|dpif(handler1)|WARN|system@ovs-system: execute ct(commit,zone=14,label=0/0x1,nat(src)),8 failed (Invalid argument) on packet
icmp,vlan_tci=0x0000,d1_src=00:00:00:f8:07:c8,d1_dst=00:00:00:fa:1e:50,nw_src=10.16.0.5,nw_dst=10.16.0.10,nw_tos=0,nw_ecn=0,nw_ttl=64,icmp_type=8,icmp_code=0
icmp_csum:f9d1
  with metadata skb_priority(0),tunnel(tun_id=0x160017000004,src=192.168.137.177,dst=192.168.137.176,ttl=64,tp_src=38881,tp_dst=7471,flags(csum|
key)),skb_mark(0),ct_state(0x21),ct_zone(0xe),ct_tuple4(src=10.16.0.5,dst=10.16.0.10,proto=1,tp_src=8,tp_dst=0),in_port(4) mtu 0
2022-06-02T23:23:31.840Z|00080|dpif(handler1)|WARN|system@ovs-system: execute ct(commit,zone=14,label=0/0x1,nat(src)),8 failed (Invalid argument) on packet
icmp,vlan_tci=0x0000,d1_src=00:00:00:f8:07:c8,d1_dst=00:00:00:fa:1e:50,nw_src=10.16.0.5,nw_dst=10.16.0.10,nw_tos=0,nw_ecn=0,nw_ttl=64,icmp_type=8,icmp_code=0
icmp_csum:15b2
  with metadata skb_priority(0),tunnel(tun_id=0x160017000004,src=192.168.137.177,dst=192.168.137.176,ttl=64,tp_src=38881,tp_dst=7471,flags(csum|
key)),skb_mark(0),ct_state(0x21),ct_zone(0xe),ct_tuple4(src=10.16.0.5,dst=10.16.0.10,proto=1,tp_src=8,tp_dst=0),in_port(4) mtu 0
2022-06-03T00:09:15.659Z|00081|dpif(handler1)|WARN|system@ovs-system: execute ct(commit,zone=14,label=0/0x1,nat(src)),8 failed (Invalid argument) on packet
icmp,vlan_tci=0x0000,d1_src=00:00:00:dc:e3:63,d1_dst=00:00:00:fa:1e:50,nw_src=10.16.63.30,nw_dst=10.16.0.10,nw_tos=0,nw_ecn=0,nw_ttl=64,icmp_type=8,icmp_code=0 icmp_csum:e5a5
  with metadata skb_priority(0),tunnel(tun_id=0x150017000004,src=192.168.137.178,dst=192.168.137.176,ttl=64,tp_src=9239,tp_dst=7471,flags(csum|
key)),skb_mark(0),ct_state(0x21),ct_zone(0xe),ct_tuple4(src=10.16.63.30,dst=10.16.0.10,proto=1,tp_src=8,tp_dst=0),in_port(4) mtu 0
2022-06-03T00:02:33.832Z|00082|dpif(handler2)|WARN|system@ovs-system: execute ct(commit,zone=14,label=0/0x1,nat(src)),8 failed (Invalid argument) on packet
icmp,vlan_tci=0x0000,d1_src=00:00:00:f8:07:c8,d1_dst=00:00:00:fa:1e:50,nw_src=10.16.0.5,nw_dst=10.16.0.10,nw_tos=0,nw_ecn=0,nw_ttl=64,icmp_type=8,icmp_code=0
icmp_csum:a819
  with metadata skb_priority(0),tunnel(tun_id=0x160017000004,src=192.168.137.177,dst=192.168.137.176,ttl=64,tp_src=38881,tp_dst=7471,flags(csum|
key)),skb_mark(0),ct_state(0x21),ct_zone(0xe),ct_tuple4(src=10.16.0.5,dst=10.16.0.10,proto=1,tp_src=8,tp_dst=0),in_port(4) mtu 0

#### ovs-vsctl show results:
0d4c4675-c9cc-440a-8c1a-878e17f81b88
  Bridge br-int
    fail_mode: secure
    datapath_type: system
    Port a2c1a8a8b83a_h
      Interface a2c1a8a8b83a_h
      Port "4fa5c4cbb1a5_h"
      Interface "4fa5c4cbb1a5_h"
    Port ovn-eef07d-0
      Interface ovn-eef07d-0
      type: stt
      options: {csum="true", key=flow, remote_ip="192.168.137.178"}
    Port ovn0
      Interface ovn0

```

```

        type: internal
    Port "04d03360e9a0_h"
        Interface "04d03360e9a0_h"
    Port eeb4d9e51b5d_h
        Interface eeb4d9e51b5d_h
    Port mirror0
        Interface mirror0
        type: internal
    Port "8e5d887ccd80_h"
        Interface "8e5d887ccd80_h"
    Port ovn-efa253-0
        Interface ovn-efa253-0
        type: stt
        options: {csum="true", key=flow, remote_ip="192.168.137.177"}
    Port "17512d5be1f1_h"
        Interface "17512d5be1f1_h"
    Port br-int
        Interface br-int
        type: internal
    ovs_version: "2.17.2"

### pinger diagnose results:
I0603 10:35:04.349404 17619 pinger.go:19]
-----
Kube-OVN:
Version:      v1.16.5
Build:        2022-04-24-08:02:50
Commit:       git-73f9d15
Go Version:   go1.17.8
Arch:         amd64
-----
I0603 10:35:04.376797 17619 config.go:166] pinger config is &{KubeConfigFile: KubeClient:0xc000493380 Port:8080 DaemonSetName:kube-system
DaemonSetName:kube-ovn-pinger Interval:5 Mode:job ExitCode:0 InternalDNS:kubernetes.default ExternalDNS: nodeName:liumengxin-ovn1-192.168.137.176 HostIP:
192.168.137.176 PodName:kube-ovn-pinger-6ftdf PodIP:10.16.0.10 PodProtocols:[IPv4] ExternalAddress: NetworkMode:kube-ovn PollTimeout:2 PollInterval:15
SystemRunDir:/var/run/openvswitch DatabaseVswitchName:Open_vSwitch DatabaseVswitchSocketRemote:unix:/var/run/openvswitch/db.sock DatabaseVswitchFileDataPath:/
etc/openvswitch/conf.db DatabaseVswitchFileLogPath:/var/log/openvswitch/ovsdb-server.log DatabaseVswitchFilePidPath:/var/run/openvswitch/ovsdb-server.pid
DatabaseVswitchFileSystemIDPath:/etc/openvswitch/system-id.conf ServiceVswitchdFileLogPath:/var/log/openvswitch/ovs-vswitchd.log ServiceVswitchdFilePidPath:/
var/run/openvswitch/ovs-vswitchd.pid ServiceOvnControllerFileLogPath:/var/log/ovn/ovn-controller.log ServiceOvnControllerFilePidPath:/var/run/ovn/ovn-
controller.pid}
I0603 10:35:04.449166 17619 exporter.go:75] liumengxin-ovn1-192.168.137.176: exporter connect successfully
I0603 10:35:04.554011 17619 ovn.go:21] ovs-vswitchd and ovsdb are up
I0603 10:35:04.651293 17619 ovn.go:33] ovn_controller is up
I0603 10:35:04.651342 17619 ovn.go:39] start to check port binding
I0603 10:35:04.749613 17619 ovn.go:135] chassis id is 1d7f3d6c-ee5-4b3c-adca-2969d9cdfd80
I0603 10:35:04.763487 17619 ovn.go:49] port in sb is [node-liumengxin-ovn1-192.168.137.176 perf-6vxkn.default kube-state-metrics-5d6885d89-4nf8h.monitoring
alertmanager-main-0.monitoring kube-ovn-pinger-6ftdf.kube-system fake-kubelet-67c55dfd89-pv86k.kube-system prometheus-k8s-0.monitoring]
I0603 10:35:04.763583 17619 ovn.go:61] ovs and ovn-sb binding check passed
I0603 10:35:05.049309 17619 ping.go:259] start to check apiserver connectivity
I0603 10:35:05.053666 17619 ping.go:268] connect to apiserver success in 4.27ms
I0603 10:35:05.053786 17619 ping.go:129] start to check pod connectivity
I0603 10:35:05.249590 17619 ping.go:159] ping pod: kube-ovn-pinger-6ftdf 10.16.0.10, count: 3, loss count 0, average rtt 16.30ms
I0603 10:35:05.354135 17619 ping.go:159] ping pod: kube-ovn-pinger-7twb4 10.16.63.30, count: 3, loss count 0, average rtt 1.81ms
I0603 10:35:05.458460 17619 ping.go:159] ping pod: kube-ovn-pinger-vh2xg 10.16.0.5, count: 3, loss count 0, average rtt 1.92ms
I0603 10:35:05.458523 17619 ping.go:83] start to check node connectivity

```

diagnose	subnet	subnet	daemonset	kube-ovn-pinger	daemonset	pod	daemonset
diagnose	IPPorts	kube-ovn-pinger	pod	IP Port			

reload

Kube-OVN

```

# kubectl ko reload
pod "ovn-central-8684dd94bd-vzgcr" deleted
Waiting for deployment "ovn-central" rollout to finish: 0 of 1 updated replicas are available...
deployment "ovn-central" successfully rolled out
pod "ovs-ovn-bsnvz" deleted
pod "ovs-ovn-m9b98" deleted
pod "kube-ovn-controller-8459db5ff4-64c62" deleted
Waiting for deployment "kube-ovn-controller" rollout to finish: 0 of 1 updated replicas are available...
deployment "kube-ovn-controller" successfully rolled out
pod "kube-ovn-cni-2klmh" deleted
pod "kube-ovn-cni-t2jz4" deleted
Waiting for daemon set "kube-ovn-cni" rollout to finish: 0 of 2 updated pods are available...
Waiting for daemon set "kube-ovn-cni" rollout to finish: 1 of 2 updated pods are available...
daemon set "kube-ovn-cni" successfully rolled out
pod "kube-ovn-pinger-ln72z" deleted
pod "kube-ovn-pinger-w8lrk" deleted
Waiting for daemon set "kube-ovn-pinger" rollout to finish: 0 of 2 updated pods are available...
Waiting for daemon set "kube-ovn-pinger" rollout to finish: 1 of 2 updated pods are available...
daemon set "kube-ovn-pinger" successfully rolled out
pod "kube-ovn-monitor-7fb67d5488-7q6zb" deleted
Waiting for deployment "kube-ovn-monitor" rollout to finish: 0 of 1 updated replicas are available...
deployment "kube-ovn-monitor" successfully rolled out

```

log

kube-ovn Kube-OVN OVN Open vSwitch log linux debug

```
# kubectl ko log all
Collecting kube-ovn logging files
Collecting ovn logging files
Collecting openvswitch logging files
Collecting linux dmesg files
Collecting linux iptables-legacy files
Collecting linux iptables-nft files
Collecting linux route files
Collecting linux link files
Collecting linux neigh files
Collecting linux memory files
Collecting linux top files
Collecting linux sysctl files
Collecting linux netstat files
Collecting linux addr files
Collecting linux ipset files
Collecting linux tcp files
Collected files have been saved in the directory /root/kubectl-ko-log
```

```
# tree kubectl-ko-log/
kubectl-ko-log/
|-- kube-ovn-control-plane
|   |-- kube-ovn
|   |   |-- kube-ovn-cni.log
|   |   |-- kube-ovn-monitor.log
|   |   '-- kube-ovn-pinger.log
|   |-- linux
|   |   |-- addr.log
|   |   |-- dmesg.log
|   |   |-- ipset.log
|   |   |-- iptables-legacy.log
|   |   |-- iptables-nft.log
|   |   |-- link.log
|   |   |-- memory.log
|   |   |-- neigh.log
|   |   |-- netstat.log
|   |   |-- route.log
|   |   |-- sysctl.log
|   |   |-- tcp.log
|   |   '-- top.log
|   |-- openvswitch
|   |   |-- ovs-vswitchd.log
|   |   '-- ovsdb-server.log
|   '-- ovn
|       |-- ovn-controller.log
|       |-- ovn-northd.log
|       |-- ovsdb-server-nb.log
|       '-- ovsdb-server-sb.log
```

perf [image]

Kube-OVN

- 1.
- 2. Hostnetwork
- 3.
- 4. OVN-NB, OVN-SB, OVN-Northd leader

image Pod `docker.io/kubeovn/test:v1.13.0`

```
# kubectl ko perf
===== Prepareing Performance Test Resources =====
pod/test-client created
pod/test-host-client created
pod/test-server created
pod/test-host-server created
service/test-server created
pod/test-client condition met
pod/test-host-client condition met
pod/test-host-server condition met
pod/test-server condition met
=====
===== Start Pod Network Unicast Performance Test =====
Size          TCP Latency    TCP Bandwidth  UDP Latency    UDP Lost Rate  UDP Bandwidth
64             82.8 us        97.7 Mbits/sec 67.6 us        (0%)           8.42 Mbits/sec
128            85.4 us        167 Mbits/sec  67.2 us        (0%)           17.2 Mbits/sec
```

```
512      85.8 us      440 Mbits/sec  68.7 us      (0%)      68.4 Mbits/sec
1k       85.1 us      567 Mbits/sec  68.7 us      (0%)      134 Mbits/sec
4k       138 us      826 Mbits/sec  78.1 us      (1.4%)    583 Mbits/sec
=====
===== Start Host Network Performance Test =====
Size      TCP Latency    TCP Bandwidth  UDP Latency    UDP Lost Rate  UDP Bandwidth
64        49.7 us      120 Mbits/sec  37.9 us      (0%)          18.6 Mbits/sec
128       49.7 us      200 Mbits/sec  38.1 us      (0%)          35.5 Mbits/sec
512       51.9 us      588 Mbits/sec  38.9 us      (0%)          142 Mbits/sec
1k        51.7 us      944 Mbits/sec  37.2 us      (0%)          279 Mbits/sec
4k        74.9 us      1.66 Gbits/sec 39.9 us      (0%)          1.20 Gbits/sec
=====
===== Start Service Network Performance Test =====
Size      TCP Latency    TCP Bandwidth  UDP Latency    UDP Lost Rate  UDP Bandwidth
64        111 us      96.3 Mbits/sec  88.4 us      (0%)          7.59 Mbits/sec
128       83.7 us      150 Mbits/sec  69.2 us      (0%)          16.9 Mbits/sec
512       87.4 us      374 Mbits/sec  75.8 us      (0%)          60.9 Mbits/sec
1k        88.2 us      521 Mbits/sec  73.1 us      (0%)          123 Mbits/sec
4k        148 us      813 Mbits/sec  77.6 us      (0.0044%)     451 Mbits/sec
=====
===== Start Pod Multicast Network Performance Test =====
Size      UDP Latency    UDP Lost Rate  UDP Bandwidth
64        0.014 ms      (0.17%)       5.80 Mbits/sec
128       0.012 ms      (0%)          11.4 Mbits/sec
512       0.016 ms      (0%)          46.1 Mbits/sec
1k        0.023 ms      (0.073%)      89.8 Mbits/sec
4k        0.035 ms      (1.3%)        126 Mbits/sec
=====
===== Start Host Multicast Network Performance =====
Size      UDP Latency    UDP Lost Rate  UDP Bandwidth
64        0.007 ms      (0%)          9.95 Mbits/sec
128       0.005 ms      (0%)          21.8 Mbits/sec
512       0.008 ms      (0%)          86.8 Mbits/sec
1k        0.013 ms      (0.045%)     168 Mbits/sec
4k        0.010 ms      (0.31%)     242 Mbits/sec
=====
===== Start Leader Recover Time Test =====
Delete ovn central nb pod
pod "ovn-central-5cb9c67d75-tlz9w" deleted
Waiting for ovn central nb pod running
===== OVN nb Recovery takes 3.305236803 s =====
Delete ovn central sb pod
pod "ovn-central-5cb9c67d75-szx4c" deleted
Waiting for ovn central sb pod running
===== OVN sb Recovery takes 3.462698535 s =====
Delete ovn central northd pod
pod "ovn-central-5cb9c67d75-zqmvy" deleted
Waiting for ovn central northd pod running
===== OVN northd Recovery takes 2.691291403 s =====
===== Remove Performance Test Resource =====
rm -f unicast-test-client.log
rm -f unicast-test-host-client.log
rm -f unicast-test-client.log
kubectl ko nbctl lb-del test-server
rm -f multicast-test-server.log
kubectl exec ovs-ovn-gxdrf -n kube-system -- ip maddr del 01:00:5e:00:00:64 dev eth0
kubectl exec ovs-ovn-h57bf -n kube-system -- ip maddr del 01:00:5e:00:00:64 dev eth0
rm -f multicast-test-host-server.log
pod "test-client" deleted
pod "test-host-client" deleted
pod "test-host-server" deleted
pod "test-server" deleted
service "test-server" deleted
=====
```

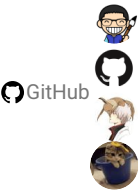
 PDF

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6.2

Kubernetes

ovs-ovn

ovn-controller

ovn-central

Kubernetes

ovn-central

ovn-central

6.2.1

```
# kubectl drain kube-ovn-worker --ignore-daemonsets --force
node/kube-ovn-worker cordoned
WARNING: ignoring DaemonSet-managed Pods: kube-system/kube-ovn-cni-zt74b, kube-system/kube-ovn-pinger-5rxfs, kube-system/kube-proxy-jpmnm, kube-system/ovs-
ovn-v2kll
evicting pod kube-system/coredns-64897985d-qsgpt
evicting pod local-path-storage/local-path-provisioner-5ddd94ff66-1lss6
evicting pod kube-system/kube-ovn-controller-8459db5ff4-94lxb
pod/kube-ovn-controller-8459db5ff4-94lxb evicted
pod/coredns-64897985d-qsgpt evicted
pod/local-path-provisioner-5ddd94ff66-1lss6 evicted
node/kube-ovn-worker drained
```

6.2.2 kubelet docker

ovs-ovn

ovn-central

```
systemctl stop kubelet
systemctl stop docker
```

Containerd

containerd CRI

Pod/

ovs-ovn

containerd

ovs-ovn

1. kubelet

```
systemctl stop kubelet
```

2. ovs-ovn Pod

```
crictl rm -f $(crictl ps | grep openvswitch | awk '{print $1}')
```

3. containerd

```
systemctl stop containerd
```

6.2.3 Node

```
rm -rf /var/run/openvswitch
rm -rf /var/run/ovn
rm -rf /etc/origin/openvswitch/
rm -rf /etc/origin/ovn/
rm -rf /etc/cni/net.d/00-kube-ovn.conflist
rm -rf /etc/cni/net.d/01-kube-ovn.conflist
rm -rf /var/log/openvswitch
rm -rf /var/log/ovn
```

6.2.4 kubectl

```
kubectl delete no kube-ovn-01
```


6.2.5

ovn-sb

kube-ovn-worker

```
# kubectl ko sbctl show
Chassis "b0564934-5a0d-4804-a4c0-476c93596a17"
  hostname: kube-ovn-worker
  Encap geneve
    ip: "172.18.0.2"
    options: {csum="true"}
  Port_Binding kube-ovn-pinger-5rxfs.kube-system
Chassis "6a29de7e-d731-4eaf-bacd-2f239ee52b28"
  hostname: kube-ovn-control-plane
  Encap geneve
    ip: "172.18.0.3"
    options: {csum="true"}
  Port_Binding coredns-64897985d-nbfln.kube-system
  Port_Binding node-kube-ovn-control-plane
  Port_Binding local-path-provisioner-5ddd94ff66-h4tn9.local-path-storage
  Port_Binding kube-ovn-pinger-hf2p6.kube-system
  Port_Binding coredns-64897985d-fhwlw.kube-system
```

6.2.6

chassis

uuidChassisid

```
# kubectl ko sbctl chassis-del b0564934-5a0d-4804-a4c0-476c93596a17
# kubectl ko sbctl show
Chassis "6a29de7e-d731-4eaf-bacd-2f239ee52b28"
  hostname: kube-ovn-control-plane
  Encap geneve
    ip: "172.18.0.3"
    options: {csum="true"}
  Port_Binding coredns-64897985d-nbfln.kube-system
  Port_Binding node-kube-ovn-control-plane
  Port_Binding local-path-provisioner-5ddd94ff66-h4tn9.local-path-storage
  Port_Binding kube-ovn-pinger-hf2p6.kube-system
  Port_Binding coredns-64897985d-fhwlw.kube-system
```

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6.2.7

6.3 ovn-central

ovn-central ovn-nb ovn-sb etcd Raft ovn-central

6.3.1 ovn-central

kube-ovn-control-plane2

ovn-central

```
# kubectl -n kube-system get pod -o wide | grep central
ovn-central-6bf58cbc97-2cdhg          1/1      Running   0           21m    172.18.0.3    kube-ovn-control-plane    <none>    <none>
ovn-central-6bf58cbc97-crmfp          1/1      Running   0           21m    172.18.0.5    kube-ovn-control-plane2   <none>    <none>
ovn-central-6bf58cbc97-lxmpl          1/1      Running   0           21m    172.18.0.4    kube-ovn-control-plane3   <none>    <none>
```

ovn-nb

ID

```
# kubectl ko nb status
1b9a
Name: OVN_Northbound
Cluster ID: 32ca (32ca07fb-739b-4257-b510-12fa18e7cce8)
Server ID: 1b9a (1b9a5d76-e69b-410c-8085-39943d0cd38c)
Address: tcp:[172.18.0.3]:6643
Status: cluster member
Role: leader
Term: 1
Leader: self
Vote: self

Last Election started 2135194 ms ago, reason: timeout
Last Election won: 2135188 ms ago
Election timer: 5000
Log: [135, 135]
Entries not yet committed: 0
Entries not yet applied: 0
Connections: <-d64b ->d64b <-4984 ->4984
Disconnections: 0
Servers:
  4984 (4984 at tcp:[172.18.0.4]:6643) next_index=135 match_index=134 last msg 1084 ms ago
  1b9a (1b9a at tcp:[172.18.0.3]:6643) (self) next_index=2 match_index=134
  d64b (d64b at tcp:[172.18.0.5]:6643) next_index=135 match_index=134 last msg 1084 ms ago
status: ok
```

kube-ovn-control-plane2

IP 172.18.0.5

ID d64b

ovn-nb

```
# kubectl ko nb kick d64b
started removal
```

```
# kubectl ko nb status
1b9a
Name: OVN_Northbound
Cluster ID: 32ca (32ca07fb-739b-4257-b510-12fa18e7cce8)
Server ID: 1b9a (1b9a5d76-e69b-410c-8085-39943d0cd38c)
Address: tcp:[172.18.0.3]:6643
Status: cluster member
Role: leader
Term: 1
Leader: self
Vote: self

Last Election started 2297649 ms ago, reason: timeout
Last Election won: 2297643 ms ago
Election timer: 5000
Log: [136, 136]
Entries not yet committed: 0
Entries not yet applied: 0
Connections: <-4984 ->4984
Disconnections: 2
Servers:
  4984 (4984 at tcp:[172.18.0.4]:6643) next_index=136 match_index=135 last msg 1270 ms ago
  1b9a (1b9a at tcp:[172.18.0.3]:6643) (self) next_index=2 match_index=135
status: ok
```

ovn-sb

ovn-sb

ID

```
kubectl ko sb status
3722
Name: OVN_Southbound
Cluster ID: d4bd (d4bd37a4-0400-499f-b4df-b4fd389780f0)
Server ID: 3722 (3722d5ae-2ced-4820-a6b2-8b744d11fb3e)
Address: tcp:[172.18.0.3]:6644
Status: cluster member
Role: leader
Term: 1
Leader: self
Vote: self

Last Election started 2395317 ms ago, reason: timeout
Last Election won: 2395316 ms ago
Election timer: 5000
Log: [130, 130]
Entries not yet committed: 0
Entries not yet applied: 0
Connections: <-e9f7 ->e9f7 <-6e84 ->6e84
Disconnections: 0
Servers:
  e9f7 (e9f7 at tcp:[172.18.0.5]:6644) next_index=130 match_index=129 last msg 1006 ms ago
  6e84 (6e84 at tcp:[172.18.0.4]:6644) next_index=130 match_index=129 last msg 1004 ms ago
  3722 (3722 at tcp:[172.18.0.3]:6644) (self) next_index=2 match_index=129
status: ok
```

kube-ovn-control-plane2 IP 172.18.0.5 ID e9f7 ovn-sb

```
# kubectl ko sb kick e9f7
started removal
```

```
# kubectl ko sb status
3722
Name: OVN_Southbound
Cluster ID: d4bd (d4bd37a4-0400-499f-b4df-b4fd389780f0)
Server ID: 3722 (3722d5ae-2ced-4820-a6b2-8b744d11fb3e)
Address: tcp:[172.18.0.3]:6644
Status: cluster member
Role: leader
Term: 1
Leader: self
Vote: self

Last Election started 2481636 ms ago, reason: timeout
Last Election won: 2481635 ms ago
Election timer: 5000
Log: [131, 131]
Entries not yet committed: 0
Entries not yet applied: 0
Connections: <-6e84 ->6e84
Disconnections: 2
Servers:
  6e84 (6e84 at tcp:[172.18.0.4]:6644) next_index=131 match_index=130 last msg 642 ms ago
  3722 (3722 at tcp:[172.18.0.3]:6644) (self) next_index=2 match_index=130
status: ok
```

ovn-central

ovn-central

NODE_IPS

```
kubectl label node kube-ovn-control-plane2 kube-ovn/role-
kubectl scale deployment -n kube-system ovn-central --replicas=2
kubectl set env deployment/ovn-central -n kube-system NODE_IPS="172.18.0.3,172.18.0.4"
kubectl rollout status deployment/ovn-central -n kube-system
```

ovn-central

OVS-OVN

```
# kubectl set env daemonset/ovs-ovn -n kube-system OVN_DB_IPS="172.18.0.3,172.18.0.4"
daemonset.apps/ovs-ovn env updated
# kubectl delete pod -n kube-system -lapp=ovs
pod "ovs-ovn-4f6jc" deleted
```

```
pod "ovs-ovn-csn2w" deleted
pod "ovs-ovn-mpbmb" deleted
```

kube-ovn-controller

```
# kubectl set env deployment/kube-ovn-controller -n kube-system OVN_DB_IPS="172.18.0.3,172.18.0.4"
deployment.apps/kube-ovn-controller env updated

# kubectl rollout status deployment/kube-ovn-controller -n kube-system
Waiting for deployment "kube-ovn-controller" rollout to finish: 1 of 3 updated replicas are available...
Waiting for deployment "kube-ovn-controller" rollout to finish: 2 of 3 updated replicas are available...
deployment "kube-ovn-controller" successfully rolled out
```

kube-ovn-control-plane2

```
rm -rf /etc/origin/ovn
```

Kubernetes

6.3.2 ovn-central

Kubernetes

ovn-central

```
/etc/origin/ovn      ovnnb_db.db  ovnsb_db.db
```

```
rm -rf /etc/origin/ovn
```

ovn-central

ovn-central

```
# kubectl ko nb status
1b9a
Name: OVN_Northbound
Cluster ID: 32ca (32ca07fb-739b-4257-b510-12fa18e7cce8)
Server ID: 1b9a (1b9a5d76-e69b-410c-8085-39943d0cd38c)
Address: tcp:[172.18.0.3]:6643
Status: cluster member
Role: leader
Term: 44
Leader: self
Vote: self

Last Election started 1855739 ms ago, reason: timeout
Last Election won: 1855729 ms ago
Election timer: 5000
Log: [147, 147]
Entries not yet committed: 0
Entries not yet applied: 0
Connections: ->4984 <-4984
Disconnections: 0
Servers:
  4984 (4984 at tcp:[172.18.0.4]:6643) next_index=147 match_index=146 last msg 367 ms ago
  1b9a (1b9a at tcp:[172.18.0.3]:6643) (self) next_index=140 match_index=146
status: ok

# kubectl ko sb status
3722
Name: OVN_Southbound
Cluster ID: d4bd (d4bd37a4-0400-499f-b4df-b4fd389780f0)
Server ID: 3722 (3722d5ae-2ced-4820-a6b2-8b744d11fb3e)
Address: tcp:[172.18.0.3]:6644
Status: cluster member
Role: leader
Term: 33
Leader: self
Vote: self

Last Election started 1868589 ms ago, reason: timeout
Last Election won: 1868579 ms ago
Election timer: 5000
Log: [142, 142]
Entries not yet committed: 0
```

```
Entries not yet applied: 0
Connections: ->6e84 <-6e84
Disconnections: 0
Servers:
  6e84 (6e84 at tcp:[172.18.0.4]:6644) next_index=142 match_index=141 last msg 728 ms ago
  3722 (3722 at tcp:[172.18.0.3]:6644) (self) next_index=134 match_index=141
status: ok
```

ovn-central

NODE_IPS

```
kubectl label node kube-ovn-control-plane2 kube-ovn/role=master
kubectl scale deployment -n kube-system ovn-central --replicas=3
kubectl set env deployment/ovn-central -n kube-system NODE_IPS="172.18.0.3,172.18.0.4,172.18.0.5"
kubectl rollout status deployment/ovn-central -n kube-system
```

ovn-central

ovs-ovn

```
# kubectl set env daemonset/ovs-ovn -n kube-system OVN_DB_IPS="172.18.0.3,172.18.0.4,172.18.0.5"
daemonset.apps/ovs-ovn env updated
# kubectl delete pod -n kube-system -lapp=ovs
pod "ovs-ovn-4f6jc" deleted
pod "ovs-ovn-csn2w" deleted
pod "ovs-ovn-mpbmb" deleted
```

kube-ovn-controller

```
# kubectl set env deployment/kube-ovn-controller -n kube-system OVN_DB_IPS="172.18.0.3,172.18.0.4,172.18.0.5"
deployment.apps/kube-ovn-controller env updated

# kubectl rollout status deployment/kube-ovn-controller -n kube-system
Waiting for deployment "kube-ovn-controller" rollout to finish: 1 of 3 updated replicas are available...
Waiting for deployment "kube-ovn-controller" rollout to finish: 2 of 3 updated replicas are available...
deployment "kube-ovn-controller" successfully rolled out
```



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6.3.3

6.4 OVN

6.4.1

kubectl backup

```
# kubectl ko nb backup
tar: Removing leading '/' from member names
backup ovn-nb db to /root/ovnnb_db.060223191654183154.backup

# kubectl ko sb backup
tar: Removing leading '/' from member names
backup ovn-sb db to /root/ovnsb_db.060223191654183154.backup
```

6.4.2

/var/log/ovn/ovn-northd.log

```
* ovn-northd is not running
ovsdb-server: ovsdb error: error reading record 2739 from OVN_Northbound log: record 2739 advances commit index to 6308 but last log index is 6307
* Starting ovsdb-nb
```

OVN_Northbound

OVN_Southbound

OVN_Northbound

ovn-nb

```
# kubectl ko nb status
9182
Name: OVN_Northbound
Cluster ID: e75f (e75fa340-49ed-45ab-990e-26cb865ebc85)
Server ID: 9182 (9182e8dd-b5b0-4dd8-8518-598cc1e374f3)
Address: tcp:[10.0.128.61]:6643
Status: cluster member
Role: leader
Term: 1454
Leader: self
Vote: self

Last Election started 1732603 ms ago, reason: timeout
Last Election won: 1732587 ms ago
Election timer: 1000
Log: [7332, 12512]
Entries not yet committed: 1
Entries not yet applied: 1
Connections: ->f080 <-f080 <-e631 ->e631
Disconnections: 1
Servers:
  f080 (f080 at tcp:[10.0.129.139]:6643) next_index=12512 match_index=12510 last msg 63 ms ago
  9182 (9182 at tcp:[10.0.128.61]:6643) (self) next_index=10394 match_index=12510
  e631 (e631 at tcp:[10.0.131.173]:6643) next_index=12512 match_index=0
```

```
kubectl ko nb kick e631
```

```
mv /etc/origin/ovn/ovnnb_db.db /tmp
```

ovn-central Pod

```
kubectl delete pod -n kube-system ovn-central-xxxx
```

6.4.3

leader

ovn-central

ovn-central ovn-central

```
kubectl scale deployment -n kube-system ovn-central --replicas=0
```

ovsdb-tool cluster-to-standalone

ovn-central NODE_IPS /etc/origin/ovn

docker

```
docker run -it -v /etc/origin/ovn:/etc/ovn kubeovn/kube-ovn:v1.16.5 bash
cd /etc/ovn/
ovsdb-tool cluster-to-standalone ovnnb_db_standalone.db ovnnb_db.db
ovsdb-tool cluster-to-standalone ovnsb_db_standalone.db ovnsb_db.db
```

containerd docker ctr

```
ctr -n k8s.io image pull docker.io/kubeovn/kube-ovn:v1.16.5
ctr -n k8s.io run --rm -t \
  --mount type=bind,src=/etc/origin/ovn,dst=/etc/ovn,options=rbind:rw \
  docker.io/kubeovn/kube-ovn:v1.16.5 ovn-recover bash
cd /etc/ovn/
ovsdb-tool cluster-to-standalone ovnnb_db_standalone.db ovnnb_db.db
ovsdb-tool cluster-to-standalone ovnsb_db_standalone.db ovnsb_db.db
exit
```

ovn-central Pod ovsdb-tool cluster-to-standalone kubectl cp

ovn-central

```
mv /etc/origin/ovn/ovnnb_db.db /tmp
mv /etc/origin/ovn/ovnsb_db.db /tmp
```

ovnnb_db.db ovnsb_db.db ovn-central NODE_IPS /etc/origin/ovn/

```
mv /etc/origin/ovn/ovnnb_db_standalone.db /etc/origin/ovn/ovnnb_db.db
mv /etc/origin/ovn/ovnsb_db_standalone.db /etc/origin/ovn/ovnsb_db.db
```

ovn-central

```
kubectl scale deployment -n kube-system ovn-central --replicas=3
kubectl rollout status deployment/ovn-central -n kube-system
```



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6.4.4

6.6 Join CIDR

Join CIDR

Join CIDR Pod

6.6.1 Join

```
kubectl patch subnet join --type='json' -p '[{"op": "replace", "path": "/metadata/finalizers", "value": []}]'
kubectl delete subnet join
```

6.6.2

```
kubectl annotate node ovn.kubernetes.io/allocated=false --all --overwrite
```

6.6.3 Join

kube-ovn-controller Join

```
kubectl edit deployment -n kube-system kube-ovn-controller
```

```
args:
- --node-switch-cidr=100.51.0.0/16
```

kube-ovn-controller join

```
kubectl delete pod -n kube-system -lapp=kube-ovn-controller
```

Join

# kubectl get subnet												
NAME	PROVIDER	VPC	PROTOCOL	CIDR	PRIVATE	NAT	DEFAULT	GATEWAYTYPE	V4USED	V4AVAILABLE	V6USED	V6AVAILABLE
EXCLUDEIPS												
join	ovn	ovn-cluster	IPv4	100.51.0.0/16	false	false	false	distributed	2	65531	0	0
["100.51.0.1"]												
ovn-default	ovn	ovn-cluster	IPv4	10.17.0.0/16	false	true	true	distributed	5	65528	0	0
["10.17.0.1"]												

6.6.4 ovn0

ovn0 kube-ovn-cni

```
kubectl delete pod -n kube-system -l app=kube-ovn-cni
```

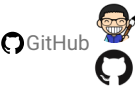
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6.6.5

6.7

kube-ovn.yaml

```
vi kube-ovn.yaml
# ...
- name: kube-ovn-controller
  image: "docker.io/kubeovn/kube-ovn:v1.16.5"
  imagePullPolicy: IfNotPresent
  args:
    - /kube-ovn/start-controller.sh
    - --v=3
# ...
#
```

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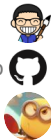
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6.7.1

6.8

Kube-OVN

6.8.1

1.
 - Pod Pod
 - Pod
 - Pod Service
 - Pod
 - Pod
 - kube-ovn-pinger
2.
 - Pod
 - kube-ovn-cni CNI
 - ovs-ovn OVS
 - kube-ovn-controller
 - ovn-central OVN
 - dmesg
 - netstat -s
3. CPU, IO
- kubectl ko logs

6.8.2 Pod IP

Pod Running kubectl describe Pod duplicate IPv4 address <ip> found on logical switch port <port>

1. Pod IP Pod IP IP IP Pod
2. kube-ovn-controller IP
3. kube-ovn-controller IP
4. kubectl ko nbctl show OVN IP
5. OVN Kubernetes IP kubectl ko nbctl lsp-del <port>

6.8.3 Pod ping gateway failed

Pod Running kubectl describe Pod network <ip> with gateway <gw ip> is not ready for interface eth0 after 30 checks

1. kubectl ko sbctl show Pod
2. ovn-central ovs-ovn ovn-central ovs-ovn

3. Pod
4. Underlay Underlay

6.8.4 Pod

VPC Pod

1. kubectl ko trace OVN ACL
2. ACL ACL
3. ACL Subnet stats
4. Subnet Spec
5. kube-ovn-controller

6.8.5 Pod IP CIDR

Pod IP CIDR

1. /etc/cni/net.d/ Kube-OVN CNI
2.
3. kubelet Pod

6.8.6 Debug Pod

kubectl debug Pod ContainerCreating Pod Event network not ready no address allocated

debug Pod yaml yaml Annotation

```
ovn.kubernetes.io/ip_address
ovn.kubernetes.io/mac_address
ovn.kubernetes.io/allocated
ovn.kubernetes.io/routed
```

debug Pod yaml debug Pod

6.8.7 ARM

ARM Offload

netstat

```
# netstat -us
IcmpMsg:
  InType0: 22
  InType3: 24
  InType8: 117852
  OutType0: 117852
```

```
OutType3: 29
OutType8: 22
Udp:
  3040636 packets received
  0 packets to unknown port received.
  4 packet receive errors
  602 packets sent
  0 receive buffer errors
  0 send buffer errors
  InCsumErrors: 4
UdpLite:
IpExt:
  InBcastPkts: 10244
  InOctets: 4446320361
  OutOctets: 1496815600
  InBcastOctets: 3095950
  InNoECTPkts: 7683903
```

InCsumErrors

tx offload TCP

```
ethtool -K eth0 tx off
```

CentOS 7 4.19.90-25.16.v2101

6.8.8 Pod Service

Pod Service dmesg

```
netlink: Unknown conntrack attr (type=6, max=5)
openvswitch: netlink: Flow actions may not be safe on all matching packets.
```

OVS NAT

- 1. OVS
- 2. Overlay kube-ovn-controller --enable-lb=false OVN LB kube-proxy Service

6.8.9 ovn-central

v1.11.x 1 Pod OVN NB SB Kube-OVN ovsdb-server/compact

v1.13 ENABLE_COMPACT false dist/images/install.sh start-db.sh ovn-central true

```
- name: ENABLE_COMPACT
  value: "false"
```

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 Support

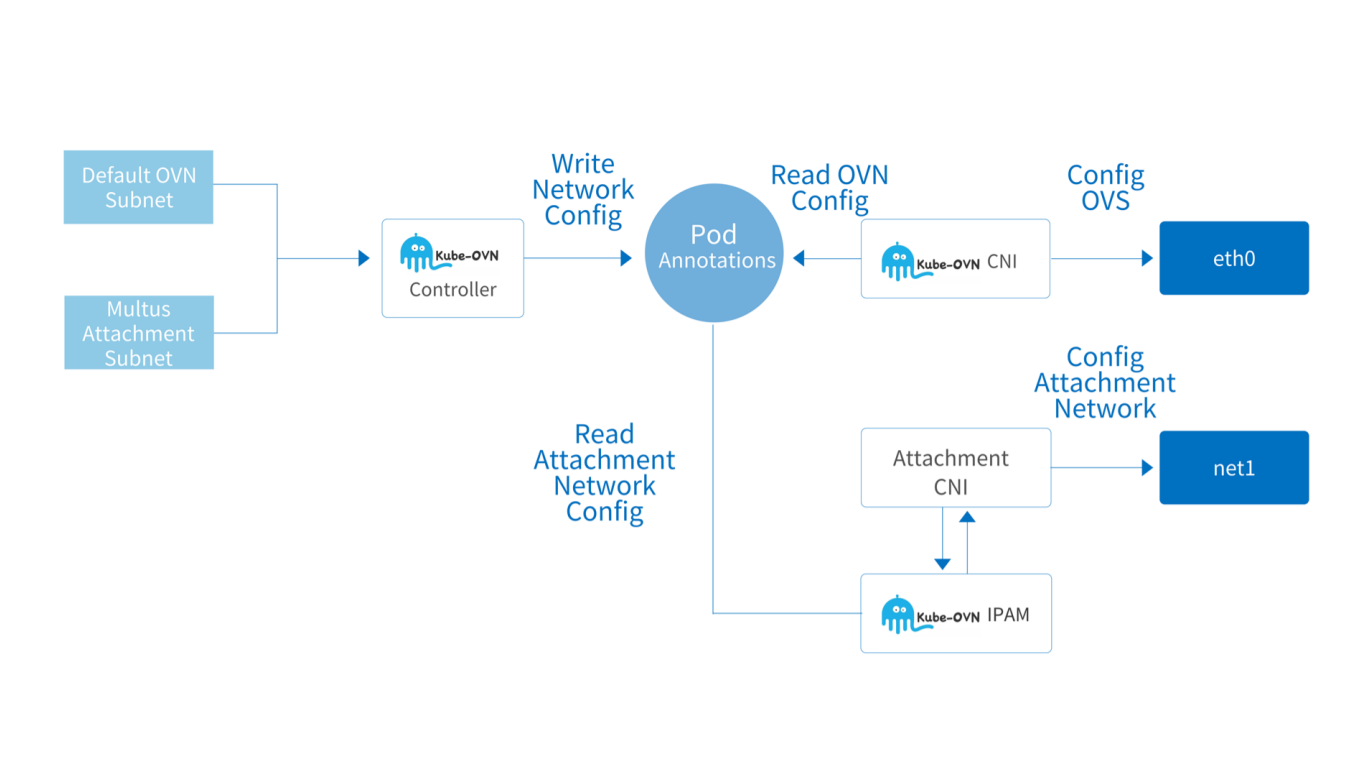
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6.8.10



	Kube-OVN	IP	eth0	OVN	net1	CNI	net1	multus-cni	NetworkAttachmentDefinition
Pod	kube-ovn-controller	Pod	Pod	Pod	annotation	Subnet	IP	Pod	Pod annotation
	CNI	kube-ovn-cni	ipam	kube-ovn-cni	Pod annotation		CNI		CNI

7.1.2

NetworkAttachmentDefinition spec multus defaultConfDir CNI

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: macvlan-conf-2
```

kube-ovn-controller NetworkAttachmentDefinition provider spec kube-ovn-controller provider spec Kube-OVN IPAM

7.1.3

Kube-OVN Multus

[Kube-OVN](#) [Multus how to use](#) [Kube-OVN](#) [Multus-CNI](#)

CNI IPAM

[Kube-OVN](#) [CNI](#)

NETWORKATTACHMENTDEFINITION

[macvlan](#) [ipam](#) [kube-ovn](#)

```
# macvlan
sudo modprobe macvlan
```

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: macvlan
  namespace: default
spec:
  config: '{
    "cniVersion": "0.3.0",
    "type": "macvlan",
    "master": "eth0",
    "mode": "bridge",
    "ipam": {
      "type": "kube-ovn",
      "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
      "provider": "macvlan.default"
    }
  }'
```

- spec.config.ipam.type: kube-ovn kube-ovn
- server_socket: Kube-OVN socket /run/openvswitch/kube-ovn-daemon.sock
- provider: NetworkAttachmentDefinition <name>.<namespace> Kube-OVN Subnet
- master:

Info

provider Underlay ProviderNetwork

KUBE-OVN SUBNET

Kube-OVN Subnet cidrBlock exclude_ips provider NetworkAttachmentDefinition <name>.<namespace> macvlan Subnet

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: macvlan
spec:
  protocol: IPv4
  provider: macvlan.default
  cidrBlock: 172.17.0.0/16
  gateway: 172.17.0.1
  excludeIps:
    - 172.17.0.0..172.17.0.10
```

gateway private nat provider ovn attachment network

Pod

Pod annotation k8s.v1.cni.cncf.io/networks NetworkAttachmentDefinition <namespace>/<name>

```
apiVersion: v1
kind: Pod
metadata:
  name: samplepod
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/networks: default/macvlan
spec:
  containers:
    - name: samplepod
      command: ["/bin/ash", "-c", "trap : TERM INT; sleep infinity & wait"]
      image: docker.io/library/alpine:edge
```

IP Pod

IP Pod <networkAttachmentName>.<networkAttachmentNamespace>.kubernetes.io/ip_address annotation

```
apiVersion: v1
kind: Pod
metadata:
  name: static-ip
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/networks: default/macvlan
    ovn.kubernetes.io/ip_address: 10.16.0.15
    ovn.kubernetes.io/mac_address: 00:00:00:53:6B:B6
```

```

    macvlan.default.kubernetes.io/ip_address: 172.17.0.100
    macvlan.default.kubernetes.io/mac_address: 00:00:00:53:6B:BB
spec:
  containers:
  - name: static-ip
    image: docker.io/library/nginx:alpine

```

IP

ippool `<networkAttachmentName>.<networkAttachmentNamespace>.kubernetes.io/ip_pool` annotation

```

apiVersion: apps/v1
kind: Deployment
metadata:
  namespace: default
  name: static-workload
  labels:
    app: static-workload
spec:
  replicas: 2
  selector:
    matchLabels:
      app: static-workload
  template:
    metadata:
      labels:
        app: static-workload
      annotations:
        k8s.v1.cni.cncf.io/networks: default/macvlan
        ovn.kubernetes.io/ip_pool: 10.16.0.15,10.16.0.16,10.16.0.17
        macvlan.default.kubernetes.io/ip_pool: 172.17.0.200,172.17.0.201,172.17.0.202
    spec:
      containers:
      - name: static-workload
        image: docker.io/library/nginx:alpine

```

macvlan Pod

macvlan Pod Pod annotation default-route

```

apiVersion: v1
kind: Pod
metadata:
  name: samplepod-route
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/networks: '[{"name": "macvlan", "namespace": "default", "default-route": ["172.17.0.1"]}]'
spec:
  containers:
  - name: samplepod-route
    command: ["/bin/ash", "-c", "trap : TERM INT; sleep infinity & wait"]
    image: docker.io/library/alpine:edge

```

macvlan Pod

macvlan Pod annotation `v1.multus-cni.io/default-network` NetworkAttachmentDefinition `<namespace>/<name>`

```

apiVersion: v1
kind: Pod
metadata:
  name: samplepod-macvlan
  namespace: default
  annotations:
    v1.multus-cni.io/default-network: default/macvlan
spec:
  containers:
  - name: samplepod-macvlan
    command: ["/bin/ash", "-c", "trap : TERM INT; sleep infinity & wait"]
    image: docker.io/library/alpine:edge

```

KUBE-OVN SUBNET PROVIDER OVN

provider ovn subnet IP provider ovn Kube-OVN Subnet cidrBlock exclude_ips Subnet

```

apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: macvlan
spec:
  protocol: IPv4
  provider: ovn

```

```
cidrBlock: 172.17.0.0/16
gateway: 172.17.0.1
excludeIps:
- 172.17.0.0..172.17.0.10
```

Pod

provider ovn subnet IP Pod annotation k8s.v1.cni.cncf.io/networks

<networkAttachmentName>.<networkAttachmentNamespace>.kubernetes.io/logical_switch

```
apiVersion: v1
kind: Pod
metadata:
  name: samplepod
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/networks: default/macvlan
    macvlan.default.kubernetes.io/logical_switch: macvlan
spec:
  containers:
  - name: samplepod
    command: ["/bin/ash", "-c", "trap : TERM INT; sleep infinity & wait"]
    image: docker.io/library/alpine:edge
```

- k8s.v1.cni.cncf.io/networks: NetworkAttachmentDefinition <namespace>/<name>
- macvlan.default.kubernetes.io/logical_switch:

- <networkAttachmentName>.<networkAttachmentNamespace>.kubernetes.io/logical_switch provider
- ovn subnet ipam IP Pod IP macvlan Pod
- macvlan Pod

Kube-OVN

Kube-OVN

NETWORKATTACHMENTDEFINITION

provider ovn

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: attachnet
  namespace: default
spec:
  config: '{
    "cniVersion": "0.3.0",
    "type": "kube-ovn",
    "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
    "provider": "attachnet.default.ovn"
  }'
```

- spec.config.type: kube-ovn CNI Kube-OVN
- server_socket: Kube-OVN socket /run/openvswitch/kube-ovn-daemon.sock
- provider: NetworkAttachmentDefinition <name>.<namespace>.ovn Kube-OVN Subnet ovn

KUBE-OVN SUBNET

Kube-OVN provider NetworkAttachmentDefinition <name>.<namespace>.ovn ovn Kube-OVN Subnet

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: attachnet
spec:
  protocol: IPv4
  provider: attachnet.default.ovn
  cidrBlock: 172.17.0.0/16
  gateway: 172.17.0.1
  excludeIps:
  - 172.17.0.0..172.17.0.10
```

Pod

Pod annotation k8s.v1.cni.cncf.io/networks NetworkAttachmentDefinition <namespace>/<name>

```
apiVersion: v1
kind: Pod
metadata:
  name: samplepod
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/networks: default/attachnet
spec:
  containers:
  - name: samplepod
    command: ["/bin/ash", "-c", "trap : TERM INT; sleep infinity & wait"]
    image: docker.io/library/alpine:edge
```

IP/MAC

NetworkAttachmentDefinition IP/MAC k8s.v1.cni.cncf.io/networks interface

annotation <nadName>.<nadNamespace>.kubernetes.io/ip_address.<interfaceName> <nadName>.<nadNamespace>.kubernetes.io/mac_address.<interfaceName> - IP/MAC

Kube-OVN Pod <networkAttachmentName>.<networkAttachmentNamespace>.ovn.kubernetes.io/routes annotation

```
apiVersion: v1
kind: Pod
metadata:
  name: custom-routes-attach
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/networks: default/attachnet
    attachnet.default.ovn.kubernetes.io/routes: |
      [{
        "dst": "192.168.0.101/24",
        "gw": "172.17.0.254"
      }, {
        "gw": "172.17.0.254"
      }]
spec:
  containers:
  - name: custom-routes-attach
    command: ["/bin/ash", "-c", "trap : TERM INT; sleep infinity & wait"]
    image: docker.io/library/alpine:edge
```

dst

Deployment DaemonSet StatefulSet Annotation .spec.template.metadata.annotations

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: custom-routes-attach
  labels:
    app: nginx
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
      annotations:
        k8s.v1.cni.cncf.io/networks: default/attachnet
        attachnet.default.ovn.kubernetes.io/routes: |
          [{
            "dst": "192.168.0.101/24",
            "gw": "172.17.0.254"
          }, {
            "gw": "172.17.0.254"
          }]
    spec:
      containers:
      - name: nginx
        image: docker.io/library/nginx:alpine
```

KUBE-OVN SUBNET PROVIDER OVN

provider ovn subnet IP provider ovn Kube-OVN Subnet cidrBlock exclude_ips Subnet

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: attachnet
spec:
  protocol: IPv4
  provider: ovn
  cidrBlock: 172.17.0.0/16
  gateway: 172.17.0.1
  excludeIps:
    - 172.17.0.0..172.17.0.10
```

Pod

provider ovn subnet IP Pod annotation k8s.v1.cni.cncf.io/networks

<networkAttachmentName>.<networkAttachmentNamespace>.ovn.kubernetes.io/logical_switch

```
apiVersion: v1
kind: Pod
metadata:
  name: samplepod
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/networks: default/attachnet
    attachnet.default.ovn.kubernetes.io/logical_switch: attachnet
spec:
  containers:
    - name: samplepod
      command: ["/bin/ash", "-c", "trap : TERM INT; sleep infinity & wait"]
      image: docker.io/library/alpine:edge
```

- k8s.v1.cni.cncf.io/networks: NetworkAttachmentDefinition <namespace>/<name>
- attachnet.default.ovn.kubernetes.io/logical_switch:

- <networkAttachmentName>.<networkAttachmentNamespace>.ovn.kubernetes.io/logical_switch provider
- Kube-OVN Pod IP Pod IP macvlan Pod Kube-OVN Pod
- Kube-OVN Pod <networkAttachmentName>.<networkAttachmentNamespace>.ovn.kubernetes.io/routes annotation

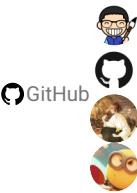
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🕒 2022 5 20



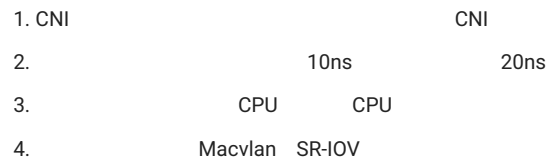
7.1.4

7.2

Kube-OVN

Kube-OVN

7.2.1



7.2.2

Overlay

- Kubernetes: 1.22.0
- OS: CentOS 7
- Kube-OVN: 1.8.0 *Overlay*
- CPU: Intel(R) Xeon(R) E-2278G
- Network: 2*10Gbps, xmit_hash_policy=layer3+4

```
qperf -t 60 <server ip> -ub -oo msg_size:1 -vu tcp_lat tcp_bw udp_lat udp_bw 1 tcp/udp
```

Type	tcp_lat (us)	udp_lat (us)	tcp_bw (Mb/s)	udp_bw(Mb/s)
Kube-OVN Default	25.7	22.9	27.1	1.59
Kube-OVN Optimized	13.9	12.9	27.6	5.57
HOST Network	13.1	12.4	28.2	6.02

Overlay Underlay

Kube-OVN Overlay Underlay

Environment:

- Kubernetes: 1.22.0
- OS: CentOS 7
- Kube-OVN: 1.8.0
- CPU: AMD EPYC 7402P 24-Core Processor
- Network: Intel Corporation Ethernet Controller XXV710 for 25GbE SFP28

```
qperf -t 60 <server ip> -ub -oo msg_size:1 -vu tcp_lat tcp_bw udp_lat udp_bw
```

Type	tcp_lat (us)	udp_lat (us)	tcp_bw (Mb/s)	udp_bw(Mb/s)
Kube-OVN Overlay	15.2	14.6	23.6	2.65
Kube-OVN Underlay	14.3	13.8	24.2	3.46
HOST Network	16.6	15.4	24.8	2.64

```
qperf -t 60 <server ip> -ub -oo msg_size:1K -vu tcp_lat tcp_bw udp_lat udp_bw
```

Type	tcp_lat (us)	udp_lat (us)	tcp_bw (Gb/s)	udp_bw(Gb/s)
Kube-OVN Overlay	16.5	15.8	10.2	2.77
Kube-OVN Underlay	15.9	14.5	9.6	3.22
HOST Network	18.1	16.6	9.32	2.66

```
qperf -t 60 <server ip> -ub -oo msg_size:4K -vu tcp_lat tcp_bw udp_lat udp_bw
```

Type	tcp_lat (us)	udp_lat (us)	tcp_bw (Gb/s)	udp_bw(Gb/s)
Kube-OVN Overlay	34.7	41.6	16.0	9.23
Kube-OVN Underlay	32.6	44	15.1	6.71
HOST Network	35.9	45.9	14.6	5.59



7.2.3

CPU

CPU

CPU

```
cpupower frequency-set -g performance
```

```
# ethtool -g eno1
Ring parameters for eno1:
Pre-set maximums:
RX:                4096
RX Mini:           0
RX Jumbo:          0
TX:                4096
Current hardware settings:
RX:                255
RX Mini:           0
RX Jumbo:          0
TX:                255
```

```
ethtool -G eno1 rx 4096
ethtool -G eno1 tx 4096
```


tuned

tuned profile

tuned-adm profile network-latency

tuned-adm profile network-throughput

irqbalance CPU CPU

OVN LB

OVN L2 LB conntrack recirculate CPU 20% CPU Overlay kube-proxy Service
Pod-to-Pod kube-ovn-controller

```
command:
- /kube-ovn/start-controller.sh
args:
...
- --enable-lb=false
...
```

Underlay kube-proxy iptables ipvs LB Service

conntrack

OVN LB Service Service NetworkPolicy Subnet A Pod Subnet B Service
kube-ovn-controller --skip-conntrack-dst-cidrs conntrack

--skip-conntrack-dst-cidrs="10.17.0.0/16,169.254.169.245/32"

FastPath

network ns netfilter 20% CPU netfilter FastPath netfilter CPU

netfilter iptables ipvs nftables

FastPath

insmod kube_ovn_fastpath.ko FastPath dmesg

```
# dmesg
...
[619631.323788] init_module,kube_ovn_fastpath_local_out
[619631.323798] init_module,kube_ovn_fastpath_post_routing
[619631.323800] init_module,kube_ovn_fastpath_pre_routing
[619631.323801] init_module,kube_ovn_fastpath_local_in
...
```

OVS

OVS flow 10% CPU x86 CPU popcnt sse4.2 flow CPU
5%

FastPath

CPU

```
cat /proc/cpuinfo | grep popcnt
cat /proc/cpuinfo | grep sse4_2
```

CENTOS

```
yum install -y gcc kernel-devel-$(uname -r) python3 autoconf automake libtool rpm-build openssl-devel
```

OVS RPM :

```
git clone -b branch-3.5 --depth=1 https://github.com/openvswitch/ovs.git
cd ovs
curl -s https://github.com/kubeovn/ovs/commit/2d2c83c26d4217446918f39d5cd5838e9ac27b32.patch | git apply
./boot.sh
./configure --with-linux=/lib/modules/$(uname -r)/build CFLAGS="-g -O2 -mpopcnt -msse4.2"
make rpm-fedora-kmod
cd rpm/rpmbuild/RPMS/x86_64/
```

RPM

```
rpm -i openvswitch-kmod-3.5.1-1.el7.x86_64.rpm
```

Kube-OVN OVS

UBUNTU

```
apt install -y autoconf automake libtool gcc build-essential libssl-dev
```

OVS

```
apt install -y autoconf automake libtool gcc build-essential libssl-dev

git clone -b branch-3.5 --depth=1 https://github.com/openvswitch/ovs.git
cd ovs
curl -s https://github.com/kubeovn/ovs/commit/2d2c83c26d4217446918f39d5cd5838e9ac27b32.patch | git apply
./boot.sh
./configure --prefix=/usr/ --localstatedir=/var --enable-ssl --with-linux=/lib/modules/$(uname -r)/build
make -j `nproc`
make install
make modules_install

cat > /etc/depmod.d/openvswitch.conf << EOF
override openvswitch * extra
override vport-* * extra
EOF

depmod -a
cp debian/openvswitch-switch.init /etc/init.d/openvswitch-switch
/etc/init.d/openvswitch-switch force-reload-kmod
```

Kube-OVN OVS

STT

 **Warning**

OpenVswitch 3.6 [STT](#) [Tunnel](#)

		Geneve	Vxlan	UDP		UDP	TCP		TCP	offload		TCP
	CPU	TCP										
STT		TCP			TCP		TCP		TCP		TCP	
STT			OVS	OVS								

STT

```
kubect1 set env daemonset/ovs-ovn -n kube-system TUNNEL_TYPE=stt
kubect1 delete pod -n kube-system -lapp=ovs
```

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 GitHub







7.2.4

7.3 FastPath

Profile **Netfilter**

20%

CPU

FastPath

Netfilter

CPU

FastPath

7.3.1

```
git clone --depth=1 https://github.com/kubeovn/kube-ovn.git
```

7.3.2

RPM CentOS RHEL

```
yum install -y kernel-devel-$(uname -r) gcc elfutils-libelf-devel
```

DEB Ubuntu Debian

```
apt install -y linux-headers-$(uname -r) build-essential
```

7.3.3

3.x

```
cd kube-ovn/fastpath/3.x
make all
```

4.x ~ 6.x

```
cd kube-ovn/fastpath/4.x-6.x
make all
```

7.3.4

kube_ovn_fastpath.ko

```
insmod kube_ovn_fastpath.ko
```

dmesg

```
# dmesg
[619631.323788] init_module,kube_ovn_fastpath_local_out
[619631.323798] init_module,kube_ovn_fastpath_post_routing
[619631.323800] init_module,kube_ovn_fastpath_pre_routing
[619631.323801] init_module,kube_ovn_fastpath_local_in
```

```
rmmod kube_ovn_fastpath.ko
```



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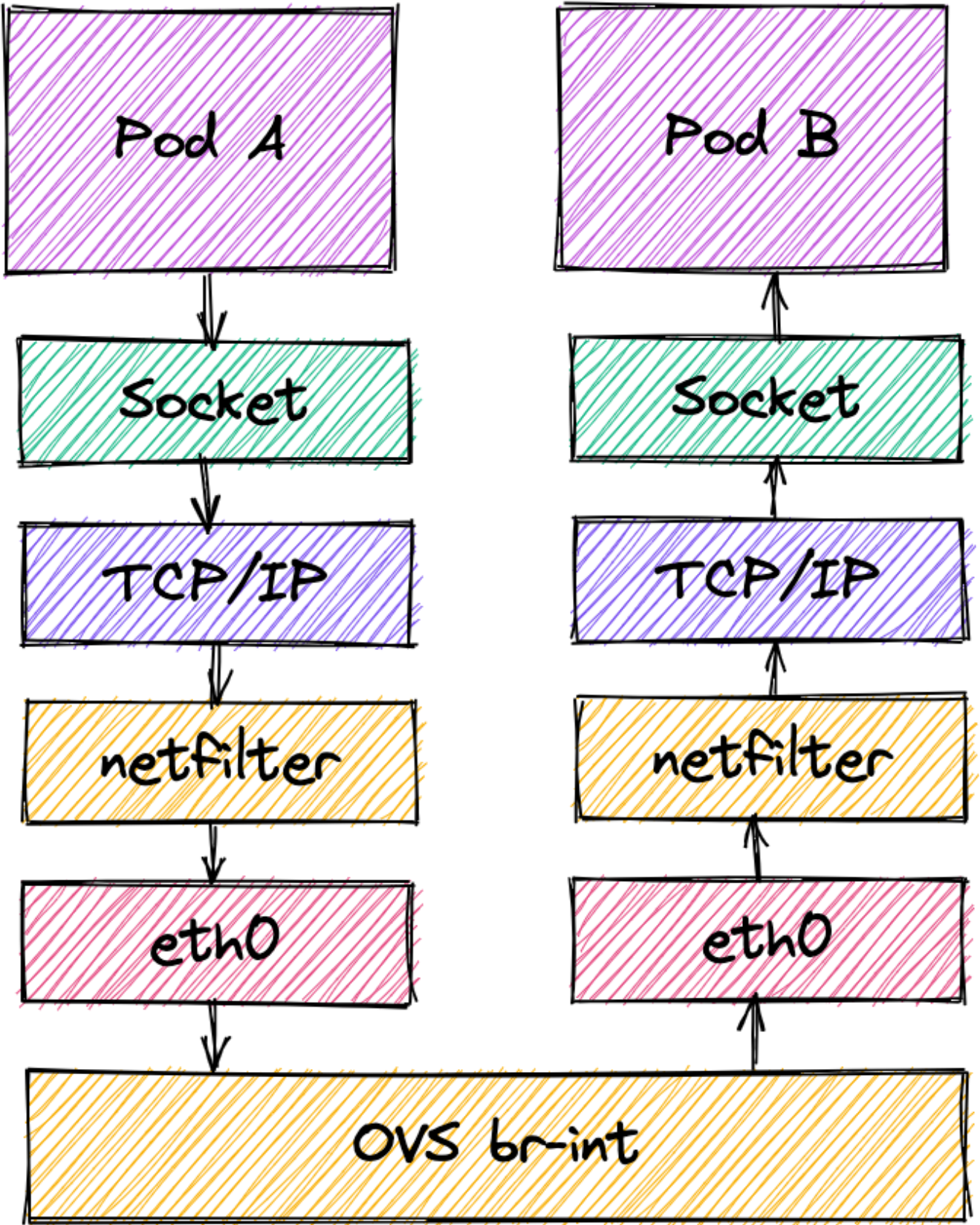
7.3.5

7.4 eBPF TCP

5G Pod TCP Intel [istio-tcpip-bypass](#) Pod eBPF TCP/IP socket

7.4.1

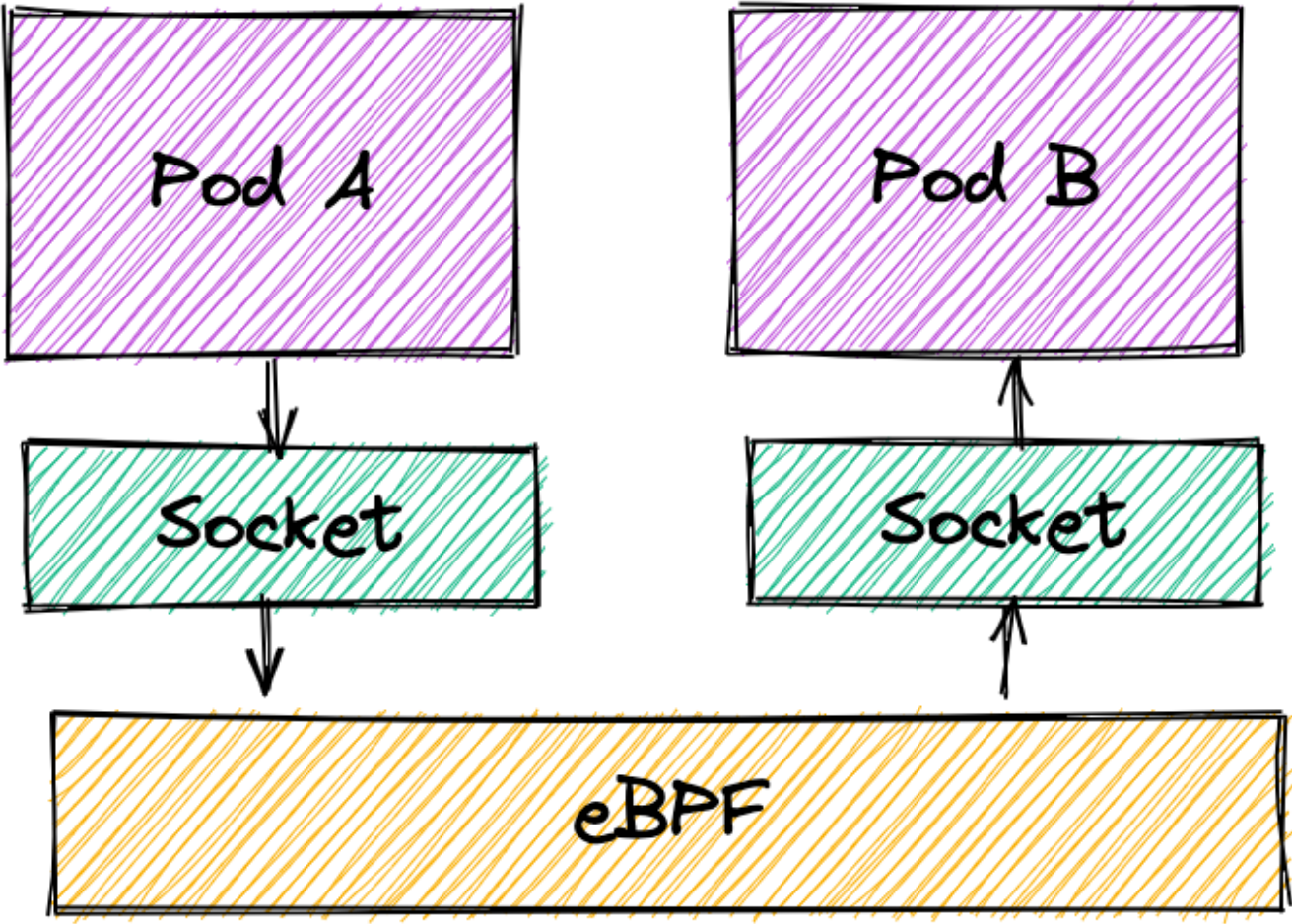
Pod TCP TCP/IP, netfilter OVS



istio-tcpip-bypass

TCP

socket



TCP Service Mesh Service Mesh

Tanzu Service Mesh Acceleration using eBPF

7.4.2

eBPF Ubuntu 20.04 Linux 5.4.0-74-generic

7.4.3

Pod nodeSelector

```
# kubectl create deployment perf --image=kubeovn/perf:dev --replicas=2
deployment.apps/perf created
# kubectl get pod -o wide
NAME                READY   STATUS    RESTARTS   AGE   IP            NODE   NOMINATED NODE   READINESS GATES
perf-7697bc6ddf-b2cpv 1/1     Running   0           28s   100.64.0.3    sealos <none>          <none>
perf-7697bc6ddf-p2xpt 1/1     Running   0           28s   100.64.0.2    sealos <none>          <none>
```

Pod qperf server Pod qperf client

```
# kubectl exec -it perf-7697bc6ddf-b2cpv sh
/ # qperf

# kubectl exec -it perf-7697bc6ddf-p2xpt sh
/ # qperf -t 60 100.64.0.3 -ub -oo msg_size:1:16K:*4 -vu tcp_lat tcp_bw
```

istio-tcpip-bypass

```
kubectl apply -f https://raw.githubusercontent.com/intel/istio-tcpip-bypass/main/bypass-tcpip-daemonset.yaml
```


perf client

```
# kubectl exec -it perf-7697bc6ddf-p2xpt sh
/ # qperf -t 60 100.64.0.3 -ub -oo msg_size:1:16K:*4 -vu tcp_lat tcp_bw
```

7.4.4

TCP	40% ~ 60%	1024	40% ~ 80%	
Packet Size (byte)	eBPF tcp_lat (us)	Default tcp_lat (us)	eBPF tcp_bw (Mb/s)	Default tcp_bw(Mb/s)
1	20.2	44.5	1.36	4.27
4	20.2	48.7	5.48	16.7
16	19.6	41.6	21.7	63.5
64	18.8	41.3	96.8	201
256	19.2	36	395	539
1024	18.3	42.4	1360	846
4096	16.5	62.6	4460	2430
16384	20.2	58.8	9600	6900

512

eBPF

TCP

eBPF

eBPF TCP

7.4.5

1. [istio-tcpip-bypass](#)

2. [Deep Dive TCP/IP Bypass with eBPF in Service Mesh](#)

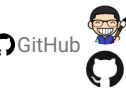
3. [Tanzu Service Mesh Acceleration using eBPF](#)
- PDF

Slack

Support

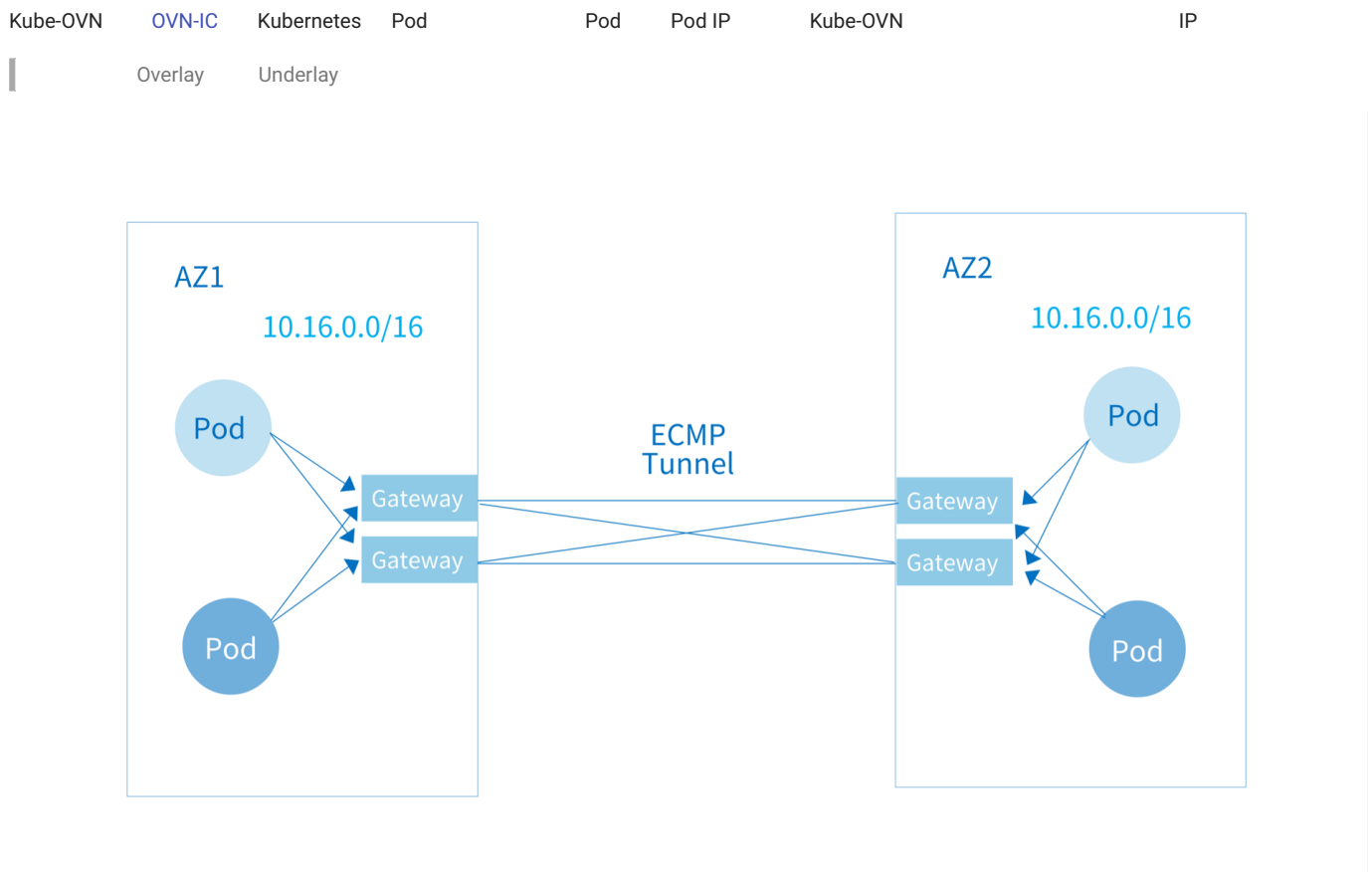
2025 9 10

2022 9 27



7.4.6

7.5 OVN-IC



 Limitation

OVN-IC	Pod IP	Service DNS	NetworkPolicy	Istio
--------	--------	-------------	---------------	-------

7.5.1

```
1.1.11.16 install.sh
```

ENABLE_IC=true

deployment ovn-ic-controller

2. CIDR

3. kube-ovn-controller IP

4. IP

5. VPC VPC

7.5.2 OVN-IC

1

1 Kube-OVN v1.11.16

"	"	"	"	Deployment	master	master	1	master
---	---	---	---	------------	--------	--------	---	--------

```
install-ovn-ic.sh
```

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/dist/images/install-ic-server.sh
```

TS_NUM ECMP Path

```
sed 's/VERSION=.*//VERSION=v1.16.5/' dist/images/install-ic-server.sh | TS_NUM=3 bash
```

```
deployment.apps/ovn-ic-server created
Waiting for deployment spec update to be observed...
Waiting for deployment "ovn-ic-server" rollout to finish: 0 out of 3 new replicas have been updated...
Waiting for deployment "ovn-ic-server" rollout to finish: 0 of 3 updated replicas are available...
Waiting for deployment "ovn-ic-server" rollout to finish: 1 of 3 updated replicas are available...
Waiting for deployment "ovn-ic-server" rollout to finish: 2 of 3 updated replicas are available...
deployment "ovn-ic-server" successfully rolled out
OVN IC Server installed Successfully
```

```
kubect1 ko icsbctl show
```

```
kubect1 ko icsbctl show
availability-zone az0
gateway 059b5c54-c540-4d77-b009-02d65f181a02
  hostname: kube-ovn-worker
  type: geneve
  ip: 172.18.0.3
  port ts-az0
    transit switch: ts
    address: ["00:00:00:B4:8E:BE 169.254.100.97/24"]
gateway 74ee4b9a-ba48-4a07-861e-1a8e4b9f905f
  hostname: kube-ovn-worker2
  type: geneve
  ip: 172.18.0.2
  port ts1-az0
    transit switch: ts1
    address: ["00:00:00:19:2E:F7 169.254.101.90/24"]
gateway 7e2428b6-344c-4dd5-a0d5-972c1ccec581
  hostname: kube-ovn-control-plane
  type: geneve
  ip: 172.18.0.4
  port ts2-az0
    transit switch: ts2
    address: ["00:00:00:EA:32:BA 169.254.102.103/24"]
availability-zone az1
gateway 034da7cb-3826-4318-81ce-6a877a9bf285
  hostname: kube-ovn1-worker
  type: geneve
  ip: 172.18.0.6
  port ts-az1
    transit switch: ts
    address: ["00:00:00:25:3A:B9 169.254.100.51/24"]
gateway 2531a683-283e-4fb8-a619-bdbcb33539b8
  hostname: kube-ovn1-worker2
  type: geneve
  ip: 172.18.0.5
  port ts1-az1
    transit switch: ts1
    address: ["00:00:00:52:87:F4 169.254.101.118/24"]
gateway b0efb0be-e5a7-4323-ad4b-317637a757c4
  hostname: kube-ovn1-control-plane
  type: geneve
  ip: 172.18.0.8
  port ts2-az1
    transit switch: ts2
    address: ["00:00:00:F6:93:1A 169.254.102.17/24"]
```

2

kube-ovn-controller IP OVN-IC

docker OVN-IC

```
docker run --name=ovn-ic-db -d --env "ENABLE_OVN_LEADER_CHECK=false" --network=host --privileged -v /etc/ovn:/etc/ovn -v /var/run/ovn:/var/run/ovn -v /var/log/ovn:/var/log/ovn kubeovn/kube-ovn:v1.16.5 bash start-ic-db.sh
```

containerd docker

```
ctr -n k8s.io run -d --env "ENABLE_OVN_LEADER_CHECK=false" --net-host --privileged --mount="type=bind,src=/etc/ovn/,dst=/etc/ovn,options=rbind:rw" --mount="type=bind,src=/var/run/ovn,dst=/var/run/ovn,options=rbind:rw" --mount="type=bind,src=/var/log/ovn,dst=/var/log/ovn,options=rbind:rw" docker.io/kubeovn/kube-ovn:v1.16.5 ovn-ic-db bash start-ic-db.sh
```

7.5.3

VPC Subnet CIDR OVN-IC Subnet CIDR

kube-system Namespace ovn-ic-config ConfigMap

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: ovn-ic-config
  namespace: kube-system
data:
  enable-ic: "true"
  az-name: "az1"
  ic-db-host: "192.168.65.3"
  ic-nb-port: "6645"
  ic-sb-port: "6646"
  gw-nodes: "az1-gw"
  auto-route: "true"
```

- enable-ic:
- az-name:
- ic-db-host: OVN-IC
- ic-nb-port: OVN-IC 6645
- ic-sb-port: OVN-IC 6646
- gw-nodes:
- auto-route:

ovn-ic-config ConfigMap ConfigMap ConfigMap

ovn-ic ts

```
# ovn-ic-sbctl show
availability-zone az1
gateway deee03e0-af16-4f45-91e9-b50c3960f809
  hostname: az1-gw
  type: geneve
  ip: 192.168.42.145
  port ts-az1
  transit switch: ts
  address: ["00:00:00:50:AC:8C 169.254.100.45/24"]
availability-zone az2
gateway e94cc831-8143-40e3-a478-90352773327b
  hostname: az2-gw
  type: geneve
  ip: 192.168.42.149
  port ts-az2
  transit switch: ts
  address: ["00:00:00:07:4A:59 169.254.100.63/24"]
```

```
# kubectl ko nbctl lr-route-list ovn-cluster
IPv4 Routes
10.42.1.1 169.254.100.45 dst-ip (learned)
10.42.1.3 100.64.0.2 dst-ip
10.16.0.2 100.64.0.2 src-ip
10.16.0.3 100.64.0.2 src-ip
10.16.0.4 100.64.0.2 src-ip
10.16.0.6 100.64.0.2 src-ip
10.17.0.0/16 169.254.100.45 dst-ip (learned)
100.65.0.0/16 169.254.100.45 dst-ip (learned)
```

1 Pod ping 2 Pod IP

Subnet disableInterConnection

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
```

```

name: no-advertise
spec:
  cidrBlock: 10.199.0.0/16
  disableInterConnection: true

```

7.5.4

CIDR

kube-system Namespace ovn-ic-config ConfigMap auto-route false

```

apiVersion: v1
kind: ConfigMap
metadata:
  name: ovn-ic-config
  namespace: kube-system
data:
  enable-ic: "true"
  az-name: "az1"
  ic-db-host: "192.168.65.3"
  ic-nb-port: "6645"
  ic-sb-port: "6646"
  gw-nodes: "az1-gw"
  auto-route: "false"

```

```

[root@az1 ~]# kubectl ko nbctl show
switch a391d3a1-14a0-4841-9836-4bd930c447fb (ts)
  port ts-az1
    type: router
    router-port: az1-ts
  port ts-az2
    type: remote
    addresses: ["00:00:00:4B:E2:9F 169.254.100.31/24"]

```

```

[root@az2 ~]# kubectl ko nbctl show
switch da6138b8-de81-4908-abf9-b2224ec4edf3 (ts)
  port ts-az2
    type: router
    router-port: az2-ts
  port ts-az1
    type: remote
    addresses: ["00:00:00:FB:2A:F7 169.254.100.79/24"]

```

az1 az2 169.254.100.31 az2 az1 169.254.100.79

az1 CIDR 10.16.0.0/24 az2 CIDR 10.17.0.0/24

az1 az2

```
kubectl ko nbctl lr-route-add ovn-cluster 10.17.0.0/24 169.254.100.31
```

az2 az1

```
kubectl ko nbctl lr-route-add ovn-cluster 10.16.0.0/24 169.254.100.79
```

7.5.5 OVN-IC

1

1 Kube-OVN v1.11.16

1

2

OVN-IC Raft 3

OVN-IC leader

docker

```
docker run --name=ovn-ic-db -d --env "ENABLE_OVN_LEADER_CHECK=false" --network=host --privileged -v /etc/ovn:/etc/ovn -v /var/run/ovn:/var/run/ovn -v /var/log/ovn:/var/log/ovn -e LOCAL_IP="192.168.65.3" -e NODE_IPS="192.168.65.3,192.168.65.2,192.168.65.1" kubeovn/kube-ovn:v1.16.5 bash start-ic-db.sh
```

containerd

```
ctr -n k8s.io run -d --env "ENABLE_OVN_LEADER_CHECK=false" --net-host --privileged --mount="type=bind,src=/etc/ovn/,dst=/etc/ovn,options=rbind:rw" --mount="type=bind,src=/var/run/ovn,dst=/var/run/ovn,options=rbind:rw" --mount="type=bind,src=/var/log/ovn,dst=/var/log/ovn,options=rbind:rw" --env="NODE_IPS="192.168.65.3,192.168.65.2,192.168.65.1" --env="LOCAL_IP="192.168.65.3" docker.io/kubeovn/kube-ovn:v1.16.5 ovn-ic-db bash start-ic-db.sh
```

- LOCAL_IP IP
- NODE_IPS OVN-IC IP
- OVN-IC follower

docker

```
docker run --name=ovn-ic-db -d --network=host --privileged -v /etc/ovn:/etc/ovn -v /var/run/ovn:/var/run/ovn -v /var/log/ovn:/var/log/ovn -e LOCAL_IP="192.168.65.2" -e NODE_IPS="192.168.65.3,192.168.65.2,192.168.65.1" -e LEADER_IP="192.168.65.3" kubeovn/kube-ovn:v1.16.5 bash start-ic-db.sh
```

containerd

```
ctr -n k8s.io run -d --net-host --privileged --mount="type=bind,src=/etc/ovn/,dst=/etc/ovn,options=rbind:rw" --mount="type=bind,src=/var/run/ovn,dst=/var/run/ovn,options=rbind:rw" --mount="type=bind,src=/var/log/ovn,dst=/var/log/ovn,options=rbind:rw" --env="NODE_IPS="192.168.65.3,192.168.65.2,192.168.65.1" --env="LOCAL_IP="192.168.65.2" --env="LEADER_IP="192.168.65.3" docker.io/kubeovn/kube-ovn:v1.16.5 ovn-ic-db bash start-ic-db.sh
```

- LOCAL_IP IP
- NODE_IPS OVN-IC IP
- LEADER_IP: OVN-IC leader IP
- ovn-ic-config OVN-IC

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: ovn-ic-config
  namespace: kube-system
data:
  enable-ic: "true"
  az-name: "az1"
  ic-db-host: "192.168.65.3,192.168.65.2,192.168.65.1"
  ic-nb-port: "6645"
  ic-sb-port: "6646"
  gw-nodes: "az1-gw"
  auto-route: "true"
```

7.5.6 ECMP

1

ECMP ECMP path 3 ECMP path

```
kubectrl edit deployment ovn-ic-server -n kube-system
```

'TS_NUM' TS_NUM ECMP Path

7.5.7

ovn-ic-config Configmap

```
kubectrl -n kube-system delete cm ovn-ic-config
```

ts

```
kubectrl ko nbctl ls-del ts
```

7.5.8 az-name

kubectl edit ovn-ic-config configmap az-name ovn-cni pod 10

ovn-appctl -t ovn-controller inc-engine/recompute

7.5.9

ovn-ic-config Configmap

kubectl -n kube-system delete cm ovn-ic-config

ts

kubectl ko nbctl ls-del ts

OVN-IC

docker

docker stop ovn-ic-db
docker rm ovn-ic-db

containerd

ctr -n k8s.io task kill ovn-ic-db
ctr -n k8s.io containers rm ovn-ic-db

deployment ovn-ic-server

kubectl delete deployment ovn-ic-server -n kube-system

master DB

rm -f /etc/origin/ovn/ovn_ic_nb_db.db
rm -f /etc/origin/ovn/ovn_ic_sb_db.db

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7.5.10

7.6 Submariner

Submariner	Kubernetes	Pod	Service	Kube-OVN	
OVN-IC	Submariner	Kube-OVN	Kube-OVN	Service	Submariner

7.6.1

- Service CIDR CIDR

7.6.2 Submariner

Note

Submariner v0.22.0 nftables Kube-OVN Submariner

```
kubectl create namespace submariner-operator
kubectl create configmap submariner-global --namespace=submariner-operator --from-literal=use-nftables=false
```

```
subctl
```

```
curl -Ls https://get.submariner.io | bash
export PATH=$PATH:~/local/bin
echo export PATH=$PATH:~/local/bin >> ~/.profile
```

```
kubeconfig      submariner-broker
```

```
subctl deploy-broker
```

```
cluster0      CIDR   10.16.0.0/16 join    CIDR   100.64.0.0/16 cluster1      CIDR   11.16.0.0/16 join    CIDR   100.68.0.0/16
```

```
kubeconfig cluster0      broker      :
```

```
subctl join broker-info.subm --clusterid cluster0 --clustercidr 100.64.0.0/16,10.16.0.0/16 --natt=false --cable-driver vxlan --health-check=false
kubectl label nodes cluster0 submariner.io/gateway=true
```

```
kubeconfig cluster1      broker      :
```

```
subctl join broker-info.subm --clusterid cluster1 --clustercidr 100.68.0.0/16,11.16.0.0/16 --natt=false --cable-driver vxlan --health-check=false
kubectl label nodes cluster1 submariner.io/gateway=true
```

```
join      gateway, routeagent pod      ,      submariner-operator      clusterrole      :
```

```
- apiGroups:
- "apps"
resources:
- daemonsets
verbs:
- create
- get
- list
- watch
- update
```

```
subnet ovn-default      centralized      gatewayNode
```

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: ovn-default
spec:
  gatewayType: centralized
  gatewayNode: "node1,node2" # submariner gateway
```

```
submariner      gateway      gatewayNode
```


Pod IP

subctl

```
subctl show all
subctl diagnose all
```

Submariner Submariner

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7.6.3

7.7 Overlay

Underlay		Pod	IP
VPC	Overlay	Pod IP	IP

7.7.1

- `ip_forward`
- `iptables` `forward` `Drop`
- `ct` `INVALID`

7.7.2

<code>natOutgoing</code>	<code>false</code>	<code>nat</code>	Pod IP
--------------------------	--------------------	------------------	--------

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: routed
spec:
  protocol: IPv4
  cidrBlock: 10.166.0.0/16
  default: false
  excludeIps:
    - 10.166.0.1
  gateway: 10.166.0.1
  gatewayType: distributed
  natOutgoing: false
```

Pod	Kubernetes
-----	------------

```
ip route add 10.166.0.0/16 via 192.168.2.10 dev eth0
```

10.166.0.0/16	192.168.2.10	Kubernetes
---------------	--------------	------------

IP	Keepalived	VIP	VIP
Subnet	<code>gatewayType</code>	<code>centralized</code>	<code>gatewayNode</code> IP

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: routed
spec:
  protocol: IPv4
  cidrBlock: 10.166.0.0/16
  default: false
  excludeIps:
    - 10.166.0.1
  gateway: 10.166.0.1
  gatewayType: centralized
  gatewayNode: "node1"
  natOutgoing: false
```

<code>nat</code>	VPC NAT
------------------	---------

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7.7.3

7.8 Overlay

Overlay	tunnel

7.8.1

Kube-OVN	Kubernetes Node IP
IFACE=eth1	
ens[a-z0-9]*, eth[a-z0-9]*	
kube-ovn-cni DaemonSet	
args: - --iface=eth1	
annotation ovn.kubernetes.io/tunnel_interface	annotation iface annotation
kubect1 annotate node no1 ovn.kubernetes.io/tunnel_interface=ethx	

7.8.2

Kube-OVN		IP	tunnel
- IP - Overlay			
1. Node	IP		
2. Subnet	nodeNetwork		
3. kube-ovn-daemon	IP	OVS	
4. Pod	nodeNetwork	Pod	OVS IP
ovn.kubernetes.io/node_networks JSON key value IP			
kubect1 annotate node <node-name> ovn.kubernetes.io/node_networks='{ "storage": "192.168.100.10", "app": "172.16.0.10" }'			
- storage IP 192.168.100.10 IP - app IP 172.16.0.10 IP			
IP			

spec.nodeNetwork

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: storage-subnet
spec:
  protocol: IPv4
  cidrBlock: 10.100.0.0/16
  gateway: 10.100.0.1
  nodeNetwork: storage
```

storage-subnet	Pod	storage	tunnel
nodeNetwork	IP	IFACE	IP

OVS IP

OVS IP

```
ovs-vsctl get open . external-ids:ovn-encap-ip
```

IP

```
"192.168.1.10,192.168.100.10,172.16.0.10"
```

IP

```
ovs-vsctl get open . external-ids:ovn-encap-ip-default
```

POD IP

Pod Pod OVS IP

```
ovs-vsctl --columns=external_ids find interface external-ids:iface-id="<pod-name>.<namespace>"
```

nodeNetwork encap-ip

```
external_ids : {encap-ip="192.168.100.10", iface-id="test-pod.default", ...}
```

1.

- eth0 IP 192.168.1.10 192.168.1.11
- eth1 IP 10.10.10.10 10.10.10.11

```
kubect1 annotate node node1 ovn.kubernetes.io/node_networks='{"storage": "10.10.10.10"}'
kubect1 annotate node node2 ovn.kubernetes.io/node_networks='{"storage": "10.10.10.11"}'
```

2.

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: storage-net
spec:
  protocol: IPv4
  cidrBlock: 10.200.0.0/16
  gateway: 10.200.0.1
  nodeNetwork: storage
```

```
namespaces:
- storage-namespace
```

3. POD

```
apiVersion: v1
kind: Pod
metadata:
  name: storage-pod
  namespace: storage-namespace
spec:
  containers:
  - name: app
    image: docker.io/library/nginx:alpine
```

Pod eth1

- 1.
- 2. IP IP
- 3. Overlay Underlay Underlay Underlay
- 4. nodeNetwork IP
- 5. Pod nodeNetwork Pod
- 6. ovn.kubernetes.io/node_networks Pod

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7.9 BGP

Kube-OVN	Pods	Subnets	Services	EIPs	IP	BGP	IP	kube-ovn-speaker
Pod	Subnet	annotation						
EIP	BGP		VPC NAT Gateway			EIPs		

7.9.1 kube-ovn-speaker

kube-ovn-speaker [GoBGP](#)

kube-ovn-speaker

```
kubectl label nodes speaker-node-1 ovn.kubernetes.io/bgp=true
kubectl label nodes speaker-node-2 ovn.kubernetes.io/bgp=true
```

kube-ovn-speaker ECMP

yaml:

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/yamls/speaker.yaml
```

yaml

```
- --neighbor-address=10.32.32.254
- --neighbor-ipv6-address=2409:AB00:AB00:2000::AFB:8AFE
- --neighbor-as=65030
- --cluster-as=65000
```

```
- --neighbor-address=10.32.32.252,10.32.32.253
- --neighbor-ipv6-address=2409:AB00:AB00:2000::AFB:8AFC,2409:AB00:AB00:2000::AFB:8AFD
- --neighbor-as=65030
- --cluster-as=65000
```

- neighbor-address: BGP Peer
- neighbor-as: BGP Peer AS
- cluster-as: AS

yaml:

```
kubectl apply -f speaker.yaml
```

7.9.2 Pod/Subnet

BGP Subnet natOutgoing false Pod IP

annotation

```
kubectl annotate pod sample ovn.kubernetes.io/bgp=true
kubectl annotate subnet ovn-default ovn.kubernetes.io/bgp=true
```

annotation

```
kubectl annotate pod sample ovn.kubernetes.io/bgp-
kubectl annotate subnet ovn-default ovn.kubernetes.io/bgp-
```

BGP

7.9.3 ClusterIP Service

Service

ClusterIP

kube-ovn-speaker

--announce-cluster-ip

true

BGP

annotation

kubectl annotate service sample ovn.kubernetes.io/bgp=true

annotation

kubectl annotate service sample ovn.kubernetes.io/bgp-

7.9.4 EIPs

EIPs

VPC NAT Gateway

BGP

BGP Sidecar

VPC NAT Gateway

BGP

BGP Speaker Sidecar

NetworkAttachmentDefinition

NAD

VPC

Subnet

Sidecar

Kubernetes API

EIPs

VPC

CoreDNS

NAD

NetworkAttachmentDefinition

Subnet

provider

{nadName}.{nadNamespace}.ovn

apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
 name: api-ovn-nad
 namespace: default
spec:
 config: '{
 "cniVersion": "0.3.0",
 "type": "kube-ovn",
 "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
 "provider": "api-ovn-nad.default.ovn"
 }'

apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
 name: vpc-apiserver-subnet
spec:
 protocol: IPv4
 cidrBlock: 100.100.100.0/24
 provider: api-ovn-nad.default.ovn

ovn-vpc-nat-config

ConfigMap

apiNadProvider

BGP Speaker

:

apiVersion: v1
kind: ConfigMap
metadata:
 name: ovn-vpc-nat-config
 namespace: kube-system
data:
 apiNadProvider: api-ovn-nad.default.ovn # What NetworkAttachmentDefinition provider to use so that the sidecar
can access the K8S API, as it can't by default due to VPC segmentation
 bgpSpeakerImage: docker.io/kubeovn/kube-ovn:v1.13.0 # Sets the BGP speaker image used
 image: docker.io/kubeovn/vpc-nat-gateway:v1.13.0

ovn-default

provider

provider: api-ovn-nad.default.ovn

VPC NAT Gateway

BGP

kind: VpcNatGateway
apiVersion: kubeovn.io/v1
metadata:
 name: vpc-natgw
spec:
 vpc: vpc1
 subnet: net1
 lanIp: 10.0.1.10
 bgpSpeaker:
 enabled: true
 asn: 65500
 remoteAsn: 65000
 neighbors:
 - 100.127.4.161
 - fd:01::1
 enableGracefulRestart: true # Optional


```
routerId: 1.1.1.1      # Optional
holdTime: 1m          # Optional
password: "password123" # Optional
extraArgs:             # Optional, passed directly to the BGP speaker
  - -v5               # Enables verbose debugging of the BGP speaker sidecar
selector:
  - "kubernetes.io/os: linux"
externalSubnets:
  - ovn-vpc-external-network # Network on which we'll speak BGP and receive/send traffic to the outside world
                                # BGP neighbors need to be on that network
```

BGP EIP

```
kubectl annotate eip sample ovn.kubernetes.io/bgp=true
```

7.9.5

kube-ovn-speaker :

• Cluster:

speaker Pod IPs/Subnet CIDRs IP CIDR Pod speaker Pod

Pod Subnet

• Local:

Pod IPs Pod Cluster Pod

:

Local kube-ovn-speaker Pod speaker

Cluster Pod/Subnet annotation ovn.kubernetes.io/bgp

• ovn.kubernetes.io/bgp=cluster ovn.kubernetes.io/bgp=true Cluster

• ovn.kubernetes.io/bgp=local Local

Service kube-proxy ClusterIP Service Cluster

7.9.6 BGP

kube-ovn-speaker BGP

• announce-cluster-ip: Service false

• auth-password: BGP peer

• holdtime: BGP 90

• graceful-restart: BGP Graceful Restart

• graceful-restart-time: BGP Graceful restart time RFC4724 3

• graceful-restart-deferral-time: BGP Graceful restart deferral time RFC4724 4.1

• passivemode: Speaker passive peer

• ebgp-multihop: ebgp ttl 1

7.9.7 BGP routes debug

```
# show peer neighbor
gobgp neighbor

# show announced routes to one peer
gobgp neighbor 10.32.32.254 adj-out
```

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7.9.8

7.10 MetalLB Kube-OVN Underlay

MetalLB Kubernetes Kubernetes Service

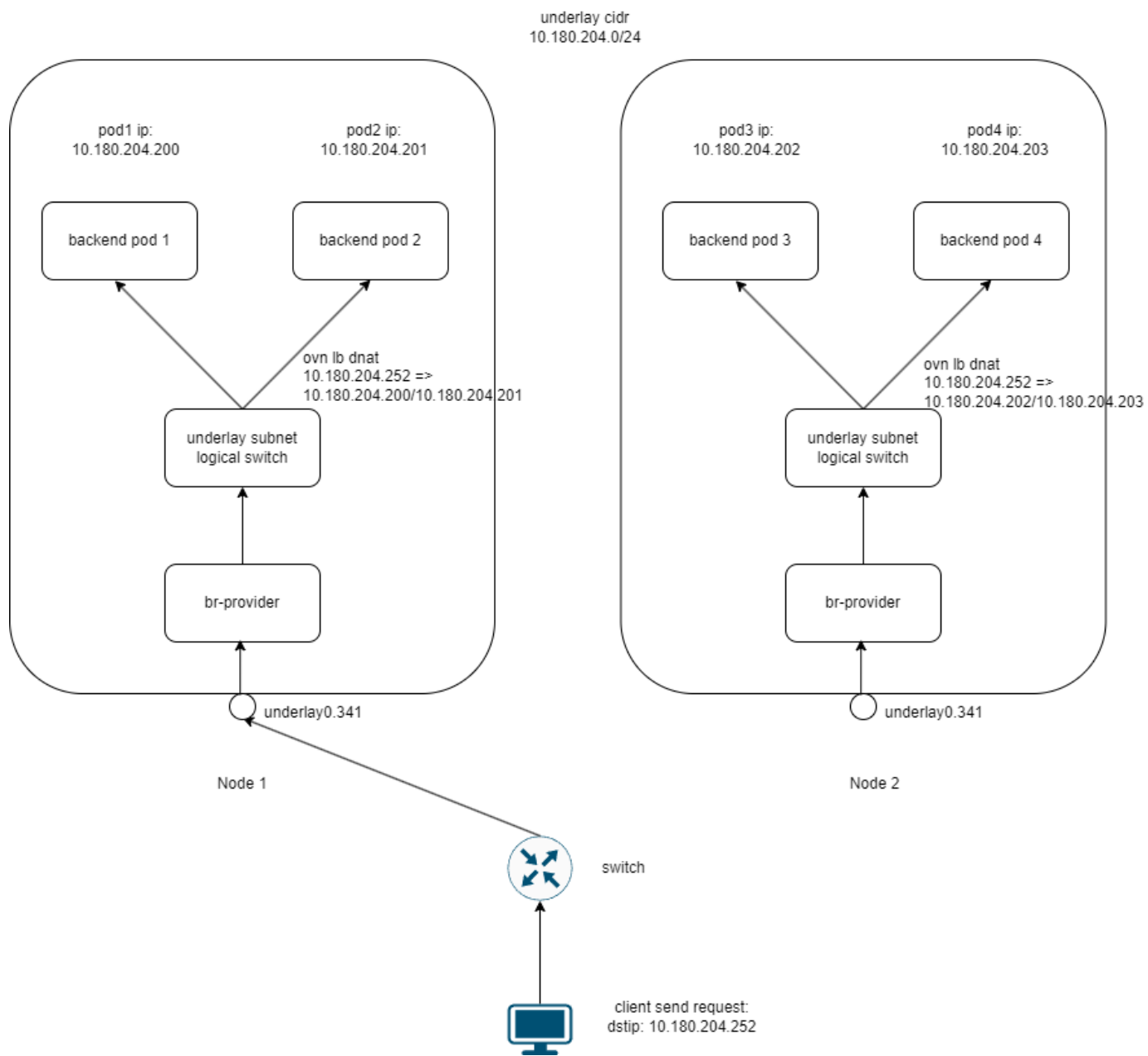
MetalLB Kube-OVN Underlay

7.10.1

Kube-OVN 1.14.0 MetalLB Underlay

- MetalLB IP
- Pod MetalLB VIP Underlay
- IP SNAT

7.10.2



1 MetalLB VIP Kube-OVN Underlay

MetalLB Kube-OVN Underlay

1. VIP 10.180.204.252 IP MetalLB L2 Node1 metallb VIP
 2. VIP underlay0.341
 3. br-provider Underlay
 4. br-provider OpenFlow OVN
 5. underlay subnet OVN ovn lb dnat
 6. OVN Pod
- 10.180.204.0/24 VIP Pod IP

7.10.3

- Kube-OVN `--enable-ovn-lb-prefer-local=true`
- Underlay `enableExternalLBAddress=true`
- Underlay `excludeIps` MetalLB IP

7.10.4

1. Kube-OVN

Kube-OVN Kube-OVN `--enable-ovn-lb-prefer-local=true` `--ls-ct-skip-dst-lport-ips=false`

```
# kube-ovn-controller Deployment
kubectl edit deployment -n kube-system kube-ovn-controller
```

```
--enable-ovn-lb-prefer-local=true
--ls-ct-skip-dst-lport-ips=false
```

2. Underlay

Underlay LoadBalancer excludeIps MetalLB IP

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: underlay-subnet
spec:
  protocol: IPv4
  provider: ovn
  cidrBlock: 10.180.204.0/24 #
  gateway: 10.180.204.1
  excludeIps:
    - 10.180.204.250
    - 10.180.204.251
    - 10.180.204.252 # MetalLB VIP 10.180.204.252
  natOutgoing: false
  enableExternalLBAddress: true # subnet cidr ip metallb vip
```

3. MetalLB

MetalLB MetalLB

```
kubectl apply -f https://raw.githubusercontent.com/metallb/metallb/v0.13.7/config/manifests/metallb-native.yaml
```

MetalLB L2

```
apiVersion: metallb.io/v1beta1
kind: IPAddressPool
metadata:
```

```

name: underlay-pool
namespace: metallb-system
spec:
  addresses:
  - 10.180.204.250-10.180.204.254 #      VIP 10.180.204.252
---
apiVersion: metallb.io/v1beta1
kind: L2Advertisement
metadata:
  name: l2-advert
  namespace: metallb-system
spec:
  ipAddressPools:
  - underlay-pool

```

4. LoadBalancer Service

LoadBalancer Service Underlay Pod

```

apiVersion: apps/v1
kind: Deployment
metadata:
  labels:
    app: nginx
  name: deploy-169402624
spec:
  replicas: 3
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      annotations:
        ovn.kubernetes.io/logical_switch: underlay-subnet
      labels:
        app: nginx
    spec:
      containers:
      - args:
        - netexec
        - --http-port
        - "80"
        image: kubeovn/agnhost:2.47
        imagePullPolicy: IfNotPresent
        name: nginx
---
apiVersion: v1
kind: Service
metadata:
  name: nginx-lb
spec:
  externalTrafficPolicy: Local
  ipFamilies:
  - IPv4
  ipFamilyPolicy: PreferDualStack
  ports:
  - port: 80
    protocol: TCP
    targetPort: 80
  selector:
    app: nginx
  type: LoadBalancer

```

7.10.5

1. Service MetalLB IP

```
kubectl get svc nginx-lb
```

EXTERNAL-IP IP 10.180.204.252

1. Service IP

```
curl http://10.180.204.252
```

1. Pod

Service endpoints Pod

```

# Service endpoints
kubectl get endpoints nginx-lb

```

```
# Pod
kubectl get pods -l app=nginx -o wide
```

1. IP

nginx Pod	IP	IP	SNAT	IP
-----------	----	----	------	----

```
kubectl exec -it $(kubectl get pods -l app=nginx -o name | head -n1) -- cat /var/log/nginx/access.log
```

7.10.6

IP

MetalLB	Underlay	CIDR	Underlay	excludeIps	IP
---------	----------	------	----------	------------	----

MetalLB	Kube-OVN Underlay	underlay0.341	VLAN	VLAN	ARP	MetalLB VIP
---------	-------------------	---------------	------	------	-----	-------------

- Kube-OVN	--enable-ovn-lb-prefer-local=true	- Service	externalTrafficPolicy: Local
------------	-----------------------------------	-----------	------------------------------

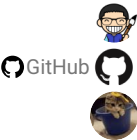
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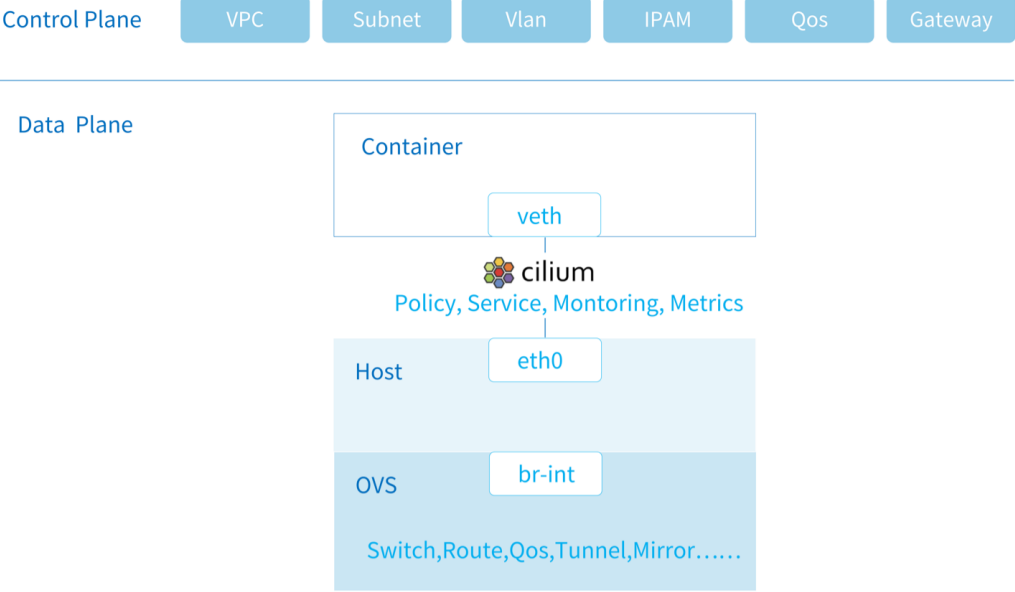
7.10.7

7.11 Cilium

Cilium eBPF Kube-OVN CNI Chaining Kube-OVN eBPF

Cilium Kube-OVN

-
- Hubble



7.11.1

1. Linux 4.19 eBPF
2. Helm Cilium Helm [Installing Helm](#)

7.11.2 Kube-OVN

Cilium Kube-OVN networkpolicy CNI

install.sh

```
ENABLE_NP=false
CNI_CONFIG_PRIORITY=10
```

kube-ovn-controller networkpolicy

```
args:
- --enable-np=false
```

kube-ovn-cni CNI

```
args:
- --cni-conf-name=10-kube-ovn.conflist
```

```
mv /etc/cni/net.d/01-kube-ovn.conf /etc/cni/net.d/10-kube-ovn.conf
```

7.11.3 Cilium

Cilium generic-veth

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: cni-configuration
  namespace: kube-system
data:
  cni-config: |-
    {
      "name": "generic-veth",
      "cniVersion": "0.3.1",
      "plugins": [
        {
          "type": "kube-ovn",
          "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
          "ipam": {
            "type": "kube-ovn",
            "server_socket": "/run/openvswitch/kube-ovn-daemon.sock"
          }
        },
        {
          "type": "portmap",
          "snat": true,
          "capabilities": {"portMappings": true}
        },
        {
          "type": "cilium-cni"
        }
      ]
    }
}
```

```
kubectl apply -f chaining.yaml
```

Helm Cilium

```
helm repo add cilium https://helm.cilium.io/
helm install cilium cilium/cilium --version 1.11.6 \
--namespace kube-system \
--set cni.chainingMode=generic-veth \
--set cni.customConf=true \
--set cni.configMap=cni-configuration \
--set tunnel=disabled \
--set enableIPv4Masquerade=false \
--set devices="eth+ ovn0 geneve_sys_6081 vxlan_sys_4789" \
--set enableIdentityMark=false
```

Cilium

```
# cilium status
  /--\
 /--\_--\
 \--\_--\
  \--\_--\
 /--\_--\
 \--\_--\
  \--\_--\
   \--\

DaemonSet          cilium          Desired: 2, Ready: 2/2, Available: 2/2
Deployment          cilium-operator Desired: 2, Ready: 2/2, Available: 2/2
Containers:
  cilium            Running: 2
  cilium-operator   Running: 2

Cluster Pods:      8/11 managed by Cilium
Image versions
  cilium            quay.io/cilium/cilium:v1.11.6: 2
  cilium-operator   quay.io/cilium/operator-generic:v1.11.6: 2
```



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7.11.4

7.12 Cilium NetworkPolicy

Kube-OVN	Cilium	Cilium
Cilium	Cilium	Cilium L3 L4

7.12.1

Pod

namespace test yaml test namespace label app=test Pod Pod

```
apiVersion: apps/v1
kind: Deployment
metadata:
  labels:
    app: test
  name: test
  namespace: test
spec:
  replicas: 1
  selector:
    matchLabels:
      app: test
  strategy:
    rollingUpdate:
      maxSurge: 25%
      maxUnavailable: 25%
    type: RollingUpdate
  template:
    metadata:
      labels:
        app: test
    spec:
      containers:
        - image: docker.io/library/nginx:alpine
          imagePullPolicy: IfNotPresent
          name: nginx
```

yaml default namespace label app=dynamic Pod Pod

```
apiVersion: apps/v1
kind: Deployment
metadata:
  labels:
    app: dynamic
  name: dynamic
  namespace: default
spec:
  replicas: 2
  selector:
    matchLabels:
      app: dynamic
  strategy:
    rollingUpdate:
      maxSurge: 25%
      maxUnavailable: 25%
    type: RollingUpdate
  template:
    metadata:
      creationTimestamp: null
      labels:
        app: dynamic
    spec:
      containers:
        - image: docker.io/library/nginx:alpine
          imagePullPolicy: IfNotPresent
          name: nginx
```

Pod Label :

```
# kubectl get pod -o wide --show-labels
NAME                                READY   STATUS    RESTARTS   AGE   IP            NODE                                NOMINATED NODE   READINESS GATES   LABELS
dynamic-7d8d7874f5-9v5c4           1/1     Running   0           28h   10.16.0.35    kube-ovn-worker                    <none>            <none>             app=dynamic, pod-
dynamic-7d8d7874f5-8z2n            1/1     Running   0           28h   10.16.0.36    kube-ovn-control-plane             <none>            <none>             app=dynamic, pod-
dynamic-7d8d7874f5-6dsg6           1/1     Running   0           7h20m 10.16.0.2     kube-ovn-control-plane             <none>            <none>             app=dynamic, pod-
dynamic-7d8d7874f5-6dsg6           1/1     Running   0           7h20m 10.16.0.2     kube-ovn-control-plane             <none>            <none>             app=dynamic, pod-
dynamic-7d8d7874f5-6dsg6           1/1     Running   0           7h20m 10.16.0.2     kube-ovn-control-plane             <none>            <none>             app=dynamic, pod-
```

dynamic-7d8d7874f5-tjgtp template-hash=7d8d7874f5	1/1	Running	0	7h46m	10.16.0.42	kube-ovn-worker	<none>	<none>	app=dynamic, pod-
label=test1-77b6764857-swq4k template-hash=77b6764857	1/1	Running	0	3h43m	10.16.0.12	kube-ovn-worker	<none>	<none>	app=test1, pod-
# Pod test-54c98bc466-mft5s template-hash=54c98bc466	1/1	Running	0	8h	10.16.0.41	kube-ovn-worker	<none>	<none>	app=test, pod-

L3

yaml CiliumNetworkPolicy :

```
apiVersion: "cilium.io/v2"
kind: CiliumNetworkPolicy
metadata:
  name: "l3-rule"
  namespace: test
spec:
  endpointSelector:
    matchLabels:
      app: test
  ingress:
    - fromEndpoints:
      - matchLabels:
          app: dynamic
```

default namespace Pod Pod test namespace Pod

default namespace :

```
# kubectl exec -it dynamic-7d8d7874f5-9v5c4 -- bash
bash-5.0# ping -c 3 10.16.0.41
PING 10.16.0.41 (10.16.0.41): 56 data bytes

--- 10.16.0.41 ping statistics ---
3 packets transmitted, 0 packets received, 100% packet loss
```

test namespace Pod :

```
# kubectl exec -it -n test dynamic-7d8d7874f5-6dsg6 -- bash
bash-5.0# ping -c 3 10.16.0.41
PING 10.16.0.41 (10.16.0.41): 56 data bytes
64 bytes from 10.16.0.41: seq=0 ttl=64 time=2.558 ms
64 bytes from 10.16.0.41: seq=1 ttl=64 time=0.223 ms
64 bytes from 10.16.0.41: seq=2 ttl=64 time=0.304 ms

--- 10.16.0.41 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.223/1.028/2.558 ms
```

Cilium CiliumNetworkPolicy Namespace Cilium

Namespace Pod Namespace Pod

Namespace Namespace

CiliumNetworkPolicy namespace :

```
ingress:
- fromEndpoints:
- matchLabels:
    app: dynamic
    k8s:io.kubernetes.pod.namespace: default # Namespace Pod
```

CiliumNetworkPolicy :

```
# kubectl get cnp -n test -o yaml l3-rule
apiVersion: cilium.io/v2
kind: CiliumNetworkPolicy
metadata:
  name: l3-rule
  namespace: test
spec:
  endpointSelector:
    matchLabels:
      app: test
  ingress:
    - fromEndpoints:
      - matchLabels:
```

```

    app: dynamic
  - matchLabels:
    app: dynamic
    k8s:io.kubernetes.pod.namespace: default

```

default namespace Pod Pod :

```

# kubectl exec -it dynamic-7d8d7874f5-9v5c4 -n test -- bash
bash-5.0# ping -c 3 10.16.0.41
PING 10.16.0.41 (10.16.0.41): 56 data bytes
64 bytes from 10.16.0.41: seq=0 ttl=64 time=2.383 ms
64 bytes from 10.16.0.41: seq=1 ttl=64 time=0.115 ms
64 bytes from 10.16.0.41: seq=2 ttl=64 time=0.142 ms

--- 10.16.0.41 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.115/0.880/2.383 ms

```

Kubernetes	networkpolicy	Cilium	Namespace	Namespace	Namespace	Pod	Pod	Pod
Kube-OVN	Kube-OVN	k8s	Namespace	Pod	Pod	Namespace	Namespace	Pod

L4

yaml L4 :

```

apiVersion: "cilium.io/v2"
kind: CiliumNetworkPolicy
metadata:
  name: "l4-rule"
  namespace: test
spec:
  endpointSelector:
    matchLabels:
      app: test
  ingress:
    - fromEndpoints:
      - matchLabels:
          app: dynamic
    toPorts:
      - ports:
          - port: "80"
            protocol: TCP

```

Namespace Pod

```

# kubectl exec -it -n test dynamic-7d8d7874f5-6dsg6 -- bash
bash-5.0# ping -c 3 10.16.0.41
PING 10.16.0.41 (10.16.0.41): 56 data bytes

--- 10.16.0.41 ping statistics ---
3 packets transmitted, 0 packets received, 100% packet loss
bash-5.0#
bash-5.0# curl 10.16.0.41:80
<html>
<head>
  <title>Hello World!</title>
  <link href='//fonts.googleapis.com/css?family=Open+Sans:400,700' rel='stylesheet' type='text/css'>
  <style>
    body {
      background-color: white;
      text-align: center;
      padding: 50px;
      font-family: "Open Sans", "Helvetica Neue", Helvetica, Arial, sans-serif;
    }
    #logo {
      margin-bottom: 40px;
    }
  </style>
</head>
<body>
  <h1>Hello World!</h1>
  <h3>Links found</h3>
  <h3>I am on test-54c98bc466-mft5s</h3>
  <h3>Cookie =</h3>
  <b>KUBERNETES</b> listening in 443 available at tcp://10.96.0.1:443<br />
  <h3>my name is hanhouchao!</h3>
  <h3> RequestURI=' /'</h3>
</body>
</html>

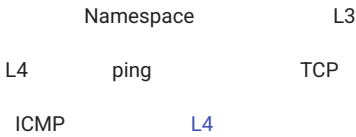
```

Namespace

Pod

```
# kubectl exec -it -n test label-test1-77b6764857-swq4k -- bash
bash-5.0# ping -c 3 10.16.0.41
PING 10.16.0.41 (10.16.0.41): 56 data bytes

--- 10.16.0.41 ping statistics ---
3 packets transmitted, 0 packets received, 100% packet loss
bash-5.0#
bash-5.0# curl -v 10.16.0.41:80 --connect-timeout 10
* Trying 10.16.0.41:80...
* After 10000ms connect time, move on!
* connect to 10.16.0.41 port 80 failed: Operation timed out
* Connection timeout after 10001 ms
* Closing connection 0
curl: (28) Connection timeout after 10001 ms
```



L7

chaining L7 Cilium [Generic Veth Chaining](#)

[issue 12454](#)

PDF

Slack

Support

2026 7 4

2022 8 2

GitHub

+1

7.13 Cilium

Kube-OVN	Cilium	Cilium
Cilium	Hubble	Hubble

7.13.1 Hubble

Cilium	Hubble	Hubble
helm	Hubble	

```
helm upgrade cilium cilium/cilium --version 1.11.6 \
  --namespace kube-system \
  --reuse-values \
  --set hubble.relay.enabled=true \
  --set hubble.ui.enabled=true
```

Hubble	cilium status	Hubble
--------	---------------	--------

```
# cilium status
/---\
/---\ Cilium:      OK
\---\ Operator:   OK
/---\ Hubble:     OK
\---\ ClusterMesh: disabled

Deployment      hubble-relay      Desired: 1, Ready: 1/1, Available: 1/1
Deployment      cilium-operator    Desired: 2, Ready: 2/2, Available: 2/2
DaemonSet      cilium             Desired: 2, Ready: 2/2, Available: 2/2
Deployment      hubble-ui          Desired: 1, Ready: 1/1, Available: 1/1
Containers:     cilium             Running: 2
                hubble-ui          Running: 1
                hubble-relay       Running: 1
                cilium-operator    Running: 2

Cluster Pods:   16/17 managed by Cilium
Image versions  hubble-relay      quay.io/cilium/hubble-relay:v1.11.6@sha256:fd9034a2d04d5b973f1e8ed44f230ea195b89c37955ff32e34e5aa68f3ed675a: 1
                cilium-operator    quay.io/cilium/operator-generic:v1.11.6@sha256:9f6063c7bcaede801a39315ec7c166309f6a6783e98665f6693939cf1701bc17: 2
                cilium             quay.io/cilium/cilium:v1.11.6@sha256:f7f93c26739b6641a3fa3d76b1e1605b15989f25d06625260099e01c8243f54c: 2
                hubble-ui          quay.io/cilium/hubble-ui:v0.9.0@sha256:0ef04e9a29212925da6bdfd0ba5b581765e41a01f1cc30563cef9b30b457fea0: 1
                hubble-ui          quay.io/cilium/hubble-ui-backend:v0.9.0@sha256:000df6b76719f607a9edefb9af94dfd1811a6f1b6a8a9c537cba90bf12df474b: 1

apple@bogon cilium %
```

Hubble	Hubble CLI :
--------	--------------

```
curl -L --fail --remote-name-all https://github.com/cilium/hubble/releases/download/v0.10.0/hubble-linux-amd64.tar.gz
sudo tar xzvfC hubble-linux-amd64.tar.gz /usr/local/bin
```

7.13.2

Cilium

cilium connectivity test	Cilium	cilium-test	Namespace	cilium-test
cilium-test namespace				

```
# kubectl get all -n cilium-test
NAME                                READY   STATUS    RESTARTS   AGE
pod/client-7df6cfbf7b-z5t2j         1/1     Running   0           21s
pod/client2-547996d7d8-nvgxg        1/1     Running   0           21s
pod/echo-other-node-d79544ccf-h14gg 2/2     Running   0           21s
pod/echo-same-node-5d466d5444-m17tc 2/2     Running   0           21s

NAME                                TYPE          CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
service/echo-other-node             NodePort      10.109.58.126   <none>            8080:32269/TCP   21s
service/echo-same-node              NodePort      10.108.70.32    <none>            8080:32490/TCP   21s

NAME                                READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/client              1/1     1             1           21s
deployment.apps/client2            1/1     1             1           21s
deployment.apps/echo-other-node     1/1     1             1           21s
deployment.apps/echo-same-node      1/1     1             1           21s

NAME                                DESIRED   CURRENT   READY   AGE
```

```

replicaset.apps/client-7df6cfbf7b      1      1      1      21s
replicaset.apps/client2-547996d7d8     1      1      1      21s
replicaset.apps/echo-other-node-d79544ccf 1      1      1      21s
replicaset.apps/echo-same-node-5d466d5444 1      1      1      21s

```

7.13.3

Cilium kube-system namespace Cilium pod hubble observe

```

# kubectl get pod -n kube-system -o wide
NAME                                READY  STATUS   RESTARTS  AGE    IP             NODE                NOMINATED NODE  READINESS GATES
cilium-d6h56                        1/1    Running  0          2d20h  172.18.0.2     kube-ovn-worker     <none>          <none>
cilium-operator-5887f78bbb-c7sb2    1/1    Running  0          2d20h  172.18.0.2     kube-ovn-worker     <none>          <none>
cilium-operator-5887f78bbb-wj8gt    1/1    Running  0          2d20h  172.18.0.3     kube-ovn-control-plane <none>          <none>
cilium-tq5xb                        1/1    Running  0          2d20h  172.18.0.3     kube-ovn-control-plane <none>          <none>
kube-ovn-pinger-71gk8               1/1    Running  0          21h    10.16.0.19     kube-ovn-control-plane <none>          <none>
kube-ovn-pinger-msvcn               1/1    Running  0          21h    10.16.0.18     kube-ovn-worker     <none>          <none>

# kubectl exec -it -n kube-system cilium-d6h56 -- bash
root@kube-ovn-worker:/home/cilium# hubble observe --from-namespace kube-system
Jul 29 03:24:25.551: kube-system/kube-ovn-pinger-msvcn:35576 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: ACK, PSH)
Jul 29 03:24:25.561: kube-system/kube-ovn-pinger-msvcn:35576 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: RST)
Jul 29 03:24:25.561: kube-system/kube-ovn-pinger-msvcn:35576 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: ACK, RST)
Jul 29 03:24:25.572: kube-system/kube-ovn-pinger-msvcn:35578 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: SYN)
Jul 29 03:24:25.572: kube-system/kube-ovn-pinger-msvcn:35578 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: ACK)
Jul 29 03:24:25.651: kube-system/kube-ovn-pinger-msvcn:35578 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: ACK, PSH)
Jul 29 03:24:25.661: kube-system/kube-ovn-pinger-msvcn:35578 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: RST)
Jul 29 03:24:25.661: kube-system/kube-ovn-pinger-msvcn:35578 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: ACK, RST)
Jul 29 03:24:25.761: kube-system/kube-ovn-pinger-msvcn:52004 -> 172.18.0.3:6443 to-stack FORWARDED (TCP Flags: ACK, PSH)
Jul 29 03:24:25.779: kube-system/kube-ovn-pinger-msvcn -> kube-system/kube-ovn-pinger-71gk8 to-stack FORWARDED (ICMPv4 EchoRequest)
Jul 29 03:24:25.779: kube-system/kube-ovn-pinger-msvcn <- kube-system/kube-ovn-pinger-71gk8 to-endpoint FORWARDED (ICMPv4 EchoReply)
Jul 29 03:24:25.866: kube-system/hubble-ui-7596f7ff6f-7j6f2:55836 <- kube-system/hubble-relay-959988db5-zc5vv:4245 to-stack FORWARDED (TCP Flags: ACK)
Jul 29 03:24:25.866: kube-system/hubble-ui-7596f7ff6f-7j6f2:55836 <- kube-system/hubble-relay-959988db5-zc5vv:80 to-endpoint FORWARDED (TCP Flags: ACK)
Jul 29 03:24:25.866: kube-system/hubble-ui-7596f7ff6f-7j6f2:55836 -> kube-system/hubble-relay-959988db5-zc5vv:4245 to-stack FORWARDED (TCP Flags: ACK)
Jul 29 03:24:25.866: kube-system/hubble-ui-7596f7ff6f-7j6f2:55836 -> kube-system/hubble-relay-959988db5-zc5vv:4245 to-endpoint FORWARDED (TCP Flags: ACK)
Jul 29 03:24:25.975: kube-system/kube-ovn-pinger-71gk8 -> kube-system/kube-ovn-pinger-msvcn to-endpoint FORWARDED (ICMPv4 EchoRequest)
Jul 29 03:24:25.975: kube-system/kube-ovn-pinger-71gk8 <- kube-system/kube-ovn-pinger-msvcn to-stack FORWARDED (ICMPv4 EchoReply)
Jul 29 03:24:25.979: kube-system/kube-ovn-pinger-msvcn -> 172.18.0.3 to-stack FORWARDED (ICMPv4 EchoRequest)
Jul 29 03:24:26.037: kube-system/coredns-6d4b75cb6d-lbgjg:36430 -> 172.18.0.3:6443 to-stack FORWARDED (TCP Flags: ACK)
Jul 29 03:24:26.282: kube-system/kube-ovn-pinger-msvcn -> 172.18.0.2 to-stack FORWARDED (ICMPv4 EchoRequest)

```

Hubble Relay Hubble

Hubble API Hubble Service kubectl port-forward deployment/hubble-relay -n kube-system 4245:4245

kubectl port-forward

hubble status

```

# hubble status
Healthcheck (via localhost:4245): Ok
Current/Max Flows: 8,190/8,190 (100.00%)
Flows/s: 22.86
Connected Nodes: 2/2

```

hubble observe

cilium-test

```

apple@bogon ovn-test % hubble observe
Jul 28 08:00:07.033: cilium-test/client-7df6cfbf7b-qn7q6:56906 (ID:15432) -> kube-system/coredns-6d4b75cb6d-b444j:53 (ID:11307) to-endpoint FORWARDED (UDP)
Jul 28 08:00:07.033: cilium-test/client-7df6cfbf7b-qn7q6:56906 (ID:15432) <- kube-system/coredns-6d4b75cb6d-b444j:53 (ID:11307) to-stack FORWARDED (UDP)
Jul 28 08:00:07.095: 100.64.0.3:43037 (world) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: SYN)
Jul 28 08:00:07.095: 100.64.0.3:43037 (world) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: SYN, ACK)
Jul 28 08:00:07.096: 100.64.0.3:43037 (world) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.096: 100.64.0.3:43037 (world) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.096: 100.64.0.3:43037 (world) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.097: 100.64.0.3:43037 (world) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.249: 100.64.0.2:33419 (host) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: SYN)
Jul 28 08:00:07.249: 100.64.0.2:33419 (host) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: SYN, ACK)
Jul 28 08:00:07.250: 100.64.0.2:33419 (host) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.251: 100.64.0.2:33419 (host) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.251: 100.64.0.2:33419 (host) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.251: 100.64.0.2:33419 (host) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.323: cilium-test/client2-547996d7d8-jgblc:34927 (ID:14804) <- 172.18.0.3:31514 (kube-apiserver) to-endpoint FORWARDED (TCP Flags: SYN, ACK)
Jul 28 08:00:07.323: cilium-test/client2-547996d7d8-jgblc:34927 (ID:14804) -> 172.18.0.3:31514 (kube-apiserver) to-stack FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.324: 100.64.0.2:34927 (world) -> cilium-test/echo-same-node-5d466d5444-4vllm:8080 (ID:15976) to-endpoint FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.324: cilium-test/client2-547996d7d8-jgblc:34927 (ID:14804) -> 172.18.0.3:31514 (kube-apiserver) to-stack FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.324: 100.64.0.2:34927 (world) -> cilium-test/echo-same-node-5d466d5444-4vllm:8080 (ID:15976) to-endpoint FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.324: 100.64.0.2:34927 (world) <- cilium-test/echo-same-node-5d466d5444-4vllm:8080 (ID:15976) to-stack FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.324: cilium-test/client2-547996d7d8-jgblc:34927 (ID:14804) <- 172.18.0.3:31514 (kube-apiserver) to-endpoint FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.324: cilium-test/client2-547996d7d8-jgblc:34927 (ID:14804) -> 172.18.0.3:31514 (kube-apiserver) to-stack FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.324: 100.64.0.2:34927 (world) -> cilium-test/echo-same-node-5d466d5444-4vllm:8080 (ID:15976) to-endpoint FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.340: kube-system/hubble-relay-959988db5-zc5vv:46938 (ID:53347) -> 172.18.0.2:4244 (host) to-stack FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.340: kube-system/hubble-relay-959988db5-zc5vv:46938 (ID:53347) -> 172.18.0.3:4244 (kube-apiserver) to-stack FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.340: kube-system/hubble-relay-959988db5-zc5vv:46938 (ID:53347) <- 172.18.0.2:4244 (host) to-endpoint FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.341: kube-system/hubble-relay-959988db5-zc5vv:46938 (ID:53347) <- 172.18.0.3:4244 (kube-apiserver) to-endpoint FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.409: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: SYN)
Jul 28 08:00:07.409: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: SYN)
Jul 28 08:00:07.409: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: SYN, ACK)
Jul 28 08:00:07.410: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: SYN, ACK)
Jul 28 08:00:07.410: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.410: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.410: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.411: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.411: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.412: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.412: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.412: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.412: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK, FIN)
apple@bogon ovn-test %

```

hubble observe

hubble help observe

Hubble CLI

7.13.4 UI

cilium status

Hubble UI

Hubble

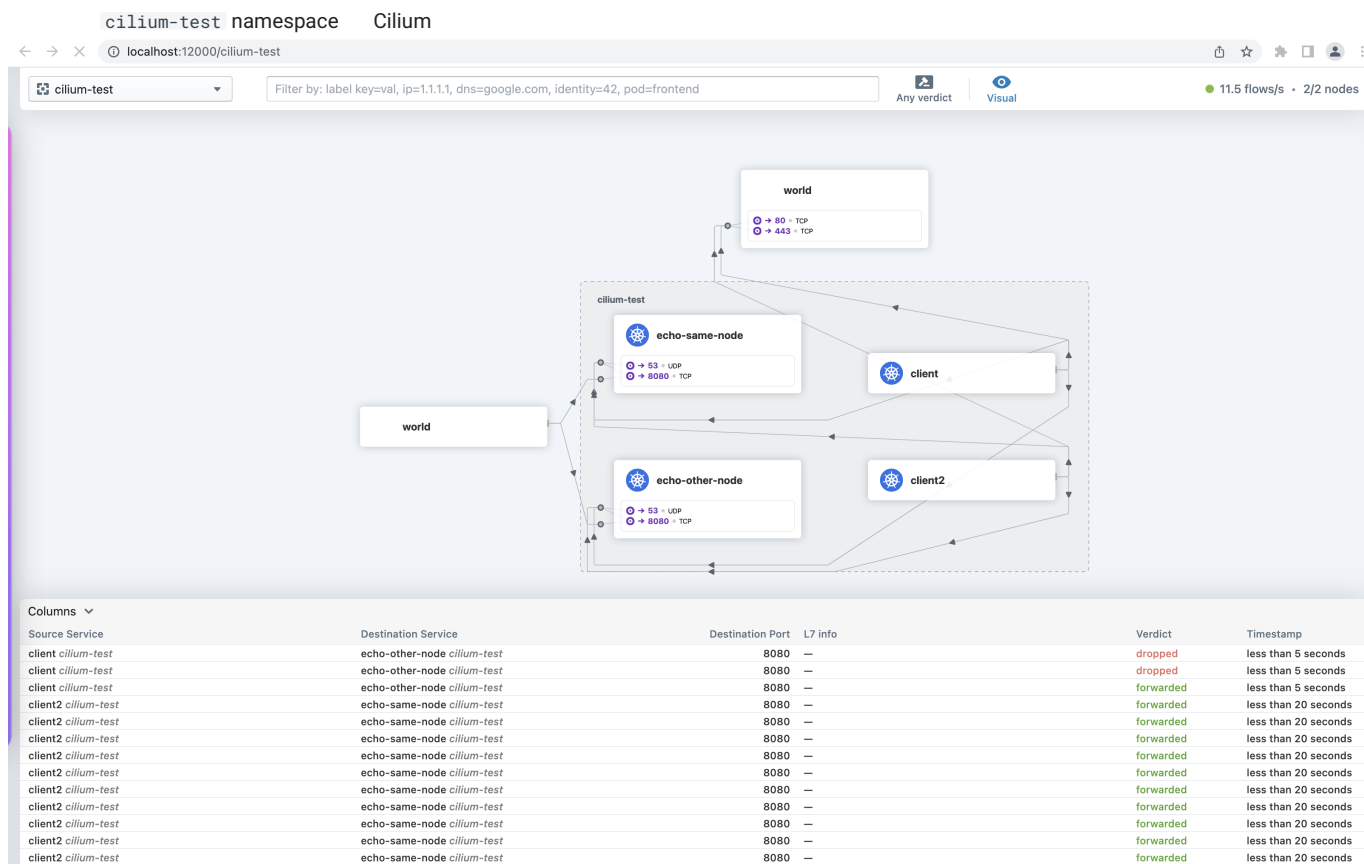
UI

cilium hubble ui

hubble-ui service

Hubble UI

http://localhost:12000 UI



7.13.5 Hubble

Hubble Pod Hubble

hubble.metrics.enabled :

```
helm upgrade cilium cilium/cilium --version 1.11.6 \
--namespace kube-system \
--reuse-values \
--set hubble.relay.enabled=true \
--set hubble.ui.enabled=true \
--set hubble.metrics.enabled="{dns,drop,tcp,flow,icmp,http}"
```

kube-system namespace hubble-metrics Endpoints Hubble :

```
# curl 172.18.0.2:9091/metrics
# HELP hubble_drop_total Number of drops
# TYPE hubble_drop_total counter
hubble_drop_total{protocol="ICMPv6",reason="Unsupported L3 protocol"} 2
# HELP hubble_flows_processed_total Total number of flows processed
# TYPE hubble_flows_processed_total counter
hubble_flows_processed_total{protocol="ICMPv4",subtype="to-endpoint",type="Trace",verdict="FORWARDED"} 335
hubble_flows_processed_total{protocol="ICMPv4",subtype="to-stack",type="Trace",verdict="FORWARDED"} 335
hubble_flows_processed_total{protocol="ICMPv6",subtype="",type="Drop",verdict="DROPPED"} 2
hubble_flows_processed_total{protocol="TCP",subtype="to-endpoint",type="Trace",verdict="FORWARDED"} 8282
hubble_flows_processed_total{protocol="TCP",subtype="to-stack",type="Trace",verdict="FORWARDED"} 6767
hubble_flows_processed_total{protocol="UDP",subtype="to-endpoint",type="Trace",verdict="FORWARDED"} 1642
hubble_flows_processed_total{protocol="UDP",subtype="to-stack",type="Trace",verdict="FORWARDED"} 1642
# HELP hubble_icmp_total Number of ICMP messages
# TYPE hubble_icmp_total counter
hubble_icmp_total{family="IPv4",type="EchoReply"} 335
hubble_icmp_total{family="IPv4",type="EchoRequest"} 335
hubble_icmp_total{family="IPv4",type="RouterSolicitation"} 2
# HELP hubble_tcp_flags_total TCP flag occurrences
# TYPE hubble_tcp_flags_total counter
hubble_tcp_flags_total{family="IPv4",flag="FIN"} 2043
hubble_tcp_flags_total{family="IPv4",flag="RST"} 301
hubble_tcp_flags_total{family="IPv4",flag="SYN"} 1169
hubble_tcp_flags_total{family="IPv4",flag="SYN-ACK"} 1169
```

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7.13.6

7.14

Kube-OVN

7.14.1

```
kind: Subnet
apiVersion: kubeovn.io/v1
metadata:
  name: external
spec:
  cidrBlock: 172.31.0.0/16
  gatewayType: centralized
  natOutgoing: false
  externalEgressGateway: 192.168.0.1
  policyRoutingTableID: 1000
  policyRoutingPriority: 1500
```

- natOutgoing: false
- externalEgressGateway
- policyRoutingTableID TableID
- policyRoutingPriority

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7.14.2

7.15 VIP IP

VIP	IP	IP	VIP	kube-ovn	IP	POD	IP	IP	VIP	Openstack neutron	Allowed-Address-Pairs
AAP		Openstack	octavia			POD	IP	aliyun	terway	IP	VIP
POD	IP	VIP	IP					OVN	Switch LB	IP LB	VIP
Switch LB Rule			VIP								VIP
											OVN

• Allowed-Address-Pairs VIP

• Switch LB rule VIP

• Pod VIP IP

7.15.1 1. Allowed-Address-Pairs VIP

IP	Pod				
• Kubernetes	Kubernetes	Kubernetes	Underlay	Subnet	
• LB	Subnet	IP	Pod		
VIP	Allowed-Address-Pairs	IP	IP		
• Keepalived	IP				

1.1 VIP

IP

IP

yaml

```
apiVersion: kubeovn.io/v1
kind: Vip
metadata:
  name: vip-dynamic-01
spec:
  subnet: ovn-default
  type: ""
```

• subnet: Subnet IP

• type: ipam ip switch_lb_vip vip switch lb vip ip

VIP

# kubectl get vip									
NAME	V4IP	PV4IP	MAC	PMAC	V6IP	PV6IP	SUBNET	READY	
vip-dynamic-01	10.16.0.12		00:00:00:F0:DB:25				ovn-default	true	

VIP 10.16.0.12 IP

1.2 VIP

VIP

IP

yaml

```
apiVersion: kubeovn.io/v1
kind: Vip
metadata:
  name: static-vip01
spec:
  subnet: ovn-default
  v4ip: "10.16.0.121"
```

• subnet: Subnet IP

• v4ip: IP subnet CIDR

VIP

```
# kubectl get vip
NAME          V4IP      PV4IP  MAC          PMAC  V6IP  PV6IP  SUBNET    READY
static-vip01  10.16.0.121  10.16.0.121  00:00:00:F0:DB:26  00:00:00:F0:DB:26  00:00:00:F0:DB:26  00:00:00:F0:DB:26  ovn-default  true
```

VIP IP

1.3 Pod VIP AAP

Pod annotation VIP AAP Pod labels VIP selector

Pod annotation VIP ovn.kubernetes.io/aaps: vip-aap,vip-aap2,vip-aap3

AAP Pod AAP Pod VIP subnet Port

1.3.1 VIP AAP

```
apiVersion: kubeovn.io/v1
kind: Vip
metadata:
  name: vip-aap
spec:
  subnet: ovn-default
  namespace: default
  selector:
    - "app: aap1"
```

VIP

- namespace: AAP VIP VIP AAP
 - selector: AAP VIP Pod namespace Pod key: value AND
- VIP Port

```
# kubectl ko nbctl show ovn-default
switch e32e1d3b-c539-45f4-ab19-be4e33a061f6 (ovn-default)
  port aap-vip
    type: virtual
```

```
apiVersion: v1
kind: Pod
metadata:
  name: busybox
  annotations:
    ovn.kubernetes.io/aaps: vip-aap
  labels:
    app: aap1
spec:
  containers:
    - name: busybox
      image: busybox
      command: ["sleep", "3600"]
      securityContext:
        capabilities:
          add:
            - NET_ADMIN
```

AAP

```
# kubectl ko nbctl list logical_switch_port aap-vip
_uuid      : cd930750-0533-4f06-a6c0-217ddac73272
addresses  : []
dhcpv4_options : []
dhcpv6_options : []
dynamic_addresses : []
enabled    : []
external_ids : {ls=ovn-default, vendor=kube-ovn}
ha_chassis_group : []
mirror_rules : []
name       : aap-vip
options    : {virtual-ip="10.16.0.100", virtual-parents="busybox.default"}
parent_name : []
port_security : []
tag        : []
tag_request : []
type       : virtual
up         : false
```

virtual-ip VIP IP virtual-parents AAP Pod Port

Pod

```
# kubectl exec -it busybox -- ip addr add 10.16.0.100/16 dev eth0
# kubectl exec -it busybox01 -- ip addr show eth0
35: eth0@if36: <BROADCAST,MULTICAST,UP,LOWER_UP,M-DOWN> mtu 1400 qdisc noqueue
    link/ether 00:00:00:e2:ab:0c brd ff:ff:ff:ff:ff:ff
    inet 10.16.0.7/16 brd 10.16.255.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet 10.16.0.100/16 scope global secondary eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::200:ff:fee2:ab0c/64 scope link
        valid_lft forever preferred_lft forever
```

Pod	IP	VIP	IP	subnet	Pod	IP
-----	----	-----	----	--------	-----	----

7.15.2.2. Switch LB rule vip

```
apiVersion: kubeovn.io/v1
kind: Vip
metadata:
  name: slr-01
spec:
  subnet: ovn-default
  type: switch_lb_vip
```

- subnet: Subnet IP
- type: ipam ip switch_lb_vip vip switch lb vip ip

7.15.3.3. Pod VIP IP

v1.12

IP

```
apiVersion: kubeovn.io/v1
kind: Vip
metadata:
  name: pod-use-vip
spec:
  subnet: ovn-default
  type: ""
```

annotation	VIP	Pod
------------	-----	-----

```
apiVersion: v1
kind: Pod
metadata:
  name: static-ip
  annotations:
    ovn.kubernetes.io/vip: pod-use-vip # vip
  namespace: default
spec:
  containers:
    - name: static-ip
      image: docker.io/library/nginx:alpine
```

3.1 StatefulSet Kubevirt VM VIP

StatefulSet	VM	Pod	VIP
VM	VIP	kube-ovn-controller	keep-vm-ip true Kubevirt VM

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 Support

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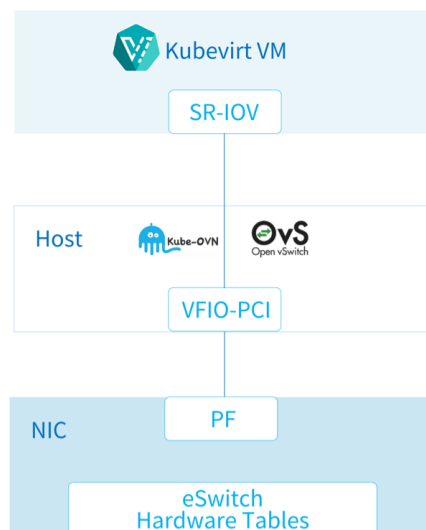


+2

7.15.4

7.16 Mellanox Offload

Kube-OVN Processing (ASAP²) OVS OVS CPU eSwitch CPU OVS Mellanox CPU Accelerated Switching And Packet



Note

2022

7.16.1

- Mellanox CX5/CX6/CX7/BlueField ASAP²
- CentOS 8 Stream Linux 5.7
- dp_hash hash OVN LB
- bond

7.16.2 SR-IOV Device Plugin

Mellanox offload SR-IOV Device Plugin [sriov-network-operator](#)

SR-IOV Device Plugin

SR-IOV

ID 84:00.0 84.00.1

```
# lspci -nn | grep ConnectX-5
84:00.0 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5] [15b3:1017]
84:00.1 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5] [15b3:1017]
```


ID

```
# ls -l /sys/class/net/ | grep 84:00.0
lrwxrwxrwx 1 root root 0 Feb 4 16:16 enp132s0f0np0 -> ../../devices/pci0000:80/0000:80:08.0/0000:84:00.0/net/enp132s0f0np0
# ls -l /sys/class/net/ | grep 84:00.1
lrwxrwxrwx 1 root root 0 Feb 4 16:16 enp132s0f1np1 -> ../../devices/pci0000:80/0000:80:08.0/0000:84:00.1/net/enp132s0f1np1
```

bond

```
enp132s0f0np0 enp132s0f1np1 bond1
```

```
# ip link show enp132s0f0np0 | grep bond
160: enp132s0f0np0: <BROADCAST,MULTICAST,SLAVE,UP,LOWER_UP> mtu 1500 qdisc mq master bond1 state UP mode DEFAULT group default qlen 1000
# ip link show enp132s0f1np1 | grep bond
169: enp132s0f1np1: <BROADCAST,MULTICAST,SLAVE,UP,LOWER_UP> mtu 1500 qdisc mq master bond1 state UP mode DEFAULT group default qlen 1000
```

bond VF

```
ifenslave -d bond1 enp132s0f0np0
ifenslave -d bond1 enp132s0f1np1
echo 0 > /sys/class/net/enp132s0f0np0/device/sriov_numvfs
echo 0 > /sys/class/net/enp132s0f1np1/device/sriov_numvfs
ip link set enp132s0f0np0 down
ip link set enp132s0f1np1 down
```

OVS

SMFS DMFS

- SMFS (software-managed flow steering)
- DMFS (device-managed flow steering)

sysfs devlink API

```
# sysfs
echo <smfs|dmfs> > /sys/class/net/enp132s0f0np0/compat/devlink/steering_mode
echo <smfs|dmfs> > /sys/class/net/enp132s0f1np1/compat/devlink/steering_mode
# devlink
devlink dev param set pci/84:00.0 name flow_steering_mode value smfs cmode runtime
devlink dev param set pci/84:00.1 name flow_steering_mode value smfs cmode runtime
```

VF

```
# cat /sys/class/net/enp132s0f0np0/device/sriov_totalvfs
127
# cat /sys/class/net/enp132s0f1np1/device/sriov_totalvfs
127
```

VF

```
# echo '4' > /sys/class/net/enp132s0f0np0/device/sriov_numvfs
# echo '4' > /sys/class/net/enp132s0f1np1/device/sriov_numvfs
# ip link show enp132s0f0np0
160: enp132s0f0np0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc mq state DOWN mode DEFAULT group default qlen 1000
    link/ether 08:c0:eb:74:c3:4a brd ff:ff:ff:ff:ff:ff
    vf 0 link/ether 00:00:00:00:00:00 brd ff:ff:ff:ff:ff:ff, spoof checking off, link-state disable, trust off, query_rss off
    vf 1 link/ether 00:00:00:00:00:00 brd ff:ff:ff:ff:ff:ff, spoof checking off, link-state disable, trust off, query_rss off
    vf 2 link/ether 00:00:00:00:00:00 brd ff:ff:ff:ff:ff:ff, spoof checking off, link-state disable, trust off, query_rss off
    vf 3 link/ether 00:00:00:00:00:00 brd ff:ff:ff:ff:ff:ff, spoof checking off, link-state disable, trust off, query_rss off
# ip link show enp132s0f1np1
169: enp132s0f1np1: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc mq state DOWN mode DEFAULT group default qlen 1000
    link/ether 08:c0:eb:74:c3:4b brd ff:ff:ff:ff:ff:ff
    vf 0 link/ether 00:00:00:00:00:00 brd ff:ff:ff:ff:ff:ff, spoof checking off, link-state disable, trust off, query_rss off
    vf 1 link/ether 00:00:00:00:00:00 brd ff:ff:ff:ff:ff:ff, spoof checking off, link-state disable, trust off, query_rss off
    vf 2 link/ether 00:00:00:00:00:00 brd ff:ff:ff:ff:ff:ff, spoof checking off, link-state disable, trust off, query_rss off
    vf 3 link/ether 00:00:00:00:00:00 brd ff:ff:ff:ff:ff:ff, spoof checking off, link-state disable, trust off, query_rss off
# ip link set enp132s0f0np0 up
# ip link set enp132s0f1np1 up
```

VF ID

```
# lspci -nn | grep ConnectX-5 | grep Virtual
84:00.2 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
84:00.3 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
84:00.4 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
```

```
84:00.5 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
84:00.6 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
84:00.7 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
84:01.0 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
84:01.1 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
```

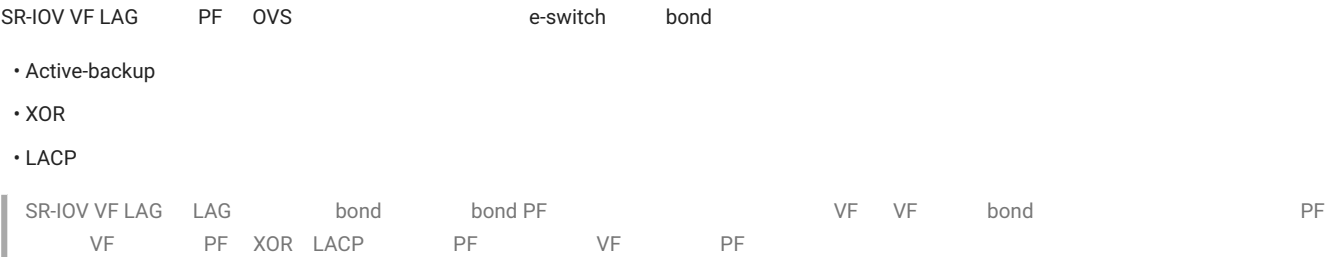
VF

```
echo 0000:84:00.2 > /sys/bus/pci/drivers/mlx5_core/unbind
echo 0000:84:00.3 > /sys/bus/pci/drivers/mlx5_core/unbind
echo 0000:84:00.4 > /sys/bus/pci/drivers/mlx5_core/unbind
echo 0000:84:00.5 > /sys/bus/pci/drivers/mlx5_core/unbind
echo 0000:84:00.6 > /sys/bus/pci/drivers/mlx5_core/unbind
echo 0000:84:00.7 > /sys/bus/pci/drivers/mlx5_core/unbind
echo 0000:84:01.0 > /sys/bus/pci/drivers/mlx5_core/unbind
echo 0000:84:01.1 > /sys/bus/pci/drivers/mlx5_core/unbind
```

eSwitch

```
devlink dev eswitch set pci/0000:84:00.0 mode switchdev
devlink dev eswitch set pci/0000:84:00.1 mode switchdev
ethtool -K enp132s0f0np0 hw-tc-offload on
ethtool -K enp132s0f1np1 hw-tc-offload on
```

SR-IOV VF



LACP

```
modprobe bonding mode=802.3ad
ip link set enp132s0f0np0 master bond1
ip link set enp132s0f1np1 master bond1
ip link set enp132s0f0np0 up
ip link set enp132s0f1np1 up
ip link set bond1 up
```

bond

VF

```
echo 0000:84:00.2 > /sys/bus/pci/drivers/mlx5_core/bind
echo 0000:84:00.3 > /sys/bus/pci/drivers/mlx5_core/bind
echo 0000:84:00.4 > /sys/bus/pci/drivers/mlx5_core/bind
echo 0000:84:00.5 > /sys/bus/pci/drivers/mlx5_core/bind
echo 0000:84:00.6 > /sys/bus/pci/drivers/mlx5_core/bind
echo 0000:84:00.7 > /sys/bus/pci/drivers/mlx5_core/bind
echo 0000:84:01.0 > /sys/bus/pci/drivers/mlx5_core/bind
echo 0000:84:01.1 > /sys/bus/pci/drivers/mlx5_core/bind
```

NetworkManager

NetworkManager

```
systemctl stop NetworkManager
systemctl disable NetworkManager
```

DEVICE PLUGIN

VF Pod VF SR-IOV Device Plugin

SR-IOV Configmap

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: sriovdp-config
  namespace: kube-system
data:
  config.json: |
```

```
{
  "resourceList": [{
    "resourcePrefix": "mellanox.com",
    "resourceName": "cx5_sriov_switchdev",
    "selectors": {
      "vendors": ["15b3"],
      "devices": ["1018"],
      "drivers": ["mlx5_core"]
    }
  }]
}
```

SR-IOV Device Plugin ConfigMap ConfigMap name sriovdp-config

- selectors: VF
- vendors:
- devices:
- drivers:

selectors pciAddresses acpiIndexes VF [SR-IOV ConfigMap](#)

SR-IOV

```
kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/sriov-network-device-plugin/v3.6.2/deployments/sriovdp-daemonset.yaml
```

SR-IOV Kubernetes Node

```
kubectl describe node kube-ovn-01 | grep mellanox

mellanox.com/cx5_sriov_switchdev: 8
mellanox.com/cx5_sriov_switchdev: 8
mellanox.com/cx5_sriov_switchdev 0      0
```

sriov-network-operator SR-IOV Device Plugin

[node-feature-discovery](#)

```
kubectl apply -k https://github.com/kubernetes-sigs/node-feature-discovery/deployment/overlays/default?ref=v0.11.3
```

offload annotation:

```
kubectl label nodes [offloadNicNode] feature.node.kubernetes.io/network-sriov.capable=true
```

Operator

```
git clone --depth=1 https://github.com/kubeovn/sriov-network-operator.git
kubectl apply -k sriov-network-operator/deploy
```

Operator

```
# kubectl get -n kube-system all | grep sriov
NAME                                READY    STATUS    RESTARTS    AGE
pod/sriov-network-config-daemon-bf9nt 1/1      Running   0           8s
pod/sriov-network-operator-54d7545f65-296gb 1/1      Running   0           10s

NAME                                DESIRED    CURRENT    READY    UP-TO-DATE    AVAILABLE    NODE
SELECTOR
daemonset.apps/sriov-network-config-daemon 1          1          1        1              1            beta.kubernetes.io/os=linux,feature.node.kubernetes.io/network-sriov.capable=true 8s

NAME                                READY    UP-TO-DATE    AVAILABLE    AGE
deployment.apps/sriov-network-operator 1/1      1             1           10s

NAME                                DESIRED    CURRENT    READY    AGE
replicaset.apps/sriov-network-operator-54d7545f65 1          1          1       10s
```

SriovNetworkNodeState node1 Mellanox

```
# kubectl get sriovnetworknodestates.sriovnetwork.openshift.io -n kube-system node1 -o yaml
apiVersion: sriovnetwork.openshift.io/v1
kind: SriovNetworkNodeState
spec: ...
```

```
status:
  interfaces:
  - deviceID: "1017"
    driver: mlx5_core
    mtu: 1500
    pciAddress: "0000:5f:00.0"
    totalVfs: 8
    vendor: "15b3"
    linkSeed: 25000Mb/s
    linkType: ETH
    mac: 08:c0:eb:f4:85:bb
    name: ens41f0np0
  - deviceID: "1017"
    driver: mlx5_core
    mtu: 1500
    pciAddress: "0000:5f:00.1"
    totalVfs: 8
    vendor: "15b3"
    linkSeed: 25000Mb/s
    linkType: ETH
    mac: 08:c0:eb:f4:85:bb
    name: ens41f1np1
```

SriovNetworkNodePolicy

nicSelector

```
apiVersion: srionetwork.openshift.io/v1
kind: SrionetworkNodePolicy
metadata:
  name: policy
  namespace: kube-system
spec:
  nodeSelector:
    feature.node.kubernetes.io/network-srionetwork.capable: "true"
  eSwitchMode: switchdev
  numVfs: 3
  nicSelector:
    pfNames:
    - ens41f0np0
    - ens41f1np1
  resourceName: cx_srionetwork_switchdev
```

SrionetworkNodeState

status

```
# kubectl get srionetworknodestates.srionetwork.openshift.io -n kube-system node1 -o yaml

...
spec:
  interfaces:
  - eSwitchMode: switchdev
    name: ens41f0np0
    numVfs: 3
    pciAddress: 0000:5f:00.0
    vfGroups:
    - policyName: policy
      vfRange: 0-2
      resourceName: cx_srionetwork_switchdev
  - eSwitchMode: switchdev
    name: ens41f1np1
    numVfs: 3
    pciAddress: 0000:5f:00.1
    vfGroups:
    - policyName: policy
      vfRange: 0-2
      resourceName: cx_srionetwork_switchdev
status:
  interfaces:
  - Vfs:
    - deviceID: 1018
      driver: mlx5_core
      pciAddress: 0000:5f:00.2
      vendor: "15b3"
    - deviceID: 1018
      driver: mlx5_core
      pciAddress: 0000:5f:00.3
      vendor: "15b3"
    - deviceID: 1018
      driver: mlx5_core
      pciAddress: 0000:5f:00.4
      vendor: "15b3"
    deviceID: "1017"
    driver: mlx5_core
    linkSeed: 25000Mb/s
    linkType: ETH
    mac: 08:c0:eb:f4:85:ab
    mtu: 1500
    name: ens41f0np0
    numVfs: 3
    pciAddress: 0000:5f:00.0
    totalVfs: 3
    vendor: "15b3"
  - Vfs:
```

```
- deviceID: 1018
  driver: mlx5_core
  pciAddress: 0000:5f:00.5
  vendor: "15b3"
- deviceID: 1018
  driver: mlx5_core
  pciAddress: 0000:5f:00.6
  vendor: "15b3"
- deviceID: 1018
  driver: mlx5_core
  pciAddress: 0000:5f:00.7
  vendor: "15b3"
deviceID: "1017"
driver: mlx5_core
linkSeed: 25000Mb/s
linkType: ETH
mac: 08:c0:eb:f4:85:bb
mtu: 1500
name: ens41f1np1
numVfs: 3
pciAddress: 0000:5f:00.1
totalvfs: 3
vendor: "15b3"
```

VF

```
# lspci -nn | grep ConnectX
5f:00.0 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5] [15b3:1017]
5f:00.1 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5] [15b3:1017]
5f:00.2 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
5f:00.3 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
5f:00.4 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
5f:00.5 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
5f:00.6 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
5f:00.7 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
```

PF

```
# cat /sys/class/net/ens41f0np0/compat/devlink/mode
switchdev
```

7.16.3 Multus-CNI

SR-IOV Device Plugin

ID

Multus-CNI

Kube-OVN

Multus-CNI

Multus-CNI

```
kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/multus-cni/v4.0.2/deployments/multus-daemonset-thick.yml
```

multus

Thin

Thick

SR-IOV

Thick

NetworkAttachmentDefinition

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: sriov
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/resourceName: mellanox.com/cx5_sriov_switchdev
spec:
  config: '{
    "cniVersion": "0.3.1",
    "name": "kube-ovn",
    "plugins":[
      {
        "type": "kube-ovn",
        "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
        "provider": "sriov.default.ovn"
      },
      {
        "type": "portmap",
        "capabilities":{
          "portMappings":true
        }
      }
    ]
  }'
```

• provider:

NetworkAttachmentDefinition

{(name).(namespace).ovn

7.16.4 Overlay

Kube-OVN

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/dist/images/install.sh
```

IFACE			IP						
<pre>ENABLE_MIRROR=\${ENABLE_MIRROR:-false} HW_OFFLOAD=\${HW_OFFLOAD:-true} ENABLE_LB=\${ENABLE_LB:-false} IFACE="bond1"</pre>									
#	SR-IOV	Device Plugin	bond	IFACE	bond1	bond	IFACE	enp132s0f0np0	enp132s0f1np1

Kube-OVN

```
bash install.sh
```

VF Pod

yamlVFPod:

```
apiVersion: v1
kind: Pod
metadata:
  name: nginx-overlay
  annotations:
    v1.multus-cni.io/default-network: default/sriov
    sriov.default.ovn.kubernetes.io/logical_switch: ovn-default
spec:
  containers:
  - name: nginx-overlay
    image: docker.io/library/nginx:alpine
  resources:
    requests:
      mellanox.com/cx5_sriov_switchdev: '1'
    limits:
      mellanox.com/cx5_sriov_switchdev: '1'
```

- v1.multus-cni.io/default-network: NetworkAttachmentDefinition {namespace}/{name}
- sriov.default.ovn.kubernetes.io/logical_switch: Pod Pod

7.16.5 Underlay

Kube-OVN

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/dist/images/install.sh
```

IFACE		IP	
<pre>ENABLE_MIRROR=\${ENABLE_MIRROR:-false} HW_OFFLOAD=\${HW_OFFLOAD:-true} ENABLE_LB=\${ENABLE_LB:-false} IFACE=""</pre>			
#	Underlay	IFACE	PF
		IFACE	K8s
			PF

Kube-OVN

```
bash install.sh
```

VF Pod

yamlVFPod:

```

apiVersion: kubeovn.io/v1
kind: ProviderNetwork
metadata:
  name: underlay-offload
spec:
  defaultInterface: bond1

---
apiVersion: kubeovn.io/v1
kind: Vlan
metadata:
  name: vlan0
spec:
  id: 0
  provider: underlay-offload

---
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: vlan0
spec:
  protocol: IPv4
  provider: ovn
  cidrBlock: 10.10.204.0/24
  gateway: 10.10.204.254
  vlan: vlan0
  excludeIps:
    - 10.10.204.1..10.10.204.100

---
apiVersion: v1
kind: Pod
metadata:
  name: nginx-underlay
  annotations:
    k8s.v1.cni.cncf.io/networks: '[{
      "name": "sriov",
      "namespace": "default",
      "default-route": ["10.10.204.254"]
    }]'
    sriov.default.ovn.kubernetes.io/logical_switch: vlan0
spec:
  containers:
    - name: nginx-underlay
      image: docker.io/library/nginx:alpine
      resources:
        requests:
          mellanox.com/cx5_sriov_switchdev: '1'
        limits:
          mellanox.com/cx5_sriov_switchdev: '1'

```

• v1.multus-cni.io/default-network: NetworkAttachmentDefinition {namespace}/{name}

multus VF Pod VF Pod VF Pod multus

yaml VF Pod:

```

apiVersion: v1
kind: Pod
metadata:
  name: nginx-underlay-noVF
  annotations:
    ovn.kubernetes.io/logical_switch: vlan0
spec:
  containers:
    - name: nginx-underlay-noVF
      image: docker.io/library/nginx:alpine

```

VF Pod ovs-kernel e-switch

7.16.6

Pod ovs-ovn

```

# ovs-appctl dpctl/dump-flows -m type=offloaded
ufid:91cc45de-e7e9-4935-8f82-1890430b0f66, skb_priority(0/0),skb_mark(0/0),ct_state(0/0x23),ct_zone(0/0),ct_mark(0/0),ct_label(0/0x1),recirc_id(0),dp_hash(0/0),in_port(5b45c61b307e_h),packet_type(ns=0/0,id=0/0),eth(src=00:00:00:c5:6d:4e,dst=00:00:00:e7:16:ce),eth_type(0x0800),ipv4(src=0.0.0.0/0.0.0.0,dst=0.0.0.0/0.0.0.0,proto=0/0,tos=0/0,ttl=0/0,frag=no), packets:941539, bytes:62142230, used:0.260s, offloaded:yes, dp:tc, actions:54235e5753b8_h
ufid:e00768d7-e652-4d79-8182-3291d852b791, skb_priority(0/0),skb_mark(0/0),ct_state(0/0x23),ct_zone(0/0),ct_mark(0/0),ct_label(0/0x1),recirc_id(0),dp_hash(0/0),in_port(54235e5753b8_h),packet_type(ns=0/0,id=0/0),eth(src=00:00:00:e7:16:ce,dst=00:00:00:c5:6d:4e),eth_type(0x0800),ipv4(src=0.0.0.0/0.0.0.0,dst=0.0.0.0/0.0.0.0,proto=0/0,tos=0/0,ttl=0/0,frag=no), packets:82386659, bytes:115944854173, used:0.260s, offloaded:yes, dp:tc, actions:5b45c61b307e_h

```

offloaded:yes, dp:tc

 PDF

 Slack

 Support

 2025 9 10

 2022 5 24



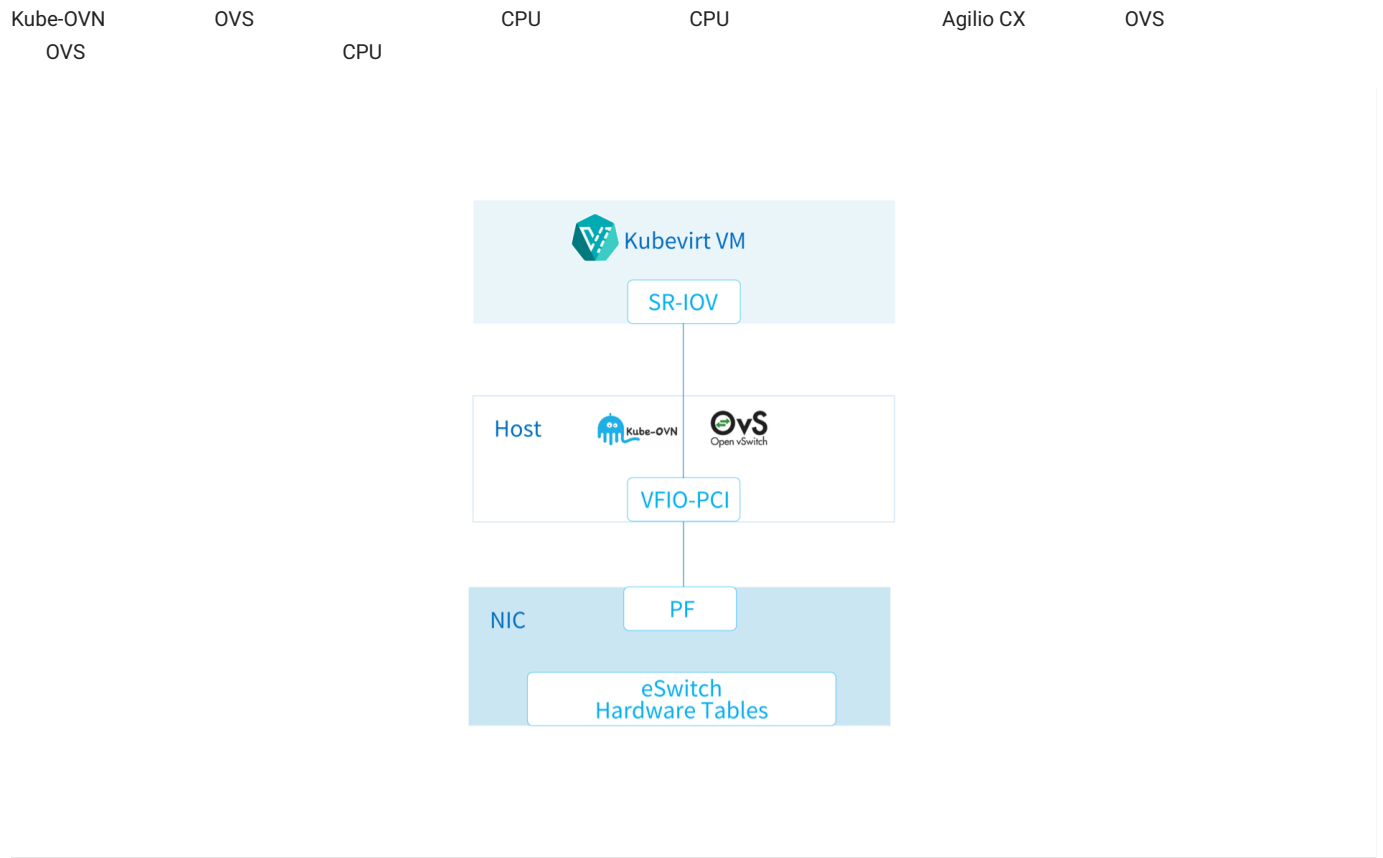




GitHub

7.16.7

7.17 Offload



Note

2022

7.17.1

- Agilio CX
- CentOS 8 Stream Linux 5.7
- dp_hash hash OVN LB

7.17.2 SR-IOV

[Agilio Open vSwitch TC User Guide](#)

```
#!/bin/bash
DEVICE=${1}
DEFAULT_ASSY=scan
ASSY=${2:-${DEFAULT_ASSY}}
APP=${3:-flower}

if [ "x${DEVICE}" = "x" -o ! -e /sys/class/net/${DEVICE} ]; then
    echo Syntax: ${0} device [ASSY] [APP]
    echo
    echo This script associates the TC Offload firmware
    echo with a Netronome SmartNIC.
    echo
    echo device: is the network device associated with the SmartNIC
```

```
echo ASSY: defaults to ${DEFAULT_ASSY}
echo APP: defaults to flower. flower-next is supported if updated
echo     firmware has been installed.
exit 1
fi

# It is recommended that the assembly be determined by inspection
# The following code determines the value via the debug interface
if [ "${ASSY}x" = "scanx" ]; then
    ethtool -W ${DEVICE} 0
    DEBUG=$(ethtool -w ${DEVICE} data /dev/stdout | strings)
    SERIAL=$(echo "${DEBUG}" | grep "^SN:")
    ASSY=$(echo ${SERIAL} | grep -oE AMDA[0-9]{4})
fi

PCIADDR=$(basename $(readlink -e /sys/class/net/${DEVICE}/device))
FWDIR="/lib/firmware/netronome"

# AMDA0081 and AMDA0097 uses the same firmware
if [ "${ASSY}" = "AMDA0081" ]; then
    if [ ! -e ${FWDIR}/${APP}/nic_AMDA0081.nffw ]; then
        ln -sf nic_AMDA0097.nffw ${FWDIR}/${APP}/nic_AMDA0081.nffw
    fi
fi

FW="${FWDIR}/pci-${PCIADDR}.nffw"
ln -sf "${APP}/nic-${ASSY}.nffw" "${FW}"

# insert distro-specific initramfs section here...
```

```
./agilio-tc-fw-select.sh ens47np0 scan
rmmod nfp
modprobe nfp
```

VF VF

```
# cat /sys/class/net/ens3/device/sriov_totalvfs
65

# echo 4 > /sys/class/net/ens47/device/sriov_numvfs
```

7.17.3 SR-IOV Device Plugin

VF

Pod

VF

SR-IOV Device Plugin

SR-IOV Configmap

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: sriovdp-config
  namespace: kube-system
data:
  config.json: |
    {
      "resourceList": [{
        "resourcePrefix": "corigine.com",
        "resourceName": "agilio_sriov",
        "selectors": {
          "vendors": ["19ee"],
          "devices": ["6003"],
          "drivers": ["nfp_netvf"]
        }
      }]
    }
  
```

SR-IOV Network Device Plugin intel/sriov-network-device-plugin k8snetworkplumbingwg

kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/sriov-network-device-plugin/master/deployments/sriovdp-daemonset.yaml

SR-IOV Kubernetes Node

```
kubectl describe no containerserver | grep corigine

corigine.com/agilio_sriov: 4
corigine.com/agilio_sriov: 4
corigine.com/agilio_sriov 0          0
```

7.17.4 Multus-CNI

SR-IOV Device Plugin	ID	Multus-CNI	Kube-OVN	Multus-CNI
Multus-CNI				
<pre>kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/multus-cni/master/deployments/multus-daemonset.yml</pre>				

NetworkAttachmentDefinition

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: default
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/resourceName: corigine.com/agilio_sriov
spec:
  config: '{
    "cniVersion": "0.3.1",
    "name": "kube-ovn",
    "plugins":[
      {
        "type":"kube-ovn",
        "server_socket":"/run/openvswitch/kube-ovn-daemon.sock",
        "provider": "default.default.ovn"
      },
      {
        "type":"portmap",
        "capabilities":{"
          "portMappings":true
        }
      }
    ]
  }'
```

• provider: NetworkAttachmentDefinition {name}.{namespace}.ovn

7.17.5 Kube-OVN

<pre>wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/dist/images/install.sh</pre>	
--	--

IFACE IP

```
ENABLE_MIRROR=${ENABLE_MIRROR:-false}
HW_OFFLOAD=${HW_OFFLOAD:-true}
ENABLE_LB=${ENABLE_LB:-false}
IFACE="ensp01"
```

Kube-OVN

```
bash install.sh
```

7.17.6 VF Pod

yaml	VF	Pod:
------	----	------

```
apiVersion: v1
kind: Pod
metadata:
  name: nginx
  namespace: default
  annotations:
    v1.multus-cni.io/default-network: default/default
spec:
  containers:
    - name: nginx
      image: docker.io/library/nginx:alpine
      resources:
        requests:
          corigine.com/agilio_sriov: '1'
```

```
limits:
  corigine.com/agilio_sriov: '1'
```

• v1.multus-cni.io/default-network: NetworkAttachmentDefinition {namespace}/{name}

Pod ovs-ovn

```
# ovs-appctl dpctl/dump-flows -m type=offloaded
ufid:91cc45de-e7e9-4935-8f82-1890430b0f66, skb_priority(0/0),skb_mark(0/0),ct_state(0/0x23),ct_zone(0/0),ct_mark(0/0),ct_label(0/0x1),recirc_id(0),dp_hash(0/0),in_port(5b45c61b307e_h),packet_type(ns=0/0,id=0/0),eth(src=00:00:00:c5:6d:4e,dst=00:00:00:e7:16:ce),eth_type(0x0800),ipv4(src=0.0.0.0/0.0.0.0,dst=0.0.0.0/0.0.0.0,proto=0/0,tos=0/0,ttl=0/0,frag=no), packets:941539, bytes:62142230, used:0.260s, offloaded:yes, dp:tc, actions:54235e5753b8_h
ufid:e00768d7-e652-4d79-8182-3291d852b791, skb_priority(0/0),skb_mark(0/0),ct_state(0/0x23),ct_zone(0/0),ct_mark(0/0),ct_label(0/0x1),recirc_id(0),dp_hash(0/0),in_port(54235e5753b8_h),packet_type(ns=0/0,id=0/0),eth(src=00:00:00:e7:16:ce,dst=00:00:00:c5:6d:4e),eth_type(0x0800),ipv4(src=0.0.0.0/0.0.0.0,dst=0.0.0.0/0.0.0.0,proto=0/0,tos=0/0,ttl=0/0,frag=no), packets:82386659, bytes:115944854173, used:0.260s, offloaded:yes, dp:tc, actions:5b45c61b307e_h
```

offloaded:yes, dp:tc

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 GitHub



7.18 Offload

Kube-OVN	OVS		CPU	CPU		metaScale	OVS
OVS		CPU					

 **Note**

1.

2024

2.

1.11

Kube-OVN

7.18.1

- metaScale
- MCR
- BIOS SR-IOV VT-d

7.18.2

hw-offload Kube-OVN

1.

```
wget https://raw.githubusercontent.com/yunsilicon/kube-ovn/release-1.11/dist/images/install.sh
```

1.

/opt/ovs-config/ovs-dpdk-config

```
# specify log level for ovs dpdk, the value is info or dbg, default is info
VLOG=info
# specify nic offload, the value is true or false, default is true
HW_OFFLOAD=true
# specify cpu mask for ovs dpdk, not specified by default
CPU_MASK=0x02
# specify socket memory, not specified by default
SOCKET_MEM="2048,2048"
# specify encap IP
ENCAP_IP=6.6.6.200/24
# specify pci device
DPDK_DEV=0000:b3:00.0
# specify mtu, default is 1500
PF_MTU=1500
# specify bond name if bond enabled, not specified by default
BR_PHY_BOND_NAME=bond0
```

1. Kube-OVN

```
bash install.sh
```

SR-IOV

1. metaScale ID b3:00.0:

```
[root@k8s-master ~]# lspci -d 1f67:
b3:00.0 Ethernet controller: Device 1f67:1111 (rev 02)
b3:00.1 Ethernet controller: Device 1f67:1111 (rev 02)
```

1. ID p3p1

```
ls -l /sys/class/net/ | grep b3:00:0
lrwxrwxrwx 1 root root 0 May  7 16:30 p3p1 -> ../../devices/pci0000:b2/0000:b2:00.0/0000:b3:00.0/net/p3p1
```

1. VF

```
cat /sys/class/net/p3p1/device/sriov_totalvfs
512
```

1. VF

```
echo '10' > /sys/class/net/p3p1/device/sriov_numvfs
```

1. VF

```
lspci -d 1f67:
b3:00.0 Ethernet controller: Device 1f67:1111 (rev 02)
b3:00.1 Ethernet controller: Device 1f67:1111 (rev 02)
b3:00.2 Ethernet controller: Device 1f67:1112
b3:00.3 Ethernet controller: Device 1f67:1112
b3:00.4 Ethernet controller: Device 1f67:1112
b3:00.5 Ethernet controller: Device 1f67:1112
b3:00.6 Ethernet controller: Device 1f67:1112
b3:00.7 Ethernet controller: Device 1f67:1112
b3:01.0 Ethernet controller: Device 1f67:1112
b3:01.1 Ethernet controller: Device 1f67:1112
b3:01.2 Ethernet controller: Device 1f67:1112
b3:01.3 Ethernet controller: Device 1f67:1112
```

1. switchdev

```
devlink dev eswitch set pci/0000:b3:00.0 mode switchdev
```

SR-IOV Device Plugin

1. SR-IOV ConfigMap

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: sriovdp-config
  namespace: kube-system
data:
  config.json: |
    {
      "resourceList": [{
        "resourceName": "xsc_sriov",
        "resourcePrefix": "yunsilicon.com",
        "selectors": {
          "vendors": ["1f67"],
          "devices": ["1012", "1112"]
        }
      }]
    }
```

1. SR-IOV Device Plugin DevicePlugin

2. SR-IOV

```
# kubectl describe node <node name> | grep yunsilicon.com/xsc_sriov
yunsilicon.com/xsc_sriov: 10
yunsilicon.com/xsc_sriov: 10
yunsilicon.com/xsc_sriov 0 0
```

Multus-CNI

1. Multus-CNI Multus-CNI

```
kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/multus-cni/master/deployments/multus-daemonset.yml
```

1. NetworkAttachmentDefinition

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
```

```

metadata:
  name: sriov-net1
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/resourceName: yunsilicon.com/xsc_sriov
spec:
  config: '{
    "cniVersion": "0.3.1",
    "name": "kube-ovn",
    "plugins": [
      {
        "type": "kube-ovn",
        "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
        "provider": "sriov-net1.default.ovn"
      },
      {
        "type": "portmap",
        "capabilities": {
          "portMappings": true
        }
      }
    ]
  }'

```

SR-IOV Pod

```

apiVersion: v1
kind: Pod
metadata:
  name: nginx
  annotations:
    v1.multus-cni.io/default-network: default/sriov-net1
spec:
  containers:
    - name: nginx
      image: docker.io/library/nginx:alpine
      resources:
        requests:
          yunsilicon.com/xsc_sriov: '1'
        limits:
          yunsilicon.com/xsc_sriov: '1'

```

Offload

Pod `ovs-ovn`

```

ovs-appctl dpctl/dump-flows type=offloaded
flow-dump from pmd on cpu core: 9
ct_state(-new+est-rel+rpl+trk),ct_mark(0/0x3),recirc_id(0x2d277),in_port(15),packet_type(ns=0,id=0),eth(src=00:00:00:9d:fb:1a,dst=00:00:00:ce:cf:b9),eth_type(0x0800),ipv4(dst=10.16.0.14,frag=no), packets:6, bytes:588, used:7.276s, actions:ct(zone=4,nat),recirc(0x2d278)
ct_state(-new+est-rel+rpl+trk),ct_mark(0/0x3),recirc_id(0x2d275),in_port(8),packet_type(ns=0,id=0),eth(src=00:00:00:ce:cf:b9,dst=00:00:00:9d:fb:1a),eth_type(0x0800),ipv4(dst=10.16.0.18,frag=no), packets:5, bytes:490, used:7.434s, actions:ct(zone=6,nat),recirc(0x2d276)
ct_state(-new+est-rel+rpl+trk),ct_mark(0/0x1),recirc_id(0x2d276),in_port(8),packet_type(ns=0,id=0),eth(src=00:00:00:ce:cf:b9,dst=00:00:00:9d:fb:1a/01:00:00:00:00:00),eth_type(0x0800),ipv4(frag=no), packets:5, bytes:490, used:7.434s, actions:15
recirc_id(0),in_port(15),packet_type(ns=0,id=0),eth(src=00:00:00:9d:fb:1a/01:00:00:00:00:00,dst=00:00:00:ce:cf:b9),eth_type(0x0800),ipv4(dst=10.16.0.14/255.192.0.0,frag=no), packets:6, bytes:588, used:7.277s, actions:ct(zone=6,nat),recirc(0x2d277)
recirc_id(0),in_port(8),packet_type(ns=0,id=0),eth(src=00:00:00:ce:cf:b9/01:00:00:00:00:00,dst=00:00:00:9d:fb:1a),eth_type(0x0800),ipv4(dst=10.16.0.18/255.192.0.0,frag=no), packets:6, bytes:588, used:7.434s, actions:ct(zone=4,nat),recirc(0x2d275)
ct_state(-new+est-rel+rpl+trk),ct_mark(0/0x1),recirc_id(0x2d278),in_port(15),packet_type(ns=0,id=0),eth(dst=00:00:00:ce:cf:b9/01:00:00:00:00:00),eth_type(0x0800),ipv4(frag=no), packets:6, bytes:588, used:7.277s, actions:8

```

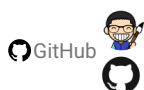

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7.18.3

7.19 Offload

 **Note**

2024

7.19.1

- 2200E
- HADOS
- BIOS SR-IOV

7.19.2

SR-IOV

1. 2200E vendor ID 1f47 ID 00:0a.0 00:0b.0 2200E

```
lspci | grep 1f47
00:0a.0 Ethernet controller: Device 1f47:1001 (rev 10)
00:0b.0 Ethernet controller: Device 1f47:1001 (rev 10)
```

1. VF

```
cat /sys/bus/pci/devices/0000\:00\:0a.0/sriov_totalvfs
256
```

1. VF VF

```
echo 7 > /sys/bus/pci/devices/0000\:00\:0a.0/sriov_numvfs
```

1. VF

```
lspci | grep 1f47
00:0a.0 Ethernet controller: Device 1f47:1001 (rev 10)
00:0a.1 Ethernet controller: Device 1f47:110f (rev 10)
00:0a.2 Ethernet controller: Device 1f47:110f (rev 10)
00:0a.3 Ethernet controller: Device 1f47:110f (rev 10)
00:0a.4 Ethernet controller: Device 1f47:110f (rev 10)
00:0a.5 Ethernet controller: Device 1f47:110f (rev 10)
00:0a.6 Ethernet controller: Device 1f47:110f (rev 10)
00:0a.7 Ethernet controller: Device 1f47:110f (rev 10)
00:0b.0 Ethernet controller: Device 1f47:1001 (rev 10)
```

SR-IOV Device Plugin

1. SR-IOV Configmap SR-IOV Device Plugin VF Pod

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: sriovdp-config
  namespace: kube-system
data:
  config.json: |
    {
      "resourceList": [{
        "resourceName": "sriov_dpu",
        "resourcePrefix": "yusur.tech",
        "selectors": {
          "vendors": ["1f47"],
          "devices": ["110f"]
        }
      }]
    }
```



```
]
}
```

1. SR-IOV Device Plugin

```
kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/sriov-network-device-plugin/v3.6.2/deployments/sriovdp-daemonset.yaml
```

1. SR-IOV kubernetes Node

```
kubectl describe node node1 | grep yusur
yusur.tech/sriov_dpu: 7
yusur.tech/sriov_dpu: 7
yusur.tech/sriov_dpu: 0 0
```

7.19.3 Multus-CNI

Multus-CNI Kube-OVN SRIOV Device ID

```
kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/multus-cni/v4.0.2/deployments/multus-daemonset-thick.yml
```

NetworkAttachmentDefinition

```
apiVersion:
  "k8s.cni.cncf.io/v1"
kind:
  NetworkAttachmentDefinition
metadata:
  name: test
  namespace: kube-system
  annotations:
    k8s.v1.cni.cncf.io/resourceName: yusur.tech/sriov_dpu
spec:
  config: '{
    "cniVersion": "0.3.1",
    "name": "kube-ovn",
    "plugins": [
      {
        "type": "kube-ovn",
        "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
        "provider": "test.kube-system.ovn"
      },
      {
        "type": "portmap",
        "capabilities": {
          "portMappings": true
        }
      }
    ]
  }'
```

• provider: NetworkAttachmentDefinition {name}.{namespace}.ovn

7.19.4 Kube-OVN

1.

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.12/dist/images/install.sh
```

1. IFACE IP

```
ENABLE_MIRROR=${ENABLE_MIRROR:-false}
HW_OFFLOAD=${HW_OFFLOAD:-true}
ENABLE_LB=${ENABLE_LB:-false}
IFACE="p0"
```

1. kube-ovn

```
bash install.sh
```

VF	pod
yaml	VF Pod

```
apiVersion: v1
kind: Pod
metadata:
  name: nginx
  namespace: default
  annotations:
    v1.multus-cni.io/default-network: kube-system/test
spec:
  containers:
    - name: nginx
      image: docker.io/library/nginx:alpine
      resources:
        requests:
          yusur.tech/sriov_dpu: '1'
        limits:
          yusur.tech/sriov_dpu: '1'
```

• v1.multus-cni.io/default-network: NetworkAttachmentDefinition {namespace}/{name}

Offload

Pod ovs-ovn

```
# ovs-appctl dpctl/dump-flows -m type=offloaded
ufid:67c2e10f-92d4-4574-be70-d072815ff166, skb_priority(0/0),skb_mark(0/0),ct_state(0/0x23),ct_zone(0/0),ct_mark(0/0),ct_label(0/0),recirc_id(0),dp_hash(0/0),in_port(d85b161b6840_h),packet_type(ns=0/0,id=0/0),eth(src=0a:c9:1c:70:01:09,dst=8a:18:a4:22:b7:7d),eth_type(0x0800),ipv4(src=10.0.1.10,dst=10.0.1.6,proto=6,tos=0/0x3,ttl=0/0,frag=no),tcp(src=60774,dst=9001), packets:75021, bytes:109521630, offload_packets:75019, offload_bytes:109521498, used: 3.990s, offloaded:yes, dp:tc, actions:set(tunnel(tun_id=0x5,dst=192.168.201.12,ttl=64,tp_dst=6081,geneve({class=0x102,type=0x80,len=4,0xa0006})), flags(csum|key))), genev_sys_6081
ufid:7940666e-a0bd-42a5-8116-1e84e81bb338, skb_priority(0/0),tunnel(tun_id=0x5,src=192.168.201.12,dst=192.168.201.11,ttl=0/0,tp_dst=6081,geneve({class=0x102,type=0x80,len=4,0x6000a})), flags(+key)),skb_mark(0/0),ct_state(0/0),ct_zone(0/0),ct_mark(0/0),ct_label(0/0),recirc_id(0),dp_hash(0/0),in_port(genev_sys_6081),packet_type(ns=0/0,id=0/0),eth(src=8a:18:a4:22:b7:7d,dst=0a:c9:1c:70:01:09),eth_type(0x0800),ipv4(src=10.0.1.6,dst=10.0.1.10,proto=6,tos=0/0,ttl=0/0,frag=no),tcp(src=9001,dst=60774), packets:6946, bytes:459664, offload_packets:6944, offload_bytes:459532, used: 4.170s, dp:tc, offloaded:yes, actions:d85b161b6840_h
```

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7.19.5

7.20 DPDK

Kube-OVN OVS-DPDK KubeVirt DPDK

KubeVirt OVS-DPDK patchVhostuser implementation KubeVirt KVM Device Plugin OVS-DPDK

7.20.1

- DPDK
- Hugepages

7.20.2 DPDK

driverctl DPDK

```
driverctl set-override 0000:00:0b.0 uio_pci_generic
```

7.20.3

OVS-DPDK Kube-OVN

```
kubectl label nodes <node> ovn.kubernetes.io/ovs_dp_type="userspace"
```

OVS-DPDK /opt/ovs-config ovs-dpdk-config

```
ENCAP_IP=192.168.122.193/24
DPDK_DEV=0000:00:0b.0
```

- ENCAP_IP:
- DPDK_DEV: PCI ID

7.20.4 Kube-OVN

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/dist/images/install.sh
```

DPDK

```
bash install.sh --with-hybrid-dpdk
```

7.20.5

vhostuser OVS-DPDK

KVM Device Plugin KVM Device Plugin

```
kubectl apply -f https://raw.githubusercontent.com/kubevirt/kubernetes-device-plugins/master/manifests/kvm-ds.yml
```

NetworkAttachmentDefinition

```
apiVersion: k8s.cni.cncf.io/v1
kind: NetworkAttachmentDefinition
metadata:
  name: ovn-dpdk
  namespace: default
spec:
  config: >-
    {
      "cniVersion": "0.3.0",
      "type": "kube-ovn",
```

```

    "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
    "provider": "ovn-dpdk.default.ovn",
    "vhost_user_socket_volume_name": "vhostuser-sockets",
    "vhost_user_socket_name": "sock"
}

```

Dockerfile VM

```

FROM quay.io/kubevirt/virt-launcher:v0.46.1

# wget http://cloud.centos.org/centos/7/images/CentOS-7-x86_64-GenericCloud.qcow2
COPY CentOS-7-x86_64-GenericCloud.qcow2 /var/lib/libvirt/images/CentOS-7-x86_64-GenericCloud.qcow2

```

```

apiVersion: v1
kind: ConfigMap
metadata:
  name: vm-config
data:
  start.sh: |
    chmod u+w /etc/libvirt/qemu.conf
    echo "hugetlbfs_mount = \"/dev/hugepages\" >> /etc/libvirt/qemu.conf
    virtlogd &
    libvirtd &

    mkdir /var/lock

    sleep 5

    virsh define /root/vm/vm.xml
    virsh start vm

  tail -f /dev/null
  vm.xml: |
    <domain type='kvm'>
      <name>vm</name>
      <uuid>4a9b3f53-fa2a-47f3-a757-dd87720d9d1d</uuid>
      <memory unit='KiB'>2097152</memory>
      <currentMemory unit='KiB'>2097152</currentMemory>
      <memoryBacking>
        <hugepages>
          <page size='2' unit='M' nodeset='0' />
        </hugepages>
      </memoryBacking>
      <vcpu placement='static'>2</vcpu>
      <cputune>
        <shares>4096</shares>
        <vcpupin vcpu='0' cpuset='4' />
        <vcpupin vcpu='1' cpuset='5' />
        <emulatorpin cpuset='1,3' />
      </cputune>
      <os>
        <type arch='x86_64' machine='pc'>hvm</type>
        <boot dev='hd' />
      </os>
      <features>
        <acpi />
        <apic />
      </features>
      <cpu mode='host-model'>
        <model fallback='allow' />
        <topology sockets='1' cores='2' threads='1' />
        <numa>
          <cell id='0' cpus='0-1' memory='2097152' unit='KiB' memAccess='shared' />
        </numa>
      </cpu>
      <on_reboot>restart</on_reboot>
      <devices>
        <emulator>/usr/libexec/qemu-kvm</emulator>
        <disk type='file' device='disk'>
          <driver name='qemu' type='qcow2' cache='none' />
          <source file='/var/lib/libvirt/images/CentOS-7-x86_64-GenericCloud.qcow2' />
          <target dev='vda' bus='virtio' />
        </disk>

        <interface type='vhostuser'>
          <mac address='00:00:00:0A:30:89' />
          <source type='unix' path='/var/run/vm/sock' mode='server' />
          <model type='virtio' />
          <driver queues='2'>
            <host mrg_rxbuf='off' />
          </driver>
        </interface>
        <serial type='pty'>
          <target type='isa-serial' port='0'>
            <model name='isa-serial' />
          </target>
        </serial>
        <console type='pty'>
          <target type='serial' port='0' />
        </console>
      </devices>
    </domain>

```

```

    </console>
    <channel type='unix'>
      <source mode='bind' path='/var/lib/libvirt/qemu/channel/target/domain-1-vm/org.qemu.guest_agent.0' />
      <target type='virtio' name='org.qemu.guest_agent.0' state='connected' />
      <alias name='channel0' />
      <address type='virtio-serial' controller='0' bus='0' port='1' />
    </channel>

  </devices>
</domain>
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: vm-deployment
  labels:
    app: vm
spec:
  replicas: 1
  selector:
    matchLabels:
      app: vm
  template:
    metadata:
      labels:
        app: vm
    annotations:
      k8s.v1.cnf.cncf.io/networks: default/ovn-dpdk
      ovn-dpdk.default.ovn.kubernetes.io/ip_address: 10.16.0.96
      ovn-dpdk.default.ovn.kubernetes.io/mac_address: 00:00:00:0A:30:89
  spec:
    nodeSelector:
      ovn.kubernetes.io/ovs_dp_type: userspace
    securityContext:
      runAsUser: 0
    volumes:
      - name: vhostuser-sockets
        emptyDir: {}
      - name: xml
        configMap:
          name: vm-config
      - name: hugepage
        emptyDir:
          medium: HugePages-2Mi
      - name: libvirt-runtime
        emptyDir: {}
    containers:
      - name: vm
        image: vm-vhostuser:latest
        command: ["bash", "/root/vm/start.sh"]
        securityContext:
          capabilities:
            add:
              - NET_BIND_SERVICE
              - SYS_NICE
              - NET_RAW
              - NET_ADMIN
          privileged: false
          runAsUser: 0
        resources:
          limits:
            cpu: '2'
            devices.kubevirt.io/kvm: '1'
            memory: '8784969729'
            hugepages-2Mi: 2Gi
          requests:
            cpu: 666m
            devices.kubevirt.io/kvm: '1'
            ephemeral-storage: 50M
            memory: '4490002433'
        volumeMounts:
          - name: vhostuser-sockets
            mountPath: /var/run/vm
          - name: xml
            mountPath: /root/vm/
          - mountPath: /dev/hugepages
            name: hugepage
          - name: libvirt-runtime
            mountPath: /var/run/libvirt

```

Pod

```

# virsh set-user-password vm root 12345
Password set successfully for root in vm

# virsh console vm
Connected to domain 'vm'
Escape character is ^] (Ctrl + ])

CentOS Linux 7 (Core)
Kernel 3.10.0-1127.el7.x86_64 on an x86_64

```

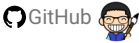
```
localhost login: root
Password:
Last login: Fri Feb 25 09:52:54 on ttyS0
```

```
ip link set eth0 mtu 1400
ip addr add 10.16.0.96/16 dev eth0
ip ro add default via 10.16.0.1
ping 114.114.114.114
```



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7.20.6

7.21 OpenStack

	OpenStack	Kubernetes		OpenStack Neutron	OVN	Kube-OVN
OVN	OpenStack	Kubernetes				

7.21.1

OVN-IC	Kubernetes	OpenStack	Kubernetes
--------	------------	-----------	------------

- OpenStack Kubernetes CIDR
-
- IP
- Kubernetes OpenStack VPC

OVN-IC

OVN-IC

```
docker run --name=ovn-ic-db -d --network=host -v /etc/ovn:/etc/ovn -v /var/run/ovn:/var/run/ovn -v /var/log/ovn:/var/log/ovn kubeovn/kube-ovn:v1.16.5 bash start-ic-db.sh
```

Kubernetes

kube-system Namespace ovn-ic-config ConfigMap

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: ovn-ic-config
  namespace: kube-system
data:
  enable-ic: "true"
  az-name: "az1"
  ic-db-host: "192.168.65.3"
  ic-nb-port: "6645"
  ic-sb-port: "6646"
  gw-nodes: "az1-gw"
  auto-route: "true"
```

- enable-ic:
- az-name:
- ic-db-host: OVN-IC
- ic-nb-port: OVN-IC 6645
- ic-sb-port: OVN-IC 6646
- gw-nodes:
- auto-route:

OpenStack

Kubernetes

```
# openstack router create router0
# openstack router list
+-----+-----+-----+-----+-----+
| ID | Name | Status | State | Project |
+-----+-----+-----+-----+-----+
| d5b38655-249a-4192-8046-71aa4d2b4af1 | router0 | ACTIVE | UP | 98a29ab7388347e7b5ff8bdd181ba4f9 |
+-----+-----+-----+-----+-----+
```

OpenStack OVN

```
ovn-nbctl set NB_Global . name=op-az
```

OVN-IC

OVN-IC

```
/usr/share/ovn/scripts/ovn-ctl --ovn-ic-nb-db=tcp:192.168.65.3:6645 \
--ovn-ic-sb-db=tcp:192.168.65.3:6646 \
--ovn-northd-nb-db=unix:/run/ovn/ovnnb_db.sock \
--ovn-northd-sb-db=unix:/run/ovn/ovnsb_db.sock \
start_ic
```

- `ovn-ic-nb-db` `ovn-ic-sb-db` : OVN-IC
- `ovn-northd-nb-db` `ovn-northd-sb-db` : OVN

```
ovs-vsctl set open_vswitch . external_ids:ovn-is-interconn=true
```

OpenStack OVN

ts router0

```
ovn-nbctl lrp-add router0 lrp-router0-ts 00:02:ef:11:39:4f 169.254.100.73/24
ovn-nbctl lsp-add ts lsp-ts-router0 -- lsp-set-addresses lsp-ts-router0 router \
-- lsp-set-type lsp-ts-router0 router \
-- lsp-set-options lsp-ts-router0 router-port=lrp-router0-ts
ovn-nbctl lrp-set-gateway-chassis lrp-router0-ts {gateway chassis} 1000
ovn-nbctl set NB_Global . options:ic-route-adv=true options:ic-route-learn=true
```

Kubernetes

```
# ovn-nbctl lr-route-list router0
IPv4 Routes
      10.0.0.22          169.254.100.34 dst-ip (learned)
      10.16.0.0/16       169.254.100.34 dst-ip (learned)
```

router0

Kubernetes Pod

7.21.2 OVN

OpenStack Kubernetes OVN VPC Subnet

Kube-OVN OVN OpenStack Neutron OVN OpenStack networking-ovn Neutron

Neutron

Neutron /etc/neutron/plugins/ml2/ml2_conf.ini

```
[ovn]
...
ovn_nb_connection = tcp:[192.168.137.176]:6641,tcp:[192.168.137.177]:6641,tcp:[192.168.137.178]:6641
ovn_sb_connection = tcp:[192.168.137.176]:6642,tcp:[192.168.137.177]:6642,tcp:[192.168.137.178]:6642
ovn_l3_scheduler = OVN_L3_SCHEDULER
```

- `ovn_nb_connection` `ovn_sb_connection` : Kube-OVN `ovn-central`

OVS

```
ovs-vsctl set open . external-ids:ovn-remote=tcp:[192.168.137.176]:6642,tcp:[192.168.137.177]:6642,tcp:[192.168.137.178]:6642
ovs-vsctl set open . external-ids:ovn-encap-type=geneve
ovs-vsctl set open . external-ids:ovn-encap-ip=192.168.137.200
```

- `external-ids:ovn-remote` : Kube-OVN `ovn-central`
- `ovn-encap-ip` : IP

Kubernetes OpenStack

Kubernetes OpenStack OpenStack Pod



Note

```
kube-ovn-controller args --enable-external-vpc=true
```

OpenStack

```
# openstack router list
+-----+-----+-----+-----+-----+
| ID | Name | Status | State | Project |
+-----+-----+-----+-----+-----+
| 22040ed5-0598-4f77-bffd-e7fd4db47e93 | router0 | ACTIVE | UP | 62381a21d569404aa236a5dd8712449c |
+-----+-----+-----+-----+-----+

# openstack network list
+-----+-----+-----+
| ID | Name | Subnets |
+-----+-----+-----+
| cd59e36a-37db-4c27-b709-d35379a7920f | provider | 01d73d9f-fdaa-426c-9b60-aa34abbfacae |
+-----+-----+-----+

# openstack subnet list
+-----+-----+-----+-----+
| ID | Name | Network | Subnet |
+-----+-----+-----+-----+
| 01d73d9f-fdaa-426c-9b60-aa34abbfacae | provider-v4 | cd59e36a-37db-4c27-b709-d35379a7920f | 192.168.1.0/24 |
+-----+-----+-----+-----+

# openstack server list
+-----+-----+-----+-----+-----+-----+
| ID | Name | Status | Networks | Image | Flavor |
+-----+-----+-----+-----+-----+-----+
| 8433d622-a8d6-41a7-8b31-49abfd64f639 | provider-instance | ACTIVE | provider=192.168.1.61 | ubuntu | m1 |
+-----+-----+-----+-----+-----+-----+
```

Kubernetes VPC

```
# kubectl get vpc
NAME                                STANDBY SUBNETS
neutron-22040ed5-0598-4f77-bffd-e7fd4db47e93 true ["neutron-cd59e36a-37db-4c27-b709-d35379a7920f"]
ovn-cluster                         true ["join", "ovn-default"]
```

neutron-22040ed5-0598-4f77-bffd-e7fd4db47e93 OpenStack VPC

Kube-OVN VPC Subnet Pod

VPC, Subnet Namespace net2 Pod:

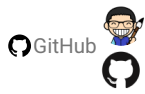
```
apiVersion: v1
kind: Namespace
metadata:
  name: net2
---
apiVersion: kubeovn.io/v1
kind: Vpc
metadata:
  creationTimestamp: "2021-06-20T13:34:11Z"
  generation: 2
  labels:
    ovn.kubernetes.io/vpc_external: "true"
  name: neutron-22040ed5-0598-4f77-bffd-e7fd4db47e93
  resourceVersion: "583728"
  uid: 18d4c654-f511-4def-a3a0-a6434d237c1e
spec:
  namespaces:
    - net2
---
kind: Subnet
apiVersion: kubeovn.io/v1
metadata:
  name: net2
spec:
  vpc: neutron-22040ed5-0598-4f77-bffd-e7fd4db47e93
  namespaces:
    - net2
  cidrBlock: 12.0.1.0/24
  natOutgoing: false
---
apiVersion: v1
kind: Pod
metadata:
  name: ubuntu
```

```
namespace: net2
spec:
  containers:
  - image: docker.io/kubeovn/kube-ovn:v1.8.0
    command:
    - "sleep"
    - "604800"
    imagePullPolicy: IfNotPresent
    name: ubuntu
  restartPolicy: Always
```



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7.21.3

7.22 IPsec

Kube-OVN v1.13.0 OVN/OVS IPsec Geneve/Vxlan/STT

7.22.1

- UDP 500 IKE UDP 4500 NAT-T
- xfrm / af_key strongSwan/libreswan

7.22.2

kube-ovn-cni CertificateSigningRequest kube-ovn-controller approve kube-ovn-cni IPsec ipsec

7.22.3 IPsec

kube-ovn-controller Deployment kube-ovn-cni DaemonSet --enable-ovn-ipsec=false --enable-ovn-ipsec=true

ENABLE_OVN_IPSEC=true

7.22.4 cert-manager

cert-manager IPsec --cert-manager-ipsec-cert=true kube-ovn-cni cert-manager IPsec
Kube-OVN CA

```
args:
- --enable-ovn-ipsec=true
- --cert-manager-ipsec-cert=true
```

7.22.5

- kube-ovn-cni ovs-appctl ipsec/show IPsec SA SPI
- kubect1 get csr kube-ovn-cni CSR IPsec CSR approve controller signer.go
- IPsec false IPsec SA kube-ovn-cni Pod



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7.22.6

7.23 OVN

	Pod	GRE/ERSPAN
	Kube-OVN	v1.12

7.23.1 Multus-CNI

Multus-CNI	Multus
------------	--------

7.23.2

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: attachnet
  namespace: default
spec:
  config: |
    {
      "cniVersion": "0.3.1",
      "type": "kube-ovn",
      "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
      "provider": "attachnet.default.ovn"
    }

```

provider

<NAME>.<NAMESPACE>.ovn

7.23.3 Underlay

	MTU	LSP/Pod	Underlay
Underlay			

```
apiVersion: kubeovn.io/v1
kind: ProviderNetwork
metadata:
  name: net1
spec:
  defaultInterface: eth1
---
apiVersion: kubeovn.io/v1
kind: Vlan
metadata:
  name: vlan1
spec:
  id: 0
  provider: net1
---
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: subnet1
spec:
  protocol: IPv4
  cidrBlock: 172.19.0.0/16
  excludeIps:
    - 172.19.0.2..172.19.0.20
  gateway: 172.19.0.1
  vlan: vlan1
  provider: attachnet.default.ovn

```

provider

provider

7.23.4 Pod

Pod

```
apiVersion: v1
kind: Pod

```

```

metadata:
  name: pod1
  annotations:
    k8s.v1.cni.cncf.io/networks: default/attachnet
spec:
  containers:
    - name: bash
      image: docker.io/kubeovn/kube-ovn:v1.16.5
      args:
        - bash
        - -c
        - sleep infinity
  securityContext:
    privileged: true

```

Pod IP

```

$ kubectl get ips | grep pod1
pod1.default          10.16.0.12  00:00:00:FF:34:24  kube-ovn-worker  ovn-default
pod1.default.attachnet.default.ovn  172.19.0.21  00:00:00:A0:30:68  kube-ovn-worker  subnet1

```

IP 172.19.0.21

7.23.5 OVN

OVN

```

kubectl ko nbctl mirror-add mirror1 gre 99 from-lport 172.19.0.21
kubectl ko nbctl lsp-attach-mirror coredns-787d4945fb-gpnkb.kube-system mirror1

```

coredns-787d4945fb-gpnkb.kube-system OVN LSP <POD_NAME>.<POD_NAMESPACE>

OVN

```

ovn-nbctl mirror-add <NAME> <TYPE> <INDEX> <FILTER> <IP>

NAME - add a mirror with given name
TYPE - specify TYPE 'gre' or 'erspan'
INDEX - specify the tunnel INDEX value
        (indicates key if GRE, erspan_idx if ERSPAN)
FILTER - specify FILTER for mirroring selection
        ('to-lport' / 'from-lport')
IP - specify Sink / Destination i.e. Remote IP

ovn-nbctl mirror-del [NAME]          remove mirrors
ovn-nbctl mirror-list                print mirrors

ovn-nbctl lsp-attach-mirror PORT MIRROR attach source PORT to MIRROR
ovn-nbctl lsp-detach-mirror PORT MIRROR detach source PORT from MIRROR

```

7.23.6 Pod

Pod

```

root@pod1:/kube-ovn# ip link add mirror1 type gretap local 172.19.0.21 key 99 dev net1
root@pod1:/kube-ovn# ip link set mirror1 up

```

Pod

```

root@pod1:/kube-ovn# tcpdump -i mirror1 -nnve
tcpdump: listening on mirror1, link-type EN10MB (Ethernet), snapshot length 262144 bytes
05:13:30.328008 00:00:00:a3:f5:e2 > 00:00:00:97:0f:6e, ethertype ARP (0x0806), length 42: Ethernet (len 6), IPv4 (len 4), Request who-has 10.16.0.7 tell 10.16.0.4, length 28
05:13:30.559167 00:00:00:a3:f5:e2 > 00:00:00:89:d5:cc, ethertype IPv4 (0x0800), length 212: (tos 0x0, ttl 64, id 57364, offset 0, flags [DF], proto UDP (17), length 198)
10.16.0.4.53 > 10.16.0.6.50472: 34511 NXDomain*- 0/1/1 (170)
05:13:30.559343 00:00:00:a3:f5:e2 > 00:00:00:89:d5:cc, ethertype IPv4 (0x0800), length 212: (tos 0x0, ttl 64, id 57365, offset 0, flags [DF], proto UDP (17), length 198)
10.16.0.4.53 > 10.16.0.6.45177: 1659 NXDomain*- 0/1/1 (170)
05:13:30.560625 00:00:00:a3:f5:e2 > 00:00:00:89:d5:cc, ethertype IPv4 (0x0800), length 200: (tos 0x0, ttl 64, id 57367, offset 0, flags [DF], proto UDP (17), length 186)
10.16.0.4.53 > 10.16.0.6.43848: 2636*- 0/1/1 (158)
05:13:30.562774 00:00:00:a3:f5:e2 > 00:00:00:89:d5:cc, ethertype IPv4 (0x0800), length 191: (tos 0x0, ttl 64, id 57368, offset 0, flags [DF], proto UDP (17), length 177)
10.16.0.4.53 > 10.16.0.6.37755: 48737 NXDomain*- 0/1/1 (149)
05:13:30.563523 00:00:00:a3:f5:e2 > 00:00:00:89:d5:cc, ethertype IPv4 (0x0800), length 187: (tos 0x0, ttl 64, id 57369, offset 0, flags [DF], proto UDP (17), length 173)
10.16.0.4.53 > 10.16.0.6.53887: 45519 NXDomain*- 0/1/1 (145)
05:13:30.564940 00:00:00:a3:f5:e2 > 00:00:00:89:d5:cc, ethertype IPv4 (0x0800), length 201: (tos 0x0, ttl 64, id 57370, offset 0, flags [DF], proto UDP (17),

```

```
length 187)
10.16.0.4.53 > 10.16.0.6.40846: 25745 NXDomain*- 0/1/1 (159)
05:13:30.565140 00:00:00:a3:f5:e2 > 00:00:00:89:d5:cc, ethertype IPv4 (0x0800), length 201: (tos 0x0, ttl 64, id 57371, offset 0, flags [DF], proto UDP (17),
length 187)
10.16.0.4.53 > 10.16.0.6.45214: 61875 NXDomain*- 0/1/1 (159)
05:13:30.566023 00:00:00:a3:f5:e2 > 00:00:00:55:e4:4e, ethertype IPv4 (0x0800), length 80: (tos 0x0, ttl 64, id 45937, offset 0, flags [DF], proto UDP (17),
length 66)
10.16.0.4.44116 > 172.18.0.1.53: 16025+ [1au] AAAA? kube-ovn.io. (38)
```

7.23.7

1. ERSPAN OVN Linux 4.14 ERSPAN IPv6 Linux 4.16
2. OVN

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7.23.8

7.24 DNS Kube-OVN

NodeLocal DNSCache DaemonSet DNS DNS Kube-OVN

7.24.1 DNS

Kubernetes DNS

Kubernetes Nodelocaldns cache

```
#!/bin/bash

localdns=169.254.20.10
domain=cluster.local
kubedns=10.96.0.10

wget https://raw.githubusercontent.com/kubernetes/kubernetes/master/cluster/addons/dns/nodelocaldns/nodelocaldns.yaml
sed -i "s/___PILLAR__LOCAL__DNS___/$localdns/g; s/___PILLAR__DNS__DOMAIN___/$domain/g; s/___PILLAR__DNS__SERVER___/g; s/___PILLAR__CLUSTER__DNS___/$kubedns/g"
nodelocaldns.yaml

kubectl apply -f nodelocaldns.yaml
```

kubelet /var/lib/kubelet/config.yaml clusterDNS DNS IP 169.254.20.10 kubelet

Kube-OVN DNS

Kubernetes Nodelocal DNSCache Kube-OVN

UNDERLAY SUBNET U2O

Underlay Subnet DNS U2O kubectl edit subnet {your subnet} spec.u2oInterconnection = true , Overlay Subnet

KUBE-OVN-CONTROLLER DNS IP

```
kubectl edit deployment kube-ovn-controller -n kube-system
```

spec.template.spec.containers.args --node-local-dns-ip=169.254.20.10

POD

Pod /etc/resolv.conf nameserver DNS IP Pod nameserver DNS ClusterIP u2o Pod Pod

7.24.2 DNS

Pod Pod DNS 169.254.20.10

```
# kubectl exec -it pod1 -- nslookup github.com
Server: 169.254.20.10
Address: 169.254.20.10:53

Name: github.com
Address: 20.205.243.166
```

DNS ovn0 DNS DNS

```
# tcpdump -i any port 53

06:20:00.441889 659246098c56_h P ifindex 17 00:00:00:73:f1:06 ethertype IPv4 (0x0800), length 75: 10.16.0.2.40230 > 169.254.20.10.53: 1291+ A? baidu.com. (27)
06:20:00.441889 ovn0 In ifindex 7 00:00:00:50:32:cd ethertype IPv4 (0x0800), length 75: 10.16.0.2.40230 > 169.254.20.10.53: 1291+ A? baidu.com. (27)
06:20:00.441950 659246098c56_h P ifindex 17 00:00:00:73:f1:06 ethertype IPv4 (0x0800), length 75: 10.16.0.2.40230 > 169.254.20.10.53: 1611+ AAAA? baidu.com. (27)
06:20:00.441950 ovn0 In ifindex 7 00:00:00:50:32:cd ethertype IPv4 (0x0800), length 75: 10.16.0.2.40230 > 169.254.20.10.53: 1611+ AAAA? baidu.com. (27)
06:20:00.442203 ovn0 Out ifindex 7 00:00:00:52:99:d8 ethertype IPv4 (0x0800), length 145: 169.254.20.10.53 > 10.16.0.2.40230: 1611* 0/1/0 (97)
```

```
06:20:00.442219 659246098c56_h Out ifindex 17 00:00:00:ea:b3:5e ethertype IPv4 (0x0800), length 145: 169.254.20.10.53 > 10.16.0.2.40230: 1611* 0/1/0 (97)
06:20:00.442273 ovn0 Out ifindex 7 00:00:00:52:99:d8 ethertype IPv4 (0x0800), length 125: 169.254.20.10.53 > 10.16.0.2.40230: 1291* 2/0/0 A 39.156.66.10, A
110.242.68.66 (77)
06:20:00.442278 659246098c56_h Out ifindex 17 00:00:00:ea:b3:5e ethertype IPv4 (0x0800), length 125: 169.254.20.10.53 > 10.16.0.2.40230: 1291* 2/0/0 A 39.
156.66.10, A 110.242.68.66 (77)
```

7.24.3



NetworkPolicy NetworkPolicy DNS IP 169.254.20.10 CIDR NetworkPolicy DNS Pod

NetworkPolicy

Pod DNS NetworkPolicy

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: allow-local-dns-and-node-cidr
  namespace: default #
spec:
  podSelector: {} # Pod
  policyTypes:
    - Ingress
    - Egress
  egress:
    # DNS
    - to:
      - ipBlock:
          cidr: 169.254.20.10/32
        # CIDR CIDR
    - to:
      - ipBlock:
          cidr: 10.0.0.0/8 # CIDR
  ingress:
    # DNS
    - from:
      - ipBlock:
          cidr: 169.254.20.10/32
        # CIDR CIDR
    - from:
      - ipBlock:
          cidr: 10.0.0.0/8 # CIDR
```

- 169.254.20.10/32 DNS IP
- 10.0.0.0/8 CIDR

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7.24.4

7.25 VPC NAT

7.25.1

VPC	Overlay	natOutgoing	Subnet	Pod	SNAT	IP		Pod	SNAT
NAT		CIDR	IP	SNAT					

7.25.2

subnet.SpecnatOutgoingnatOutgoingPolicyRules

```
spec:
  natOutgoing: true
  natOutgoingPolicyRules:
    - action: forward
      match:
        srcIPs: 10.0.11.0/30,10.0.11.254
    - action: nat
      match:
        srcIPs: 10.0.11.128/26
        dstIPs: 114.114.114.114,8.8.8.8
```

NAT

1. IP 10.0.11.0/30 10.0.11.254 SNAT

2. IP 10.0.11.128/26 IP 114.114.114.114 8.8.8.8 SNAT

actionmatchaction, actionforwardnatforwardSNAT, natSNATnatOutgoingPolicyRules

SNAT

matchsrcIPsdstIPsIPIP match.srcIPsmatch.dstIPsCIDRIP

matchnatOutgoingPolicyRulesaction

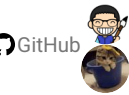
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8.

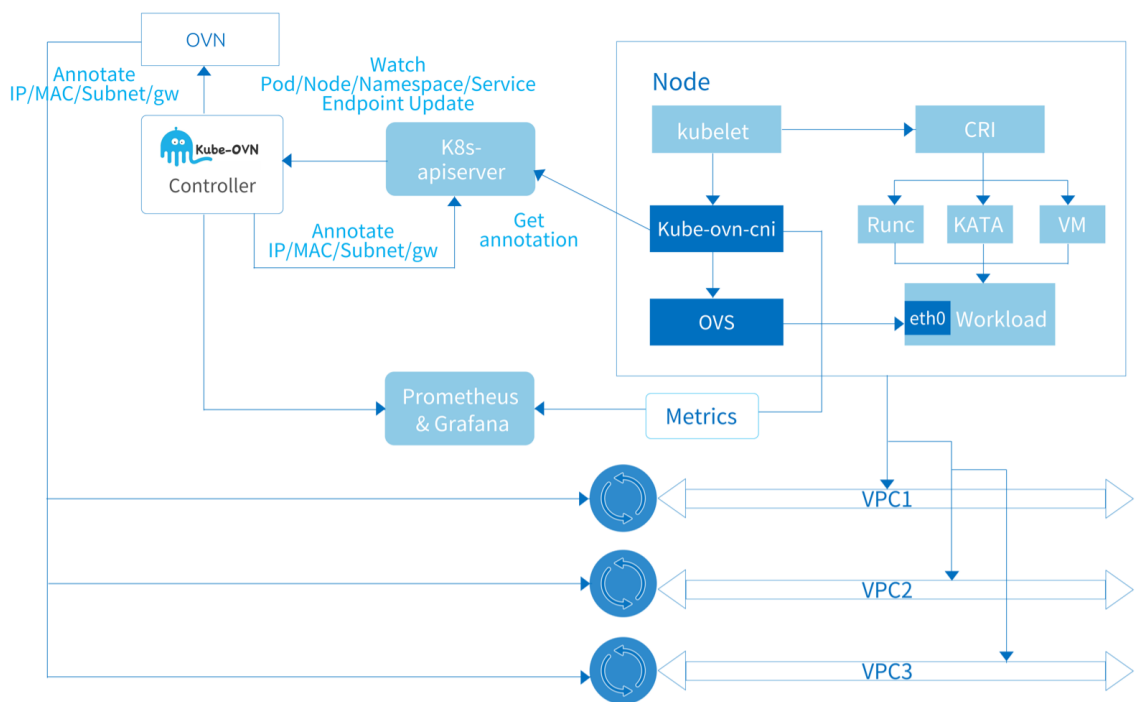
8.1

Kube-OVN									
Kube-OVN	Kubernetes	OVN	SDN		Kube-OVN	OVN	Kubernetes	CNI Service	Networkpolicy
SDN		VPC	QoS	ACL					
Kube-OVN			Cilium	Submariner	Prometheus	KubeVirt			

8.1.1

Kube-OVN

- OVN/OVS
- Agent
-



OVN/OVS

OVN/OVS	Kube-OVN	OVN/OVS	SDN	Kube-OVN	ovn-architecture(7)
OVN	Kube-OVN	OVN	Kubernetes		
OVN/OVS	Kubernetes				

OVN-CENTRAL

ovn-central Deployment OVN ovn-nb, ovn-sb, ovn-northd

- ovn-nb API kube-ovn-controller ovn-nb
- ovn-sb ovn-nb
- ovn-northd ovn-nb ovn-sb

ovn-central Raft

OVS-OVN

ovs-ovn DaemonSet Pod openvswitch, ovssdb, ovn-controller ovn-central Agent

Agent

Kube-OVN OVN Kubernetes

KUBE-OVN-CONTROLLER

Deployment Kubernetes OVN Kube-OVN kube-ovn-controller OVN

Pod Service Endpoint Node NetworkPolicy VPC Subnet Vlan ProviderNetwork

Pod kube-ovn-controller Pod IPAM ovn-central ACL kube-ovn-controller

CIDR Pod annotation annotation kube-ovn-cni

KUBE-OVN-CNI

DaemonSet CNI OVS

DaemonSet kube-ovn kubelet kube-ovn-cni CNI kube-ovn-cni /opt/cni/bin

kube-ovn-cni

- 1. ovn-controller vswitchd
- 2. CNI add/del
 - a. veth OVS
 - b. OVS
 - c. iptables/ipset/route
- 3. QoS.
- 4. ovn0
- 5. Vlan/Underlay/EIP
- 6.

Kube-OVN

KUBE-OVN-SPEAKER

DaemonSet Pod IP

BGP

KUBE-OVN-PINGER

DaemonSet OVS Kube-OVN

KUBE-OVN-MONITOR

Deployment OVN Kube-OVN

KUBECTL-KO

kubectl

kubectl

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8.1.2

8.2 What's Next

This document lists the features merged into the master branch for the next minor release.

8.2.1 Post-v1.15.0

- NetworkPolicy now supports provider-scoped policies for multi-network pods using the `ovn.kubernetes.io/policy-for` annotation. [#6223](#)
- Support static IP/MAC for multiple interfaces on the same logical switch. [#6060](#)
- MetalLB underlay integration now supports IPv6 and dual-stack. [#6159](#)
- KubeVirt live-migration multi-chassis options now apply to all VM NICs, not just the primary one. [#6241](#)
- Add human-readable descriptions to all Kube-OVN CRD fields for better `kubectl explain` output. [#6133](#) [#6147](#)
- Support BGP and EVPN (L3VPN) for VPC Egress Gateway, running FRR in the egress gateway Pod. [#6224](#)
- Support per-pod NIC granular DHCP control via annotations, overriding subnet-level DHCP settings. [#6475](#)
- Extend SecurityGroup API with tiers, larger priority range (1-16384), localAddress and port matches. [#6330](#)
- Use BigInt for SubnetStatus IP count fields to support large address ranges. [#6403](#)
- Improve hairpin logic to support FIP, SNAT from any CIDR within the VPC and LoadBalancers. [#6445](#)
- VPC NAT Gateway
 - Support user-defined annotations on NAT gateway Pod template. [#6256](#)
 - Allow SNAT EIP to FIP EIP traffic. [#6329](#)
 - Allow any EIP to share external subnet gateway in one native VLAN. [#6426](#)
- VPC Egress Gateway
 - Support custom resources and bandwidth limits. [#6407](#)
 - Accept integer Mbps values and `M`, `Mi`, `G`, or `Gi` bit-rate quantities for per-replica bandwidth limits. [#7123](#)
- Interconnection
 - Add vendor ID to transit switches to avoid conflicts with other OVN controllers. [#6186](#)
- Reliability
 - OpenFlow synchronization: detect and recover from stale or missing OVS flows automatically. [#6117](#)
 - OVN DB: back up Raft header before rejoining cluster to improve recovery. [#6106](#)
- Performance
 - Strip `managedFields` from informer cache to reduce memory usage. [#6119](#)
 - Add field selectors to informer factory to reduce API server load. [#6091](#)
- Security
 - Replace wildcard RBAC verbs with explicit verb lists. [#6233](#)
 - Specify ephemeral storage limits for containers. [#6259](#)
- Helm Chart
 - Make DaemonSet update strategy configurable via `values.yaml`. [#6136](#)
 - Introduce `extraEnv` variable for all components. [#6142](#)
 - Add `affinity` and `nodeSelector` support for ovs-ovn and ovs-ovn-dpdk DaemonSets. [#6308](#)
 - Add `external-gateway-config-ns` option for controller. [#6211](#)
 - Introduce ServiceMonitor and labels to services. [#6340](#)

8.2.2 Post-v1.14.0

- ACL log supports ratelimiting. [#5938](#)

- Subnet with centralized gateway now supports nodeSelectors. [#5956](#)
- Overlay encapsulation NIC selection. [#5946](#)
- Performance: skip conntrack for specific dst CIDRs. [#5821](#)
- NetworkPolicy supports `lax` mode which only deny traffic type of TCP, UDP and SCTP. That means ARP, ICMP and DHCP traffic are always allowed. [#5745](#)
- Remove internal-port type interface code. [#5794](#)
- IPPool
- Multiple IPPools now can bind to the same Namespace. [#5731](#)
- Pods in a bound namespace will only get IPs from the bound pool(s), not other ranges in the subnet. [#5731](#)
- IPPool will create an AddressSet that can be work with VPC Policy Route and ACL. [#5920](#)
- `AdminNetworkPolicy` now supports specify egress peers using FQDNs. [#5703](#)
- Using ARP for IPv4 network ready check: now you don't need ACL allow rules for gateway to make Pod running. [#5716](#)
- Non-primary CNI mode: you can run Kube-OVN as the secondary only network, without annoying unused annotations and logical switch port allocations. [#5618](#)
- VPC NAT Gateway:
 - No default EIP mode: the secondary interface can initialize without a default EIP to avoid the waste. [#5605](#)
 - Custom routes: you can control the route rules within the vpc-nat-gateway Pods to control traffic paths. [#5608](#)
 - Gratuitous ARP: VPC NAT Gateway automatically sends gratuitous ARP packets during initialization to accelerate network convergence. [#5607](#)
 - Healthchecks for static endpoints in `SwitchLBRules`: SLR with both selector or endpoints key can support healthchecks. [#5435](#)
- Underlay
 - Node Selectors for `ProviderNetwork`: instead of adding/removing nodes to the `ProviderNetwork` one by one, you can use node selectors to simplify the workflow. [#5518](#)
 - Different `NetworkProvider`s can now share the same VLAN. [#5471](#)
 - Auto create VLAN sub-interfaces. [#5966](#)
 - Auto move VLAN sub-interfaces to OVS bridges. [#5949](#)
 - Adding `pod_name` and `pod_namespace` labels to interface metrics. [#5463](#)
- IPsec
 - Support `cert-manager` to issue certificates. [#5365](#)
 - Request new certificate if current certificate is not trusted. [#5710](#)
- kubectl-ko
 - Collect IPsec and xFRM information. [#5472](#)
 - Replace `Endpoint` with `EndpointSlice`. [#5425](#)
 - NetworkAttachment caching: reduce APIServer load in large-scale deployments with Multus. [#5386](#)
 - Upgrade `ovs` to 3.5 and `ovn` to 25.03. [#5537](#)



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8.2.3

8.3

Kube-OVN

8.3.1

Kube-OVN

1. default
- Pod IP
- CIDR
- 10.16.0.0/16
- 10.16.0.1
2. join
- Node Pod
- ,
- CIDR
- 100.64.0.0/16
- 100.64.0.1

```
POD_CIDR="10.16.0.0/16"
POD_GATEWAY="10.16.0.1"
JOIN_CIDR="100.64.0.0/16"
EXCLUDE_IPS=""
```

EXCLUDE_IP POD_CIDR 192.168.10.20..192.168.10.30

Overlay Service CIDR

Join

8.3.2 Service

kube-proxy iptables Kube-OVN Kube-OVN Service CIDR

```
SVC_CIDR="10.96.0.0/12"
```

kube-ovn-controller Deployment

```
args:
- --service-cluster-ip-range=10.96.0.0/12
```

8.3.3 Overlay

Kube-OVN Kubernetes Node IP

```
IFACE=eth1
```

ens[a-z0-9]*,eth[a-z0-9]*

kube-ovn-cni DaemonSet

```
args:
- --iface=eth1
```

annotation ovn.kubernetes.io/tunnel_interface annotation iface annotation

```
kubect1 annotate node no1 ovn.kubernetes.io/tunnel_interface=ethx
```

Overlay

8.3.4 MTU

Overlay	Kube-OVN	MTU	MTU	Overlay	Pod	MTU	MTU - 100	Underlay	Pod
MTU									
Overlay	MTU	kube-ovn-cni	DaemonSet						
<pre>args: - --mtu=1333</pre>									

8.3.5

Kube-OVN	mirror0	tcpdump							
<pre>ENABLE_MIRROR=true</pre>									
kube-ovn-cni	DaemonSet	:							
<pre>args: - --enable-mirror=true</pre>									

8.3.6 LB

Underlay	kube-proxy ClusterIP	Service kube-proxy	OVN kube-proxy	L2 LB	ClusterIP	Service	LB
<pre>ENABLE_LB=false</pre>							
kube-ovn-controller Deployment							
<pre>args: - --enable-lb=false</pre>							

LB

Kube-OVN v1.12.0	subnet crd	spec	enableLb	Kube-OVN	LB	LB	kube-ovn-controller		
Deployment	enable-lb	load-balancer		enableLb		load-balancer	v1.12.0	enableLb	

8.3.7 NetworkPolicy

Kube-OVN	OVN	ACL	NetworkPolicy	NetworkPolicy	Cilium Chain	eBPF	NetworkPolicy	Kube-OVN
NetworkPolicy								
ENABLE_NP=false								
kube-ovn-controller Deployment								
args: - --enable-np=false								

NetworkPolicy

8.3.8 EIP SNAT

EIP	SNAT	kube-ovn-controller							
-----	------	---------------------	--	--	--	--	--	--	--

```
ENABLE_EIP_SNAT=false
```

kube-ovn-controller Deployment

```
args:
- --enable-eip-snat=false
```

EIP SNAT EIP SNAT

8.3.9 Load Balancer Service

VPC Load Balancer Service LoadBalancer Service

```
ENABLE_LB_SVC=true
```

kube-ovn-controller Deployment

```
args:
- --enable-lb-svc=true
```

8.3.10 ECMP

ECMP ECMP kube-ovn-controller Deployment :

```
args:
- --enable-ecmp=true
```

Kube-OVN v1.12.0 subnet crd spec enableEcmp ECMP ECMP kube-ovn-controller
Deployment enable-ecmp v1.12.0

8.3.11 Kubevirt VM

Kubevirt VM kube-ovn-controller StatefulSet Pod IP VM

1.10.6 kube-ovn-controller Deployment

```
args:
- --keep-vm-ip=false
```

8.3.12 CNI

Kube-OVN /opt/cni/bin CNI /etc/cni/net.d CNI 01-kube-ovn.conflist CNI

```
CNI_CONF_DIR="/etc/cni/net.d"
CNI_BIN_DIR="/opt/cni/bin"
CNI_CONFIG_PRIORITY="01"
```

kube-ovn-cni DaemonSet Volume

```
volumes:
- name: cni-conf
  hostPath:
    path: "/etc/cni/net.d"
- name: cni-bin
  hostPath:
    path: "/opt/cni/bin"
...
args:
- --cni-conf-name=01-kube-ovn.conflist
```

8.3.13

Kube-OVN

Overlay

Geneve

Vxlan

STT

TUNNEL_TYPE="vxlan"

ovs-ovn DaemonSet

env:
- name: TUNNEL_TYPE
 value: "vxlan"

STT

ovs

8.3.14 SSL

OVN DB

API

SSL

:

ENABLE_SSL=true

SSL

8.3.15 contrack CIDR

contrack

NAT/

kube-ovn-controller

CIDR

SKIP_CONTRACK_DST_CIDRS="10.10.0.0/16,fd00:10::/64"

args:
- --skip-contrack-dst-cidrs=10.10.0.0/16,fd00:10::/64

8.3.16 OVN IPsec

Kube-OVN

OVN

IPSec

Geneve/Vxlan/STT

ENABLE_OVN_IPSEC=true

daemon

args:
- --enable-ovn-ipsec=true

OVN IPsec

8.3.17 AdminNetworkPolicy / DNS

AdminNetworkPolicy

ANP/CNP

DNS

DNSNameResolver

CRD

ENABLE_ANP=true
ENABLE_DNS_NAME_RESOLVER=true

args:
- --enable-anp=true
- --enable-dns-name-resolver=true

8.3.18 VPC NAT Gateway

VPC NAT Gateway controller

```
ENABLE_NAT_GW=true
```

ovn-vpc-nat-gw-config ConfigMap enable-vpc-nat-gw

8.3.19 ovn-northd

ovn-northd

```
OVN_NORTHD_N_THREADS=4
```

1 CPU

8.3.20 NetworkPolicy

Kube-OVN standard / lax NetworkPolicy

```
NP_ENFORCEMENT=standard
```

```
args:
- --np-enforcement=standard
```

NetworkPolicy ovn.kubernetes.io/network_policy_enforcement annotation [NetworkPolicy](#)

8.3.21 VXLAN TX checksum offload

VXLAN checksum offload

```
SET_VXLAN_TX_OFF=true
```

```
args:
- --set-vxlan-tx-off=true
```

8.3.22 KubeVirt

KubeVirt

```
ENABLE_LIVE_MIGRATION_OPTIMIZE=true
```

8.3.23 OVN LB endpoint

OVN LoadBalancer endpoint externalTrafficPolicy=Local

```
ENABLE_OVN_LB_PREFER_LOCAL=true
```

```
args:
- --enable-ovn-lb-prefer-local=true
```

8.3.24 ConfigMap

ConfigMap external-gw-config kube-system

```
EXTERNAL_GATEWAY_CONFIG_NS=kube-system
```

8.3.25 Non-Primary CNI

Kube-OVN	attachment network	Pod	IP	CNI	controller	daemon
<pre>args: - --non-primary-cni-mode=true</pre>						
Helm	cni_conf.NON_PRIMARY_CNI=true	chart v1	/cni.nonPrimaryCNI=true	chart v2	Non-Primary CNI	

8.3.26 ip

kube-ovn-controller/kube-ovn-cni/kube-ovn-monitor	ip	0.0.0.0	ip
ENABLE_BIND_LOCAL_IP=true			
kube-ovn-monitor	pod ip		
# netstat -tunlp grep kube-ovn tcp 0 0 172.18.0.5:10661 0.0.0.0:* LISTEN 2612/./kube-ovn-mon			
deployment	daemonSet		
env: - name: ENABLE_BIND_LOCAL_IP value: "false"			

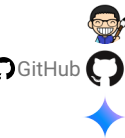
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8.3.27

8.4

Kube-OVN	Minor	Patch	Minor	OVN/OVS	API	Patch	Bug	API
----------	-------	-------	-------	---------	-----	-------	-----	-----

8.4.1

Kube-OVN backport	master, release-1.12	release-1.11	release-1.12	Bug
release-1.11	backport	Bug		

8.4.2

Minor	Patch	Bug	Bug
-------	-------	-----	-----

8.4.3 Patch

Patch	hack/release.sh			
1.	Build			
2.	tag	Docker Hub		
3.	tag	Github		
4.				
5.				
6.	Release Note PR			
7.	Release Note ()			
8.	Merge github action	Release Note PR		
9.	Github Release			
10.	Github Release	Release	v1.12.12	Release Note Release

8.4.4 Minor

Minor				
1.	Github	release-1.13 ()		
2.	VERSION, dist/images/install.sh, charts/kube-ovn/values.yaml	charts/kube-ovn/Chart.yaml	Minor	v1.14.0 ()
3.	tag	Docker Hub ()		
4.	tag	Github ()		
5.	v1.13	mkdocs.yml	version	branch ()
6.	Release Note PR			
7.	Release Note ()			
8.	Merge github action	Release Note PR		
9.	Github Release			
10.	Github Release	Release	v1.13.0	Release Note Release
11.	VERSION	Patch	v1.13.1	

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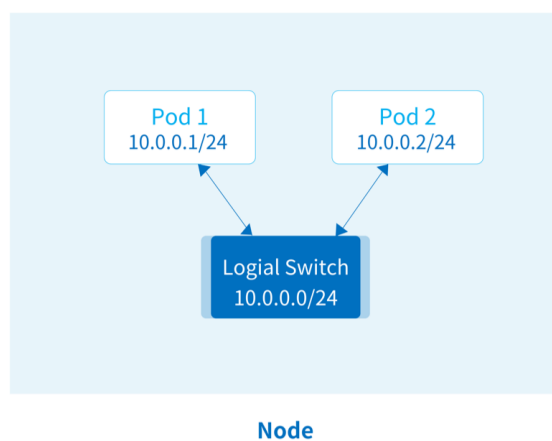


8.4.5

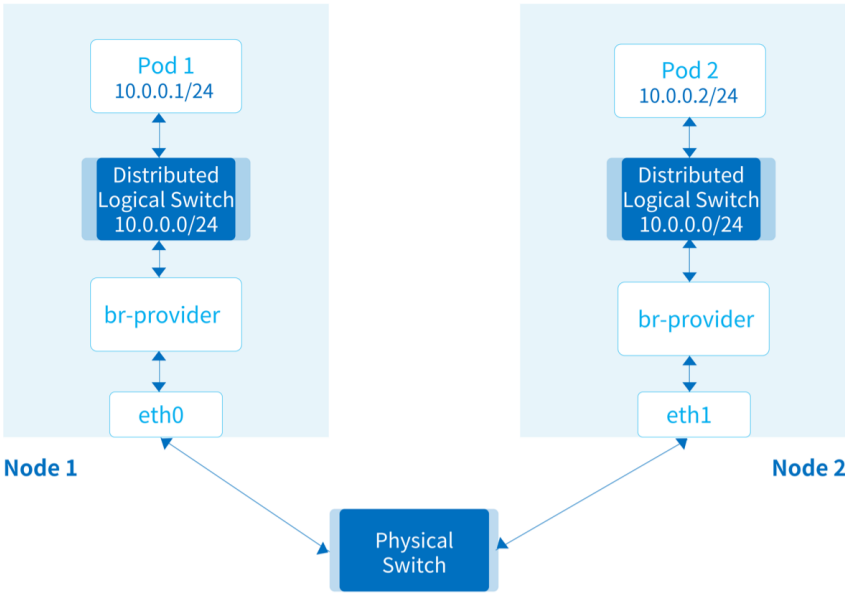
8.5 Underlay

Underlay

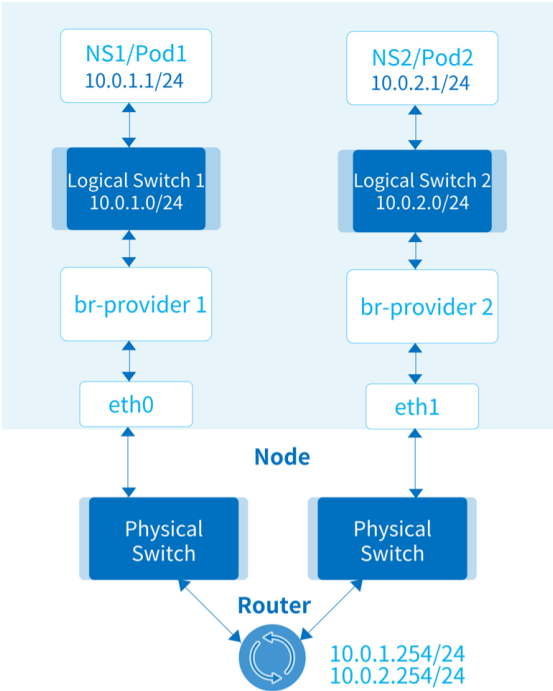
8.5.1



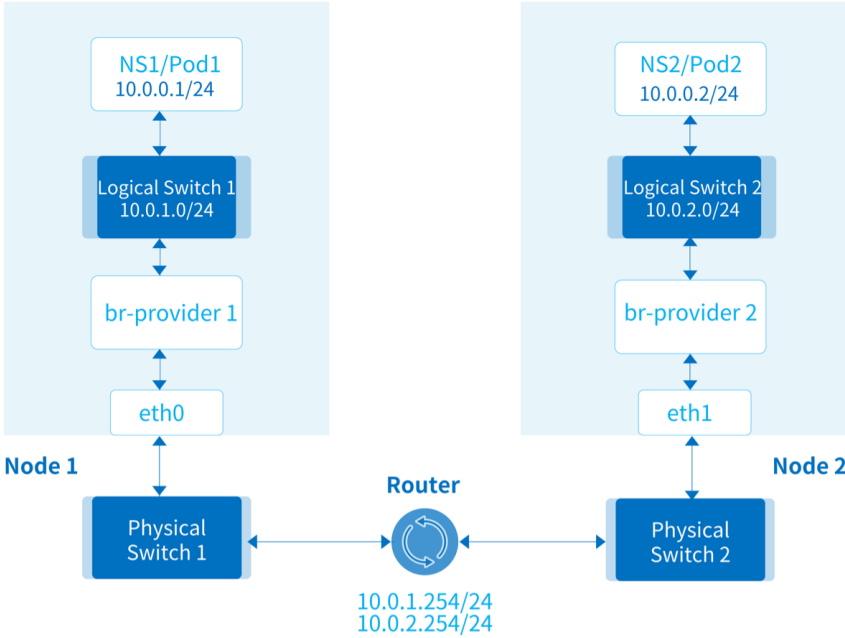
8.5.2



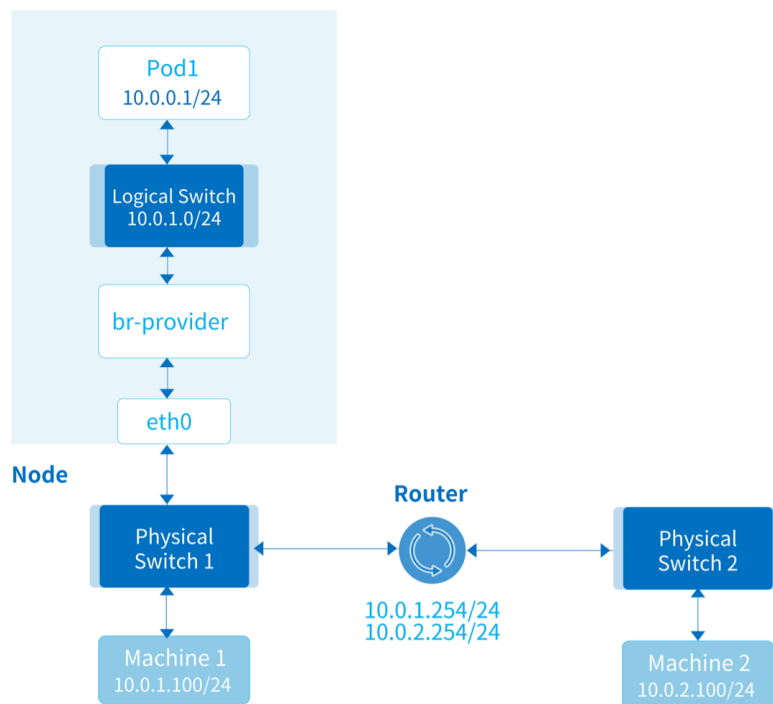
8.5.3



8.5.4

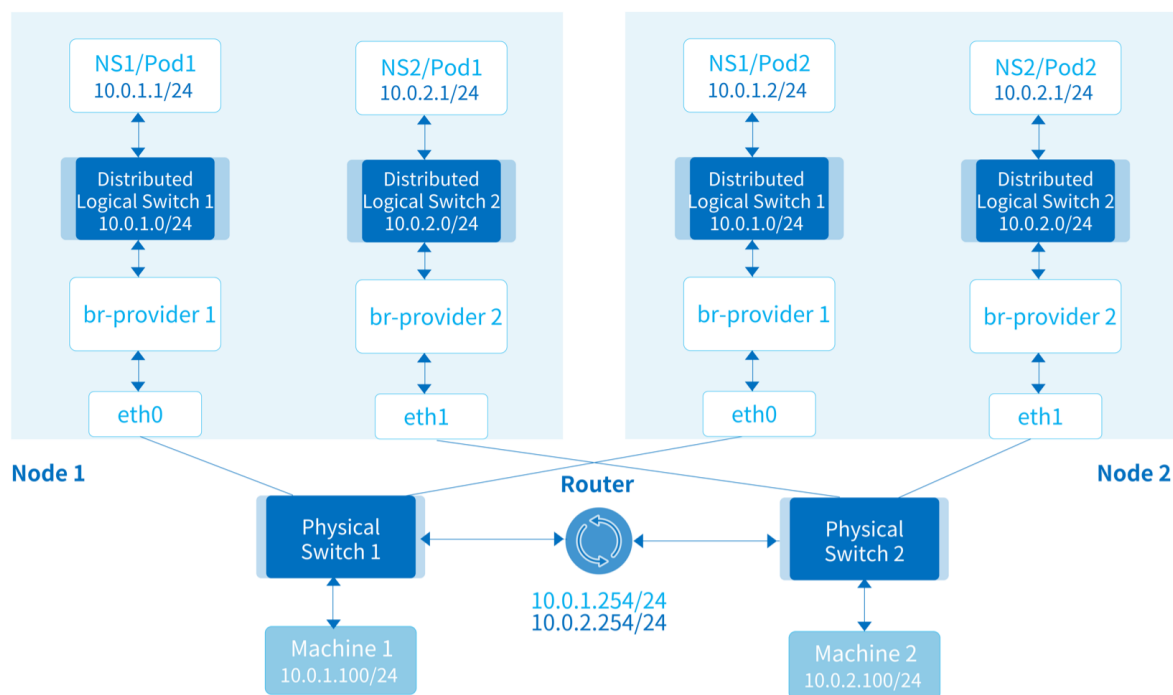


8.5.5

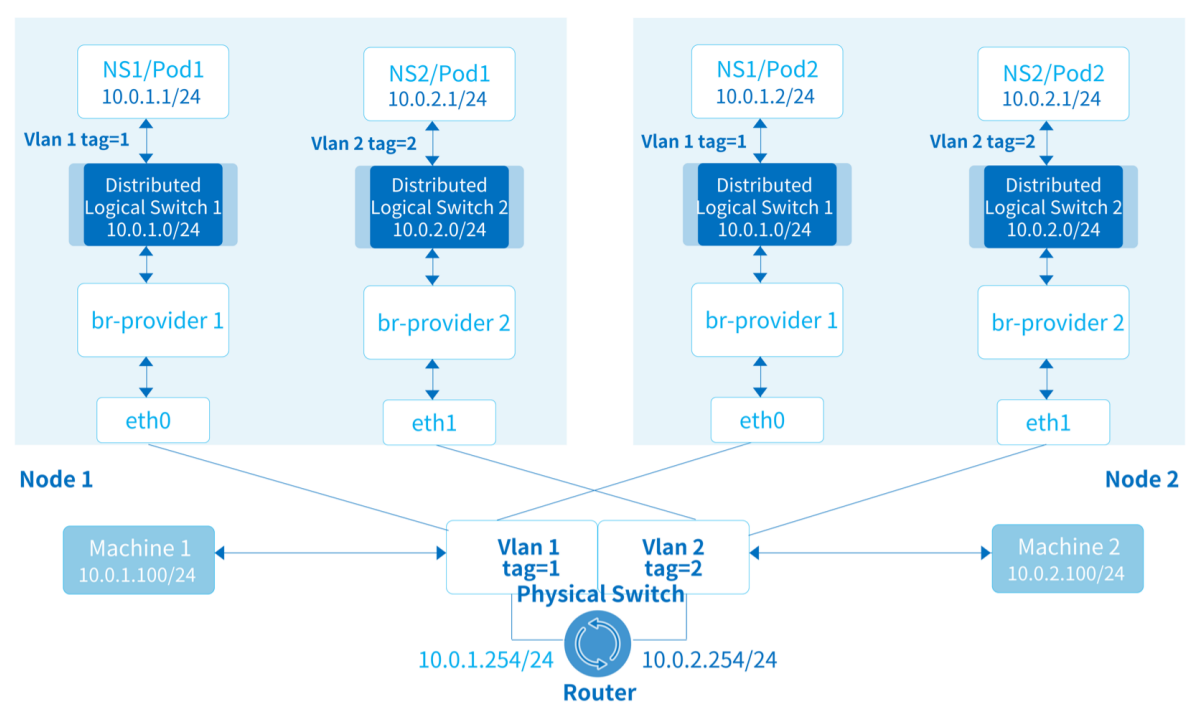


Pod

8.5.6 Vlan Tag



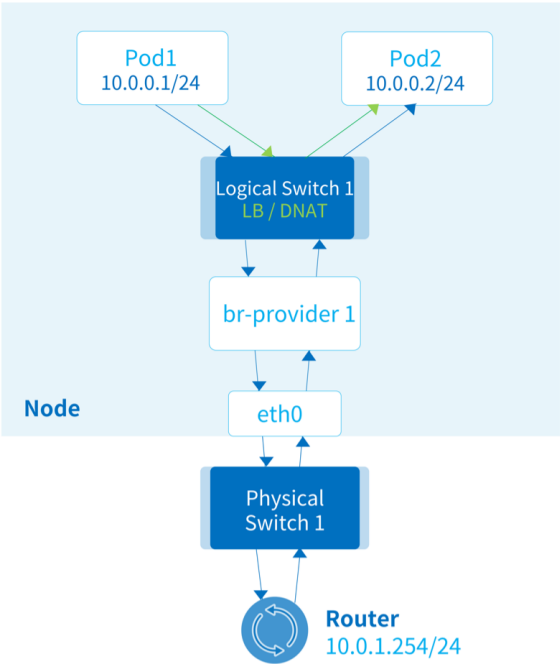
8.5.7 VLAN



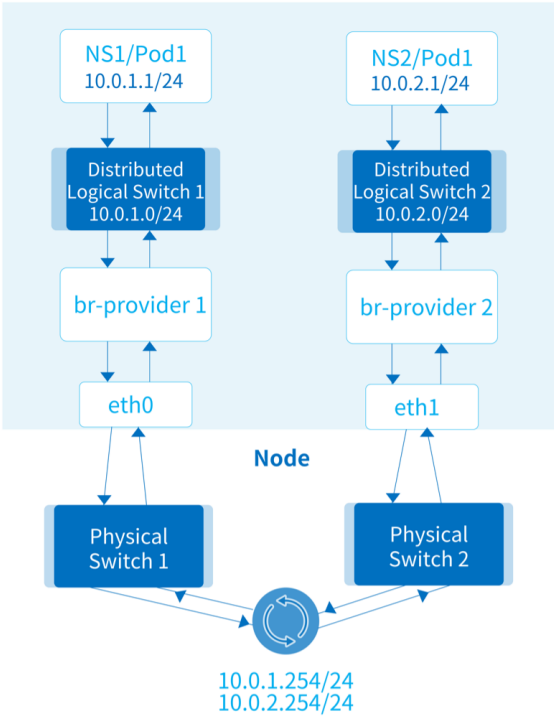
8.5.8 Pod Service IP

Kube-OVN	Kubernetes Service DNAT	Service IP	Pod Endpoint	Service IP IP	Pod MAC	Service IP MAC	MAC	MAC
----------	----------------------------	---------------	-----------------	------------------	------------	-------------------	-----	-----

Service Pod



Service Pod



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8.5.9

8.6 Iptables

Kube-OVN ipset iptables VPC Overlay NAT

ipset

IPv4/IPv6				
ovn40services/ovn60services	hash:net	Service		
ovn40subnets/ovn60subnets	hash:net	Overlay	NodeLocal DNS IP	
ovn40subnets-nat/ovn60subnets-nat	hash:net	NatOutgoing	Overlay	
ovn40subnets-distributed-gw/ovn60subnets-distributed-gw	hash:net	Overlay		
ovn40other-node/ovn60other-node	hash:net	IP		
ovn40local-pod-ip-nat/ovn60local-pod-ip-nat	hash:ip			
ovn40subnets-nat-policy	hash:net	natOutgoingPolicyRules		
ovn40natpr-418e79269dc5-dst	hash:net	natOutgoingPolicyRules	rule	dstIPs
ovn40natpr-418e79269dc5-src	hash:net	natOutgoingPolicyRules	rule	srcIPs

iptables IPv4

filter	INPUT	-m set --match-set ovn40services src -j ACCEPT	k8s Service Pod	--
filter	INPUT	-m set --match-set ovn40services dst -j ACCEPT		--
filter	INPUT	-m set --match-set ovn40subnets src -j ACCEPT		--
filter	INPUT	-m set --match-set ovn40subnets dst -j ACCEPT		--
filter	FORWARD	-m set --match-set ovn40services src -j ACCEPT		--
filter	FORWARD	-m set --match-set ovn40services dst -j ACCEPT		--
filter	FORWARD	-m set --match-set ovn40subnets src -j ACCEPT		--
filter	FORWARD	-m set --match-set ovn40subnets dst -j ACCEPT		--
filter	FORWARD	-s 10.16.0.0/16 -m comment --comment "ovn- subnet-gateway,ovn- default"	subnet	10.16.0.0/16 subnet cidr comment ovn-subnet-gateway iptables subnet ovn-default subnet
filter	FORWARD	-d 10.16.0.0/16 -m comment --comment "ovn- subnet-gateway,ovn- default"	subnet	
filter	OUTPUT	-p udp -m udp --dport 6081 -j MARK --set-xmark 0x0	SNAT	UDP: bad checksum on VXLAN interface
nat	PREROUTING	-m comment --comment "kube-ovn prerouting rules" -j OVN-PREROUTING	OVN-PREROUTING	--
nat	POSTROUTING	-m comment --comment "kube-ovn postrouting rules" -j OVN- POSTROUTING	OVN-POSTROUTING	--
nat	OVN- PREROUTING	-i ovn0 -m set --match-set ovn40subnets src -m set -- match-set ovn40services dst -j MARK --set-xmark 0x4000/0x4000	Pod Service masquerade	LB
nat	OVN- PREROUTING	-p tcp -m addrtype --dst- type LOCAL -m set --match-		kube-proxy ipvs

		set KUBE-NODE-PORT-LOCAL-TCP dst -j MARK --set-xmark 0x80000/0x80000	ExternalTrafficPolicy Local Service TCP			
nat	OVN-PREROUTING	-p udp -m addrtype --dst-type LOCAL -m set --match-set KUBE-NODE-PORT-LOCAL-UDP dst -j MARK --set-xmark 0x80000/0x80000	ExternalTrafficPolicy Local Service UDP			
nat	OVN-POSTROUTING	-m set --match-set ovn40services src -m set --match-set ovn40subnets dst -m mark --mark 0x4000/0x4000 -j SNAT --to-source	Service IP Overlay Pod IP	kube-proxy	ipvs	
nat	OVN-POSTROUTING	-m mark --mark 0x4000/0x4000 -j MASQUERADE	SNAT	--		
nat	OVN-POSTROUTING	-m set --match-set ovn40subnets src -m set --match-set ovn40subnets dst -j MASQUERADE	Pod Service SNAT	--		
nat	OVN-POSTROUTING	-m mark --mark 0x80000/0x80000 -m set --match-set ovn40subnets-distributed-gw dst -j RETURN	ExternalTrafficPolicy Local Service Endpoint SNAT	--		
nat	OVN-POSTROUTING	-m mark --mark 0x80000/0x80000 -j MASQUERADE	ExternalTrafficPolicy Local Service Endpoint SNAT	--		
nat	OVN-POSTROUTING	-p tcp -m tcp --tcp-flags SYN NONE -m conntrack --ctstate NEW -j RETURN	Pod IP	SNAT	--	
nat	OVN-POSTROUTING	-s 10.16.0.0/16 -m set ! --match-set ovn40subnets dst -j SNAT --to-source 192.168.0.101	Pod NatOutgoing SNAT	IP	10.16.0.0/16 IP	192.168.0.101
nat	OVN-POSTROUTING	-m set --match-set ovn40subnets-nat src -m set ! --match-set ovn40subnets dst -j MASQUERADE	Pod NatOutgoing	SNAT	--	
nat	OVN-POSTROUTING	-m set --match-set ovn40subnets-nat-policy src -m set ! --match-set ovn40subnets dst -j OVN-NAT-POLICY	Pod natOutgoingPolicyRules SNAT	natOutgoingPolicyRules NAT-POLICY		OVN
nat						

	OVN- POSTROUTING	-m mark --mark 0x90001/0x90001 -j MASQUERADE --random- fully	OVN-NAT-POLICY 0x90001/0x90001	tag SNAT
nat	OVN- POSTROUTING	-m mark --mark 0x90002/0x90002 -j RETURN	OVN-NAT-POLICY 0x90002/0x90002	, tag SNAT
nat	OVN-NAT-POLICY	-s 10.0.11.0/24 -m comment --comment natPolicySubnet-net1 -j OVN-NAT-PSUBNET- aa98851157c5	10.0.11.0/24 net1 CIDR OVN-NAT- PSUBNET-aa98851157c5 natOutgoingPolicyRules	
nat	OVN-NAT- PSUBNET- xxxxxxxxxxxx	-m set --match-set ovn40natpr-418e79269dc5- src src -m set --match-set ovn40natpr-418e79269dc5- dst dst -j MARK --set-xmark 0x90002/0x90002	418e79269dc5 natOutgoingPolicyRules ID status.natOutgoingPolicyRules[index].Rule srcIPs ovn40natpr-418e79269dc5- src dstIPs ovn40natpr-418e79269dc5- dst tag 0x90002	
mangle	OVN-OUTPUT	-d 10.241.39.2/32 -p tcp -m tcp --dport 80 -j MARK --set- xmark 0x90003/0x90003	kubelet tproxy	
mangle	OVN- PREROUTING	-d 10.241.39.2/32 -p tcp -m tcp --dport 80 -j TPROXY -- on-port 8102 --on-ip 172.18.0.3 --tproxy-mark 0x90004/0x90004	kubelet tproxy	

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8.7

8.7.1

Kube-OVN	Github	/	Github	Issue	PR	Maintainer	Review	Github Action
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8.7.2

Kube-OVN

Go

Go Modules

G0111MODULE="on"

golangci-lint

local-installation

Kube-OVN

Docker buildx

Docker

buildx:

docker buildx create --use

8.7.3

Kube-OVN

git clone https://github.com/kubeovn/kube-ovn.git

cd kube-ovn

make release

ARM

make release-arm

8.7.4 base

OVS/OVN

base

base

Dockerfile

dist/images/Dockerfile.base

build x86 base image

make base-amd64

build arm base image

make base-arm64

8.7.5 E2E

Kube-OVN

:

• KIND

Kubernetes

go install sigs.k8s.io/kind@latest

• jinjanator

:

pip install jinjanator

• Ginkgo

go install github.com/onsi/ginkgo/v2/ginkgo;

go get github.com/onsi/gomega/...

E2E

make kind-init

make kind-install

make e2e

Underlay E2E

```
make kind-init
make kind-install-underlay
make e2e-underlay-single-nic
```

ovn vpc nat gw eip, fip, snat, dnat

```
make kind-init
make kind-install
make ovn-vpc-nat-gw-conformance-e2e
```

iptables vpc nat gw eip, fip, snat, dnat

```
make kind-init
make kind-install-vpc-nat-gw
make iptables-vpc-nat-gw-conformance-e2e
```

loadbalancer service

```
make kind-init
make kind-install-lb-svc
make kube-ovn-lb-svc-conformance-e2e
```

```
make kind-clean
```

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8.8 OVS/OVN

OVN/OVS	SDN	Kubernetes	Kube-OVN	Kube-OVN	OVN/OVS
OVN/OVS	Kube-OVN				
• 4228eab1d7	vswitchd ofport_usage				
• 54056ea65d	timer				
• 6b4dcb311f	fdb				
• f627b7721e	hairpin	fdb			
• 3f3e3a436f	ovsdb-tool	join-cluster	Server ID		
• a6cb8215a8	QoS				
• d4d76ddb2e	ovsdb-tool	fix-cluster			
• ffd2328d4a	netdev	CPU			
• d088c5d8c2	ovs-router	kube-ipvs0			
• 1b31f07dc6					
• 54b7678229	ovs-sandbox	docker run			
• 9ee66bd91b					
• e889d46924	Underlay	resubmit			
• f9e97031b5	ovn-controller	Kube-OVN	localnet	GARP	
• 78cade0187	conntrack				
• 85aa6263ad	northd	DNS IP	conntrack		
• 34dc3e3fcf	lflow	lport	conntrack		
• a297b840c2		DNAT	lsp		
• 03e35ed9c5	ovn-controller				
• e7d3ba53cd	ACL	DNS IP	conntrack		
• 9286e1fd57					
• e5916eb53a		lr-lb DNAT			
• e4e6ea9c5f	BFD LRP				
• e76880e792	northd	nb	version_compatibility		
• 477695a010	northd	localnet	lrp	arp/nd	
• 2209d9a335	ovn-controller	LSP			
• 52b727b331		ovn-northd			
• 53b9837f7f	LB	ACL			
• 737d9f932e	GW				
• sbrec-delete-null-check	ovn-controller	sbrec			
• 20626ea909	LB	ACL			
• a2d9ff3ccd	Deb				



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8.8.1

8.9

Kube-OVN	OVN/OVS	Geneve	Vxlan	STT	OVN
OVN Architecture Design Decision					

8.9.1 Geneve

Geneve	Kube-OVN	OVN	Offload	Geneve	24bit
datapath		datapath	32768		
Mellanox	OVS	Geneve	5.4	backport	
UDP	TCP over UDP	TCP	CPU		

8.9.2 Vxlan

Vxlan	OVN	Offload	OVN	datapath	4096	datapath
datapath	4096	inport	ACL			
Mellanox	OVS	Vxlan				
UDP	TCP over UDP	TCP	CPU			

8.9.3 STT

Warning

OpenVswitch3.6STTTunnel

STT	OVN	TCP	TCP	TCP	OVN	datapath
	OVS					
	OVS					

8.9.4

- [VXLAN vs GENEVE: Understand The Difference](#)
- [OVN FAQ](#)
- [What is Geneve](#)

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8.9.5

8.10 Kube-OVN

Kube-OVN

8.10.1 ovn-monitor

OVN

Gauge	kube_ovn_ovn_status	OVN	(2) follower	(1) leader, (0)
Gauge	kube_ovn_failed_req_count	OVN		
Gauge	kube_ovn_log_file_size_bytes	OVN		
Gauge	kube_ovn_db_file_size_bytes	OVN		
Gauge	kube_ovn_chassis_info	OVN chassis	(1)	(0)
Gauge	kube_ovn_db_status	OVN	, (1)	(0)
Gauge	kube_ovn_logical_switch_info	OVN logical switch	(1)	logical switch
Gauge	kube_ovn_logical_switch_external_id	OVN logical switch external_id	(1)	external-id
Gauge	kube_ovn_logical_switch_port_binding	OVN logical switch logical switch port		(1)
Gauge	kube_ovn_logical_switch_tunnel_key	OVN logical switch tunnel key		
Gauge	kube_ovn_logical_switch_ports_num	OVN logical switch logical port		
Gauge	kube_ovn_logical_switch_port_info	OVN logical switch port	(1)	
Gauge	kube_ovn_logical_switch_port_tunnel_key	OVN logical switch port tunnel key		
Gauge	kube_ovn_cluster_enabled	(1) OVN		(0) OVN
Gauge	kube_ovn_cluster_role		(1)	
Gauge	kube_ovn_cluster_status		(1)	
Gauge	kube_ovn_cluster_term	RAFT term		
Gauge	kube_ovn_cluster_leader_self		leader (1)	(0)
Gauge	kube_ovn_cluster_vote_self		leader (1)	(0)
Gauge	kube_ovn_cluster_election_timer	election timer		
Gauge	kube_ovn_cluster_log_not_committed	commit RAFT		
Gauge	kube_ovn_cluster_log_not_applied	apply RAFT		
Gauge	kube_ovn_cluster_log_index_start	RAFT		
Gauge	kube_ovn_cluster_log_index_next	RAFT		
Gauge	kube_ovn_cluster_inbound_connections_total			
Gauge	kube_ovn_cluster_outbound_connections_total			
Gauge	kube_ovn_cluster_inbound_connections_error_total			
Gauge	kube_ovn_cluster_outbound_connections_error_total			

8.10.2 ovs-monitor

ovsdb vswitchd

Gauge	ovs_status	OVS	(1)	(0)
Gauge	ovs_info	OVS	(1)	
Gauge	failed_req_count	OVS		
Gauge	log_file_size_bytes	OVS		
Gauge	db_file_size_bytes	OVS		
Gauge	datapath	Datapath	(1)	
Gauge	dp_total	OVS	datapath	
Gauge	dp_if	Datapath	(1)	
Gauge	dp_if_total	datapath	port	
Gauge	dp_flows_total	Datapath	flow	
Gauge	dp_flows_lookup_hit	Datapath	flow	
Gauge	dp_flows_lookup_missed	Datapath	flow	
Gauge	dp_flows_lookup_lost	Datapath	userspace	
Gauge	dp_masks_hit	Datapath	mask	
Gauge	dp_masks_total	Datapath	mask	
Gauge	dp_masks_hit_ratio	Datapath	mask	
Gauge	interface	OVS	(1)	
Gauge	interface_admin_state		(0) down, (1) up, (2)	
Gauge	interface_link_state		(0) down, (1) up, (2)	
Gauge	interface_mac_in_use	OVS Interface	MAC	
Gauge	interface_mtu	OVS Interface	MTU	
Gauge	interface_of_port	OVS Interface	OpenFlow Port ID	
Gauge	interface_if_index	OVS Interface	Index	
Gauge	interface_tx_packets	OVS Interface		
Gauge	interface_tx_bytes	OVS Interface		
Gauge	interface_rx_packets	OVS Interface		
Gauge	interface_rx_bytes	OVS Interface		
Gauge	interface_rx_crc_err	OVS Interface		
Gauge	interface_rx_dropped	OVS Interface		
Gauge	interface_rx_errors	OVS Interface		
Gauge	interface_rx_frame_err	OVS Interface		
Gauge	interface_rx_missed_err	OVS Interface	miss	
Gauge	interface_rx_over_err	OVS Interface	overrun	
Gauge	interface_tx_dropped	OVS Interface		

Gauge	interface_tx_errors	OVS Interface
Gauge	interface_collisions	OVS interface

8.10.3 kube-ovn-pinger

Gauge	pinger_ovs_up	OVS	
Gauge	pinger_ovs_down	OVS	
Gauge	pinger_ovn_controller_up	ovn-controller	
Gauge	pinger_ovn_controller_down	ovn-controller	
Gauge	pinger_inconsistent_port_binding	OVN-SB portbinding	OVS interface
Gauge	pinger_apiserver_healthy	kube-ovn-pinger	apiserver
Gauge	pinger_apiserver_unhealthy	kube-ovn-pinger	apiserver
Histogram	pinger_apiserver_latency_ms	kube-ovn-pinger	apiserver
Gauge	pinger_internal_dns_healthy	kube-ovn-pinger	
Gauge	pinger_internal_dns_unhealthy	kube-ovn-pinger	
Histogram	pinger_internal_dns_latency_ms	kube-ovn-pinger	
Gauge	pinger_external_dns_healthy	kube-ovn-pinger	
Gauge	pinger_external_dns_unhealthy	kube-ovn-pinger	
Histogram	pinger_external_dns_latency_ms	kube-ovn-pinger	
Histogram	pinger_pod_ping_latency_ms	kube-ovn-pinger	ping Pod
Gauge	pinger_pod_ping_lost_total	kube-ovn-pinger	ping Pod
Gauge	pinger_pod_ping_count_total	kube-ovn-pinger	ping Pod
Histogram	pinger_node_ping_latency_ms	kube-ovn-pinger	ping Node
Gauge	pinger_node_ping_lost_total	kube-ovn-pinger	ping Node
Gauge	pinger_node_ping_count_total	kube-ovn-pinger	ping Node
Histogram	pinger_external_ping_latency_ms	kube-ovn-pinger	ping
Gauge	pinger_external_ping_lost_total	kube-ovn-pinger	ping

8.10.4 kube-ovn-controller

kube-ovn-controller

Histogram	rest_client_request_latency_seconds	apiserver
Counter	rest_client_requests_total	apiserver
Counter	lists_total	API list
Summary	list_duration_seconds	API list
Summary	items_per_list	API list
Counter	watches_total	API watch
Counter	short_watches_total	API watch
Summary	watch_duration_seconds	API watch
Summary	items_per_watch	API watch
Gauge	last_resource_version	resource version
Histogram	ovs_client_request_latency_milliseconds	OVN
Gauge	subnet_available_ip_count	IP
Gauge	subnet_used_ip_count	IP

8.10.5 kube-ovn-cni

kube-ovn-cni

Histogram	cni_op_latency_seconds	CNI
Counter	cni_wait_address_seconds_total	CNI
Counter	cni_wait_connectivity_seconds_total	CNI
Counter	cni_wait_route_seconds_total	CNI
Histogram	rest_client_request_latency_seconds	apiserver
Counter	rest_client_requests_total	apiserver
Counter	lists_total	API list
Summary	list_duration_seconds	API list
Summary	items_per_list	API list
Counter	watches_total	API watch
Counter	short_watches_total	API watch
Summary	watch_duration_seconds	API watch
Summary	items_per_watch	API watch
Gauge	last_resource_version	resource version
Histogram	ovs_client_request_latency_milliseconds	OVN

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8.10.6

8.11 Kube-OVN

Kube-OVN Kube-OVN CRD CRD

8.11.1 Condition

type	String	
status	String	True False Unknown
reason	String	
message	String	
observedGeneration	Int64	
lastUpdateTime	Time	
lastTransitionTime	Time	

CRD Status Condition

8.11.2

Subnet

SUBNET

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	Subnet
metadata	ObjectMeta	Kubernetes	
spec	SubnetSpec	Subnet	
status	SubnetStatus	Subnet	

SubnetSpec

default	Bool				
vpc	String	VPC	ovn-cluster		
protocol	String	IP	IPv4 IPv6 Dual		
namespaces	[]String	namespace			
cidrBlock	String	10.16.0.0/16			
gateway	String	CIDRBlock			
excludelpls	[]String				
provider	String	OVN Subnet	NetworkAttachmentDefinition . Kube-OVN		
gatewayType	String	Overlay	distributed centralized		
gatewayNode	String	centralized			
natOutgoing	Bool	NAT	externalEgressGateway		
externalEgressGateway	String		natOutgoing		
policyRoutingPriority	Uint32				
policyRoutingTableID	Uint32	TableID			
mtu	Uint32	MTU			
private	Bool				
allowSubnets	[]String				
vlan	String	Vlan			
vips	[]String	virtual lsp virtual-ip			
logicalGateway	Bool				
disableGatewayCheck	Bool	Pod			
disableInterConnection	Bool				
enableDHCP	Bool	lsp dhcp			
dhcpV4Options	String	lsp dhcpv4_options	DHCP_Options		
dhcpV6Options	String	lsp dhcpv6_options	DHCP_Options		
enableIPv6RA	Bool	lrp	ipv6_ra_configs		
ipv6RAConfigs	String	lrp	ipv6_ra_configs		
acls	[]Acl	logical-switch	acls		
allowEWTraffic	Bool				
natOutgoingPolicyRules	[]NatOutgoingPolicyRule	NAT			
u2oInterconnectionIP	String	Underlay/Overlay	IP		
u2oInterconnection	Bool	Overlay/Underlay			
enableLb	*Bool	logical-switch	load-balancer		
enableEcmp	Bool	ECMP			

enableMulticastSnoop	Bool
enableExternalLBAddress	Bool
routeTable	String
namespaceSelectors	[]LabelSelector

Acl

direction	String	Acl	from-lport	to-lport
priority	Int	Acl	0	32767
match	String	Acl		
action	String	Acl	allow-related	allow-stateless allow drop reject

NatOutgoingPolicyRule

match	NatOutGoingPolicyMatch
action	String

NatOutGoingPolicyMatch

srcIPs	String	IP
dstIPs	String	IP

SubnetStatus

conditions	[]SubnetCondition		Condition	
v4availableIPs	BigInt	IPv4 IP	IPv6	float64
v4availableIPrange	String	IPv4		
v4usingIPs	BigInt	IPv4 IP		
v4usingIPrange	String	IPv4		
v6availableIPs	BigInt	IPv6 IP		
v6availableIPrange	String	IPv6		
v6usingIPs	BigInt	IPv6 IP		
v6usingIPrange	String	IPv6		
activateGateway	String			
dhcpV4OptionsUUID	String	lsp dhcpv4_options	DHCP_Options	
dhcpV6OptionsUUID	String	lsp dhcpv6_options	DHCP_Options	
u2oInterconnectionIP	String	Overlay/Underlay	IP	
u2oInterconnectionMAC	String	Overlay/Underlay	MAC	
u2oInterconnectionVPC	String	Overlay/Underlay	VPC	
natOutgoingPolicyRules	[]NatOutgoingPolicyRuleStatus	NAT		
mcastQuerierIP	String	IP		
mcastQuerierMAC	String	MAC		

IP

IP

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	IP
metadata	ObjectMeta	Kubernetes	
spec	IPSpec	IP	

IPSpec

podName	String	Pod
namespace	String	Pod Namespace
subnet	String	IP Subnet
attachSubnets	[]String	IP
nodeName	String	Pod
ipAddress	String	IP v4IP v6IP
v4IpAddress	String	IPv4 IP
v6IpAddress	String	IPv6 IP
attachIps	[]String	IP IP
macAddress	String	Pod MAC
attachMacs	[]String	IP MAC
containerID	String	Pod Container ID
podType	String	Pod StatefulSet VirtualMachine

Vpc

VPC

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	Vpc
metadata	ObjectMeta	Kubernetes	
spec	VpcSpec	Vpc	
status	VpcStatus	Vpc	

VpcSpec

defaultSubnet	String	
namespaces	[]String	Vpc
staticRoutes	[]StaticRoute	
policyRoutes	[]PolicyRoute	
vpcPeerings	[]VpcPeering	VPC
enableExternal	Bool	
extraExternalSubnets	[]String	
enableBfd	Bool	BFD ()
bfdPort	BFDPort	BFD

StaticRoute

policy	String		
cidr	String		
nextHopIP	String	IP	
ecmpMode	String	ECMP	
bfdId	String	BFD ID	
routeTable	String		

PolicyRoute

priority	Int		
match	String		
action	String	allow	drop reroute
nextHopIP	String	IP	action reroute

VpcPeering

remoteVpc	String	VPC
localConnectIP	String	IP

BFDPort

enabled	Bool	BFD	
ip	String	BFD	IP
nodeSelector	LabelSelector	BFD LRP	

VpcStatus

conditions	[]VpcCondition	Vpc	Condition
standby	Bool	VPC	
default	Bool	VPC	
defaultLogicalSwitch	String		
router	String		
tcpLoadBalancer	String	TCP	
udpLoadBalancer	String	UDP	
sctpLoadBalancer	String	SCTP	
tcpSessionLoadBalancer	String	TCP	
udpSessionLoadBalancer	String	UDP	
sctpSessionLoadBalancer	String	SCTP	
subnets	[]String	VPC	
vpcPeerings	[]String	VPC	
enableExternal	Bool		
extraExternalSubnets	[]String		
enableBfd	Bool	BFD	

8.11.3 Underlay

Vlan

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	Vlan
metadata	ObjectMeta	Kubernetes	
spec	VlanSpec	Vlan	
status	VlanStatus	Vlan	

VLANSPEC

id	Int	Vlan tag	0~4096
provider	String	Vlan	ProviderNetwork

VLANSTATUS

subnets	[]String	Vlan	
conflict	Bool		
conditions	[]VlanCondition	Vlan	Condition

ProviderNetwork

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	ProviderNetwork
metadata	ObjectMeta	Kubernetes	
spec	ProviderNetworkSpec	ProviderNetwork	
status	ProviderNetworkStatus	ProviderNetwork	

PROVIDERNETWORKSPEC

defaultInterface	String					
customInterfaces	[]CustomInterface					
nodeSelector	LabelSelector	OVS	matchLabels	matchExpressions	nodeSelector	excludeNodes
excludeNodes	[]String					
exchangeLinkName	Bool	OVS				

CustomInterface

interface	String	Underlay
nodes	[]String	

PROVIDERNETWORKSTATUS

ready	Bool		
readyNodes	[]String		
notReadyNodes	[]String		
vlan	[]String	Vlan	
conditions	[]ProviderNetworkCondition	ProviderNetwork	Condition

8.11.4

SecurityGroup

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	SecurityGroup
metadata	ObjectMeta	Kubernetes	
spec	SecurityGroupSpec	SecurityGroup	
status	SecurityGroupStatus	SecurityGroup	

SECURITYGROUPSPEC

ingressRules	[]SecurityGroupRule
egressRules	[]SecurityGroupRule
allowSameGroupTraffic	Bool

SecurityGroupRule

ipVersion	String	IP	ipv4	ipv6
protocol	SgProtocol		all	icmp tcp udp
priority	Int		1-200	
remoteType	SgRemoteType		address	securityGroup
remoteAddress	String			
remoteSecurityGroup	String			
portRangeMin	Int		1	
portRangeMax	Int		65535	
policy	SgPolicy		allow	drop

SECURITYGROUPSTATUS

portGroup	String	
allowSameGroupTraffic	Bool	
ingressMd5	String	MD5
egressMd5	String	MD5
ingressLastSyncSuccess	Bool	
egressLastSyncSuccess	Bool	

8.11.5

IP

Vip

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	Vip
metadata	ObjectMeta	Kubernetes	
spec	VipSpec	Vip	
status	VipStatus	Vip	

VIPSPEC

namespace	String	VIP
subnet	String	VIP
type	String	VIP
v4ip	String	IPv4
v6ip	String	IPv6
macAddress	String	MAC
selector	[]String	
attachSubnets	[]String	

VIPSTATUS

conditions	[]VipCondition	VIP	Condition
type	String	VIP	
v4ip	String	IPv4	
v6ip	String	IPv6	
mac	String	MAC	

SwitchLBRule

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	SwitchLBRule
metadata	ObjectMeta	Kubernetes	
spec	SwitchLBRuleSpec	SwitchLBRule	
status	SwitchLBRuleStatus	SwitchLBRule	

SWITCHLBRULESPEC

vip	String	IP
namespace	String	
selector	[]String	
endpoints	[]String	
sessionAffinity	String	
ports	[]SwitchLBRulePort	

SwitchLBRulePort

name	String
port	Int32
targetPort	Int32
protocol	String

SWITCHLBRULESTATUS

conditions	[]SwitchLBRuleCondition	SwitchLBRule	Condition
ports	String	SwitchLBRule	
service	String	SwitchLBRule	service

8.11.6 QoS IP

QoSPolicy

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	QoSPolicy
metadata	ObjectMeta	Kubernetes	
spec	QoSPolicySpec	QoSPolicy	

QOSPOLICYSPEC

bandwidthLimitRules	QoSPolicyBandwidthLimitRules
shared	Bool
bindingType	QoSPolicyBindingType

IPPool

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	IPPool
metadata	ObjectMeta	Kubernetes	
spec	IPPoolSpec	IPPool	
status	IPPoolStatus	IPPool	

IPPOOLSPEC

subnet	String	
namespaces	[]String	
ips	[]String	IP

IPPOOLSTATUS

v4AvailableIPs	BigInt	IPv4	IP
v4AvailableIPRange	String	IPv4	IP
v4UsingIPs	BigInt	IPv4	IP
v4UsingIPRange	String	IPv4	IP
v6AvailableIPs	BigInt	IPv6	IP
v6AvailableIPRange	String	IPv6	IP
v6UsingIPs	BigInt	IPv6	IP
v6UsingIPRange	String	IPv6	IP
conditions	[]IPPoolCondition	IP	Condition

8.11.7 NAT IP

IptablesEIP

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	IptablesEIP
metadata	ObjectMeta	Kubernetes	
spec	IptablesEIPSpec	IptablesEIP	
status	IptablesEIPStatus	IptablesEIP	

IPTABLESEIPSPEC

v4ip	String	IPv4
v6ip	String	IPv6
macAddress	String	MAC
natGwDp	String	NAT
qosPolicy	String	QoS
externalSubnet	String	

IPTABLESEIPSTATUS

ready	Bool	IptablesEIP		
ip	String	IptablesEIP	IP	IPv4
redo	String	IptablesEIP CRD		
nat	String	IptablesEIP	fip snat dnat	
qosPolicy	String	QoS		
conditions	[]IptablesEIPCondition	IptablesEIP	Condition	

OvnEip

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	OvnEip
metadata	ObjectMeta	Kubernetes	
spec	OvnEipSpec	OvnEip	
status	OvnEipStatus	OvnEip	

OVNEIPSPEC

externalSubnet	String	
v4Ip	String	IPv4
v6Ip	String	IPv6
macAddress	String	MAC
type	String	lrp lsp nat

IptablesFIPRule

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	IptablesFIPRule
metadata	ObjectMeta	Kubernetes	
spec	IptablesFIPRuleSpec	IptablesFIPRule	

IPTABLESFIPRULESPEC

eip	String	IP
internalIP	String	IP

OvnFip

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	OvnFip
metadata	ObjectMeta	Kubernetes	
spec	OvnFipSpec	OvnFip	
status	OvnFipStatus	OvnFip	

OVNFIPSPEC

ovnEip	String	OVN EIP
ipType	String	IP vip ip
ipName	String	IP
vpc	String	VPC
v4Ip	String	IPv4
v6Ip	String	IPv6
type	String	distributed centralized

IptablesDnatRule

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	IptablesDnatRule
metadata	ObjectMeta	Kubernetes	
spec	IptablesDnatRuleSpec	IptablesDnatRule	

IPTABLESDNATRULESPEC

eip	String	IP
externalPort	String	
protocol	String	
internalIP	String	IP
internalPort	String	

OvnDnatRule

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	OvnDnatRule
metadata	ObjectMeta	Kubernetes	
spec	OvnDnatRuleSpec	OvnDnatRule	
status	OvnDnatRuleStatus	OvnDnatRule	

OVNDNATRULESPEC

ovnEip	String	OVN EIP
ipType	String	IP vip ip
ipName	String	IP
internalPort	String	
externalPort	String	
protocol	String	
vpc	String	VPC
v4Ip	String	IPv4
v6Ip	String	IPv6

OVNDNATRULESTATUS

vpc	String	VPC	
v4Eip	String	IPv4 EIP	
v6Eip	String	IPv6 EIP	
externalPort	String		
v4Ip	String	IPv4	
v6Ip	String	IPv6	
internalPort	String		
protocol	String		
ipName	String	IP	
ready	Bool	DNAT	
conditions	[]OvnDnatRuleCondition	OVN DNAT	Condition

IptablesSnatRule

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	IptablesSnatRule
metadata	ObjectMeta	Kubernetes	
spec	IptablesSnatRuleSpec	IptablesSnatRule	

IPTABLESSNATRULESPEC

eip	String	IP
internalCIDR	String	CIDR

OvnSnatRule

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	OvnSnatRule
metadata	ObjectMeta	Kubernetes	
spec	OvnSnatRuleSpec	OvnSnatRule	
status	OvnSnatRuleStatus	OvnSnatRule	

OVNSNATRULESPEC

ovnEip	String	OVN EIP
vpcSubnet	String	VPC
ipName	String	IP
vpc	String	VPC
v4IpCidr	String	IPv4 CIDR
v6IpCidr	String	IPv6 CIDR

8.11.8 VPC

VpcNatGateway

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	VpcNatGateway
metadata	ObjectMeta	Kubernetes	
spec	VpcNatGatewaySpec	VpcNatGateway	
status	VpcNatGatewayStatus	VpcNatGateway	

VPCNATGATEWAYSPEC

vpc	String	VPC	Pod	VPC
subnet	String	VPC	Pod	
externalSubnets	[]String			
lanIp	String	VPC	Pod	IP
selector	[]String	Kubernetes Selector		
tolerations	[]Toleration	Kubernetes		
affinity	Affinity	Kubernetes		
qosPolicy	String	QoS		
bgpSpeaker	VpcBgpSpeaker	BGP speaker		
routes	[]Route			

VpcBgpSpeaker

enabled	Bool	BGP speaker
asn	Uint32	
remoteAsn	Uint32	
neighbors	[]String	BGP
holdTime	Duration	BGP
routerId	String	BGP ID
password	String	BGP
enableGracefulRestart	Bool	
extraArgs	[]String	

Route

cidr	String	CIDR
nextHopIP	String	IP

VPCNATGATEWAYSTATUS

qosPolicy	String	QoS
externalSubnets	[]String	
selector	[]String	Kubernetes Selector
tolerations	[]Toleration	Kubernetes
affinity	Affinity	Kubernetes

VpcEgressGateway

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	VpcEgressGateway
metadata	ObjectMeta	Kubernetes	
spec	VpcEgressGatewaySpec	VpcEgressGateway	
status	VpcEgressGatewayStatus	VpcEgressGateway	

VPCGRESSGATEWAYSPEC

vpc	String	VPC	VPC
bgpConf	String	BgpConf	
evpnConf	String	EvpnConf	EVPN
replicas	Int32	1	
prefix	String	<prefix><veg-name>	
image	String	kube-ovn-controller	
internalSubnet	String	VPC	
externalSubnet	String		
internalIPs	[]String	IP	v4, v6
externalIPs	[]String	IP	v4, v6
internalIPPool	String	IPPool	internalSubnet internalIPs
externalIPPool	String	IPPool	externalSubnet externalIPs
selectors	[]VpcEgressGatewaySelector	/Pod	
trafficPolicy	String	Cluster	Local VPC
bfd	VpcEgressGatewayBFDDConfig	BFD	
policies	[]VpcEgressGatewayPolicy		
nodeSelector	[]VpcEgressGatewayNodeSelector		
tolerations	[]Toleration	Kubernetes	
resources	ResourceRequirements		
bandwidth	BandwidthLimit	/	Mbps bit-rate quantity

VpcEgressGatewaySelector

namespaceSelector	LabelSelector
podSelector	LabelSelector Pod

VpcEgressGatewayBFDDConfig

enabled	Boolean	BFD	VPC	.spec.bfd.enabled=true	false
minRX	Int32	BFD	ms	1000	
minTX	Int32	BFD	ms	1000	
multiplier	Int32	BFD	3		

VpcEgressGatewayPolicy

snat	Boolean	SNAT/MASQUERADE	false
ipBlocks	[]String	CIDR	
subnets	[]String		

VpcEgressGatewayNodeSelector

matchLabels	map[String]String
matchExpressions	[]NodeSelectorRequirement
matchFields	[]NodeSelectorRequirement

BandwidthLimit

ingress	Int64	String	VPC	Mbps quantity	M Mi G Gi bit/s
egress	Int64	String	VPC	Mbps quantity	M Mi G Gi bit/s

VPCEGRESSGATEWAYSTATUS

replicas	Int32	scale
labelSelector	String	scale
ready	Boolean	
phase	String	Pending Processing Completed
internalIPs	[]String	IP
externalIPs	[]String	IP
conditions	[]Condition	
workload	VpcEgressWorkload	

VpcEgressWorkload

apiVersion	String	API
kind	String	Deployment
name	String	
nodes	[]String	

BgpConf

cluster-scoped	VpcEgressGateway.spec.bgpConf		
apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	BgpConf
metadata	ObjectMeta	Kubernetes	
spec	BgpConfSpec	BgpConf	

BGPCONFSPEC

localASN	UInt32	BGP AS Number
peerASN	UInt32	BGP AS Number
routerId	String	BGP Router ID
neighbours	[]String	BGP IP
password	String	TCP MD5
holdTime	Duration	BGP Hold time
keepaliveTime	Duration	BGP Keepalive
connectTime	Duration	BGP Connect
ebgpMultiHop	Boolean	eBGP MultiHop
gracefulRestart	Boolean	BGP Graceful Restart

EvpnConf

cluster-scoped	VpcEgressGateway.spec.evpnConf			EVPN	VNI	Route Target
apiVersion	String	Kubernetes	kubeovn.io/v1			
kind	String	Kubernetes	EvpnConf			
metadata	ObjectMeta	Kubernetes				
spec	EvpnConfSpec	EvpnConf				

EVPNCONFSPEC

vni	UInt32	EVPN VNI
routeTargets	[]String	Route Target

DNSNameResolver

cluster-scoped	enable-dns-name-resolver	ANP/CNP	DNS
apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	DNSNameResolver
metadata	ObjectMeta	Kubernetes	
spec	DNSNameResolverSpec	DNSNameResolver	
status	DNSNameResolverStatus	DNSNameResolver	

DNSNAMERESOLVERSPEC

name	String	DNS	*
------	--------	-----	---

DNSNAMERESOLVERSTATUS

resolvedNames	[]DNSNameResolverResolvedName	DNS	IP
---------------	-------------------------------	-----	----

DNSNameResolverResolvedName

dnsName	String	DNS	
resolvedAddresses	[]DNSNameResolverResolvedAddress	IP	TTL
conditions	[]Condition	Degraded	
resolutionFailures	Int32	5	TTL

DNSNameResolverResolvedAddress

ip	String	IP	
ttlSeconds	Int32	TTL	lastLookupTime + ttlSeconds
lastLookupTime	Time		

VpcDns

apiVersion	String	Kubernetes	kubeovn.io/v1
kind	String	Kubernetes	VpcDns
metadata	ObjectMeta	Kubernetes	
spec	VpcDNSSpec	VpcDns	
status	VpcDNSStatus	VpcDns	

VPCDNSSPEC

replicas	Int32	
vpc	String	VPC
subnet	String	

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8.12 Annotation

Kube-OVN

Pod

Node Annotation

Annotation

Annotation

8.12.1 Pod Annotation

Key	Value	Description				
ovn.kubernetes.io/allocated	true or false	Pod				
ovn.kubernetes.io/routed	true or false	Pod	OVN			
ovn.kubernetes.io/routes	String	Pod				
ovn.kubernetes.io/mac_address	String	Pod	Mac	Pod	Annotation	Mac
ovn.kubernetes.io/ip_address	String	Pod	IP	Pod	Annotation	IP
\<nadName>.\<nadNamespace>.kubernetes.io/ ip_address.\<interfaceName>	String	interface	IP interfaceName Multus ovn.kubernetes.io/ip_address			
\<nadName>.\<nadNamespace>.kubernetes.io/ mac_address.\<interfaceName>	String	interface	MAC interfaceName Multus			
ovn.kubernetes.io/cidr	String	Pod	CIDR			
ovn.kubernetes.io/gateway	String	Pod	Gateway			
ovn.kubernetes.io/ip_pool	IP	Pod	Workload	IP		
ovn.kubernetes.io/bgp	true , cluster , local	BGP	Pod			
ovn.kubernetes.io/snat	String	Pod	SNAT			
ovn.kubernetes.io/eip	String	Pod	EIP			
ovn.kubernetes.io/vip	String	Pod	VIP	Annotation	VIP	
ovn.kubernetes.io/aaps	String	Pod	AAPs (Additional Allowed Addresses Pairs)			
ovn.kubernetes.io/virtualmachine	String	Pod	VirtualMachineInstance			
ovn.kubernetes.io/activation_strategy	String	Pod				
ovn.kubernetes.io/logical_router	String	Pod	VPC			
ovn.kubernetes.io/layer2_forward	true or false	Pod	OVN LSP	unknown		
ovn.kubernetes.io/port_security	true or false	Pod	Port Security			
ovn.kubernetes.io/logical_switch	String	Pod				
ovn.kubernetes.io/vlan_id	Int	Pod	Vlan ID			
ovn.kubernetes.io/ingress_rate	Int	Pod	Mbits/s			
ovn.kubernetes.io/egress_rate	Int	Pod	Mbits/s			
ovn.kubernetes.io/security_groups	String	Pod	Security Group			
ovn.kubernetes.io/default_route	true or false					
ovn.kubernetes.io/provider_network	String	Pod	ProviderNetwork			
ovn.kubernetes.io/mirror	true or false	Pod				
ovn.kubernetes.io/north_gateway	String	Pod				
ovn.kubernetes.io/latency	Int	Pod	ms			
ovn.kubernetes.io/limit	Int	Pod	qdisc			
ovn.kubernetes.io/loss	Float	Pod				
ovn.kubernetes.io/jitter	Int	Pod	ms			

Key	Value	Description
ovn.kubernetes.io/generate-hash	true or false	Pod
ovn.kubernetes.io/attachmentprovider	String	Pod
ovn.kubernetes.io/ingress_burst	Int	Pod burst
ovn.kubernetes.io/vpc_nat_gw	String	VPC NAT Gateway Pod
ovn.kubernetes.io/vpc_nat_gw_init	true or false	VPC NAT Gateway Pod
ovn.kubernetes.io/vpc_nat_gw_container_restarted	true or false	VPC NAT Gateway
ovn.kubernetes.io/vpc_nat_gw_activated	true or false	VPC NAT Gateway
\<provider>.kubernetes.io/ip_pool	IP	provider IP
\<provider>.kubernetes.io/dhcp-v4-options	String	provider DHCPv4
\<provider>.kubernetes.io/port_security	true or false	provider

8.12.2 Node Annotation

Key	Value	Description
ovn.kubernetes.io/allocated	true or false	ovn0 join
ovn.kubernetes.io/mac_address	String	Node ovn0 Mac
ovn.kubernetes.io/ip_address	String	Node ovn0 IP
ovn.kubernetes.io/cidr	String	Node ovn0 join CIDR
ovn.kubernetes.io/gateway	String	Node ovn0 join Gateway
ovn.kubernetes.io/chassis	String	Node OVN-SouthBoundDB Chassis ID
ovn.kubernetes.io/port_name	String	Node ovn0 OVN-NorthboundDB LSP
ovn.kubernetes.io/logical_switch	String	Node ovn0
ovn.kubernetes.io/tunnel_interface	String	
ovn.kubernetes.io/node_networks	String	Pod attachment networks NetworkAttachmentDefinition

8.12.3 Namespace Annotation

Key	Value	Description
ovn.kubernetes.io/cidr	CIDR	Namespace CIDR
ovn.kubernetes.io/exclude_ips	excludeIPs	Namespace excludeIPs

8.12.4 Subnet Annotation

Key	Value	Description
ovn.kubernetes.io/bgp	true, cluster, local	BGP
ovn.kubernetes.io/vpc	String	VPC VPC

8.12.5 Service Annotation

Key	Value	Description
ovn.kubernetes.io/bgp	true, cluster, local	BGP Service ClusterIP Service Pod/Subnet
ovn.kubernetes.io/switch_lb_vip	String	Service Kube-OVN VIP
ovn.kubernetes.io/vpc	String	Service VPC
ovn.kubernetes.io/service_external_ip_from_subnet	true or false	Service IP
ovn.kubernetes.io/service_health_check	true or false	Service
ovn.kubernetes.io/lb_svc_img	String	

8.12.6 Networkpolicy Annotation

Key	Value	Description
ovn.kubernetes.io/enable_log	true or false	NetworkPolicy
ovn.kubernetes.io/log_acl_actions	"allow,drop,pass"	Action ACL
ovn.kubernetes.io/acl_log_meter_rate	Int	NetworkPolicy /
ovn.kubernetes.io/network_policy_enforcement	standard or lax	NetworkPolicy
ovn.kubernetes.io/network_policy_for	String	NetworkPolicy provider <nadName>.<nadNamespace>.ovn

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8.12.7

8.13

8.13.1

Bad	Good

Bad	Good
Kube-OVN1.10Kube-OVN	Kube-OVN1.10Kube-OVN



Bad	Good
wget 127.0.0.1	wget 127.0.0.1

8.13.2

yamlyaml

Bad	Good
apiVersion: kubeovn.io/v1 kind: Subnet metadata: name: attach-subnet	apiVersion: kubeovn.io/v1 kind: Subnet metadata: name: attach-subnet

bash

Bad	Good
wget 127.0.0.1	wget 127.0.0.1

#

Bad	Good
<pre>oilbeater@macdeMac-3 ~ ping 114.114.114.114 -c 3 PING 114.114.114.114 (114.114.114.114): 56 data bytes 64 bytes from 114.114.114.114: icmp_seq=0 ttl=83 time=10.429 ms 64 bytes from 114.114.114.114: icmp_seq=1 ttl=79 time=11.360 ms 64 bytes from 114.114.114.114: icmp_seq=2 ttl=76 time=10.794 ms --- 114.114.114.114 ping statistics --- 3 packets transmitted, 3 packets received, 0.0% packet loss round-trip min/avg/max/stddev = 10.429/10.861/11.360/0.383 ms</pre>	<pre># ping 114.114.114.114 -c 3 PING 114.114.114.114 (114.114.114.114): 56 data bytes 64 bytes from 114.114.114.114: icmp_seq=0 ttl=83 time=10.429 ms 64 bytes from 114.114.114.114: icmp_seq=1 ttl=79 time=11.360 ms 64 bytes from 114.114.114.114: icmp_seq=2 ttl=76 time=10.794 ms --- 114.114.114.114 ping statistics --- 3 packets transmitted, 3 packets received, 0.0% packet loss round-trip min/avg/max/stddev = 10.429/10.861/11.360/0.383 ms</pre>

#

Bad	Good
<pre># mv /etc/origin/ovn/ovnnb_db.db /tmp # mv /etc/origin/ovn/ovnsb_db.db /tmp</pre>	<pre>mv /etc/origin/ovn/ovnnb_db.db /tmp mv /etc/origin/ovn/ovnsb_db.db /tmp</pre>

8.13.3

md

Bad	Good
<pre>[](http://kubeovn.github.io/prepare)</pre>	<pre>[](./prepare.md)</pre>

Bad	Good
<pre>[Kubernetes](http://kubernetes.io)</pre>	<pre>[Kubernetes](http://kubernetes.io){: target="_blank" }</pre>

8.13.4

Bad	Good
<pre>'''bash wget 127.0.0.1 '''</pre>	<pre>'''bash wget 127.0.0.1 '''</pre>

Bad	Good
<pre>'''bash wget 127.0.0.1 '''</pre>	<pre>'''bash wget 127.0.0.1 '''</pre>

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8.13.5

9.

Kube-OVN

9.1

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- LinkedIn [Kube-OVN](#)
- Issue / [GitHub Issues](#) [GitHub Discussions](#)

9.2

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9.3

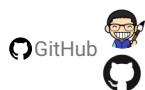
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- [USERS.md](#)



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9.4
